Perishable Capacity

Airline network planning and revenue management from first principles



Prepared for Enzo Simier

2026

Perishable Capacity: Airline Network Planning and Revenue Management from First Principles

Network Planning edition. A shorter companion to the full reference edition (24 chapters), which keeps every derivation.

Personal study edition prepared for Enzo Simier, Network Planning Analyst, Montréal. October 2026.

Written from the Network Planning Reading Library assembled on 27 October 2026: the MIT OpenCourseWare courses 16.75J **Airline Management** (Peter P. Belobaba, Spring 2006) and 1.206J **Airline Schedule Planning** (Cynthia Barnhart, Spring 2003); chapters of Kenneth Train's **Discrete Choice Methods with Simulation**; the network planning preparation handbook; Air Canada's second-quarter 2026 MD&A; the Competition Bureau's 2025 airline market study; the IATA Worldwide Airport Slot Guidelines; EUROCONTROL's think paper on punctuality; the Airbus A321 airport planning manual; and the current public sources cited in the text.

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Laurentian, the airline of the worked examples, is fictional. Its network uses real airports and aircraft types, but every demand, fare and cost figure attached to it is illustrative and invented for teaching. Nothing in this book is an Air Canada forecast, cost, target or policy. Public facts are dated October 2026 unless stated otherwise.

Set in TeX Gyre Pagella and Inter with XeLaTeX. Figures drawn with Matplotlib and TikZ. The land on the cover is sampled from NOAA's GLOBE one-kilometre land mask.

Contents

How to Use This Book **ix**
What you need to know already . . . ix
How the book is organized . . . ix
Reading paths . . . x
Conventions . . . x
The running case . . . x
About the sources . . . xi

I The Airline as a System **1**

1 The Planning Problem **2**
1.1 The planning decisions and their horizons . . . 3
1.2 Decomposition and feedback . . . 5
1.3 Why airline planning is hard . . . 7
1.4 The network planning analyst . . . 8
1.5 Incremental network contribution . . . 11
1.6 Laurentian, the running case . . . 14
1.7 Air Canada in October 2026 . . . 16
1.8 Summary . . . 18
1.9 Further reading . . . 18

2 Measuring an Airline **19**
2.1 Four objects: movements, legs, itineraries and tickets 20
2.2 Passengers per day each way . . . 21
2.3 Capacity, traffic and load factor . . . 22
2.4 Yield, unit revenue and unit cost . . . 24
2.5 Stage length . . . 27
2.6 Break-even load factor . . . 29
2.7 Reading Air Canada's second-quarter 2026 MD&A . 30
2.8 Summary . . . 34
2.9 Further reading . . . 35

3 Costs and Productivity **36**
3.1 Cost categories and their drivers . . . 37
3.2 Why unit cost falls with stage length . . . 39
3.3 Aircraft size: trip cost versus seat cost . . . 41
3.4 Utilization and productivity . . . 43
3.5 Which costs count: decision horizons . . . 45
3.6 Laurentian's cost model at work . . . 48
3.7 Air Canada's cost structure in 2026 . . . 50

3.8 Summary 52
3.9 Further reading 53

II Demand **54**

4 Air Travel Demand **55**
4.1 Markets, itineraries and legs 56
4.2 What drives demand 57
4.3 Price elasticity 58
4.4 Time, frequency and schedule delay 61
4.5 Seasonality and the weekly cycle 64
4.6 Market size, market share and the S-curve 65
4.7 Quality of service and the traffic a new nonstop creates 68
4.8 Summary 72
4.9 Further reading 73

5 How Passengers Choose **74**
5.1 Random utility and the logit share 75
5.2 What the coefficients mean to a planner 77
5.3 Close substitutes: IIA and nested logit 80
5.4 Travellers differ: business and leisure 83
5.5 Decision windows and the PODS passenger 84
5.6 Choice models inside planning tools 88
5.7 Why naive fare elasticities are wrong 90
5.8 Summary 95
5.9 Further reading 95

6 Forecasting Demand **96**
6.1 What to forecast, and at which level 97
6.2 Benchmarks and the main method families 98
6.3 Measuring accuracy honestly 100
6.4 Booking curves and pickup 103
6.5 Forecasting a new route 105
6.6 Uncertainty, scenarios and the decision 107
6.7 Summary 110
6.8 Further reading 110

7 Spill and Recapture **111**
7.1 Demand, load and spill 112
7.2 The expected spill formula 114
7.3 Load factor, demand factor and spill rate 116
7.4 Unconstraining: recovering demand from censored loads 119
7.5 What is a seat worth? 120

7.6 Recapture 122
7.7 Spill when revenue management controls the flight 123
7.8 Summary 125
7.9 Further reading 125

III Designing the Network **127**

8 Network Structure and Connectivity **128**
8.1 Point to point and hub and spoke 129
8.2 The combinatorics of connectivity 130
8.3 Economies of density and scope 132
8.4 Connecting traffic, banks and the sixth freedom . . 135
8.5 Alliances, codeshares, interline and joint ventures . 139
8.6 Freedoms of the air, bilateral agreements and Blue Sky 141
8.7 Market structure and hub premiums in Canada . . . 143
8.8 Summary 145
8.9 Further reading 146

9 Fleet Planning **147**
9.1 The fleet planning problem 148
9.2 Top-down and bottom-up models 149
9.3 Evaluating an aircraft 152
9.4 Payload-range diagrams 154
9.5 The A321XLR on Montreal–Seville 156
9.6 Owning, leasing and commonality 158
9.7 A Laurentian fleet decision 160
9.8 Summary 163
9.9 Further reading 164

10 Route Evaluation **165**
10.1 Three views of route economics 166
10.2 Feed, network contribution and displacement 169
10.3 Avoidable costs and the opportunity cost of aircraft time 171
10.4 The handbook's exercises 174
10.5 Laurentian YUL–SVQ: a full seasonal evaluation . . 177
10.6 Scenarios, sensitivity and the threshold that reverses the decision 181
10.7 The one-page decision memo 184
10.8 Summary 185
10.9 Further reading 185

11 Frequency and Schedule Design **186**
11.1 Frequency, schedule delay and the S-curve 187

11.2 Frequency versus gauge 190
11.3 Building the timetable 192
11.4 Hub banks and connections 194
11.5 Rotations, clocks and the fleet a timetable needs . . 197
11.6 Seasons, slots and airport restrictions 200
11.7 Schedule design tools and the retiming decision . . 203
11.8 Summary 206
11.9 Further reading 206

12 From Schedule to Aircraft and Crews 207
12.1 From timetable to tails and crews 208
12.2 The time-space network 209
12.3 Fleet assignment 210
12.4 A Laurentian bank solved 214
12.5 Maintenance routing and lines of flying 217
12.6 Crew pairings, pay and rostering 219
12.7 What breaks when a planner moves a flight 224
12.8 Summary 226
12.9 Further reading 227

13 Robustness and Operations Recovery 228
13.1 Where delays come from 229
13.2 Slack and propagated delay 230
13.3 Buffers versus utilization 232
13.4 Robust routing and retiming 234
13.5 Operations control and recovery 236
13.6 Measuring what passengers experience 239
13.7 Disruption in the business case 240
13.8 Summary 244
13.9 Further reading 245

IV Revenue 246

14 Revenue Management for Planners 247
14.1 One cabin, many prices 248
14.2 From restricted to branded fares 250
14.3 Booking limits and Littlewood's rule 253
14.4 What seat control does to a flight 256
14.5 Overbooking 257
14.6 Network control and bid prices 262
14.7 Willingness to pay, sell-up and dynamic pricing . . 266
14.8 Summary 270
14.9 Further reading 271

15 Where Planning Meets Revenue Management 272
15.1 Two functions, two horizons . . . 273
15.2 Bid prices: the value of a seat, departure by departure 274
15.3 Valuing an upgauge and a new leg . . . 276
15.4 RM-aware spill . . . 281
15.5 Consistent forecasts . . . 284
15.6 How revenue management reacts to a schedule change . . . 285
15.7 Questions for the revenue management team . . . 286
15.8 Summary . . . 288
15.9 Further reading . . . 289

V Practice 290

16 The Industry in 2026 291
16.1 Network planning practice and tools . . . 292
16.2 Supply constraints: engines, deliveries and the long-range narrowbody . . . 293
16.3 Carbon rules that enter route economics . . . 296
16.4 Pricing, offers and distribution . . . 300
16.5 The Canadian market in 2026 . . . 302
16.6 Air Canada in the autumn of 2026 . . . 304
16.7 What changed since 2006, and what did not . . . 306
16.8 Summary . . . 307
16.9 Further reading . . . 308

17 The Analyst's Toolkit 309
17.1 Data: objects, sources and their traps . . . 310
17.2 A reproducible workflow . . . 314
17.3 Clocks, seasons and the schedule card . . . 317
17.4 Sensitivity matrices and decision thresholds . . . 320
17.5 The one-page decision memo . . . 323
17.6 Audit checks . . . 324
17.7 The first weeks . . . 326
17.8 Summary . . . 327
17.9 Further reading . . . 327

A Mathematical Notes 328
A.1 Why unit cost falls with stage length . . . 328
A.2 The S-curve of market share . . . 330
A.3 Logit choice probabilities, IIA and the logsum . . . 332
A.4 The normal loss function and the spill formula . . . 335
A.5 The fleet assignment model in symbols . . . 338
A.6 Delay propagation and slack . . . 340

A.7 Littlewood's rule and EMSR-b 342
A.8 The cost-based overbooking limit 344
A.9 Bid prices as shadow prices 346

B Glossary **350**

C Sources and Reading List **360**
C.1 The reading library 360
C.2 Reading the older material 362
C.3 Where to go next 363

References **364**

Index **380**

How to Use This Book

This is the Network Planning edition of Perishable Capacity. It explains how an airline network is planned: how an airline decides where to fly, how often, with which aircraft and at what price, and how those decisions show up in revenue, cost and operations. Every chapter leads with the mechanism, shows it in a figure and works it through in numbers on one running case, so that you can reproduce the arithmetic with a spreadsheet. Revenue management, crew planning and operations research appear from the planner's seat: what those teams do, what they need from the network plan, and what they send back.

The full reference edition (24 chapters, 778 pages) covers the same ground with complete derivations, estimation methods and optimization models. This edition keeps the formulas a planner actually computes, states the few results it needs without proof, and collects their derivations in Appendix A. When a chapter points to the full edition, it is for the mathematics, never for something you need to follow the argument.

What you need to know already

Nothing about airlines is assumed. The economics is at the level of an applied economist: demand and elasticities, fixed and variable cost, opportunity cost. The probability is the normal distribution and an expected value. Appendix B defines every acronym and term of art, and Appendix C lists the documents the book was built from.

How the book is organized

Part I, The airline as a system, sets out the sequence of planning decisions, from fleet to operations, the measures airlines use to describe themselves (ASK, RPK, yield, unit revenue and unit cost) and the structure of airline costs.

Part II, Demand, explains what drives air travel, how travellers choose among itineraries, how demand is forecast, and what happens when demand exceeds the seats on offer (spill and recapture).

Part III, Designing the network, is the core of the planner's job: network structure and connectivity, fleet planning, route evaluation, frequency and schedule design, what happens to the schedule once

it reaches the fleet and crew planners, and how the plan holds up when it meets reality.

Part IV, Revenue, explains revenue management as the planner needs it, from fare structures to bid prices, and then the interface between planning and revenue management.

Part V, Practice, describes the industry as it stands in 2026 and gives a working toolkit: data sources, workflows and the checks that keep an analysis honest.

Reading paths

If you start the job next week, read Chapters 1, 2, 7, 8, 10, 11, 15 and 17 first. Add Chapters 4 to 6 for the demand side of a route case, Chapters 9, 12 and 13 if your desk is closer to feasibility and operations, and Chapter 14 before your first meeting with the revenue management team.

Conventions

Every chapter opens with a short list of what it covers and the documents of the reading library it draws on. Key terms are set in **bold** where they are defined. Six kinds of boxes structure the text:

- **Definition** boxes state precise meanings.
- **Result** boxes state a formula or a rule of thumb that planners use; derivations, where they matter, are in Appendix A.
- **Worked example** boxes solve a problem with numbers, usually on the running case.
- **In practice** boxes describe how the idea is used by airlines today.
- **Pitfall** boxes flag common errors.
- **Key idea** boxes isolate the one sentence to remember.

Definitions, results, examples, figures, tables and equations are numbered by chapter. Money is in Canadian dollars unless stated. Distances are in statute miles (mi) or kilometres (km) and, for aircraft performance, nautical miles (nm). Symbols are defined where they are first used.

The running case

Most worked examples use **Laurentian**, a fictional airline with its hub at Montreal-Trudeau (YUL). Laurentian's geography and aircraft are real; its demand, fares and costs are invented, chosen to be

plausible and to make each model's behaviour visible. Do not read any Laurentian number as a fact about Air Canada or any other airline.

Table 1 Laurentian's fleet. Trip cost is block hours times the hourly cost, plus a departure cost (CAD 2,000 to 7,000 by type and region) and a cost per passenger (CAD 20 domestic, 30 transborder, 50 transatlantic). All figures are illustrative.

Aircraft	Seats	Business	Premium economy	Economy	CAD per block hour
A220-300	137	12		125	8,000
737-8	169	16		153	9,000
A321XLR	182	14		168	11,000
787-9	298	30	21	247	15,500

Laurentian's most important route in this book is YUL–SVQ, Montreal to Seville, flown with the A321XLR, a narrowbody with transatlantic range. Its schedule pattern mirrors a real service that was announced publicly in October 2026 for launch in September 2027: an evening departure from Montreal arriving the next morning in Seville, three times a week at launch and four times a week from October. That public announcement is the only real element; everything Laurentian earns and spends on the route is invented. For network examples Laurentian also runs a small bank at YUL that connects two feeder flights, from Quebec City (YQB) and Boston (BOS), to two transatlantic flights, to Paris (CDG) and Seville (SVQ). Its four legs and sixteen fare products are small enough to solve by hand and rich enough to show every network effect in Chapters 8, 10 and 14.

About the sources

The reading library mixes teaching material from 2003 and 2006 with documents dated 2025 and 2026. The older material teaches durable methods: spill models, fleet assignment, bid prices and choice models have not changed in substance. Its fares, market shares, software descriptions and parameter values are history, and the text says so wherever it uses them. Current facts are dated and sourced. When the book makes a claim about practice in 2026, it cites a public source; when a claim could not be verified, it is labelled as such or left out.

PART I

The Airline as a System

An airline sells journeys but produces flights, and it commits to aircraft years before it knows who will fly. Part I sets out the sequence of decisions that turns a fleet into a flown timetable, the measures airlines use to describe capacity, traffic, revenue and cost, and the structure of the costs that every network decision has to cover.

CHAPTER

1

The Planning Problem

IN THIS CHAPTER

- The planning decisions, from fleet orders a decade ahead to recovery on the day, and when each is taken.
- Why the decisions depend on each other, why airlines still take them in sequence, and the feedback loops between the stages.
- Why the problem is hard: perishable seats, network effects, uncertainty, lumpy aircraft, competition and regulation.
- The network planning analyst's decisions and deliverables, and the handbook's core decision with its five questions.
- Incremental network contribution, and why a full flight can be the wrong use of an aircraft.
- Laurentian, the running case, its Montreal hub, and Air Canada's public context in October 2026.

DRAWS ON Belobaba, Introduction to the airline planning process; Airline schedule development; Route planning and evaluation (MIT 16.75J, 2006); Barnhart, Airline schedule planning (MIT 1.206J, 2003); handbook; Air Canada Q2 2026 MD&A

Suppose that in your first week the manager of your desk asks: should Laurentian, the fictional Montreal airline of this book, add a fourth weekly departure to Seville next October, or should that aircraft time go somewhere else? The question names one route, but the answer runs through the whole airline. The extra departure needs an aircraft free on the right days, a crew that can legally fly it, slots and gates at both ends, and customers who would not otherwise have flown on another Laurentian flight. It changes which connections the Montreal hub offers, what revenue management charges on the other three departures, and what the aircraft would have earned elsewhere. A planner who answers "the route is 85 percent full, so yes" has not answered the question.

This chapter maps the territory that question lives in, using the framework of the preparation handbook written for this job (Network Planning Handbook 2026). Its central object is the incremental

network contribution of a change, which compares two whole networks rather than two route profit statements.

1.1 The planning decisions and their horizons

Belobaba's introduction to airline planning lists five decisions (Belobaba 2006c). **Fleet planning** decides which aircraft to acquire or retire, how many and when. **Route evaluation** decides which network to operate and which city pairs to serve. **Schedule development** decides how often, when and with which aircraft each route is flown. **Pricing** decides which fare products, with which restrictions, to offer in each origin-destination market. **Revenue management** (RM) decides how many bookings to accept for each fare product on each flight. Crew scheduling, airport resources and operations control follow (Barnhart 2003).

Schedule development is itself four linked problems (Belobaba 2006f): **frequency planning** (departures per route), **timetable development** (departure and arrival times, including connections at hubs), **fleet assignment** (the aircraft type of each flight) and **aircraft rotation planning** (chaining flights into sequences each aircraft can fly, with as many aircraft leaving each station as arrive). A **flight leg** is one non-stop operation from A to B; an **itinerary** is the legs a passenger uses for one journey (Chapter 2).

What separates the decisions is time. Early decisions are large, slow to reverse and made with little information; late ones are detailed, quick and boxed in by everything decided before (Figure 1.1).

Fleet planning works years ahead because aircraft are expensive, long-lived and delivered slowly (Belobaba 2006c). Air Canada's commitments show the lead times: 787-10s ordered in September 2023 begin to arrive in 2026, eight A350-1000s announced in February 2026 are scheduled from 2030, and 30 A321XLRs arrive between 2026 and 2029 (Air Canada 2026d).

Route planning takes the fleet as given and selects markets. The economics turn on forecast demand, the share the airline can win against competitors, the effects on the rest of the network and the opportunity cost of the aircraft; the practical side on range, rotations, traffic rights and slots (Belobaba 2006i). Air Canada announced Montreal to Seville on 14 October 2026 for a first flight on 8 September 2027 (Air Canada 2026f), and the analysis came well before the announcement.

Schedule development begins a year or more ahead with frequency plans; timetables and rotations are fixed two to six months ahead and revised until departure (Belobaba 2006f). The industry

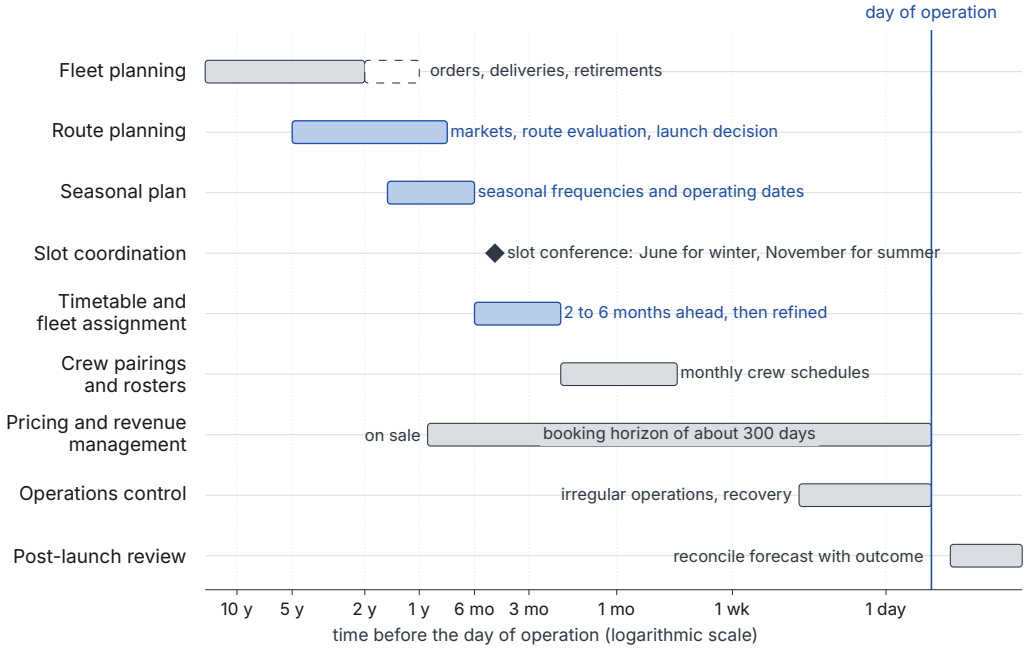


Figure 1.1 Planning decisions and their typical horizons, on a logarithmic scale of time before the day of operation. Bars are indicative: practice varies by airline, and every decision is revisited as its date approaches. Cobalt marks the decisions at the centre of a network planner's work. Sources: Belobaba's 2006 lectures on schedule development and revenue management; the IATA Worldwide Airport Slot Guidelines for slot conferences.

runs on a northern summer season from the last Sunday in March and a winter season from the last Sunday in October; slots at coordinated airports are requested at the IATA Slot Conference, in June for the following winter and November for the following summer (IATA, ACI, and WWACG 2025). **Crew scheduling** then builds pairings (legal sequences of legs from a crew base and back) and monthly rosters (Barnhart 2003).

Pricing and revenue management take over once the schedule is on sale, revising booking limits over about 300 days of bookings (Belobaba 2006l). **Operations control** owns the day itself, recovering when weather, air traffic control or maintenance break the plan (Belobaba 2006j). A **post-launch review** compares results with the approved proposal, so that the next forecast is better (Network Planning Handbook 2026).

Two properties of this timeline matter throughout. Each stage inherits the constraints of earlier ones: a timetable cannot use aircraft the fleet plan did not buy. And the cost of changing a decision rises as the day approaches, while information improves; a published

schedule creates commitments to customers, so cancelling a flight already on sale faces a different threshold from dropping a route not yet sold (Network Planning Handbook 2026). A planner asks what can still be changed, and at what cost.

1.2 Decomposition and feedback

Why not choose fleet, routes, timetable, assignments, crews and prices together? Because the joint problem is far too large: a day of 400 flights, four aircraft types and departures free to move up to 30 minutes either way in five-minute steps already gives about 10^{686} combinations of timetable and fleet assignment. Most are infeasible, which is the real difficulty: aircraft flows, connection times, maintenance and crew rules couple every flight to the others (Belobaba 2006f).

Airlines therefore **decompose** the problem into four stages: schedule design selects the legs and their times; fleet assignment chooses a type for each leg to maximize contribution; aircraft routing builds tail sequences that respect maintenance; crew scheduling covers the legs with legal pairings and rosters (Barnhart 2003). Each stage takes the previous output as fixed, and later stages send problems back upstream through feedback loops (Figure 1.2).

Most of a planner's iteration happens in these loops. A fleet assignment that spills too many passengers, or needs more aircraft of a type than exist, sends the timetable back (loop 1). Maintenance routing can force a swap of types (loop 2). Crew duty limits and overnight costs can make a timetable expensive or illegal for crews (loop 3). Delays propagate through tight turns, which argues for buffers that cost utilization (loop 4, Chapter 13). A schedule needing more aircraft than exist, or leaving some idle, revises the fleet plan (loop 5). RM's bid prices show which legs are short of seats (loop 6, Chapter 15).

Sequential planning yields a feasible plan, but not always the best: a plan that no single stage can improve alone may still be beaten by changing two stages together. The shortfall is the **value of integration**. With invented numbers: schedule design prefers 07:00 because it earns CAD 2,000 more than 08:00; fleet assignment then finds that the only aircraft free at 07:00 is a larger type costing CAD 3,500 more per trip, while a smaller one was free at 08:00. The plan loses CAD 1,500 and neither stage sees it. Integrated models address such traps (Lohatepanont and Barnhart 2004; Barnhart, Kniker, and Lohatepanont 2002; Cohn and Barnhart 2003), but inconsistent data, missing models of competition and coordination between depart-

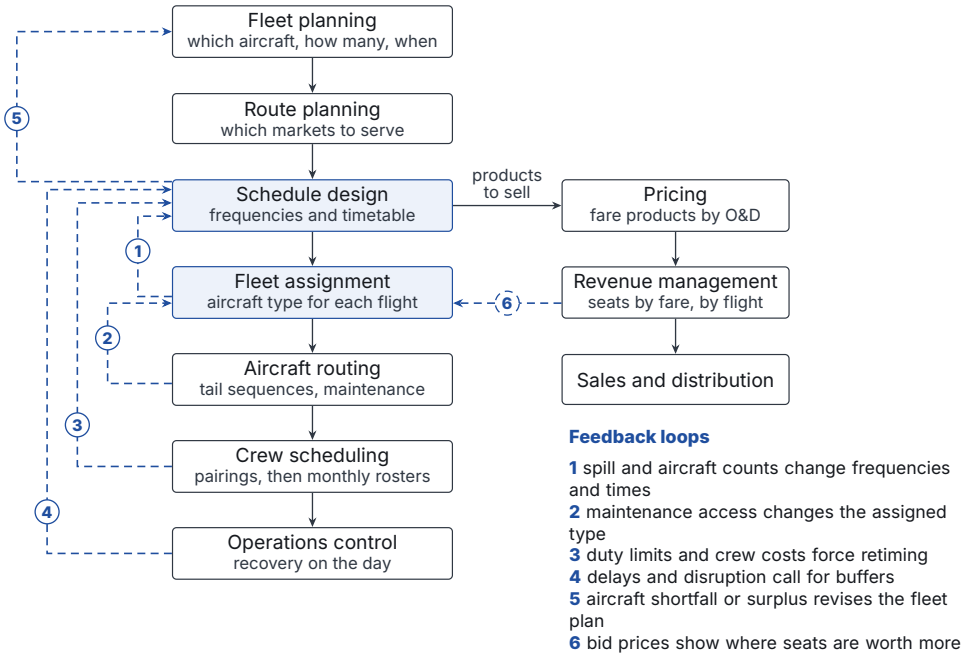


Figure 1.2 The sequential decomposition of airline planning (solid arrows) and the feedback loops that tie it together (dashed, numbered). Schedule design and fleet assignment, shaded, are the core of network planning. Adapted from the planning charts in Belobaba's 2006 introduction to the planning process and Barnhart's 2003 course on airline schedule planning.

ments still keep airlines planning in stages and iterating (Belobaba 2006c).

WORKED EXAMPLE 1.1 One route touches every stage

Laurentian's Montreal to Seville pattern mirrors the timetable Air Canada has published for its own service, as worked through in the handbook (Network Planning Handbook 2026): depart YUL 20:30, arrive Seville (SVQ) 09:30 next morning, depart 12:45, arrive Montreal 14:40, three times a week at launch and four from October, with a 182-seat A321XLR.

Timetable. With Montreal at UTC minus 4 and Seville at UTC plus 2, block time is 7 hours 0 minutes eastbound and 7 hours 55 minutes westbound, against the winds; the aircraft is back 18 hours 10 minutes after leaving. North America and Europe change clocks on different dates, and in the weeks with a five-

hour offset the same local times would leave the westbound flight an hour short.

Fleet. Only the A321XLR and the 787-9 qualify (the A220-300 and 737-8 would be at the edge of their range westbound), and the 787-9's 298 seats far exceed the illustrative mean demand of 150 to 175 per departure (Chapter 7).

Routing. Three weekly round trips occupy the aircraft for 54.5 hours, 32.4 percent of the week; four occupy 72.7 hours, 43.3 percent. The rest of the week must fit around this pattern and still reach maintenance.

Crew. The aircraft turns in Seville in 3 hours 15 minutes; the crew cannot, since the round trip exceeds 18 hours, far beyond a workable duty. Crews lay over, adding hotel nights the timetable creates, and crew planners check the pattern against the fatigue rules (Transport Canada 2024).

RM. Once on sale, every schedule change touches customers already booked.

One line in a route plan (four weekly instead of three) is at once a fleet, routing, crew, airport and inventory decision.

1.3 Why airline planning is hard

Airline planning combines six features that are each manageable alone and difficult together.

> **DEFINITION 1.1 Perishable capacity**
>
> Capacity is **perishable** when it is produced at a fixed time whether or not it is sold, cannot be stored, and is worth nothing once that time has passed. A seat is flown, empty or full, and cannot be sold after the doors close.

Perishable seats. Capacity is set before demand is known, so some flights leave seats empty while others turn customers away (Chapter 7). Once a flight is scheduled, almost all its cost is fixed: on Laurentian's illustrative model, a YUL–SVQ departure with 150 passengers costs CAD 88,000, of which only CAD 7,500 (8.5 percent) varies with passengers; average cost is CAD 587 per passenger, marginal cost CAD 50. Any customer paying more than the

marginal cost plus the seat's value to someone else is worth carrying, the root of differentiated fares and RM (Chapter 14).

Network effects. Demand is defined by origin-destination market, supply by flight leg, and each leg serves many markets (Belobaba 2006d). In Belobaba's 2006 example, one Boston to Houston flight carried bookings in over twenty markets, the local market only about 30 percent of them (historical). The value of a flight cannot be read from that flight alone.

Uncertainty. Demand per departure varies from week to week with a coefficient of variation of typically 0.2 to 0.4 (Belobaba 2006b). Costs move too: in the second quarter of 2026 Air Canada paid 50.8 percent more per litre of fuel than a year earlier (Air Canada 2026d), after suspending its full-year guidance on 30 April 2026 because fuel costs were too uncertain to forecast (The Canadian Press 2026).

Indivisibility. Aircraft come with 137, 169, 182 or 298 seats, frequencies are integers, and an extra departure needs a whole aircraft for a whole rotation, so the best plan can jump between configurations (Chapters 9 and 12).

Competition. Market share depends on frequency, timing, fares and product relative to rivals, and rises in an S-shape with frequency share, which rewards matching rivals' frequencies (Belobaba 2006d). The Competition Bureau found that Air Canada and WestJet carried 56 to 78 percent of domestic passengers at major Canadian airports in 2023 (Competition Bureau Canada 2025) (Chapters 4 and 8).

Regulation and access. Traffic rights, slots (IATA, ACI, and WWACG 2025), crew fatigue rules (Transport Canada 2024) and government approval of joint ventures (Air Canada 2026g) can each make an attractive plan infeasible. Airport restrictions need care: the handbook warns against summarizing Montreal's as "YUL closes at night", since the airport describes a 24-hour operation with category-specific restrictions (Network Planning Handbook 2026).

A seat is many products at once: it serves many markets, it expires at departure, and it comes in a block with every other seat on the aircraft.

1.4 The network planning analyst

The handbook describes the analyst's core as demand estimation, substitution, constrained optimization, opportunity cost and coun-

terfactual analysis, applied to sellable itineraries, rotations and seasonal schedules (Network Planning Handbook 2026). A good planner shows that a change makes economic sense and can be operated (Table 1.1).

Table 1.1 Decision levels of network planning (adapted from the handbook, 2026). The levels overlap in time; who owns which decision must be learned inside each airline.

Decision level	Typical question	Evidence to produce
Multi-year network and fleet	Which regions and aircraft capabilities deserve scarce capital?	Scenarios, fleet availability, demand outlook, airport and regulatory feasibility
Seasonal plan	Which routes, months, days and frequencies should operate?	Demand and contribution by period, competitors' timetables, aircraft-days required
Schedule refinement	What if we retime, upgauge or move a frequency?	Changed itineraries, revenue displacement, airport peaks, feasible rotations
In-season adjustment	Should capacity be added, cut or redeployed?	Booking outlook from RM, avoidable costs, customer commitments, feasible alternatives
Post-launch review	Why did results differ from the proposal?	A reconciled bridge of volume, fare, cost and operational variances

DEFINITION 1.2 The core decision

Among the schedules and aircraft allocations that can actually be operated, choose the one that adds the most value to the airline's whole network, measured against the best alternative that could realistically be chosen instead (Network Planning Handbook 2026).

Feasible means the plan respects aircraft, crew, airport, maintenance and regulatory constraints. **Incremental** means value is the difference between the network with and without the change. **Alternative** means the comparison is with the best other use of the same scarce resources, not with doing nothing. The handbook turns the definition into five questions (Network Planning Handbook 2026):

1. **The counterfactual.** What happens without the change? Specify a dated baseline schedule, competitor assumptions and a cost scenario.
2. **The journeys.** Which passenger journeys change? Count local, connecting, partner, diverted, stimulated and lost demand.
3. **The increment.** What is new to the airline, as opposed to traffic that merely moves between its own services?
4. **The binding resources.** Which resources become scarce? Aircraft, feeder seats, crews, slots, gates, maintenance access and cash can all bind.
5. **The alternative.** What is the best feasible alternative over the same dates and resources, with downside cases and a clear decision threshold?

A sixth habit: say what would make the recommendation wrong, for example schedule A unless revised feeder availability would cut its contribution below schedule B's. The deliverables follow: a schedule card for each proposal (both directions, local and UTC times, block times, turns, aircraft and seats by cabin, connections, the implied rotation), a contribution bridge from gross revenue to incremental network contribution, a sensitivity analysis naming the assumption that would reverse the ranking, and a short decision memo with conditions and owners of open inputs (Network Planning Handbook 2026). Chapters 10 and 17 return to them.

> **PITFALL**
>
> Answering a network question with a load factor. A route can be 85 percent full and still be the wrong use of its aircraft: its passengers may have flown the airline anyway, its connecting passengers may displace more valuable ones on feeders, or the aircraft may earn more elsewhere. Load factor describes a flight; it does not value a decision.

IN PRACTICE Network planning in 2026: organization and tools

At Air Canada, network planning sits on the commercial side: the Chief Commercial Officer's portfolio covers global network development and performance with yield management and pricing, sales, cargo, and alliance and joint-venture management (Air Canada 2025). Planning systems still follow the classical chain:

Lufthansa Systems describes its NetLine/Plan as evaluating O&D profitability with linked demand, spill and cost models (a vendor description) (Lufthansa Systems 2026), and the company confirms that Air Canada uses its NetLine/Ops ++ for operations control (Lufthansa Systems 2025a). The tentative 2026 program of the AGIFORS scheduling and strategic planning study group lists work by HEC Montréal, KLM and the University of Twente on integrated planning with bank structures, and by Air Canada on aircraft routing under uncertainty (AGIFORS 2026). Treat these as public signals; your own desk's tools and decision rights are learned on the job.

1.5 Incremental network contribution

A route profit statement reports how a flight performed under the airline's accounting rules. A planning decision needs to know how much better off the whole airline is with a change than without it. The two differ whenever the change moves passengers between services, uses seats on other flights, or takes an aircraft from another use.

DEFINITION 1.3 Incremental network contribution

For a feasible schedule s, let $R(s)$ be the passenger revenue the airline retains (net of taxes, partner settlements and other deductions), summed over every journey it carries; $O(s)$ other net revenue such as cargo and ancillaries; and $\mathrm{AC}(s)$ the operating costs the schedule causes and could avoid, summed over its flights. The **incremental network contribution** of s against a baseline b is

$$\begin{aligned}\Delta\mathrm{NC}(s) = {} & [R(s) + O(s) - \mathrm{AC}(s)] \\ & - [R(b) + O(b) - \mathrm{AC}(b)].\end{aligned} \tag{1.1}$$

Incremental fixed costs (station opening, launch marketing) and capital are deducted as the decision horizon requires (Network Planning Handbook 2026).

Revenue is counted by **journey**, so a connecting fare is counted once. Costs are **avoidable** at the decision's horizon: in a seasonal plan aircraft ownership is fixed but aircraft time is scarce, while over

a fleet plan almost everything is variable (Belobaba 2006i). And both are evaluated **across the whole network**. For one new flight, the comparison splits into the flight's own result and three deductions:

$$\begin{aligned}\Delta\mathrm{NC} = \left[\sum_j f_j\, n_j^{\mathrm{new}} + \Delta O - \mathrm{AC}_{\mathrm{new}}\right] \\ - \Delta\mathrm{AC}_{\mathrm{other}} - D_{\mathrm{other}} - \sum_j f_j\, n_j^{\mathrm{moved}}.\end{aligned} \tag{1.2}$$

The bracket is the new flight's journeys n_j^{new} at fares f_j, plus its other revenue ΔO, minus its avoidable cost: what a route profit statement shows. The deductions are the extra cost of carrying its connecting passengers on other flights, $\Delta\mathrm{AC}_{\mathrm{other}}$; the contribution of passengers they displace there, D_{other}; and the revenue of journeys n_j^{moved} that merely switch from the airline's other services. Two rules prevent the commonest errors.

RESULT 1.1 Prorates move profit between legs, not into the network

However a connecting fare is split among the legs of its itinerary (by mileage, fare ratio or a joint-venture formula), the shares add back to the whole fare. The split changes each leg's reported profit, never the network contribution of a change.

What breaks this rule is double counting: prorating a connecting fare to the new leg while also counting the full fare on the feeder, or adding a "feed value" to a simulation that already models connections (Network Planning Handbook 2026).

RESULT 1.2 Charge for a scarce aircraft once

When several feasible uses compete for one scarce resource, each evaluated against the same baseline, choose the use with the largest incremental contribution; equivalently, charge each candidate the value of the best alternative as an opportunity cost and accept it if the result is positive. Doing both counts the alternative twice and rejects candidates that are better.

The handbook accepts either explicit alternatives or a shadow price on aircraft time, provided the analyst says how the price was estimated and does not use both (Network Planning Handbook 2026); Chapters 12 and 14 show where such prices come from. The

handbook's Exercise A applies all of this; Chapter 10 takes route evaluation further.

WORKED EXAMPLE 1.2 **Exercise A from the handbook**

A candidate flight (one departure; numbers invented for teaching) carries 110 local customers at CAD 220 and 40 connecting customers at CAD 450 for their whole itineraries: CAD 24,200 plus CAD 18,000, counted once by journey. Cargo and ancillaries add CAD 3,000 net and avoidable cost is CAD 32,000, so the flight's own result is CAD 13,200. The connecting customers add CAD 2,000 of cost on feeder flights and displace CAD 4,000 of contribution there, and CAD 3,500 of the revenue is diverted from the airline's other services:

$$\Delta NC = 42{,}200 + 3{,}000 - 32{,}000 - 2{,}000 - 4{,}000 - 3{,}500 = 3{,}700.$$

The aircraft's best alternative use earns CAD 4,500, so the candidate is worth $3{,}700 - 4{,}500 = -800$ after its opportunity cost: use the aircraft elsewhere. Yet the flight carries 150 customers, 89 percent of a 169-seat 737-8, and looks profitable on its own (Figure 1.3).

The gap of CAD 800 would close if diversion were CAD 2,700, if feeder displacement were CAD 3,200, or if each connecting customer paid CAD 20 more, each with all else fixed. These are the inputs to check first. Prorating 60 percent of each connecting fare to the candidate would show its leg at CAD 6,000 and the feeders at plus CAD 1,200 instead of minus CAD 6,000, but the network total is still CAD 3,700.

PITFALL

Comparing with doing nothing. A positive incremental contribution says only that the change beats the baseline. When the change uses a scarce resource, compare it with the best feasible alternative use of that resource, over the same dates and against the same baseline.

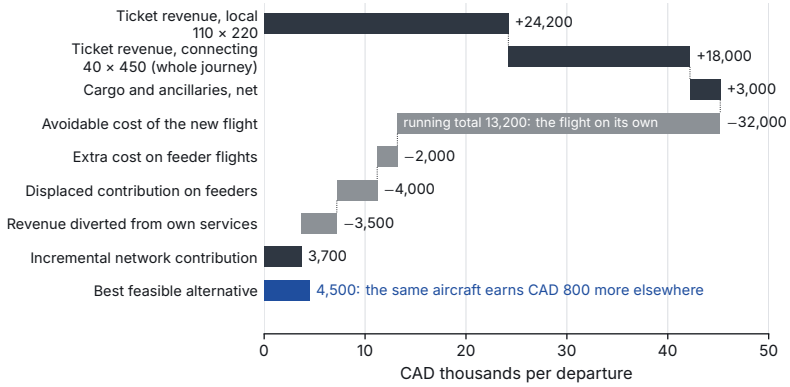


Figure 1.3 Exercise A as a contribution bridge, per departure. The flight's own result, with every connecting fare credited to it, is CAD 13,200; the effects on feeder flights and the diversion of the airline's own customers reduce the network gain to CAD 3,700, below the CAD 4,500 the same aircraft earns in its best alternative use. Numbers are the handbook's, invented for teaching.

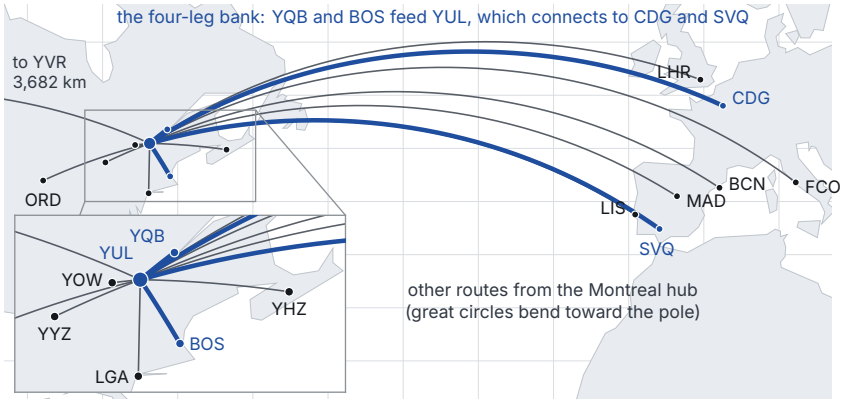


Figure 1.4 Laurentian's network: great-circle routes from the Montreal hub to the airports of the running case (equidistant cylindrical projection, on which great circles bend toward the pole). The four-leg bank is in cobalt; the inset enlarges the hub and its North American spokes; Vancouver lies off the map. Coastlines from Natural Earth (public domain).

1.6 Laurentian, the running case

Laurentian is a fictional airline hubbed at Montreal-Trudeau (YUL). Its geography and aircraft are real; its demand, fares and costs are invented to make each mechanism visible, and none is an Air Canada figure. It flies 137-seat A220-300s, 169-seat 737-8s, 182-seat A321XLRs and 298-seat 787-9s, with the cost model of Chapter 3 (Figure 1.4).

Two pieces carry most examples. The route **YUL–SVQ**, flown by the A321XLR on the pattern of Example 1.1, serves spill, route evaluation and RM (Chapters 7, 10 and 14). The **bank** at Montreal connects two feeders, from Quebec City (YQB) and Boston (BOS), to two transatlantic departures, to Paris (CDG) and Seville. Its four legs (Table 1.2) and sixteen fare products, a flexible and a restricted fare in each of eight origin-destination markets, are small enough to work by hand and rich enough to show network effects.

Table 1.2 The four legs of Laurentian's bank. BH: block hours; km: great-circle distance; Demand: mean unconstrained demand per departure over the O&D markets using the leg; DF: demand factor, demand divided by seats; Conn.: share of demand connecting to or from another leg of the bank. Demand figures are illustrative.

Leg	Aircraft	Seats	BH	km	Demand	DF	Conn.
YQB–YUL	A220-300	137	0.9	233	141	1.03	48%
BOS–YUL	A220-300	137	1.4	409	153	1.12	54%
YUL–CDG	787-9	298	6.9	5,524	317	1.06	32%
YUL–SVQ	A321XLR	182	7.0	5,556	161	0.88	30%

WORKED EXAMPLE 1.3 **The bank at a glance**

The eight markets generate 621 journeys per bank at mean demand: 470 local and 151 connecting (24.3 percent), which use two legs each. The legs therefore see $470 + 2 \times 151 = 772$ boardings against 754 seats, and three of the four legs have more demand than seats (Figure 1.5).

The journeys are worth CAD 377,639. Connecting journeys are 24.3 percent of journeys but 31.6 percent of revenue, at an average fare of CAD 790 against CAD 550 for local ones. Per seat used, the comparison reverses: a restricted Boston to Paris journey pays CAD 599 for two seats, CAD 299.50 per leg, while a restricted Montreal to Paris journey pays CAD 549 for one seat on the same transatlantic flight. Which should get the last seat on YUL–CDG depends on what the Boston passenger's feeder seat would otherwise earn: the displacement problem of Chapters 10 and 14.

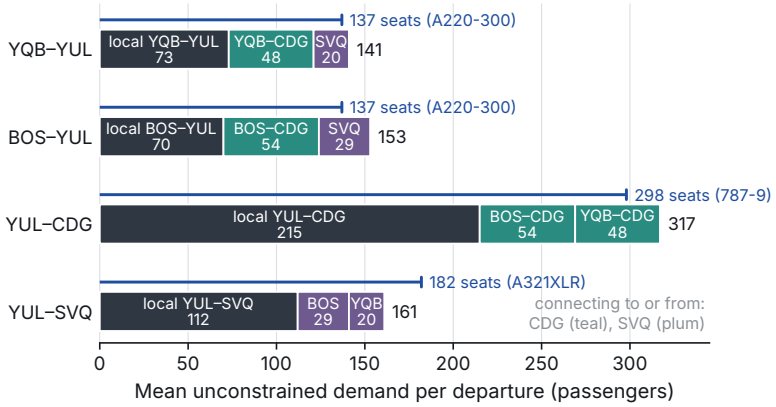


Figure 1.5 What each leg of Laurentian's bank carries: mean unconstrained demand per departure by O&D market, local in slate and connecting in colour, against the seats of the assigned aircraft (cobalt lines). Connecting traffic is coloured by the long-haul flight it uses. All figures are illustrative.

1.7 Air Canada in October 2026

These facts are public, dated and cited, and describe the job's context, not its internal workings. They were collected on 27 October 2026, before Air Canada's third-quarter results scheduled for 29 October (Network Planning Handbook 2026); check that release before relying on them.

The company. Air Canada describes itself as the largest provider of scheduled passenger services in the Canadian domestic, Canada-US transborder and international markets to and from Canada. Air Canada Express flying, bought from Jazz Aviation and PAL Airlines, feeds the mainline and Rouge, the leisure carrier; Air Canada is a founding Star Alliance member and owns Aeroplan. Its operating fleet, including regional partners, numbered 357 at 30 June 2026 (Air Canada 2026d).

Hubs and network. Air Canada develops three international hubs: Toronto Pearson, Montreal-Trudeau and Vancouver (Air Canada 2025). The handbook advises treating the shorthand (Montreal for Europe, Toronto the largest mixed network, Vancouver for the Pacific) as a hypothesis to test against schedules (Network Planning Handbook 2026). Sixth-freedom traffic, between two other countries over these hubs, is a stated growth theme: its revenue grew 9 percent year over year in the second quarter of 2026, led by the US-Atlantic and US-Pacific corridors, and premium cabins earned about 30 percent of passenger revenue (Air Canada 2026d).

A difficult 2026. After fuel prices rose sharply, Air Canada suspended its guidance on 30 April (The Canadian Press 2026) and reported a second-quarter operating loss of CAD 215 million after CAD 388 million of labour-related and legal charges (Air Canada 2026d). It reinstated guidance in August, with 2.25 to 3.25 percent growth in available seat miles for the year (Air Canada 2026c). Chapter 2 reads the quarter in detail.

Fleet transition. A220-300 deliveries continue (46 of 65 firm orders by 30 June 2026), 737 MAX 8s are moving to Rouge, and the first 787-10s are due in 2026 (Air Canada 2026d). Air Canada received its first A321XLR on 24 April 2026 (Air Canada 2026a); the type entered service on 9 June 2026 with 182 seats, 14 of them lie-flat Signature Class (Network Planning Handbook 2026; Air Canada 2026f). Of 30 aircraft, 2 were delivered by 30 June 2026, 9 are planned by the end of 2026, 19 by the end of 2027 and the last in 2029 (Air Canada 2026d). A program is not the aircraft available to a given desk in a given month (Network Planning Handbook 2026).

Montreal to Seville. The route starts on 8 September 2027 with three weekly A321XLR flights, four from 1 October 2027, departing Montreal at 20:30; Air Canada calls it the only non-stop between North America and Seville (Air Canada 2026f). Its autumn-to-spring operating periods show aircraft capability meeting seasonal demand and hub connections (Network Planning Handbook 2026). Laurentian's YUL–SVQ case copies the pattern; its economics are invented.

The United transborder agreement. On 16 October 2026 Air Canada and United announced new transborder routes for summer 2027, including Montreal to Indianapolis, and the renewal and expansion of their transborder joint business agreement, subject to regulatory approval (Air Canada 2026g). Transport Canada's consultation concerns bringing in city pairs so far carved out, among them Montreal to Chicago and Montreal to Washington (Travelweek 2026a). The agreement shapes transborder feed into Montreal; keep existing cooperation separate from changes awaiting approval (Network Planning Handbook 2026).

> **PITFALL**
>
> Reading company-wide signals as route facts. Premium revenue shares, sixth-freedom growth and a fleet program describe the airline as a whole. None establishes the profitability

of a route, the demand for a proposed service, or the aircraft available to a schedule.

1.8 Summary

- Planning is a sequence of decisions on shrinking horizons: fleet years ahead, routes and frequencies a year or more, timetables and fleet assignment two to six months, crews monthly, RM during booking, operations control on the day.
- Airlines decompose planning into schedule design, fleet assignment, routing and crew scheduling because the joint problem is too large. The sequence gives feasible plans that can miss the best one; feedback loops recover part of that value of integration.
- The core decision is the feasible schedule that adds most to the whole network against the best realistic alternative, structured by five questions: counterfactual, journeys, increment, binding resources, alternative.
- Incremental network contribution counts revenue by journey and avoidable cost by flight, across the network, against a dated baseline. Prorates move profit between legs, not into the network; a scarce aircraft is charged for once.
- In Exercise A, a flight 89 percent full and profitable on its own adds CAD 3,700 to the network and loses to an alternative worth CAD 4,500.
- Laurentian's four-leg bank at Montreal and its YUL–SVQ route, modelled on Air Canada's announced Seville service, carry the book's examples; Air Canada facts are dated and cited.

1.9 Further reading

Belobaba's introduction to the planning process (2006c) and his lectures on schedule development (2006f) and route evaluation (2006i) give this chapter's vocabulary in a few slides. Read the handbook's framework section (2026) before your first week. Barnhart and Cohn (2004) survey schedule planning and the case for integration. The full reference edition's Chapter 1 states the value of integration formally and proves the rules on prorates and opportunity cost.

CHAPTER

2

Measuring an Airline

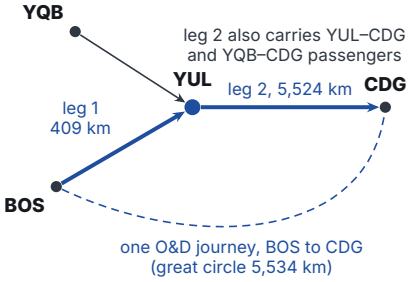
IN THIS CHAPTER

- The objects airline data count (movements, legs, itineraries, tickets), and why demand lives in origin-destination markets while supply lives on legs.
- Market size in passengers per day each way, and why it is not the load of a new flight.
- Capacity and traffic in seat-kilometres, and why the network load factor is weighted by seat-kilometres, not averaged.
- Yield, unit revenue and unit cost, and the identity that ties them together: unit passenger revenue equals yield times load factor.
- Why yield and unit cost fall with distance, what a stage-length adjustment does, and the break-even load factor with its limits.
- Air Canada's second-quarter 2026 statistics read line by line, including a figure that does not reconcile.

DRAWS ON Belobaba, Introduction to the airline planning process; Airline economics review; Operating costs and productivity measures (MIT 16.75J, 2006); handbook; Air Canada Q2 2026 MD&A

Airlines describe themselves in ratios. Air Canada's report for the second quarter of 2026 says that unit passenger revenue rose 10.5 percent on capacity up 0.3 percent, that the load factor gained 2.6 points and that unit cost rose 23.9 percent (Air Canada 2026d); a route review says a flight ran at 84 percent. Each number has a numerator, a denominator and a scope, and each hides weights. A planner needs to read them fluently, recompute them from their parts and know which questions they can answer.

Measures describe an airline; they do not decide anything (Chapter 1 judges decisions by incremental network contribution). But every contribution estimate is built from them, every forecast is reconciled against them, and every conversation with finance or revenue management uses them. This chapter builds them from the bottom up and ends by reading Air Canada's quarter with them.



One passenger, BOS to CDG via YUL	
O&D journeys	1
itineraries (paths)	1
legs flown, seats used	2
flight coupons	2
passengers carried (IATA, by flight number)	2
RPK, at flown distance	5,932
If the ticket includes the return	
tickets	1
O&D journeys	2
flight coupons	4

Figure 2.1 One passenger travelling from Boston to Paris over Montreal, and how each unit of measurement counts her. The dashed curve is the O&D journey, measured along the great circle; the cobalt arrows are the two legs she actually flies. Distances from the case data.

2.1 Four objects: movements, legs, itineraries and tickets

The handbook names the analyst's most important data skill as keeping four objects apart: an aircraft movement, a flight leg, a passenger itinerary and a ticket (Network Planning Handbook 2026). Summing rows without knowing which object a row represents double counts capacity or demand.

> **DEFINITION 2.1 The objects airline data count**
>
> An **aircraft movement** is one takeoff or one landing. A **flight leg** (segment, sector) is one non-stop operation with scheduled times; it consumes an aircraft, a crew and a slot. A **flight** is one or more legs under one flight number. An **origin-destination (O&D) market** is a directional pair of points between which passengers want to travel, and a **journey** is one passenger's trip in one direction. An **itinerary** is the sequence of legs a journey uses. A **ticket** is the commercial document, which may cover several journeys; each leg on it is a **coupon**.

Demand is defined by O&D market, but supply is produced as legs (Belobaba 2006d). Each leg supplies seats jointly to every market whose itineraries use it, and each market can be served over many hubs and airlines, so there is no well-defined supply of seats to any one market, and a market's profitability depends on how costs and revenues are allocated (Chapter 10). The accounting consequence is simple: passengers counted on legs exceed journeys by exactly the extra legs that connecting journeys use.

WORKED EXAMPLE 2.1 Boston to Paris over Montreal

A Laurentian passenger buys a restricted one-way ticket from Boston to Paris at CAD 599 and flies BOS–YUL (409 km) and YUL–CDG (5,524 km), as in Figure 2.1. She is one journey in the BOS–CDG market and one itinerary, but two leg passengers and two coupons, occupying a seat on each of two aircraft. Her revenue passenger-kilometres, measured along the legs flown, are 5,932, 7.2 percent more than the 5,534 km great circle from Boston to Paris.

Each count answers a different question. The O&D count is what a Boston-Paris demand forecast predicts; the leg counts are what fill each flight. Her fare per flown kilometre, 10.1 cents, is what enters a yield statistic, against 9.9 cents for a local YUL–CDG passenger paying CAD 549 on the same aircraft.

Across Laurentian's bank (Chapter 1), 621 journeys become 772 leg boardings, because the 151 connecting journeys count on two legs each. Adding up leg loads and calling the total "passengers" overstates demand by 24 percent.

Public reporting adds a fifth convention. Air Canada counts revenue passengers on a flight-number basis, following the IATA definition (Air Canada 2026d): a connecting passenger on two flight numbers counts twice, a through passenger under one flight number once. The handbook therefore asks analysts to record, for every table, whether its rows are tickets, passengers, coupons, legs or flights (Network Planning Handbook 2026).

> **PITFALL**
>
> Counting codeshares as flights. A flight sold under several airlines' codes appears once per marketing carrier in schedule data. Capacity counts must use operating flights only, and demand counts must not add the same passenger under each code (Network Planning Handbook 2026).

2.2 Passengers per day each way

Market sizes are quoted as a daily, directional rate, because flights are daily and directional.

DEFINITION 2.2 Passengers per day each way

The **passengers per day each way** (**PDEW**) of an O&D market over T days is its journeys in both directions divided by $2T$. When the two directions differ, report the directional rates separately.

In the handbook's example, 73,000 journeys in a year, both directions together, is $73{,}000/(2 \times 365) = 100$ PDEW (Network Planning Handbook 2026). Forgetting one of the two divisions overstates the market by a factor of two or of 365.

WORKED EXAMPLE 2.2 From PDEW to passengers on a departure

Suppose, for illustration, that a new non-stop four times a week could capture 45 percent of the 100 PDEW market. One direction produces 700 journeys a week, of which the airline carries 315. If every customer would travel on one of the four operating days, each departure carries 78.75 passengers. If only customers whose preferred day is an operating day travel, the airline carries $315 \times 4/7 = 180$ journeys a week, 45 per departure. The handbook calls the difference a missed-day penalty (Network Planning Handbook 2026).

A 100 PDEW market does not mean 100 passengers on each new flight. The rate must first pass through seasonality, day-of-week patterns, competing itineraries, the days operated, capacity and stimulation (Chapters 4 and 6).

2.3 Capacity, traffic and load factor

Airline output is measured in seat-distance, because a seat flown 5,000 km is a different product from a seat flown 300 km.

DEFINITION 2.3 Capacity, traffic and load factor

For legs l with C_l seats, N_l revenue passengers and length d_l in kilometres, **available seat-kilometres** are ASK $= \sum_l C_l d_l$ and **revenue passenger-kilometres** are RPK $= \sum_l N_l d_l$; with distances in statute miles they are **available seat-miles (ASM)** and **revenue passenger-miles (RPM)**. The **load factor** is LF = RPK/ASK, which for a single leg is passengers divided by seats.

In Belobaba's example, a 200-seat aircraft flying 1,000 miles with 140 passengers produces 200,000 ASM and 140,000 RPM, a load factor of 70 percent (Belobaba 2006c). The load factor does not depend on the unit of distance, but every measure per unit of distance does: with 1.609344 km to the mile, a cost of 7.5 cents per ASM is 4.66 cents per ASK. Air Canada reports in miles (Air Canada 2026d) and IATA in kilometres (IATA 2026d); never mix the two in one calculation (Network Planning Handbook 2026).

RESULT 2.1 The system load factor is an ASK-weighted average

The load factor of a set of legs is the average of the leg load factors N_l/C_l, each weighted by the leg's share of ASKs, $C_l d_l$/ASK. It differs from the simple average of the leg load factors whenever the legs that produce most ASKs are fuller, or emptier, than the rest.

The reason takes one line: RPK is the sum over legs of each leg's load factor times its ASKs. In many networks the long-haul legs produce most of the ASKs and are also the fullest, so the simple average understates the system figure. Passengers divided by seats is a third ratio, which weights legs by seats and ignores distance.

WORKED EXAMPLE 2.3 One bank, three load factors

On one illustrative day Laurentian's bank carries 82 passengers on YQB–YUL (137 seats, 233 km), 97 on BOS–YUL (137 seats, 409 km), 280 on YUL–CDG (298 seats, 5,524 km) and 150 on YUL–SVQ (182 seats, 5,556 km): leg load factors of 59.9, 70.8, 94.0 and 82.4 percent. Their simple average is 76.8 percent; passengers divided by seats, 609/754, is 80.8 percent; RPK divided

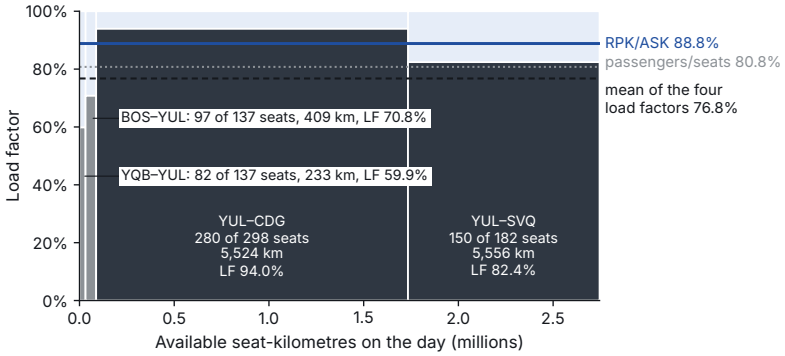


Figure 2.2 Load factors on Laurentian's bank on one illustrative day. Each bar's width is the leg's available seat-kilometres and its height the leg's load factor, so the network load factor, RPK/ASK, is the area of the bars divided by the total width. The two feeder legs contribute almost nothing to it.

by ASK is 88.8 percent. The two transatlantic legs produce 60.0 and 36.8 percent of the ASKs and the feeders 1.2 and 2.0 percent, so the feeders barely register (Figure 2.2).

None of the three numbers is wrong. RPK/ASK measures how much of the capacity produced was sold and is the one that enters unit revenue; passengers per seat describes boardings; the simple average describes nothing in particular.

PITFALL

Averaging load factors. The average of the load factors of flights of different sizes and lengths matches neither the network's capacity utilization nor its boardings. Recompute from RPK and ASK, or from passengers and seats if boardings are what you need, and say which you used (Network Planning Handbook 2026).

2.4 Yield, unit revenue and unit cost

Revenue and cost enter through three more ratios.

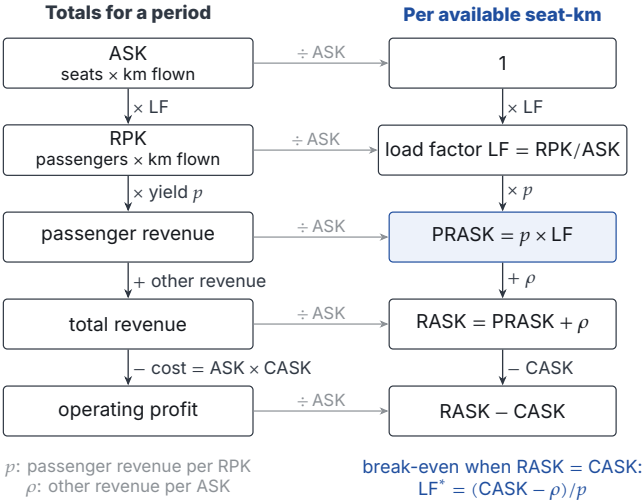


Figure 2.3 The cascade of airline measures. Totals (left) become unit measures (right) on division by ASK. Passenger revenue per ASK is the product of yield and load factor, and the operating margin per ASK is unit revenue minus unit cost; setting it to zero gives the break-even load factor of Section 2.6.

DEFINITION 2.4 **Yield, unit revenue and unit cost**

For a given period and scope, the **yield** p is passenger revenue per RPK. **Passenger revenue per ASK (PRASK)** is passenger revenue divided by ASK. **Revenue per ASK (RASK)** is total operating revenue (passenger, cargo and other) divided by ASK, so RASK = PRASK + ρ, with ρ the non-passenger revenue per ASK. **Cost per ASK (CASK)** is operating expense divided by ASK. The mile-based versions are PRASM, RASM (or TRASM) and CASM.

A ratio means something only with its scope stated: which revenues (cargo, ancillaries, loyalty, tour packages), which costs, which carriers. Air Canada's **adjusted CASM**, for example, excludes fuel, the cost of ground packages sold by its tour operator, freighter costs and items it considers unusual, and is a non-GAAP measure without standardized meaning (Air Canada 2026d). The handbook flags two common confusions: calling PRASK "yield", and comparing RASK figures built on different revenue streams (Network Planning Handbook 2026). The ratios are tied together by identities that follow from their definitions (Figure 2.3).

RESULT 2.2 Identities linking the unit measures

For a common scope and period, unit passenger revenue is yield times load factor, and operating profit Π is ASK times the margin of unit revenue over unit cost:

$$\text{PRASK} = p \times \text{LF}, \qquad \text{RASK} = p\,\text{LF} + \rho, \tag{2.1}$$

$$\Pi = \text{ASK}\,(\text{RASK} - \text{CASK}).$$

Between two periods, with g the proportional change of each quantity, PRASK growth splits in logarithms into yield growth and load factor growth, the latter being traffic growth minus capacity growth:

$$\ln(1+g_{\text{PRASK}}) = \ln(1+g_p) + \ln(1+g_{\text{RPK}}) - \ln(1+g_{\text{ASK}}). \tag{2.2}$$

The first identity simply multiplies and divides passenger revenue by RPK. Belobaba warns against reading any single term as a measure of success: a high yield is no use if the load factor is too low, a low unit cost is of little value if revenues are weak, and a high load factor may simply reflect very low fares (Belobaba 2006d). The handbook treats the identity as a reconciliation check, not a causal model (Network Planning Handbook 2026): a yield of CAD 0.20 per passenger-mile at a load factor of 0.85 must give a PRASM of 17 cents, and when reported figures do not multiply out, one was computed on a different scope or is wrong.

Belobaba's 200-seat example completes the set (historical, in dollars): revenue of 16,000 dollars and operating expense of 15,000 dollars give a yield of 11.4 cents per RPM, a unit cost of 7.5 cents per ASM and a unit revenue of 8.0 cents per ASM (Belobaba 2006c). The identity holds, $11.43 \times 0.70 = 8.0$, and the flight earns $200{,}000 \times (8.0 - 7.5)$ cents, or 1,000 dollars. In kilometres the yield is 7.10 cents, CASK 4.66 cents and RASK 4.97 cents: every number changes except the load factor and the profit.

PITFALL

Yield per passenger versus yield per kilometre. Average fare and yield move differently when the mix of trip lengths

changes: a network that adds long flights can raise its average fare while its yield falls, with no change in pricing.

2.5 Stage length

Distance shapes both sides of the unit economics. The **average stage length** is the average non-stop leg distance: aircraft miles flown divided by departures for Belobaba (Belobaba 2006d), total ASMs divided by seats dispatched for Air Canada, which weights legs by seats (Air Canada 2026d). The **average passenger trip length**, RPK divided by passengers, is usually longer, because some passengers take more than one flight. Capacity decomposes exactly as

$$\text{ASK} = \text{departures} \times \frac{\text{seats}}{\text{departure}} \times \frac{\text{ASK}}{\text{seat}},$$

frequency times average gauge times seat-weighted stage length, so capacity growth is the sum of the three growth rates, in logarithms.

Yield falls with distance because fares rise less than proportionally with distance: part of a fare pays for things that do not scale with distance (airport processes, the fixed effort of travelling), and that part shrinks per kilometre on longer trips. In Belobaba's 2006 chart of the 50 largest O&D markets, yields run from 20 to over 60 cents per mile below 300 miles to about 7 to 12 cents beyond 1,500 miles, fitted by a fare of about 64 dollars plus 6.5 cents per mile (Belobaba 2006d) (left panel of Figure 2.4; historical numbers, general shape).

Unit cost falls with distance for the same reason. Some costs come once per departure (landing and terminal charges, ground handling, the fuel and time of taxi, takeoff and climb, cycle-driven maintenance); the rest grow with distance flown. Per seat-kilometre, the per-departure part shrinks as legs lengthen, steeply at first and then slowly. A one percent longer stage lowers CASK by a percentage equal to the share of per-departure costs in the cost of a trip: close to one percent on very short legs, close to zero on long ones, never constant. Appendix A.1 derives this, and Chapter 3 builds the per-departure and distance costs from block hours and departures.

A **stage-length adjustment** restates a CASK at a reference stage length so that operations with different average distances can be compared. With a cost model, it moves the observed CASK along the model's curve: multiply by the model's CASK at the reference distance over its CASK at the actual distance. Without one, analysts often scale CASK by the ratio of the distances raised to a fixed ex-

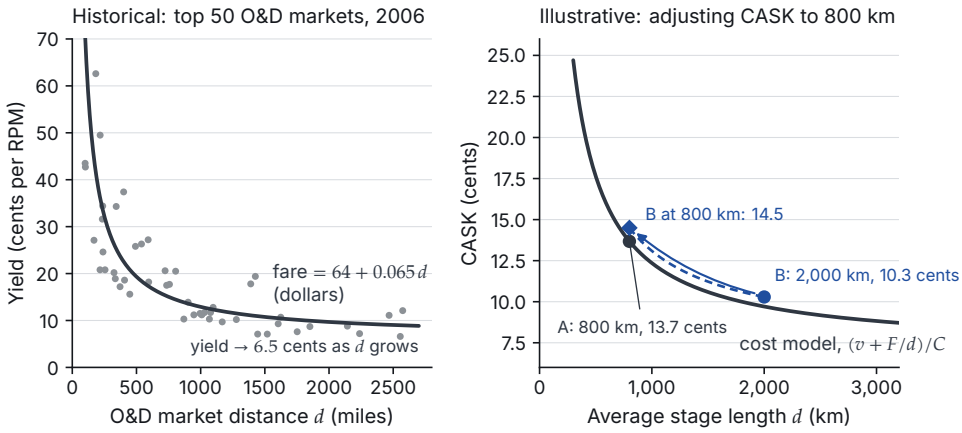


Figure 2.4 Left: yields in the 50 largest O&D markets, read from Belobaba's 2006 airline economics lecture, with a least-squares fare $f = a + b\,d$ (historical, cents per revenue passenger-mile). Right: an illustrative cost model, CASK $= (v + F/d)/C$ with F = CAD 9,000 per departure, v = CAD 12 per km and $C = 170$ seats, and carrier B moved along its cost curve to A's stage length.

ponent. Since the true elasticity changes with distance, a fixed exponent is only a local approximation, and it can decide the answer.

WORKED EXAMPLE 2.4 Which carrier is cheaper?

Two operations with the same 170-seat aircraft are compared on the illustrative cost model of the right panel of Figure 2.4. Carrier A flies an average stage of 800 km at a CASK of 13.7 cents, on the model's curve; carrier B flies 2,000 km at 10.3 cents, 6 percent above the curve. On the raw numbers B looks 24.8 percent cheaper.

Moving B along the cost curve to 800 km multiplies its CASK by the model's ratio 13.68/9.71 and gives 14.5 cents: at A's stage length B is 6.0 percent more expensive, the true difference built into the example. Fixed exponents give other answers: 0.3 makes B 13.5 cents, 1.0 percent cheaper than A, and 0.5 makes it 16.3 cents, 18.9 percent more expensive. Only 0.374, the model's own average elasticity between 800 and 2,000 km, reproduces the 6.0 percent. The adjustment decided the answer.

Distance is also only one difference between operations: longer stages come with larger aircraft and other cabin mixes, labour costs and airports, and Belobaba's 2003 data showed low-fare carriers with the lowest CASM at every stage length (Belobaba 2006e). For a network decision the adjustment is beside the point anyway: a

route's relevant cost is its own avoidable cost, built from its own block hours, departures and passengers, never a system average (Network Planning Handbook 2026).

2.6 Break-even load factor

At what load factor does an operation cover its costs? Setting the margin per ASK to zero gives the answer the handbook uses (Network Planning Handbook 2026).

RESULT 2.3 Break-even load factor

If yield p, non-passenger revenue per ASK ρ and unit cost CASK do not depend on the load factor, operating profit is zero at

$$\mathrm{LF}^{*} = \frac{\mathrm{CASK} - \rho}{p},$$

the share of seats that must be sold at the current yield for passenger revenue to cover the costs that other revenue does not.

WORKED EXAMPLE 2.5 Break-even on YUL–SVQ

Take the handbook's illustrative Seville departure in Laurentian form (Network Planning Handbook 2026): a 182-seat A321XLR flies 5,556 km with 150 passengers paying an average net fare of CAD 620, ancillaries of CAD 30 per passenger, cargo contribution of CAD 3,500, and an avoidable cost of CAD 88,000, made of CAD 80,500 per departure and CAD 50 per passenger.

Unit measures. ASK is $182 \times 5{,}556 = 1{,}011{,}192$ and RPK $150 \times 5{,}556 = 833{,}400$, a load factor of 82.4 percent. The yield is $93{,}000/833{,}400 = 11.16$ cents and PRASK $11.16 \times 0.824 = 9.20$ cents. Ancillaries and cargo add CAD 8,000, or 0.79 cents per ASK, for a RASK of 9.99 cents against a CASK of 8.70 cents. The margin of 1.29 cents per ASK is the handbook's flight contribution of CAD 13,000.

Break-even. The formula gives $(8.70 - 0.79)/11.16 = 70.9$ percent. Counting per passenger instead, each passenger adds CAD $620 + 30 - 50 = 600$, so the departure needs

$(80{,}500 - 3{,}500)/600 = 128.3$ passengers, a break-even load factor of 70.5 percent. The formula overstates by 0.4 points because the CASK and other revenue measured at 150 passengers include the passenger costs, net of ancillaries, of the passengers above break-even.

When yield moves. Compare, as the handbook does, 170 passengers at an average fare of CAD 530. Passenger revenue falls from CAD 93,000 to CAD 90,100 although the load factor rises to 93.4 percent. Each passenger now adds CAD 510, the break-even rises to 151 passengers (83.0 percent), and the contribution falls to CAD 9,700. The fuller flight earns less.

That correction is small. The example's last step shows the formula's central limitation: it treats yield as fixed, but load factor and yield are chosen together. Revenue management fills an aircraft by opening cheaper fare classes, so a higher load factor usually comes with a lower yield, and the break-even target moves with it. The handbook adds that connecting passengers have network effects a single route's ratio ignores; the break-even load factor is a diagnostic and does not replace itinerary-level analysis (Network Planning Handbook 2026). It is also an average: whether one more passenger or one more seat is worth having is a marginal question, answered with spill and bid prices (Chapters 7 and 14).

Load factor is an outcome of capacity, demand and pricing together, not an objective. The break-even load factor tells you how much room there is between them, assuming fares do not move.

2.7 Reading Air Canada's second-quarter 2026 MD&A

Air Canada's management's discussion and analysis (MD&A) for the second quarter of 2026, published on 11 August 2026, reports the quarter with the measures of this chapter (Air Canada 2026d). Its operating statistics include the regional carriers flying under capacity purchase agreements, distances are in miles, and its regional breakdown (Canada, US transborder, Atlantic, Pacific and Other) gives revenues and percentage changes but not regional capacity levels. Table 2.1 transcribes the figures used below.

Table 2.1 Air Canada, operating statistics and results for the second quarter of 2026 and 2025, as printed in the MD&A (pp. 1 and 8), including regional partners flying under capacity purchase agreements. Changes are those reported; pp means percentage points.

Statistic	Q2 2026	Q2 2025	Change
Revenue passenger miles, RPM (millions)	23,597	22,796	+3.4%
Available seat miles, ASM (millions)	26,942	26,860	+0.3%
Passenger load factor	87.5%	84.9%	+2.6 pp
Yield (cents per RPM)	23.7	22.1	+7.2%
PRASM (cents)	20.7	18.7	+10.5%
TRASM (cents)	23.3	21.0	+10.9%
CASM (cents)	24.1	19.4	+23.9%
Adjusted CASM (cents)	15.5	14.4	+7.4%
Seats dispatched (thousands)	14,644	14,478	+1.2%
Aircraft frequencies (thousands)	100.3	98.5	+1.8%
Average stage length (miles)	1,840	1,855	−0.8%
Revenue passengers carried (thousands)	12,032	11,551	+4.2%
Fuel cost per litre (cents)	132.7	88.0	+50.8%
Passenger revenue (CAD millions)	5,576	5,031	+10.9%
Cargo revenue (CAD millions)	325	253	+29%
Other revenue (CAD millions)	365	348	+5%
Operating expenses (CAD millions)	6,481	5,214	+24%
of which aircraft fuel	1,713	1,148	+49%
Operating income (CAD millions)	−215	418	

WORKED EXAMPLE 2.6 **Recompute, reconcile, interpret**

Step 1: recompute every ratio from its parts. From the levels in Table 2.1, every Q2 2025 ratio reproduces exactly. For Q2 2026, PRASM (5,576/26,942 = 20.70 cents), TRASM (23.26), CASM (24.06), adjusted CASM (15.47, from the adjusted expense of CAD 4,167 million in the MD&A's reconciliation) and stage length (1,840 miles) also reproduce. Two do not: 23,597/26,942 = 87.58 percent rounds to 87.6, not the printed 87.5, and 5,576/23,597 = 23.63 cents rounds to 23.6, not 23.7.

Step 2: locate the inconsistency. The gaps are too large to be rounding, and the printed growth of traffic (3.4 percent; the levels give 3.5) and yield (7.2; the levels give 7.1) point the same way: all the printed ratios fit Q2 2026 traffic of about 23,560 to 23,580 million RPM, none the printed 23,597. The document

does not say which is wrong. An analyst notes the discrepancy, uses the ratios with a caveat and checks the next report.

Step 3: decompose growth. With the printed ratios, $23.7 \times 0.875 = 20.74$ cents, consistent with the printed PRASM of 20.7. By (2.2), the 10.5 percent rise in PRASM splits into about 7 points of yield and 3 points of load factor: Air Canada sold its seats both fuller and dearer. Capacity barely moved: 1.8 percent more departures were offset by an average gauge 0.7 percent smaller (146.0 seats per departure against 147.0) and a stage length 0.8 percent shorter.

Step 4: margin and break-even. TRASM minus CASM is $23.26 - 24.06 = -0.80$ cents, which on 26,942 million ASMs is the reported operating loss of CAD 215 million. Other revenue per ASM is $(325 + 365)/26{,}942 = 2.56$ cents, so the break-even load factor is $(24.06 - 2.56)/23.63 = 91.0$ percent, above the 87.6 percent achieved; a year earlier it was 77.8 percent against 84.9 percent achieved. Figure 2.5 decomposes the 13.1-point rise. Fuel burned per ASM stayed at about 47.3 millilitres while fuel cost per ASM rose from 4.27 to 6.36 cents, so the fuel effect, 9.4 points of load factor at the 2025 yield, is all price. The CAD 388 million of pension, labour and legal charges add 6.5 points and other costs 5.1; higher cargo and other revenue take back 1.5 points and the 7 percent higher yield 6.4. Without the special charges, break-even would have been 84.9 percent, below the load factor achieved, and the quarter would have shown an operating profit of about CAD 173 million.

Step 5: regions. Passenger revenue was 32.7 percent Atlantic, 27.0 percent Canada, 19.3 percent US transborder, 14.0 percent Pacific and 7.0 percent other. Split with (2.2), the regions tell different stories. Transborder capacity fell 5.7 percent while traffic held, and PRASM rose 18.5 percent, about two thirds of it from yield, which the MD&A attributes to higher fares and lower industry capacity in the market (Air Canada 2026d). Pacific capacity grew 7.9 percent with a flat load factor, and PRASM rose only 3.3 percent. The pattern is consistent with capacity restraint supporting unit revenue, but cannot establish it: fares rose in every region, partly in response to the fuel shock.

The reading supports statements about the airline as a whole: unit revenue rose through fares more than load factor, unit cost

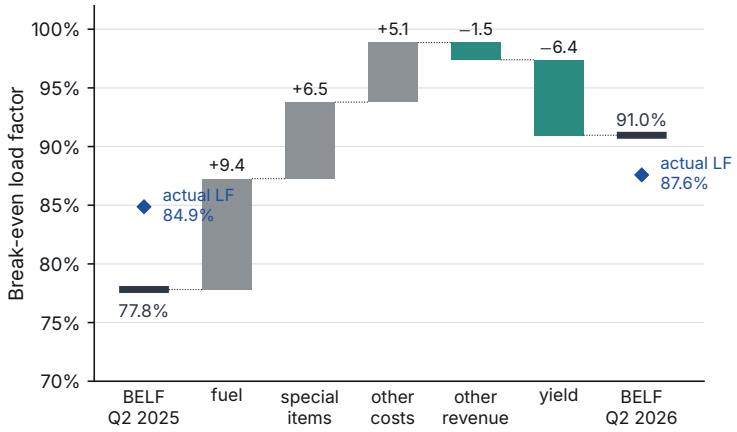


Figure 2.5 Air Canada's break-even load factor, Q2 2025 to Q2 2026, computed from the MD&A as (CASM minus other revenue per ASM) divided by yield, with the actual load factors (cobalt). The steps are applied in the order shown; a different order would split the total differently. Data: Air Canada Q2 2026 MD&A.

rose faster because of fuel and one-time charges, and the margin between break-even and actual load factor vanished. It supports nothing about routes. Regional figures give no capacity levels, so regional load factors cannot be recovered (the rounded changes place Canada's 2025 load factor anywhere between about 81 and 91 percent); regional yields are not comparable across stage lengths; and nothing separates fare changes from cabin mix or point of sale. The report's premium and sixth-freedom signals describe the company, not any route (Network Planning Handbook 2026).

Two further cautions. Passengers carried divided by seats dispatched is 82.2 percent, five points below the RPM-based load factor, partly through the weighting of Result 2.1 and partly through flight-number counting; the MD&A does not allow the two effects to be separated. And comparisons with other airlines need the same care as stage-length adjustments: the MD&A itself cautions that its non-GAAP measures may not be comparable with other companies' (Air Canada 2026d).

IN PRACTICE **Unit measures in 2026 reporting and guidance**

Airlines speak to investors in exactly these units. When Air Canada reinstated its 2026 guidance in August, it expressed it as growth in ASMs (2.25 to 3.25 percent for the year) and in adjusted CASM (5 to 6 percent), alongside adjusted EBITDA and free cash flow, with explicit fuel price and exchange-rate assumptions (Air Canada 2026c). IATA's June 2026 industry outlook forecast 2026 growth of 2.1 percent in RPKs and 1.6 percent in ASKs, with departures down 0.6 percent, and put fuel at 31.4 percent of industry costs (IATA 2026d); by the capacity decomposition of Section 2.5, more ASKs on fewer departures means larger aircraft, longer stages or both. Read each company's definitions before comparing figures across airlines.

2.8 Summary

- Keep movements, legs, itineraries and tickets apart. Demand lives in O&D markets and supply on legs; leg counts exceed journeys by the extra legs connecting journeys use. A market's PDEW is not the load of a new flight.
- The network load factor is RPK/ASK, an ASK-weighted average of leg load factors; the simple average of load factors describes nothing in particular.
- PRASK equals yield times load factor, RASK adds other revenue per ASK, and operating profit equals ASK times (RASK minus CASK). Use the identities as reconciliation checks.
- Yield and unit cost both fall with distance because fares and costs contain per-trip components. A stage-length adjustment is a model, and its assumptions can decide a comparison.
- The break-even load factor is (CASK minus other revenue per ASK) divided by yield. It assumes a fixed yield and ignores network effects: a diagnostic, not a target.
- Air Canada's Q2 2026 PRASM rose 10.5 percent, mostly through yield, while its break-even load factor rose from 77.8 to 91.0 percent, above the load factor achieved, mainly through fuel and one-time charges. One traffic figure does not reconcile; nothing speaks to routes.

2.9 Further reading

Belobaba's lectures on the planning process and airline economics (2006c, 2006d) define these measures in a few slides each. Keep the handbook's table of vocabulary and common errors (2026) at hand, and read the glossary and non-GAAP section of Air Canada's MD&A (2026d) before any results call. The full reference edition's Chapter 2 derives the load factor weighting, the stage-length elasticity and the exact break-even load factor.

CHAPTER

3

Costs and Productivity

IN THIS CHAPTER

- How airlines classify operating costs, and which activity drives each category: block hours, departures, passengers, distance or calendar time.
- A cost model for one flight, and why cost per seat-kilometre falls as stage length grows.
- Trip cost versus seat cost: why a larger aircraft is cheaper per seat but riskier per trip.
- Utilization: how working an aircraft harder spreads its ownership cost.
- Which costs count for which decision: fully allocated, avoidable and incremental cost by horizon.
- Laurentian's break-even loads, and Air Canada's reported costs in 2026.

DRAWS ON Belobaba, Operating costs and productivity measures (MIT 16.75J, 2006); handbook; Air Canada Q2 2026 MD&A

Every proposal a network planner writes ends in a comparison of revenue with cost. Cost looks like the easy half, because the airline's accounts report it to the dollar. That impression is misleading. The accounts say what the airline spent last quarter on fuel, wages and airport fees. They do not say what one more departure to Seville would cost, or what the airline would save by cancelling the Tuesday flight to Boston. Those are different numbers, and the gap between them settles a large share of network decisions.

This chapter builds a cost model for one flight from the activities that drive cost, draws its two consequences for every network (unit cost falls with distance; larger aircraft cost less per seat and more per trip), adds utilization, and ends with the question to put to every cost figure: over which horizon is it avoidable? The model feeds Chapters 9, 10 and 12.

3.1 Cost categories and their drivers

Airlines report costs in two ways (Belobaba 2006e). The **objective** classification groups costs by what is bought: labour, fuel, ownership, landing fees, commissions. The **functional** classification groups them by the activity they serve: flying the aircraft (aircraft operating cost), handling it on the ground (aircraft servicing), processing passengers and baggage (traffic servicing), passenger service on board, reservations and sales, and overhead. A pilot and a gate agent are both labour; functionally, the first moves with flying hours and the second with passengers. Planning needs the functional view, because it says what causes each cost.

> **DEFINITION 3.1 Cost driver**
>
> The **cost driver** of a cost category is the activity whose volume determines it. Five drivers matter for network decisions: **block hours** (gate departure to gate arrival), **departures** (each take-off and landing, also called a cycle), **passengers** carried, **distance** flown, and **calendar time** (costs incurred per day, month or year whatever the flying). Some costs are driven by revenue (commissions); overhead has no natural driver and is allocated by a rule.

Belobaba's lecture attaches one driver to each functional category and reports average rates for US carriers in 2003 (Table 3.1). The rates are historical; the structure is permanent.

Table 3.1 Functional cost categories, their drivers and the 2003 US average rates (adapted from Belobaba, 2006; historical values), with their treatment in Laurentian's cost model.

Functional category	Driver	2003 US average	Laurentian's model
Aircraft operating	block hours	USD 3,415 per block hour (757-200)	c_{BH} by type
Aircraft servicing	departures	USD 1,135 per departure	c_{dep} by type and region
Traffic servicing	passengers enplaned	USD 18 per passenger	in c_{pax}
Passenger service	revenue passenger miles	USD 0.015 per RPM	in c_{pax} by region
Reservations and sales	revenue	10% of revenue	netted from fares
Overhead	total cost	12% of operating expense	excluded

The same 757-200 cost USD 2,924 per block hour at one airline and USD 4,308 at another (Belobaba 2006e): a cost per block hour borrowed from another airline, or another year, is a starting point at best.

Collapsing the table into three drivers gives the cost model used throughout this book. For a departure with block time BH carrying N passengers, the **trip cost** is

$$\text{TC} = c_{\text{BH}}\,\text{BH} + c_{\text{dep}} + c_{\text{pax}}\,N, \tag{3.1}$$

where c_{BH} is the cost per block hour of the aircraft type (fuel, crew, variable maintenance, navigation charges), c_{dep} the cost per departure (landing and terminal charges, handling) and c_{pax} the cost per passenger (catering, distribution, passenger fees). The first two terms do not depend on how many people board; together they are the **capacity cost** of the departure. Laurentian's parameters are in Table 3.2. Ownership and overhead are left out on purpose: the model measures the cost a departure causes at the seasonal planning horizon (Section 3.5).

Table 3.2 Laurentian's illustrative cost parameters (CAD); departure costs for domestic and transborder / international flights. Passenger costs are CAD 20 (domestic), 30 (transborder) and 50 (transatlantic). Ranges are headline figures (Chapter 9).

Type	Seats	Per block hour	Per departure, dom. / intl	Range (nm)
A220-300	137	8,000	2,000 / 3,000	3,400
737-8	169	9,000	2,500 / 3,500	3,500
A321XLR	182	11,000	2,500 / 3,500	4,700
787-9	298	15,500	5,000 / 7,000	7,500

3.2 Why unit cost falls with stage length

Part of every flight's cost does not depend on how far it goes. Landing and handling charges are paid once per departure, and taxiing, climb and descent burn block hours that carry nobody closer to their destination. On a long flight these fixed amounts are spread over many seat-kilometres; on a short flight they dominate. To see this in numbers, approximate block time by a straight line in distance:

$$\mathrm{BH}(d) = \tau_0 + \frac{d}{s}, \tag{3.2}$$

where d is the stage length in kilometres, τ_0 a fixed allowance per flight (taxi, the slow part of climb and approach) and s the effective cruise speed. For Laurentian, $\tau_0 = 0.65$ hours and $s = 850$ km/h. The line is a smoothing device, not a timetable: real block times differ by direction with the winds (Chapter 9), so worked examples on the case's routes use their scheduled times.

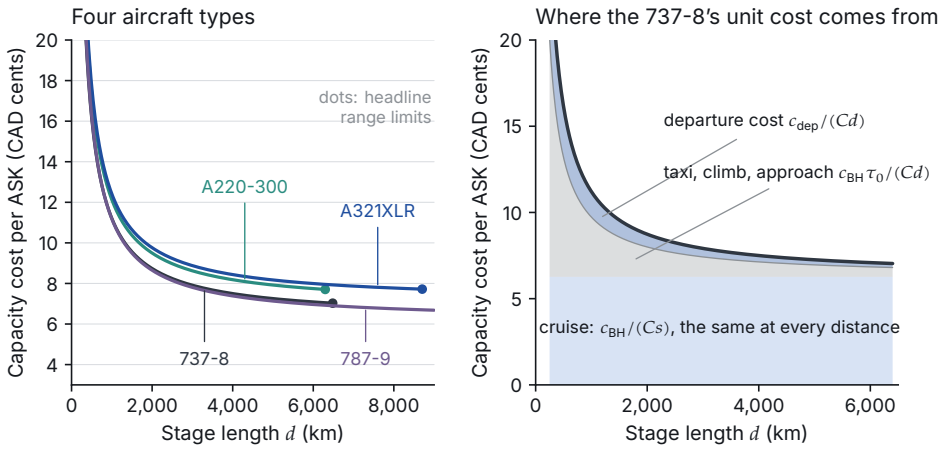


Figure 3.1 Left: capacity cost per ASK of Laurentian's four types against stage length (domestic departure costs, block time $0.65 + d/850$ hours); dots mark headline range limits. Right: the 737-8's unit cost split into the cruise cost b and the two distance-independent costs per departure, spread over d.

RESULT 3.1 Unit cost and stage length

An aircraft with C seats, cost c_{BH} per block hour and c_{dep} per departure has, on a stage of d km, a capacity cost per available seat-kilometre of

$$\text{CASK}(d) = \frac{a}{d} + b, \qquad a = \frac{c_{\text{dep}} + c_{\text{BH}}\tau_0}{C}, \qquad b = \frac{c_{\text{BH}}}{Cs}. \quad (3.3)$$

The first term is the distance-independent cost per seat (departure charge and fixed block time), spread over the distance; the second is the cruise cost per seat-kilometre. CASK falls with stage length toward b. When distance rises by 1 percent, CASK falls by a percentage equal to the share of capacity cost that does not depend on distance.

The derivation is two lines of algebra (Appendix A.1). Passenger costs add a term of the same shape, because a passenger is checked in and fed once per flight, not once per kilometre.

Figure 3.1 draws (3.3) for Laurentian's fleet. The term a, the fixed cost of a departure per seat, is similar across types (CAD 49 to 53); the types differ in the cruise cost b: 6.12 cents per ASK for the 787-9, 6.27 for the 737-8, 6.87 for the A220-300 and 7.11 for the A321XLR. The XLR is the dearest aircraft per seat-kilometre in the fleet, the

price of a low-density cabin and of the fuel and structure that give a narrowbody transatlantic range. Whether that price is worth paying is a question about trip cost and demand (Section 3.3).

WORKED EXAMPLE 3.1 One aircraft, two stage lengths

Fly Laurentian's 737-8 from Montreal to Toronto (507 km) and to Vancouver (3,682 km). The block-time line gives 1.246 and 4.982 hours, so capacity costs are $9{,}000 \times 1.246 + 2{,}500 = 13{,}714$ and $9{,}000 \times 4.982 + 2{,}500 = 47{,}338$ dollars. Dividing by 169×507 and $169 \times 3{,}682$ seat-kilometres gives CASKs of 16.0 and 7.6 cents. The Vancouver flight is 7.3 times as long, costs 3.5 times as much, and has less than half the unit cost. At 500 km, 61 percent of the capacity cost does not depend on distance, so 10 percent more distance lowers unit cost by about 6 percent; at the Vancouver distance the share is 18 percent and the fall under 2 percent.

The lesson: unit cost is partly geometry. A short-haul route is not inefficient because its CASK is high, and a network that lengthens its average stage lowers its CASK without becoming more efficient. CASK comparisons must first remove the stage-length effect (Chapter 2).

Geometry is not the whole story: in the lecture's 2003 data, US low-fare carriers sat 3 to 6 cents per ASM below the legacy carriers at every stage length (Belobaba 2006e), a gap that productivity and labour cost explain (Section 3.4).

3.3 Aircraft size: trip cost versus seat cost

Larger aircraft are cheaper per seat. Two pilots fly a 298-seat 787-9 as they fly a 137-seat A220; a departure needs one slot, one gate and one turn whatever the gauge; fuel burn grows less than proportionally with size. But a larger aircraft costs more per trip, and the trip cost is paid whether or not the extra seats fill.

DEFINITION 3.2 Trip cost, seat cost and cost per passenger

For one departure, the **trip cost** TC is the cost of operating it, the **seat cost** TC/C is the cost per seat offered, and the **cost per passenger** is the trip cost divided by the expected number of passengers carried (the expected load of Chapter 7).

The handbook's example: aircraft A costs CAD 80,000 per trip and has 180 seats; aircraft B costs CAD 105,000 and has 280 seats (Network Planning Handbook 2026). B is cheaper per seat (CAD 375 against 444), A is cheaper per trip, and neither is preferable until revenue, demand and the opportunity cost of the aircraft are known. B's 100 extra seats cost CAD 25,000 more per trip, a marginal cost of $\beta = 250$ dollars per added seat, and the planner must judge whether they will fill often enough.

RESULT 3.2 When the larger aircraft pays

Let each passenger contribute r (net fare less passenger costs), and let a larger aircraft add seats at a marginal trip cost of β per seat. Moving up to the larger aircraft raises expected contribution if and only if the added seats are filled, on average over departures, with probability at least β/r.

Each added seat earns r when it fills and costs β whether or not it does. This is the rule for the last seat in Chapter 7, applied to a block of seats; how often the seats fill depends on the whole distribution of demand, which is why the spill model is needed to apply it.

WORKED EXAMPLE 3.2 The handbook's two aircraft with uncertain demand

Let demand per departure be normal with mean μ and $k = 0.30$, and let each passenger contribute CAD 400. At $\mu = 200$, the spill formula of Chapter 7 gives expected loads of 164.7 on A and 197.5 on B: B's extra 100 seats carry 32.7 more passengers, an average fill probability of 0.327, well short of $\beta/r = 250/400 = 0.625$. B loses $400 \times 32.7 - 25{,}000 = -11{,}920$ dollars per departure against A. Per passenger carried, A costs CAD 486 and B 532 (Figure 3.2). B starts to win at $\mu = 256$ when $r = 400$, at 215 when $r = 600$, and only at 330 when $r = 300$:

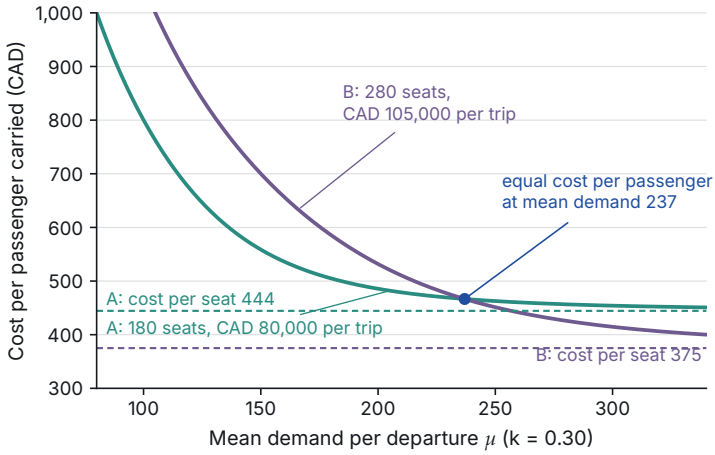


Figure 3.2 Cost per passenger carried for the handbook's aircraft A (180 seats, CAD 80,000 per trip) and B (280 seats, CAD 105,000), normal demand with $k = 0.30$. Dashed lines: costs per seat. B is always cheaper per seat, but cheaper per passenger only above a mean demand of about 237.

the more each passenger is worth, the less often the extra seats need to fill.

Seat cost is a property of the aircraft; cost per passenger is a property of the aircraft in a market. At a seven-hour block, Laurentian's A321XLR has the dearest seat in the fleet (CAD 442) and the 787-9 the cheapest (388), yet with 150 passengers a day the XLR's CAD 80,500 trip easily beats the 787-9's CAD 115,500 (Chapter 9).

3.4 Utilization and productivity

Some costs accrue with the calendar, not with flying: an aircraft's lease or depreciation, much of the pay of salaried staff, rent and insurance. The more an airline flies its assets, the thinner these costs are spread (Belobaba 2006e).

DEFINITION 3.3 Utilization and productivity

Aircraft utilization U is the number of block hours an aircraft flies per day. **Aircraft productivity** is the capacity it produces per day, in ASKs per aircraft-day: departures times average stage length times seats, or equivalently utilization times block speed (stage length divided by block time) times seats.

If an aircraft's ownership cost is O dollars a year, its ownership cost per block hour is $O/(365\,U)$: raising utilization by 10 percent cuts it by 9.1 percent. How much an aircraft can fly depends on its schedule.

RESULT 3.3 Utilization of a shuttling aircraft

An aircraft flying sectors of block time t separated by turns of length g, within an operating day of H hours, needs n block times and $n-1$ turns for n sectors, so

$$U = t\left\lfloor \frac{H+g}{t+g} \right\rfloor \approx \frac{(H+g)\,t}{t+g}. \tag{3.4}$$

Longer sectors and shorter turns raise utilization; because departures come in whole numbers, a few minutes of turn time can cost a whole sector.

In Figure 3.3, an aircraft flying one-hour sectors with 40-minute turns fits 11 sectors into the day, 11.0 block hours; with five-hour sectors it fits three, 15.0 block hours. A 787-9 owned for CAD 19.2 million a year costs CAD 3,757 per block hour at 14 hours a day and CAD 5,260 at 10.

WORKED EXAMPLE 3.3 Same aircraft, two airlines

The lecture compares the Boeing 737-500 at three US airlines in the early 2000s (Belobaba 2006e). Take two of them. Southwest's flight operating cost was 3.89 cents per seat mile, United's 8.44, and the ratio of 2.17 splits into two factors. United's cost per block hour was 1.95 times Southwest's (USD 2,974 against 1,526), driven by crew rates (927 against 388), maintenance (1,048 against 251) and ownership. Southwest produced 1.11 times as many seat miles per block hour, almost

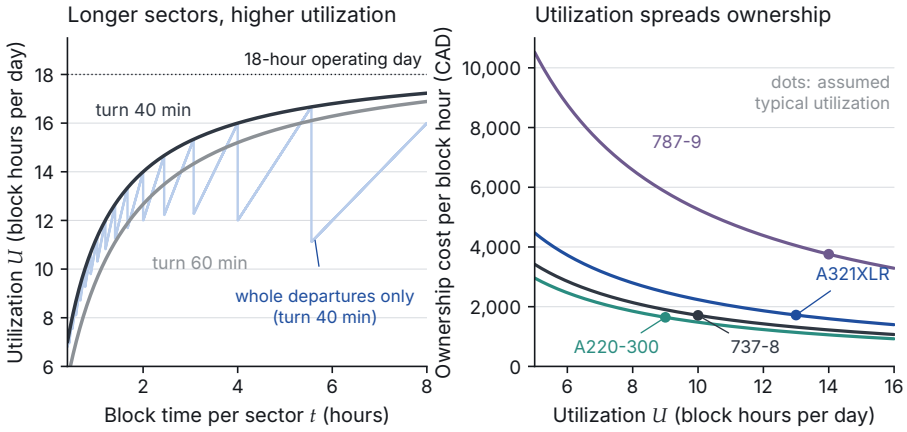


Figure 3.3 Left: utilization in an 18-hour day against sector length, for 40- and 60-minute turns; the pale sawtooth counts whole departures (40-minute turn). Right: ownership cost per block hour against utilization, with illustrative monthly ownership costs of CAD 0.45 million (A220-300), 0.52 million (737-8), 0.68 million (A321XLR) and 1.60 million (787-9).

entirely through seats (122 against 109). Utilization explains most of the ownership gap: Southwest flew its aircraft 10.2 block hours a day against 7.5, and paid 350 per block hour for ownership against 510.

The lesson: a cost per block hour is a property of an airline's whole operation (labour contracts, maintenance practice, seating density and, above all, how hard the aircraft is worked), not of the aircraft type.

Air Canada's second quarter of 2026 gives the same measures (Air Canada 2026d): 100,300 departures in the 91-day quarter, 14.6 million seats and 351 passenger aircraft at the end of June (mainline, Rouge and Express) make 146 seats per departure, 3.14 departures per aircraft-day, an average stage of 1,840 miles and about 843,000 ASMs per aircraft-day.

3.5 Which costs count: decision horizons

A cost figure answers a question only once the question says what is being decided and over what period. Cancelling a flight next Tuesday saves its fuel and landing fees, but not the pilots' salaries or the aircraft lease. Dropping a route from next summer's schedule saves crew costs too, because staffing is planned for the season. Shrinking the fleet saves the lease, but only when the lease can be ended.

DEFINITION 3.4 Fully allocated, avoidable and incremental cost

For a decision with horizon h:

- the **fully allocated cost** of a flight is its share of all the airline's operating costs, including overhead assigned by an accounting rule;
- its **avoidable cost** is the cost the airline would not incur during h if the flight were not operated; costs already contracted and not recoverable within h are **committed**;
- the **incremental cost** of a decision is the change in the airline's total cost that it causes, wherever it falls, including on other flights;
- the **opportunity cost** of a scarce resource (an aircraft, a slot, a crew) is the value of its best alternative use.

Table 3.3 adapts the handbook's cost buckets (Network Planning Handbook 2026) to three horizons: days, the seasonal schedule, and the fleet plan of several years.

Table 3.3 Which costs a flight can avoid at each horizon (adapted from the handbook). "Partly" marks costs with a variable and a fixed part, such as hotels within a fixed crew roster.

Cost bucket	Main drivers	Days	Season	Years
Fuel	block hours, weight, winds, routing	yes	yes	yes
Crew	duty time, pairings, bases, agreements	partly	yes	yes
Airport and navigation	movements, weight, passengers, distance	yes	yes	yes
Maintenance	flight hours, cycles, scheduled checks	partly	mostly	yes
Passenger service and distribution	passengers, cabin, channel	yes	yes	yes
Station and launch	staff, contracts, marketing	no	partly	yes
Aircraft ownership or lease	fleet commitments, lease terms	no	no	when leases end
Corporate overhead	shared functions	no	no	partly

RESULT 3.4 Avoidable cost grows with the horizon

Any commitment that can be undone within a short horizon can be undone within a longer one, so a flight's avoidable cost never falls as the horizon lengthens. At a given horizon, operate a flight if its revenue exceeds its avoidable cost and its contribution is at least that of the best alternative use of the same aircraft, crew and slots.

The handbook puts the consequence in two sentences worth memorizing: an accounting loss is not sufficient reason to cancel a flight whose costs are mostly unavoidable and whose contribution is positive, and a positive contribution is not sufficient reason to launch one if a better use of the aircraft exists or longer-run costs go uncovered (Network Planning Handbook 2026).

The same departure can be worth cancelling at one horizon and worth operating at another. A winter Seville departure loses about CAD 11,200 against its seasonal avoidable cost (Section 3.6): when the winter schedule is built, dropping it saves money before network effects. A week before departure its crew is rostered and paid either way, its avoidable cost shrinks to fuel, fees and passenger costs, and operating it may be the cheaper choice. At the fleet horizon the lease becomes avoidable too.

Laurentian's trip-cost model (equation (3.1)) sits at the seasonal horizon. The opportunity cost of the aircraft, often the largest cost a new route imposes, is not in the model at all; it enters through the comparison of alternatives, as displacement and network contribution in Chapter 10.

Before using any cost figure, ask which decision it is for and over what horizon: that decides which costs are avoidable, and the best alternative use of the aircraft decides the rest.

PITFALL

Applying system-average CASK to an incremental route. Take a fully allocated system CASK of 15.0 cents (illustrative). Applied to one A321XLR departure to Seville (182 seats, 5,556

km), it gives CAD 151,700, and the flight appears to lose CAD 50,700 with 150 passengers at CAD 650 (fare and ancillaries) plus CAD 3,500 of cargo; its avoidable cost is CAD 88,000, and it contributes CAD 13,000. On the A220 from Quebec City (233 km), the same average gives CAD 4,800 against an avoidable cost of CAD 11,600 with 120 passengers. The average errs in both directions: it carries costs the flight cannot avoid, and it embeds a stage length neither route has.

3.6 Laurentian's cost model at work

The trip-cost model gives the break-even load of a departure directly. Let $\mathrm{TC}_0 = c_{\mathrm{BH}}\mathrm{BH} + c_{\mathrm{dep}}$ be its capacity cost, f the average net fare allocated to the leg, a the ancillary revenue per passenger and K other contribution such as cargo.

RESULT 3.5 Break-even load from the trip-cost model

The departure breaks even with

$$N^* = \frac{\mathrm{TC}_0 - K}{f + a - c_{\mathrm{pax}}}$$

passengers, a **break-even load factor** of N^*/C: capacity cost net of cargo, divided by what each passenger brings net of the cost of carrying them.

This is the precise version of the CASK-over-yield shortcut of Chapter 2: passenger costs are netted from the fare instead of being folded into CASK. Costed with the 2003 rates of Table 3.1, a 1,000-mile, 185-seat 757 flight with 140 passengers at USD 150 breaks even at 68.5 percent by the shortcut, but at 64.4 percent once passenger costs are separated.

Table 3.4 Block hours (BH), capacity cost, CASK on capacity cost, average net fare (CAD) and break-even load factor on five Laurentian routes (illustrative). Seville includes ancillary and cargo revenue.

Route	Aircraft	BH	Capacity cost	CASK (¢)	Fare	Break-even LF
YQB–YUL	A220-300	0.9	9,200	28.8	201	37.1%
BOS–YUL	A220-300	1.4	13,200	23.6	246	44.6%
YUL–YVR	737-8	5.2	49,300	7.9	420	72.9%
YUL–CDG	787-9	6.9	113,950	6.9	675	61.2%
YUL–SVQ	A321XLR	7.0	80,500	8.0	620	70.5%

WORKED EXAMPLE 3.4 Trip costs and break-even loads on five Laurentian routes

Table 3.4 applies Result 3.5 with the running case's scheduled block times and fares. For Seville the case gives an allocated net fare of CAD 620, ancillary revenue of CAD 30 per passenger and cargo contribution of CAD 3,500 per departure. Capacity cost is $11{,}000 \times 7.0 + 3{,}500 = 80{,}500$. Each passenger brings $620 + 30 - 50 = 600$. Net of cargo the flight must cover 77,000, so $N^{*} = 77{,}000/600 = 128.3$ passengers, a load factor of 70.5 percent. With 150 passengers the trip cost is $80{,}500 + 50 \times 150 = 88{,}000$, the case's avoidable cost, and the departure contributes $150 \times 650 + 3{,}500 - 88{,}000 = 13{,}000$.

Break-even load factors span 37 to 73 percent, low on short routes, whose fares are high per kilometre, and high on long ones. They are thresholds for each route, not performance measures to compare across routes.

Demand is uncertain, so the threshold must be compared with the expected load, not with mean demand. Seville demand per departure is $\mathcal{N}(175, 42^2)$ in the peak, $\mathcal{N}(150, 40^2)$ in the shoulder and $\mathcal{N}(110, 35^2)$ in winter. The spill formula of Chapter 7 gives expected load factors of 88.7, 79.8 and 60.3 percent, and contributions per eastbound departure of CAD 19,900, 10,100 and $-11{,}200$ (Figure 3.4). A winter departure loses money before any network effect is counted.

The figure shows why break-even load factors understate risk near capacity. Below about 130 passengers spill is rare and the two lines coincide. Above it, each extra unit of mean demand adds less than one passenger, so a strong peak earns less than a mean-demand cal-

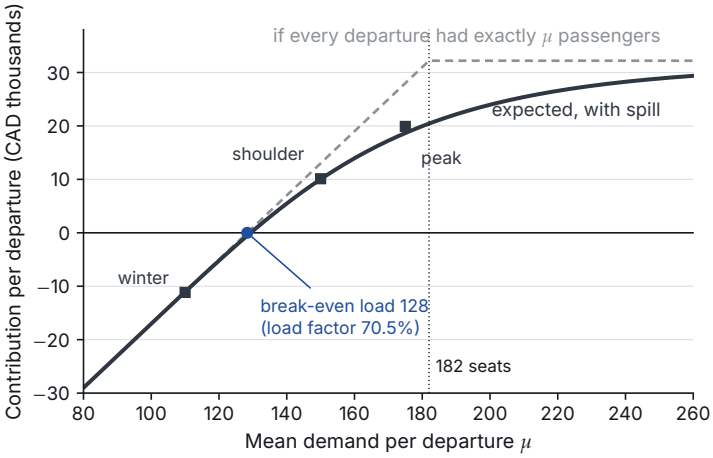


Figure 3.4 Expected contribution of an eastbound YUL–SVQ departure (A321XLR) against mean demand (CAD 600 per passenger, CAD 77,000 capacity cost net of cargo, $k = 0.27$). Dashed: every departure carries exactly its mean demand; solid: with spill. Squares mark peak, shoulder and winter, each with its own standard deviation (the peak's k-factor of 0.24 puts it slightly above the curve).

culation suggests, while a weak winter's losses are not cushioned at all.

3.7 Air Canada's cost structure in 2026

Air Canada reports its costs in the objective style, by line of expense, with two unit measures: CASM, operating expense per available seat mile, and adjusted CASM, which removes fuel, the ground package costs of Air Canada Vacations, freighter costs and special items (Air Canada 2026d). Table 3.5 reproduces the second quarter of 2026 with the driver each line mainly follows.

Table 3.5 Air Canada's operating expenses in the second quarter of 2026 (CAD millions; 26,942 million ASMs, against 26,860 million a year earlier). Amounts and total CASM from the MD&A (Air Canada 2026d), line names shortened; cents per ASM by line computed here; drivers are this book's interpretation.

Expense line	CAD M	Share	¢/ASM 2026	¢/ASM 2025	Main driver
Wages and benefits	1,740	26.8%	6.46	4.64	hours, departures, staff
Aircraft fuel	1,713	26.4%	6.36	4.27	block hours, price
Other	660	10.2%	2.45	2.25	handling, crews, rent
Depreciation and amortization	546	8.4%	2.03	1.83	fleet, time
Airport and navigation	446	6.9%	1.66	1.52	departures, weight
Aircraft maintenance	364	5.6%	1.35	1.27	hours, cycles
Sales and distribution	293	4.5%	1.09	1.02	passengers, revenue
Capacity purchase fees	232	3.6%	0.86	0.82	regional flying
Catering and onboard	180	2.8%	0.67	0.64	passengers
Ground package costs	164	2.5%	0.61	0.58	vacation sales
Information technology	143	2.2%	0.53	0.56	overhead
Total operating expense	6,481	100%	24.05	19.41	

Three lessons stand out. Fuel and labour are each about a quarter of the cost base, and fuel is the volatile one: fuel expense rose 49 percent on 0.2 percent more consumption, because the price per litre rose 51 percent despite a CAD 205 million hedging gain. Unit cost moved for reasons unrelated to the schedule: CASM rose 23.9 percent on 0.3 percent more capacity, and without CAD 360 million of one-time pension and separation charges and a CAD 28 million legal provision it would have been 22.62 cents (Air Canada 2026d). And ground packages and capacity purchase fees are not costs of Air Canada's own flying.

In kilometre terms, Air Canada's CASM was 14.9 cents per ASK. Comparing it with Laurentian's 8 cents of capacity cost on the Seville

flight is the pitfall of Section 3.5: one is fully allocated and averaged over a system with a 1,840-mile stage; the other is the avoidable cost of one long flight.

IN PRACTICE **Cost inputs to a planning model in 2026**

The fuel price was the dominant cost story of 2026. Air Canada attributes its 51 percent increase in fuel price per litre in the second quarter to conflict in the Middle East and disruption to shipping through the Strait of Hormuz (Air Canada 2026d); IATA's June 2026 outlook put fuel at about 31 percent of industry costs (IATA 2026d). When Air Canada reinstated its 2026 guidance in August, it assumed jet fuel at about CAD 1.38 per litre for the third quarter, against about CAD 0.90 in its original plan (Air Canada 2026c). Fuel should therefore enter route models as a scenario input, priced by period and burned by direction and season, not as a fixed share of cost. The handbook adds that the definitions of avoidable cost at each horizon belong to Finance, so the first question about any cost number is which costs it treats as avoidable (Network Planning Handbook 2026).

3.8 Summary

- Planning uses the functional view of costs, because each function has a driver. A flight costs $c_{\text{BH}}\text{BH} + c_{\text{dep}} + c_{\text{pax}}N$; separating what the departure causes from what each passenger causes is the first step of any break-even.
- Capacity cost per ASK is $a/d + b$: the fixed cost of a departure is spread over the stage, so unit cost falls with distance. Stage-adjust before comparing CASKs.
- Larger aircraft cost less per seat and more per trip. A larger gauge pays only if its added seats fill, on average, with probability at least their marginal cost divided by the contribution per passenger.
- Utilization spreads ownership and other calendar costs; it rises with sector length and falls with turn time.
- Avoidable cost grows with the decision horizon. Fully allocated and system-average costs answer accounting questions, not route questions; the aircraft's opportunity cost enters through its best alternative use.
- Laurentian's Seville flight breaks even at 128 passengers and loses money in a winter with mean demand of 110. Air Canada's 2026

CASM moved with fuel prices and one-time charges, not with the schedule.

3.9 Further reading

Belobaba's lecture (Belobaba 2006e) is the source of the cost taxonomy, the drivers and the 737-500 comparison. Chapter 6 of Belobaba, Odoni and Barnhart (2015) treats operating costs at textbook length, and Doganis (2019, chs. 3–4) from a commercial perspective. The handbook (Network Planning Handbook 2026) sets out the cost buckets by horizon. The full reference edition derives the stage-length elasticity, the upgauge condition and the utilization formula in its Chapter 3; Appendix A.1 gives the first.

PART II

Demand

Every plan begins with an estimate of who wants to travel, between which places, when, and at what price. This part builds that estimate: the economics of air travel demand, how travellers choose among the itineraries on offer, how demand is forecast for existing markets and new routes, and what happens when demand exceeds the seats on a flight of fixed size.

CHAPTER

4

Air Travel Demand

IN THIS CHAPTER

- Why demand lives in origin-destination markets while capacity is sold one flight leg at a time.
- What makes a market large, and how business, leisure and VFR travellers differ.
- Price and time elasticities in words and numbers: what a fare change does to revenue, and what an added frequency does to the market.
- Seasonality, and why planners never average across it.
- Market size versus market share, and the S-curve: why a carrier's share of passengers rises faster than its share of flights.
- The quality of service index and the traffic a new nonstop creates, worked through for Montreal–Seville.

DRAWS ON Belobaba, Airline economics review (MIT 16.75J, 2006); Belobaba, Introduction to PODS passenger choice model (MIT 16.75J, 2006); Boeing, Load factor analysis (1978); Boeing, Decision window path preference methodology (1993); handbook

A network planner never sells a seat, but every decision a planner makes is a forecast of who will buy one. Opening Montreal–Seville, adding a tenth daily flight to Toronto or moving the Paris departure by an hour are bets on how many people want to travel, what they will pay, and which itinerary they will choose once the competition has reacted.

Two ideas organize this chapter. Demand is generated by markets, while airlines supply flights, and one flight serves many markets. And an airline's traffic in a market is the market's size, which responds to price and travel time, times the airline's share, which responds to frequency, timing and quality relative to competitors. Chapter 5 then explains how travellers choose among itineraries, Chapter 6 projects demand forward, and Chapter 7 turns demand into passengers carried.

4.1 Markets, itineraries and legs

A market can be defined between two cities, two airports or two regions with their secondary airports (Belobaba 2006d). Montreal's traffic is concentrated at one airport, so for most of Laurentian's markets the three coincide; for London or New York they do not.

DEFINITION 4.1 O&D market, itinerary and leg demand

An **origin-destination (O&D) market** is the set of people who wish to travel from origin A to destination B in a given period, counted in one-way passenger trips, often as **passengers per day each way (PDEW)**. An **itinerary** (or path) is a sequence of flight legs, with times, that takes a passenger from A to B. **Leg demand** is the number of passengers who want a seat on a given flight leg, summed over every itinerary that uses it:

$$D_l = \sum_{j \in J} a_{lj} D_j, \tag{4.1}$$

where D_j is the demand for O&D product j and $a_{lj} = 1$ if product j uses leg l.

Each direction is a separate market, with its own travellers, seasons and points of sale. PDEW hides a trap. The handbook's example is a market of 73,000 annual passengers in both directions: divided by 365 days and two directions, 100 PDEW, yet a new daily flight should not expect 100 passengers on every departure, because demand varies by season and day of week and is shared among itineraries and carriers (Network Planning Handbook 2026).

Belobaba calls this the **dichotomy of demand and supply** (Belobaba 2006d): demand is generated market by market, while each leg departure supplies many markets jointly. On his example Boston–Houston flight, most passengers continued beyond Houston to more than twenty destinations. So the seats on a flight are not the supply to any one market, and a market's profitability cannot be stated without allocating costs and revenues by assumption (Chapter 10).

Table 4.1 Mean unconstrained demand per departure in Laurentian's four-leg bank at YUL, by O&D market (rows) and leg (columns). Each leg's demand is a column sum, as in (4.1). Illustrative.

O&D market	YQB–YUL	BOS–YUL	YUL–CDG	YUL–SVQ
YQB–YUL	73			
BOS–YUL		70		
YUL–CDG			215	
YUL–SVQ				112
YQB–CDG	48		48	
YQB–SVQ	20			20
BOS–CDG		54	54	
BOS–SVQ		29		29
Leg demand	141	153	317	161
Seats	137	137	298	182

Table 4.1 shows the dichotomy on Laurentian's bank. The YUL–CDG leg has mean demand of 317 passengers per departure for 298 seats, but only 215 of them, 68 percent, are Montreal–Paris travellers; the rest connect from Quebec City and Boston. Only about half of each feeder's demand is local. Remove the Boston flight and the Paris flight loses 54 passengers of demand, although nothing changed in the Montreal–Paris market.

> **PITFALL**
>
> Reading a leg's load as the demand of its city pair. A full YUL–CDG flight says that the leg is constrained; it says nothing direct about the Montreal–Paris market, part of which flies other carriers and paths, while much of the flight's own demand belongs to other markets.

4.2 What drives demand

Air travel is a derived demand: people fly to do business, take a holiday or see family, and a trip is taken only if its purpose is worth more than its total cost. The determinants fall into four groups (Belobaba 2006d):

- **market characteristics**: population and income at both ends, business ties, tourist appeal, and migration links that generate visits to friends and relatives;
- **the price** of travel, meaning the fares actually available;
- **the time** the journey takes from door to door, including the wait imposed by the schedule;

- **the product**: frequency and departure times, stops and connections, aircraft, cabin, reliability and loyalty programs, each relative to competitors.

The first group sets the scale of a market and changes slowly; air travel grows faster than income, with income elasticities around 1.5 for short-haul routes in developed economies and higher for long-haul travel (Smyth and Pearce 2008). The other three are what airlines control.

Travellers weigh these factors differently, and planners group them into segments. **Business** travellers make trips required by their work: the employer usually pays, the date and time are tied to a meeting, and the trip is rarely cancelled because of the fare. **Leisure** travellers pay their own money, choose whether, when and where to go, and accept an inconvenient schedule for a lower fare. **Visiting friends and relatives (VFR)** travellers sit between the two: price sensitive like leisure travellers, but tied to a destination and often to fixed dates such as holidays and family events. A Montreal–Paris flight carries all three.

The workhorse model of market demand is multiplicative (Belobaba 2006d):

$$D = M\,P^{\eta_P}\,T^{\eta_T}, \tag{4.2}$$

where D is O&D demand per period, M a market-size constant, P the average fare, T the total trip time, and $\eta_P < 0$ and $\eta_T < 0$ the price and time elasticities, constant along the curve. Each segment has its own version, because the elasticities differ.

4.3 Price elasticity

DEFINITION 4.2 Price elasticity of demand

The **price elasticity** of demand, η, is the percentage change in demand caused by a 1 percent increase in price, other things equal. Demand is **inelastic** when $-1 < \eta < 0$ and **elastic** when $\eta < -1$.

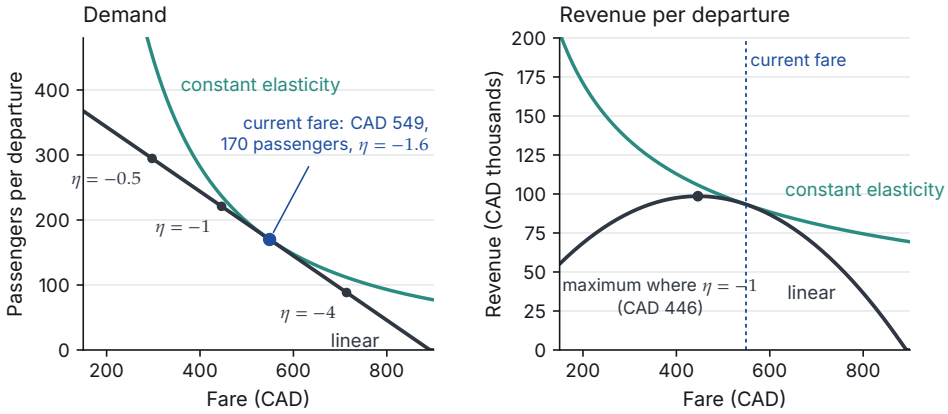


Figure 4.1 Linear and constant-elasticity demand calibrated to the same point, Laurentian's YUL–CDG restricted fare of CAD 549 with 170 passengers per departure and elasticity -1.6. Left: demand. Right: revenue per departure; the linear model peaks at CAD 446, while the constant-elasticity model says every fare cut raises revenue.

RESULT 4.1 Elasticity and revenue

A 1 percent fare increase changes revenue by about $1 + \eta$ percent, so a small fare increase raises revenue if and only if demand is inelastic. With constant-elasticity demand, as in (4.2), the elasticity is the same at every fare; with linear demand it grows in absolute value as the fare rises, and revenue peaks where $\eta = -1$.

Figure 4.1 is a warning about extrapolation. The two curves agree at the current fare and disagree far from it, and neither is right by construction. An elasticity is a local property of a market at its current fares; Belobaba's rule of thumb, that a 10 percent fare increase changes demand by ten times the elasticity in percent, is meant for such small changes (Belobaba 2006d).

WORKED EXAMPLE 4.1 A 10 percent fare increase on YUL–CDG

Laurentian sells two products on YUL–CDG: a flexible fare of CAD 1,149 with mean demand of 45 passengers per departure, mostly business travellers, and a restricted fare of CAD 549 with mean demand of 170, mostly leisure travellers. Use the lecture's illustrative elasticities, -0.8 and -1.6, with constant elasticity.

Flexible fare. At CAD 1,263.90, demand falls to $45 \times 1.1^{-0.8} = 41.70$, 7.3 percent fewer (the rule of thumb says 8). Revenue per departure rises from CAD 51,705 to CAD 52,700, by 1.9 percent.

Restricted fare. At CAD 603.90, demand falls to $170 \times 1.1^{-1.6} = 145.96$, 14.1 percent fewer (16 by the rule of thumb). Revenue falls from CAD 93,330 to CAD 88,143, by 5.6 percent.

The planner learns three things. Different segment elasticities are the economic case for different prices (Chapter 14). The size of the revenue effect depends on the functional form: a linear model puts the leisure loss at 7.6 percent. And on a flight that spills, a fare increase loses fewer carried passengers than it loses demand (Chapter 7).

Why segments differ Business travellers are less price elastic than leisure travellers: someone else pays, the trip has a fixed purpose and date, and there are few substitutes for being in Paris on Tuesday. Leisure travellers can change destination, date or mode, or stay home. Belobaba's lecture, summarizing studies available in 2006, puts typical market elasticities between −0.8 and −2.0, with business demand slightly inelastic (Belobaba 2006d). IATA's review reports median estimates from a survey for Canada's Department of Finance by Gillen, Morrison and Stewart (2004): about −0.3 for long-haul international business travel, −0.7 for short-haul business, −1.0 for long-haul international leisure and −1.5 for short-haul leisure (Smyth and Pearce 2008). Short-haul travel is more elastic because car and rail compete with it.

Market, route and carrier elasticities An elasticity depends on what is held fixed. IATA's study recommends −1.4 for a fare change on a single route, −0.8 across a whole country (a national tax) and −0.6 across a group of countries, with a multiplier of 1.2 for transatlantic markets (InterVISTAS Consulting 2007): a fare increase on one route diverts travellers to other routes and airports, and the diversion nets out as the change becomes general.

A single carrier's elasticity is larger again, because a carrier that raises its fare alone also loses share to competitors. Roughly, it is the carrier's share times the market elasticity, plus the rest times the elasticity of a carrier too small to move the market. With a market elasticity of −1.0, a 40 percent share and −3 for a small carrier (Chapter 5), that is $0.4 \times (-1.0) + 0.6 \times (-3) = -2.2$. Pricing uses the carrier's elasticity; forecasting market growth uses the market's.

PITFALL

Importing one elasticity into every decision. The handbook warns against carrying one price elasticity, or one frequency-share curve, from market to market (Network Planning Handbook 2026). An elasticity belongs to a market, a segment, a level of aggregation and a price range, and estimates from fare data are biased toward zero unless the analyst handles the fact that fares respond to demand (Chapter 5).

4.4 Time, frequency and schedule delay

Next to price, the total time a trip takes is the most important determinant of demand (Belobaba 2006d), and frequency enters through one of its components.

DEFINITION 4.3 Total trip time and schedule delay

The **total trip time** from origin to destination is

$$T = t_{\text{fixed}} + t_{\text{flight}} + t_{\text{sd}},$$

where t_{fixed} covers access, egress and processing at both airports, t_{flight} is block time plus any connection time, and t_{sd} is the **schedule delay**: the gap between when the traveller would like to leave and when a flight actually leaves.

Schedule delay is what frequency buys. Part of it is imposed by the timetable; part is suffered when the preferred flight is full, which is spill seen from the passenger's side (Douglas and Miller 1974). Belobaba models the first part as a constant divided by daily frequency (Belobaba 2006d), and the constant has a simple meaning. If preferred departure times are spread evenly over an operating day of H hours and the f flights are evenly spaced, the average traveller is a quarter of a headway, $H/(4f)$, from the nearest flight.

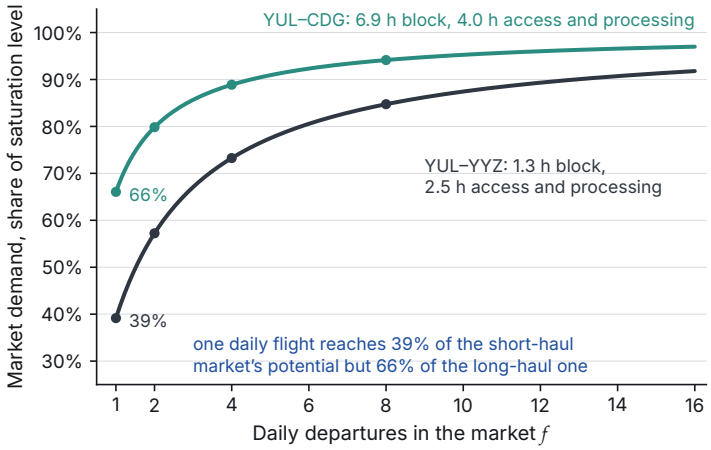


Figure 4.2 Market demand as a share of its saturation level (demand with zero schedule delay) as daily frequency grows, from (4.2) with $\eta_T = -1.2$ and $K = 4.5$ hours. The short-haul market needs many flights to approach its potential; the long-haul market is close to it with two or three.

RESULT 4.2 Schedule delay and the frequency elasticity of demand

With evenly spread preferences and evenly spaced flights, schedule delay is K/f with $K = H/4$; an 18-hour day gives $K = 4.5$ hours. If demand follows (4.2), the percentage change in market demand per 1 percent more frequency is

$$\frac{\partial \ln D}{\partial \ln f} = |\eta_T| \, \frac{K/f}{T}, \tag{4.3}$$

the absolute time elasticity times the share of schedule delay in total trip time.

Appendix A.2 gives the derivation. Frequency grows the market by cutting schedule delay, most on short-haul routes, where delay is a large share of the trip; and several connections through a hub can serve the average traveller better than one daily nonstop, because the saving in schedule delay can exceed the extra connection time. Saturation is built in: each added flight saves less waiting than the one before, so beyond some point extra flights only redistribute the market.

WORKED EXAMPLE 4.2 One more flight to Toronto or to Paris

Take a time elasticity of $\eta_T = -1.2$, in the middle of the historical range of -0.8 to -1.6 that Belobaba reports, and $K = 4.5$ hours. For YUL–YYZ assume 2.5 hours of access and processing and 1.3 hours of block time; for YUL–CDG, 4.0 and 6.9 hours.

Toronto. With 8 daily flights in the market, $T = 2.5 + 1.3 + 4.5/8 = 4.36$ hours, of which schedule delay is 12.9 percent. The frequency elasticity is $1.2 \times 0.129 = 0.155$. A ninth flight cuts T to 4.30 hours and raises market demand by $(4.30/4.36)^{-1.2} - 1 = 1.7$ percent.

Paris. With one daily flight, $T = 4.0 + 6.9 + 4.5 = 15.4$ hours, of which schedule delay is 29 percent, so the elasticity is 0.351. A second daily flight cuts T to 13.15 hours and raises demand by 20.9 percent.

The comparison is between frequencies, not between short and long haul: the second Paris flight is worth more than the ninth Toronto flight, while a second Toronto flight would raise that market by 46 percent. Eastbound transatlantic flights must leave in the evening, so their window is short and frequency matters even less than the model suggests (Chapter 11). Below daily service, a traveller who shifts to the nearest of n evenly spread weekly flights loses on average $7/(4n)$ days, 10.5 hours for four a week; leisure travellers shift trips by a day at little cost, which is why long-haul leisure routes can start below daily.

Generalized cost and the two segments Price and time combine into the **generalized cost** of a trip, $G = P + vT$ plus the cost of other inconveniences (fare restrictions, connections, unwanted departure times), where v is the traveller's value of time. The handbook lists the same ingredients as the basis of passenger choice (Network Planning Handbook 2026), and the MIT passenger simulator PODS books each simulated passenger on the path and fare with the lowest fare plus disutility costs (Belobaba 2006g).

Generalized cost explains why the segments trade price against time differently: the price elasticity is the generalized-cost elasticity times the fare's share of G, and the time elasticity the same elasticity times the share of time cost. Suppose, for illustration, that business travellers value time at CAD 90 an hour and leisure travellers at CAD

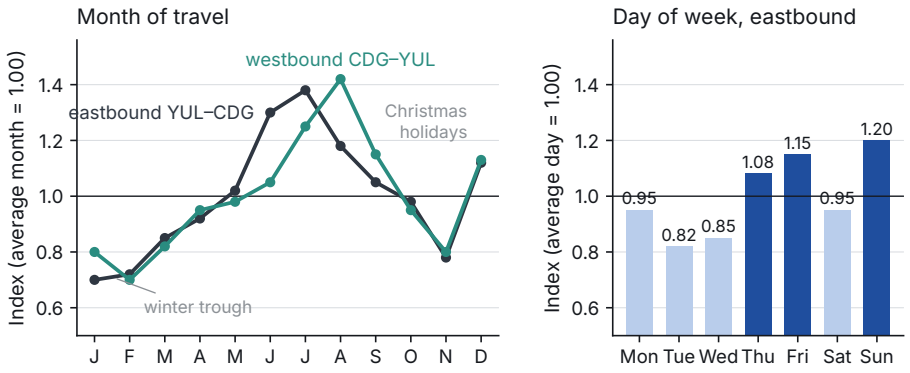


Figure 4.3 An illustrative demand profile for Laurentian's Montreal–Paris market. Left: monthly indices by direction; eastbound demand peaks in July as Montreal residents leave on holiday, westbound in August as they return and French visitors arrive. Right: the eastbound day-of-week index. The numbers are invented for teaching.

25, both with a generalized-cost elasticity of −1.6 on YUL–CDG with one daily flight ($T = 15.4$ hours). On the CAD 1,149 fare, a business traveller's G is CAD 2,535, of which the fare is 45 percent: price elasticity −0.73, time elasticity −0.87. On the CAD 549 fare, a leisure traveller's G is CAD 934, of which the fare is 59 percent: −0.94 and −0.66. Leisure trips are also more discretionary, with a larger generalized-cost elasticity, which pushes their price elasticity further out. The result is Belobaba's summary: business demand price inelastic but time elastic, leisure demand the reverse (Belobaba 2006d).

4.5 Seasonality and the weekly cycle

Demand for "the Paris flight" is not one number. It moves with the month, the day of the week, the time of day and the direction of travel, and jumps around holidays. Boeing's 1978 guide to load factor analysis insists that data be separated along these lines, one direction and one season at a time, before any statistics are computed (Boeing Commercial Airplane Company 1978). The planner answers predictable variation with the schedule, varying frequency and gauge by season and day; revenue management absorbs part of the random variation that remains; spill models price the rest (Chapter 7). A convenient model is multiplicative: expected demand on a day is average daily demand times a month index times a weekday index, each averaging one.

Figure 4.3 shows three features that matter for planning. The swing is large: if an average departure has demand of 250 passengers, a Sunday in July has $250 \times 1.38 \times 1.20 = 414$ and a Tuesday

in January $250 \times 0.70 \times 0.82 = 144$, a ratio of almost three. No single aircraft fits both, which is why airlines change frequency and gauge between the IATA summer and winter seasons (Chapter 11). The directions differ: the eastbound peak comes a month before the westbound one, so the binding direction changes during the summer. And averages mislead: an annual average PDEW or load factor mixes departures that spill heavily with departures that fly half empty, precisely the information capacity decisions need.

4.6 Market size, market share and the S-curve

An airline's traffic in a market is the product of two quantities:

$$D_a = \mathrm{MS}_a \times D_M,$$

where D_M is total market demand and MS_a is airline a's market share. Market size responds to the general level of fares and to the best trip time available. Share responds to an airline's offer relative to its competitors: frequency, departure times, path quality, fares and product (Belobaba 2006d). Frequency acts on both.

> **DEFINITION 4.4 Frequency share and market share**
>
> In a market served by nonstop carriers $b = 1, \ldots, n$ with daily frequencies f_b, the **frequency share** of carrier a is $\mathrm{FS}_a = f_a / \sum_b f_b$, and its **market share** MS_a is its fraction of the market's passengers.

The rule of thumb is that market share is approximately equal to frequency share. The empirical pattern is sharper, an **S-curve**: carriers with a high frequency share win a disproportionately high market share, and carriers with a low frequency share a disproportionately low one (Belobaba 2006d).

RESULT 4.3 The S-curve

Other things equal, carrier a's market share is

$$\mathrm{MS}_a = \frac{\mathrm{FS}_a^{\alpha}}{\sum_b \mathrm{FS}_b^{\alpha}}, \tag{4.4}$$

with an exponent α generally between 1.3 and 1.7 in the lecture's historical evidence. Equivalently, the ratio of two carriers' passengers is the ratio of their frequencies raised to the power α. In a duopoly the curve passes through equal shares at equal frequencies, with slope α there.

The ratio form is the easiest to use: with $\alpha = 1.5$, a carrier with twice its competitor's frequency carries 2.8 times its passengers, not twice as many.

If each added flight were merely another copy of the same product, share would grow in proportion to frequency ($\alpha = 1$). An exponent above one means that each flight becomes more attractive when the carrier has more of them. A carrier with more flights covers more of the time windows in which travellers will leave, and when only one carrier has a flight inside a traveller's window it wins that traveller outright (Boeing Commercial Airplane Group 1993). Frequent travellers value the option to change flights, and loyalty programs and corporate contracts favour the dominant carrier at a city. None of these is equally strong everywhere, so α belongs to a market. Formally, the S-curve is a logit choice model (Chapter 5) with utility rising in log frequency (Appendix A.2).

Market size cancels from the ratio of two carriers' shares, so a regression of log passenger ratios on log frequency ratios across markets recovers α without knowing any market's size (Figure 4.4, right: 1.49 against a true 1.5).

PITFALL

Treating frequency as if it were set at random. Carriers add flights where they expect to be strong, so a naive regression credits their unobserved advantages to frequency and produces an S-curve steeper than the truth (1.70 where the true value is 1.5, in the full reference edition's illustration). Every

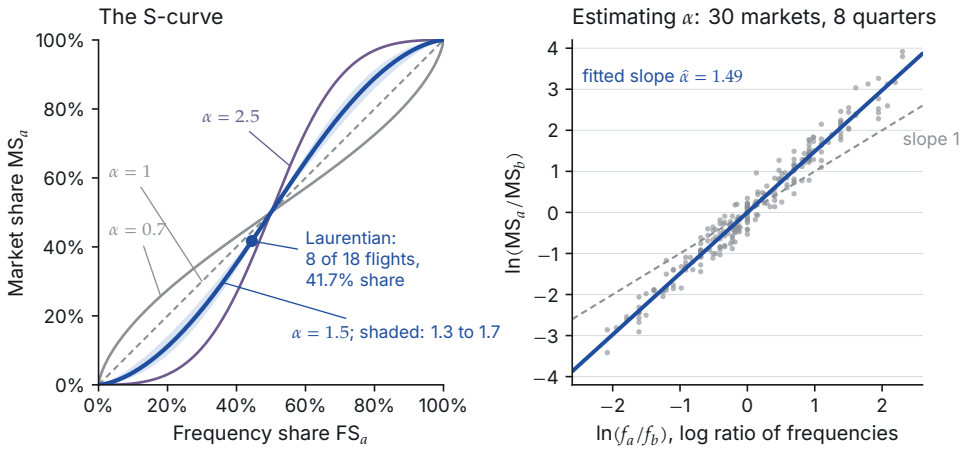


Figure 4.4 Left: the S-curve for a duopoly. The diagonal is $\alpha = 1$; the shaded band covers the lecture's range of 1.3 to 1.7, with 1.5 in cobalt; $\alpha = 2.5$ shows a steeper S and $\alpha = 0.7$ the inverse shape. Right: how α is measured, by regressing log passenger ratios on log frequency ratios across markets, here in a simulated panel of 30 duopoly markets over 8 quarters with true $\alpha = 1.5$.

added frequency then looks more valuable than it is. Ask how any S-curve parameter you are handed was estimated.

WORKED EXAMPLE 4.3 Frequency competition on YUL–YYZ

Suppose the Montreal–Toronto market has 2,400 PDEW, Laurentian operates 8 daily flights against a competitor's 10, and $\alpha = 1.5$. Laurentian's frequency share is $8/18 = 44.4$ percent and its market share is

$$\frac{0.444^{1.5}}{0.444^{1.5} + 0.556^{1.5}} = 41.7 \text{ percent},$$

1,001 PDEW, or 125 per daily flight.

Laurentian adds two flights and matches the competitor at 10. Its share rises to 50 percent, and the market grows by 0.75 percent to 2,418 PDEW because schedule delay falls (Result 4.2, with the Toronto parameters of Example 4.2). Laurentian carries 1,209 PDEW and the competitor falls from 1,399 to 1,209, so 190 of Laurentian's 208 extra passengers come from the competitor and only 18 from market growth. Each added flight brings 104 PDEW, less than the 125 average of the existing

ones but well above the 71 that the proportional rule ($\alpha = 1$) would give.

Now the competitor responds with two flights of its own, to 12. Laurentian's frequency share falls to 45.5 percent and its market share to 43.2 percent, so with a market of 2,433 PDEW it carries 1,051. Its two extra flights now add 50 PDEW between them, a quarter of the first estimate, and both carriers fly more seats in a market that has barely grown. The planner's question is never only what a frequency is worth today, but what it is worth after the competitor's best response.

The example is a small game. If two carriers split a market's revenue by the S-curve and each adds flights until the last one no longer pays, in the symmetric equilibrium they end up together spending a fraction $\alpha/2$ of the market's revenue on frequency: 65 to 85 percent for α between 1.3 and 1.7 (Appendix A.2). Hence Belobaba's observation that competitors match frequencies in nonstop markets to protect their shares (Belobaba 2006d), taken up in Chapter 11.

4.7 Quality of service and the traffic a new nonstop creates

The S-curve handles competing nonstops. Most markets are also served by connections, and a planner needs a share model that compares a daily one-stop through Madrid with a four-times-weekly nonstop.

DEFINITION 4.5 Quality of service index

A **quality of service index (QSI)** gives each itinerary i in a market a score q_i from its service level (nonstop, one-stop, online or interline connection), adjusted for elapsed time, departure time, aircraft and carrier presence, and multiplied by its number of departures. Each itinerary's predicted share is its score divided by the sum of all scores, $q_i / \sum_k q_k$; a carrier's share is the sum over its itineraries.

A QSI model is a logit share model whose weights are calibrated by judgement or from past launches rather than estimated. It inherits the logit's transparency and its main weakness: a new itinerary draws from every existing one in proportion to its share (Chap-

ter 5). Coldren and coauthors present logit itinerary share models estimated on booking data as the improvement on QSI (Coldren et al. 2003), but the structure survives in modern planning systems (see the box at the end of the chapter).

How much is a connection worth relative to a nonstop? Boeing's decision window methodology offers coefficients that depend on a path quality index and on trip length (Table 4.2) (Boeing Commercial Airplane Group 1993). At 1,000 miles an online connection is worth less than 4 percent of a nonstop; at 6,000 miles, more than three quarters. Two extra hours matter much less on a ten-hour journey.

Table 4.2 Service-based coefficients by path quality index (PQI) and trip length in statute miles (adapted from Boeing, 1993; historical). PQI 1 is a nonstop, 2 a one-stop on the same aircraft, 3 an online connection, 4 an interline connection; 5 adds further stops or time.

PQI	1,000 mi	2,000 mi	3,000 mi	4,000 mi	5,000 mi	6,000 mi
1	100.00	100.00	100.00	100.00	100.00	100.00
2	19.45	33.86	49.82	65.28	78.28	87.73
3	3.78	11.46	24.82	42.62	61.28	76.96
4	0.74	3.88	12.36	27.82	47.97	67.52
5	0.14	1.31	6.16	18.16	37.55	59.23

Market size with an outside option A share model divides a fixed market. To predict how much a new nonstop enlarges it, the model needs an outside option, the choice not to travel; without one it can only redistribute (Network Planning Handbook 2026). With air travel and the outside option in separate nests (Chapter 5), market demand is proportional to the total score of all itineraries raised to a power λ. New itineraries that win a share s raise the total score by the factor $1/(1-s)$, which gives a compact rule.

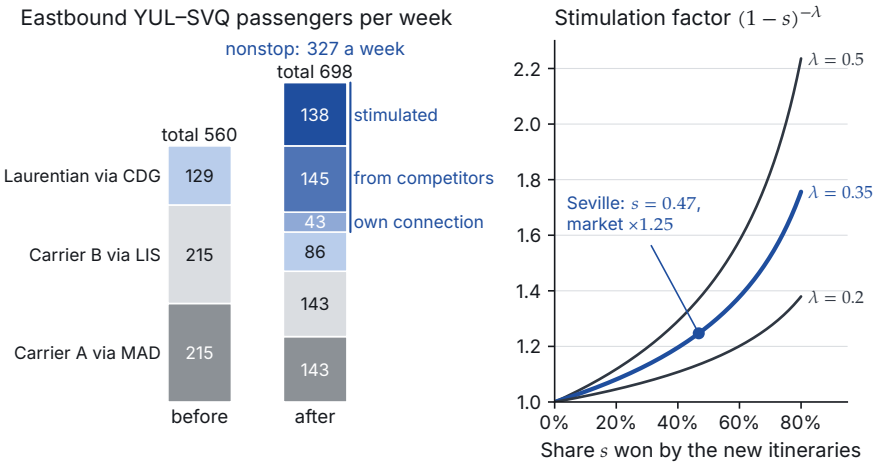


Figure 4.5 Left: the Montreal–Seville market before and after Laurentian's four-times-weekly nonstop, eastbound passengers per week; the nonstop's 327 passengers are stimulated, taken from competitors' connections, or taken from Laurentian's own connection over Paris. Right: the stimulation factor $(1-s)^{-\lambda}$ against the share won, for three values of λ.

RESULT 4.4 Stimulation by a new itinerary

If new itineraries win a share s of the enlarged market, total market demand grows by the factor

$$\frac{D_1}{D_0} = (1-s)^{-\lambda}, \tag{4.5}$$

where $\lambda \in (0, 1]$ measures how readily a better offer turns non-travellers into travellers. The new itineraries carry sD_1 passengers: $D_1 - D_0$ are **stimulated** (new to the market) and the rest are **diverted** from existing itineraries, each losing in proportion to its previous share.

A nonstop stimulates a market in proportion to how much better it is than what exists, measured by the share it wins. One that takes 10 percent of a market served by good connections barely changes it; one that takes 60 percent of a market served by poor connections can enlarge it by a third or more. The parameter λ is the hardest number in the analysis to know.

WORKED EXAMPLE 4.4 Adding a nonstop to Seville

Laurentian's YUL–SVQ mirrors a service publicly announced for launch on 8 September 2027, three times a week and four times a week from October (Network Planning Handbook 2026); every number below is invented. Before the nonstop, the market is served only by connections: assume 560 eastbound passengers a week (80 PDEW) on three daily itineraries, Carrier A over Madrid and Carrier B over Lisbon (online connections) and Laurentian over Paris with a partner's flight to Seville (an interline connection).

Scores. YUL–SVQ is 3,453 statute miles. Interpolating Table 4.2 gives about 33 for an online connection and 19 for an interline one; penalizing the long connections, take 25, 25 and 15 points per departure, against 100 for the nonstop. Weekly scores are $7\times25 = 175$ for each competitor and $7\times15 = 105$ for Laurentian, a total of 455, so the pre-launch shares are 38.5, 38.5 and 23.1 percent: 215, 215 and 129 passengers a week.

Share and stimulation. The nonstop adds $4 \times 100 = 400$ points, so its share of the enlarged market is $s = 400/855 = 46.8$ percent. With $\lambda = 0.35$, (4.5) gives a factor of $(1 - 0.468)^{-0.35} = 1.247$: the market grows to 698 passengers a week (99.8 PDEW), and the nonstop carries $0.468 \times 698 = 327$, or 82 per departure.

Sources. Of the nonstop's 327 passengers, 138 are stimulated (42 percent), 145 come from the two competitors and 43 from Laurentian's own connection over Paris. Laurentian's total traffic in the market rises from 129 to about 412 a week, an increase of 283, not 327.

Sensitivity. With $\lambda = 0.2$ the market grows by 13.5 percent and the nonstop carries 297 a week (74 per departure); with $\lambda = 0.5$, by 37.1 percent and 359 (90 per departure).

The planner learns three things. Local demand of 82 per departure is less than half of the 182 seats, so the route depends on connecting traffic over Montreal (Chapters 8 and 10); the 112 of Table 4.1 is larger because the four-leg bank's YUL–SVQ market also stands for connections from spokes outside it. The 43 passengers taken from Laurentian's own connection free seats on the over-demanded Paris flight, so their value is the displacement cost of those seats, not zero (Chapter 10). And λ should come from comparable past launches (Chapter 6).

The handbook asks that a new route case show these three sources on separate lines (Network Planning Handbook 2026), because their values differ: stimulated travellers are new revenue, travellers won from competitors are revenue the competitors will fight for, and travellers moved from the airline's own itineraries are worth only the capacity they release.

IN PRACTICE **How planners size markets and shares in 2026**

Market sizes now come from commercial datasets. OAG's air traffic product estimates O&D demand from global distribution system bookings, with hub-flow breakdowns into local, behind, beyond and bridge traffic (OAG n.d.). Cirium's traffic and fares product estimates monthly O&D passengers for more than 600 airlines from over 24 sources, fills gaps such as low-cost carriers' direct sales by regression and scaling, and is released about 55 days after the end of a month (Cirium 2026a). The handbook advises treating such totals as estimates rather than a census (Network Planning Handbook 2026); by May 2026 about one in five US travel agency transactions was settled through NDC channels (Travel Distribution News 2026), outside traditional booking data. Lufthansa Systems describes its NetLine/Plan system as chaining demand and market share, spill and recapture, and cost and revenue models (Lufthansa Systems 2026), the sequence of this chapter and Chapter 7.

4.8 Summary

- Demand is generated by O&D markets; capacity is supplied by legs, each serving many markets. A leg's load says little about any one market.
- Demand depends on income, price, trip time and the product. Business, leisure and VFR travellers weigh these differently, and demand varies predictably by season, weekday and direction; planners answer that variation with the schedule and never average it away.
- A small fare increase raises revenue only where demand is inelastic. Business demand is less price elastic and more time elastic than leisure demand; a single carrier's elasticity exceeds the market's.

- The frequency elasticity of demand is the time elasticity times the share of schedule delay in trip time, so frequency matters most where trips are short and frequency is low.
- Traffic is market size times market share. The S-curve gives the carrier with more flights a more than proportional share; a frequency must be valued after the competitor's response.
- A new nonstop that wins share s multiplies the market by $(1 - s)^{-\lambda}$; its traffic splits into stimulated passengers, passengers won from competitors and passengers moved from the airline's own itineraries, each with a different value.

4.9 Further reading

Belobaba's airline economics review (Belobaba 2006d) is the source of the demand model, trip time and the S-curve; Boeing's decision window methodology (Boeing Commercial Airplane Group 1993) is the clearest account of why frequency and timing win share. The Global Airline Industry (Belobaba, Odoni, and Barnhart 2015, ch. 3) gives a textbook treatment, and IATA reviews the elasticity evidence (Smyth and Pearce 2008; InterVISTAS Consulting 2007). The full reference edition derives the elasticity results, the S-curve and the stimulation rule, and discusses how α is estimated, in its Chapter 4; Appendix A.2 gives the S-curve and schedule-delay derivations.

CHAPTER

5

How Passengers Choose

IN THIS CHAPTER

- How a traveller's comparison of itineraries becomes a market share: random utility and the logit formula.
- What the coefficients mean to a planner: the value of time, the price of a nonstop or a better departure, and elasticities.
- Why departures of one airline, or at one time of day, are closer substitutes than logit assumes, and how nested logit and business-leisure heterogeneity correct it.
- Decision windows and the PODS passenger: how planning and revenue management simulators represent choice.
- What planning tools do with choice models, from the quality of service index to nested logit.
- Why fare elasticities read naively from airline data are wrong, and what to ask of any elasticity.

DRAWS ON Train, Discrete choice methods with simulation (2009), chapters 2, 3, 4, 6 and 13; Boeing, Decision window path preference methodology (1993, MIT 16.75J lecture notes); Belobaba, Introduction to PODS passenger choice model and ePODS educational game (MIT 16.75J, 2006); Belobaba, Airline economics review (MIT 16.75J, 2006); handbook; Air Canada, Q2 2026 MD&A; Competition Bureau (2025)

Almost every proposal a network planner makes changes the menu that travellers choose from. Moving Laurentian's evening departure to Paris by an hour, adding a second daily frequency, opening a nonstop to Seville or watching a competitor launch one: each changes which itineraries exist and what they offer in fare, elapsed time, departure time and stops. Chapter 4 sized markets and split them with the frequency-share curve and the quality of service index. To predict how a particular change moves passengers

between itineraries, between airlines, and between flying and not flying, the planner needs a model of the individual decision.

That model is the logit family of discrete choice models, the language of itinerary choice in airline planning (Coldren et al. 2003; Garrow 2010) and, for an industrial economist, McFadden's conditional logit (McFadden 1974). A planner meets it inside the share model of a planning system, the recapture rates of a spill estimate (Chapter 7) and the passenger simulators of revenue management (Chapter 14). This chapter leaves estimation to the full reference edition and keeps the mechanism, what its parameters mean in dollars, where its assumptions bite, and why fare elasticities from airline data need special care.

5.1 Random utility and the logit share

A traveller who wants to fly from Montreal to Seville compares the itineraries she can buy on fare, elapsed time, stops and connections, how far each departure is from the time she wanted, the airline (its loyalty program, its reliability) and the product (cabin, fare family, aircraft). She also has an **outside option**: not booking any of them, which covers travelling another day, on another airline, or not at all. The analyst sees most of this list and little else of what drives her: her elite status, the seat map, the meeting she cannot move.

> **DEFINITION 5.1 Random utility model**
>
> Traveller n chooses the alternative j, an itinerary or the outside option, with the highest utility $U_{nj} = V_{nj} + \varepsilon_{nj}$. The **representative utility** V_{nj} is computed from observed attributes, usually as a weighted sum $\beta_f f_{nj} + \beta_T T_{nj} + \dots$ of the fare f, the elapsed time T and others, with one **coefficient** β per attribute. The term ε_{nj} is everything the analyst does not observe, treated as random. The **choice probability** P_{ni} is the probability that i has the highest utility: among travellers who face the same observed options, the share who choose i.

The random part is not measurement noise; it is the part of travellers' tastes the model does not explain (Train 2009, ch. 2). Because a choice reveals only which alternative ranks first, **only differences in utility matter**: utilities are measured relative to the outside option, set to zero. And **the scale is arbitrary**: each coefficient is measured relative to the unobserved noise, so coefficients from different data

sets cannot be compared. Their ratios can, and Section 5.2 turns them into dollars.

Logit assumes that the unobserved terms follow the extreme-value (Gumbel) distribution, close to a normal but giving a closed form, and are independent across alternatives. The independence is what matters: a traveller's particular liking for the 18:30 is assumed to say nothing about how she feels about the 20:45 (Section 5.3 shows the cost).

RESULT 5.1 Logit choice probabilities

If the unobserved terms are independent extreme-value variables, the probability that traveller n chooses alternative i from her choice set C_n is

$$P_{ni} = \frac{e^{V_{ni}}}{\sum_{j \in C_n} e^{V_{nj}}}. \tag{5.1}$$

The probabilities lie strictly between 0 and 1 and sum to 1. Appendix A.3 derives the formula.

With two alternatives, (5.1) is an S-shaped function of the utility difference. Figure 5.1 draws it for a traveller choosing between Laurentian's nonstop and a competitor's nonstop at the same time of day, with a Laurentian preference worth 0.25 units of utility and a fare coefficient $\beta_f = -0.004$ per dollar. At equal fares Laurentian wins 56.2 percent of such travellers, and they are indifferent when Laurentian charges CAD 62.50 more $(0.25/0.004)$. Near that point each CAD 100 of fare difference moves 10 points of share; far from it the curve is flat. Price and schedule changes matter most where alternatives are close.

WORKED EXAMPLE 5.1 Shares in the Montreal–Seville market

Before Laurentian's nonstop, the Montreal–Seville market of Chapter 4 has three daily connections: Carrier A over Madrid and Carrier B over Lisbon, both online, and Laurentian over Paris with a partner's flight, an interline connection. Take the representative utility of a departure, relative to a nonstop departure, as -1.39 for an online connection and -1.90 for the

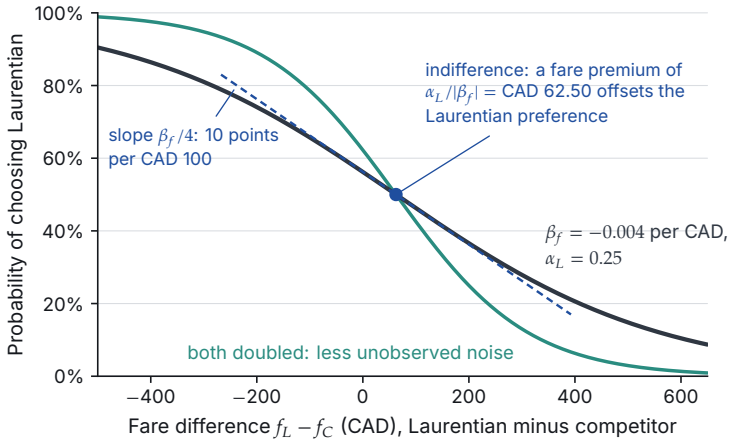


Figure 5.1 The binary logit curve: probability that a traveller chooses Laurentian's nonstop over a competitor's nonstop at the same time of day, as a function of the fare difference. The dashed tangent has slope $\beta_f/4$. The teal curve doubles both coefficients, which is what a smaller variance of unobserved utility does; the indifference point does not move, because it depends only on the ratio of the coefficients.

interline one. Then e^V is 1, 0.25 and 0.15: the relative scores 100, 25 and 15 that Chapter 4 uses.

Before. Each competitor's seven weekly departures sum to $7 \times 0.25 = 1.75$ and Laurentian's to $7 \times 0.15 = 1.05$. The denominator of (5.1) is $1.75 + 1.75 + 1.05 = 4.55$: each competitor has 38.5 percent of the market and Laurentian 23.1 percent.

After. Four weekly nonstops add $4 \times 1 = 4$, and the denominator becomes 8.55. The nonstop takes 46.8 percent, each competitor keeps 20.5 percent and the Paris connection 12.3 percent; Laurentian's total rises from 23.1 to 59.1 percent.

Reading. Every connection lost the same fraction of its share, 46.8 percent, because the common denominator scales them all down together: the nonstop draws from Laurentian's own connection exactly as from the competitors'. And with no outside option the market cannot grow; Chapter 4 adds one to estimate stimulation.

5.2 What the coefficients mean to a planner

Coefficients are measured in units of unobserved noise, which no one can interpret. Their ratios are measured in dollars.

DEFINITION 5.2 Willingness to pay and the value of time

When the fare enters utility with coefficient $\beta_f < 0$, the **money value** of attribute k is $v_k = -\beta_k/\beta_f$: the fare change that leaves utility unchanged when the attribute rises by one unit. The **value of time** is VOT $= \beta_T/\beta_f$ in dollars per hour.

Dividing utility by $-\beta_f$ turns it into money. Apart from constants such as a brand preference, the traveller compares the **generalized cost** of each itinerary,

$$\text{fare} + \text{VOT} \times \text{elapsed time} + v_d \times \text{displacement} + v_c \times \text{connections},$$

where the displacement is the gap between the departure and the time she wanted. Logit says that travellers pick the itinerary with the lowest generalized cost plus a random term, the quantity the handbook asks planners to think in (Network Planning Handbook 2026). Displacement is the individual form of the schedule delay of Chapter 4 (Koppelman, Coldren, and Parker 2008).

WORKED EXAMPLE 5.2 What travellers to Paris pay for time, a nonstop and a brand

The full reference edition fits a logit model to 4,000 simulated, illustrative shopping sessions for evening travel from Montreal to Paris. Each session offers Laurentian's 18:30 nonstop, competitor B's 20:45 nonstop, carrier C's 17:15 one-stop via London Heathrow and no booking. With an estimated fare coefficient of −0.0040 per dollar, the ratios give an hour of elapsed time worth CAD 57.0 (95 percent interval CAD 38 to 76, against a true CAD 62.50), an hour of departure displacement CAD 50.3, a connection CAD 226 on top of its longer elapsed time, and the Laurentian brand CAD 67.5.

A typical connection is 4.4 hours slower than a nonstop. At the same departure time it must therefore be about $4.4 \times 57 + 226 \approx$ CAD 476 cheaper to be equally attractive, and CAD 67.5 more than that to match Laurentian's own nonstop. Values of time are imprecise: with 500 sessions instead of 4,000, the spread of the estimate triples. Ask for intervals.

Elasticities Take a representative traveller on Montreal–Paris with utilities 0 for not booking, 0.40 for Laurentian's 18:30 at CAD 760, 0.25 for B's 20:45 at CAD 720 and −0.60 for C's connection at CAD

540. The exponentials 1, 1.492, 1.284 and 0.549 sum to 4.325, so the shares are 23.1, 34.5, 29.7 and 12.7 percent. The elasticities of the probability of i with respect to its own fare and to the fare of another alternative j are

$$\eta_{ii} = \beta_f f_i (1 - P_i), \qquad \eta_{ij} = -\beta_f f_j P_j \quad (j \neq i). \tag{5.2}$$

The factor $1 - P_i$ means that an itinerary that already wins most travellers has few left to win, so it is less elastic: at a 34.5 percent share, Laurentian's own-fare elasticity is $-0.004 \times 760 \times 0.655 = -1.99$. The cross elasticity is the same for every other alternative: a 1 percent cut in Laurentian's fare takes 1.05 percent of the probability of the competitor's nonstop and of not travelling alike. When all fares rise together, diversion between itineraries cancels, and the **market elasticity** is

$$\eta_{\text{air}} = \beta_f \bar{f} P_0, \tag{5.3}$$

where $\bar{f}$ is the expected fare of those who book and P_0 the probability of not booking: $-0.004 \times 708 \times 0.231 \approx -0.65$, a third of the itinerary elasticity. A model without an outside option implies a market that never grows or shrinks (Network Planning Handbook 2026), and the outside share is the least well measured number in airline data, because booking records contain no one who did not book.

WORKED EXAMPLE 5.3 What is a better departure time worth?

Use the Paris model of Example 5.2 to move Laurentian's nonstop from 18:30 to 19:30, everything else unchanged (Figure 5.2). Averaged over the 4,000 sessions, Laurentian's share rises from 35.2 to 35.8 percent, B's falls from 30.1 to 29.6 percent and no booking from 25.5 to 25.3 percent; 19:30 maximizes Laurentian's share. With 600 such sessions a day, consistent with the roughly 215 local passengers a day that the running case assigns to Laurentian's Paris flight, Laurentian gains 4.0 bookings and CAD 2,940 of revenue a day, about CAD 1.07 million a year. Only 1.1 of the 4.0 extra bookings are new to the market.

Travellers gain CAD 2.31 per session, about CAD 0.51 million a year, measured by the logsum, the dollar value of the best option on offer (Appendix A.3). Laurentian's revenue gain is twice the travellers' gain because most of it is a transfer from rivals.

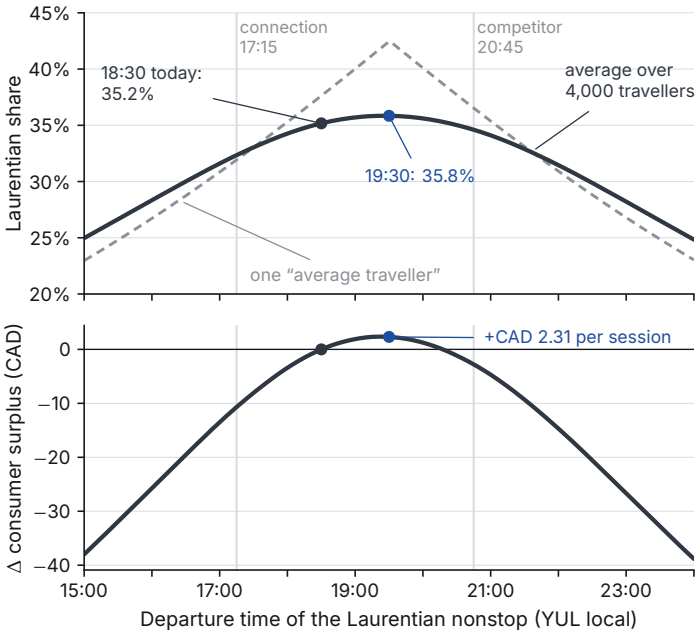


Figure 5.2 Retiming Laurentian's YUL–CDG nonstop, with the competitor at 20:45 and the connection at 17:15 unchanged. Top: Laurentian's share computed for the 4,000 sessions of Example 5.2 and averaged (dark), and computed for a single traveller with average preferences and attributes (grey). Bottom: change in expected consumer surplus per session relative to the current 18:30 departure, computed from the logsum.

PITFALL

Forecasting with an average traveller. The grey curve in Figure 5.2 applies the model once, to a traveller with the average preferred time and average fares. It predicts a gain of 4.8 points from the retiming instead of 0.67, because it puts every traveller's preferred time at 19:30. The probability at the average is not the average probability: apply a choice model to a sample of travellers or to segments, never to an average traveller.

5.3 Close substitutes: IIA and nested logit

The denominator of (5.1) is shared by every alternative, and that single fact fixes how logit substitutes.

RESULT 5.2 Independence of irrelevant alternatives

In the logit model the ratio of the probabilities of two alternatives, $P_{ni}/P_{nk} = e^{V_{ni}-V_{nk}}$, does not depend on any other alternative. A new alternative takes from each existing one the same fraction of its share, equal to the newcomer's own share.

This **independence of irrelevant alternatives** (IIA) is reasonable when all alternatives are equally different. Airline choice sets are full of close substitutes: two departures of one airline share its loyalty program and cabin, two late-evening departures share their fit with the working day, two connections share a hub. When one member of such a group improves, it draws disproportionately from its group.

PITFALL

Counting near-duplicates as alternatives. A Laurentian flight sold under a partner's code appears twice in a booking display; entered twice in a logit model, it receives two shares of the denominator, and its predicted share rises although nothing changed. Collapse codeshares, channel duplicates and near-identical connections to operated alternatives, or nest them.

WORKED EXAMPLE 5.4 A second daily frequency to Paris

Laurentian adds a 21:15 nonstop to the Paris market of Section 5.2, as convenient for the representative traveller as its 18:30, so its utility is also 0.40. The denominator rises from 4.325 to 5.816, and every existing share is multiplied by $4.325/5.816 = 0.744$. Laurentian's total share rises from 34.5 to 51.3 percent, a gain of 16.8 points, and the new flight takes 34.5 percent of its passengers from the 18:30, 29.7 percent from B, 12.7 percent from the connection and 23.1 percent from no booking.

Look at what the model assumed. The 21:15 leaves half an hour after B's 20:45 and nearly three hours after Laurentian's own flight: a close substitute for both nonstops and a poor one for not travelling. Logit nevertheless says that almost a quarter of its passengers are new to the market. Whether a second frequency mostly cannibalizes Laurentian's own flight or mostly

takes passengers from the competitor is the question the planner is paid to answer, and logit answers it by assumption.

Nested logit relaxes the assumption by grouping alternatives that share unobserved attributes into **nests**. Within a nest the alternatives are close substitutes; between nests, proportional substitution still holds. Picture a traveller choosing a nest first (Laurentian or a competitor, early or late evening, flying or not) and then an alternative within it. Each nest has a **dissimilarity parameter** λ between 0 and 1. At 1 the model collapses to logit; near 0 the members are perfect substitutes and the nest behaves like its best member. An improvement in one member draws mainly from its nest-mates (formulas in Appendix A.3). Airline research nests itineraries by time of day, carrier and level of service, and lets the data decide which structure fits (Coldren and Koppelman 2005).

WORKED EXAMPLE 5.5 The second frequency to Paris, revisited

Calibrate three nested models, with $\lambda = 0.5$ in every nest for illustration, to reproduce the same base shares exactly, and add the 21:15 again (Figure 5.3).

- **Air against no air.** Stimulation halves, from 23 to 13 percent of the new flight's passengers, and Laurentian gains 15.1 points.
- **Nests by carrier.** 62 percent of the new flight's passengers come from Laurentian's 18:30, and Laurentian gains only 8.2 points: the frequency mostly redistributes Laurentian's own customers.
- **Nests by time of day** (the 18:30 with the 17:15 connection, the 21:15 with B's 20:45). The new flight takes 56 percent of its passengers from B, and Laurentian gains 21.1 points.

The models agree perfectly on today and disagree sharply on tomorrow. Under carrier nests, the 21:15's elasticity with respect to the 18:30's fare is +2.17, against +0.65 for every other alternative: a fare cut on one Laurentian departure mostly moves passengers between Laurentian's two flights. The nesting structure is an assumption about which itineraries compete, and it decides whether a frequency is worth adding; it should be estimated, not chosen.

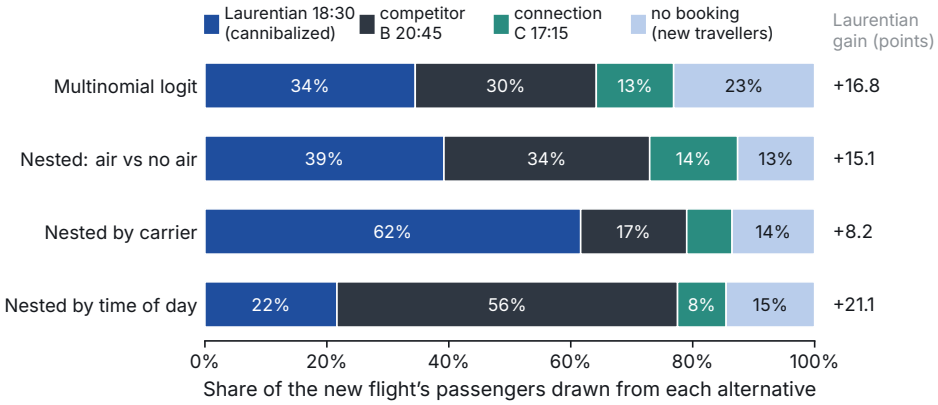


Figure 5.3 Origin of the new 21:15 flight's passengers under four choice models calibrated to the same base shares on YUL–CDG. The figures on the right are Laurentian's total gain in share points.

5.4 Travellers differ: business and leisure

Nested logit groups alternatives; it says nothing about differences between travellers. A consultant flying on her client's account and a family going on holiday weigh fare, time and flexibility very differently, and a model with one fare coefficient averages them into a traveller who does not exist.

Mixed logit lets the coefficients vary across travellers: each traveller is a logit chooser with her own coefficients, and an itinerary's market share is the average of the logit shares over the distribution of tastes. With a few discrete segments, such as business and leisure, it is called **latent class** logit. Travellers who like an itinerary resemble one another, so an improvement draws from the alternatives those same travellers would otherwise choose: a new cheap itinerary draws mainly from the other cheap ones, and IIA no longer holds. Any random utility model can be approximated by a mixed logit (McFadden and Train 2000); with continuous taste distributions its shares are computed by simulation, the estimation team's concern (Train 2009, ch. 6).

WORKED EXAMPLE 5.6 Business and leisure travellers to Paris

Suppose 30 percent of YUL–CDG shopping sessions are business travellers, with a fare coefficient of −0.0010 per dollar, and 70 percent leisure travellers, with −0.0070; fares average CAD 900 for Laurentian, 800 for B and 600 for the connection. Business travellers are 52.5 percent of Laurentian's passengers but

only 10.9 percent of the connection's (Table 5.1). A single logit fitted to 200,000 simulated sessions from this population finds a fare coefficient of −0.00453 per dollar and reproduces today's shares. The two models part company as soon as something changes.

Fare response. At CAD 900, Laurentian's own-fare elasticity is −2.59 in the two-segment model (the average of −0.42 for business and −4.99 for leisure travellers, weighted by their shares of Laurentian's passengers) and −2.79 in the single logit. At CAD 1,500 the two-segment model keeps 11.9 percent of the market for Laurentian and the single logit 3.0 percent, because business travellers keep flying (Figure 5.4, left). A single logit undervalues the high end of the fare structure, the part that fences and revenue management protect (Chapter 14).

Entry. A low-fare carrier opens a nonstop at CAD 560 that appeals to leisure travellers and wins 28.4 percent of the market. In the two-segment model it takes 22.0 percent of Laurentian's share, 27.5 percent of B's, 33.9 percent of the connection's and 34.1 percent of the no-booking share. A single logit calibrated to the same 28.4 percent takes 28.4 percent from everyone (Figure 5.4, right), overstating Laurentian's loss by almost 30 percent.

Table 5.1 Shares by segment in the two-segment YUL–CDG model of Example 5.6 at average fares of CAD 900 (Laurentian), 800 (B) and 600 (connection). Illustrative values.

Segment	No booking	Laurentian	Competitor B	Connection C
Business (30%)	4.4%	53.4%	32.4%	9.8%
Leisure (70%)	17.0%	20.8%	28.0%	34.2%
All travellers	13.2%	30.6%	29.3%	26.9%

The same base shares can hide very different travellers, and a model that averages them mispredicts exactly the events planners care about: a competitor's entry, a fare change, a new product at either end of the market.

5.5 Decision windows and the PODS passenger

Logit treats time of day as one more attribute. In the early 1990s Boeing took a more literal view: a traveller has a window of time in which a trip is acceptable, and only itineraries inside it are con-

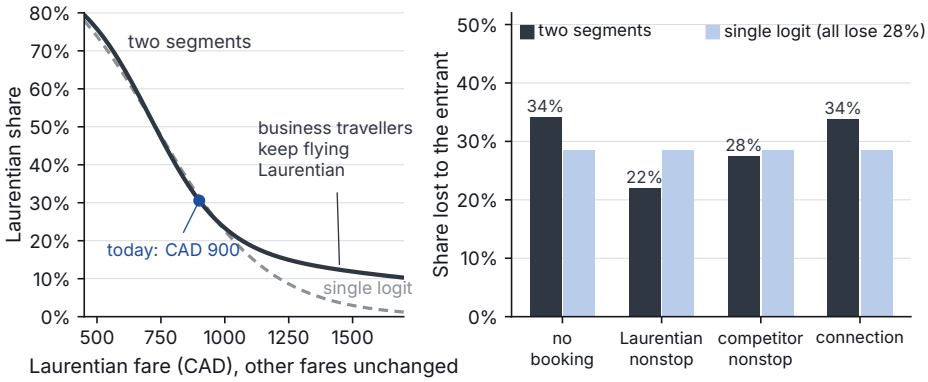


Figure 5.4 Business and leisure heterogeneity on YUL–CDG. Left: Laurentian's share as its fare changes, in the two-segment model and in a single logit fitted to the same travellers. Right: share of each alternative's base share lost to a low-fare entrant that wins 28.4 percent of the market in both models.

sidered at all (Boeing Commercial Airplane Group 1993). The passenger model of PODS, which Belobaba's 2006 lecture calls the recognized state of the art in simulating revenue management in competitive markets, descends from it (Belobaba 2006h). All parameter values in this section are historical.

> **DEFINITION 5.3 Decision window and path quality index**
>
> A traveller's **decision window** runs from the earliest departure she will accept to the latest arrival, on her preferred day. Its width is the market's **Delta-T** (local arrival time minus local departure time of the best service) plus her **schedule tolerance**. A path is considered only if it fits entirely within the window. The **path quality index** (PQI) is 1 for a nonstop plus one point for each stop, connection and interline connection.

The model places and sizes each traveller's window from time-of-day demand and a distribution of tolerances, records which paths fit (the window's **coverage state**), and sends travellers with no path to re-plan. Among the airlines with a path in the window it allocates shares from each airline's image (reliability, marketing programs, service, passenger environment) and the quality of its best path, through a market share equation that is a logit with utilities equal to the logarithms of Boeing's preference coefficients (Boeing Commercial Airplane Group 1993). Path quality matters less as trips get longer: a one-stop path is worth a fifth of a nonstop at 1,000 miles and almost nine-tenths at 6,000 (Chapter 4).

WORKED EXAMPLE 5.7 Decision windows on YUL–CDG

Apply the method to Montreal–Paris with equal airline images. Laurentian's 18:30 arrives at 07:24 local time, a Delta-T of 12.9 hours; B's 20:45 takes 13.1 hours and C's connection 16.5, so the connection fits only windows with at least 3.6 hours of tolerance. With illustrative distributions of earliest departure times (mean 17:30) and tolerances (mean 4.5 hours), 14.1 percent of 200,000 simulated travellers find no path and must re-plan. Laurentian wins 44.8 percent of travellers, B 38.6 percent and C 2.5 percent.

Adding Laurentian's 21:15 raises its share by 10.9 points, to 55.7 percent: 10.0 points from B and 0.9 from travellers who had no path, nothing from the connection. The new flight carries 18.9 percent of travellers, but 8.0 points of that come from the 18:30, because travellers with both Laurentian flights in their window simply split between them. The model behaves like a nested logit in which paths outside the window are excluded, equal paths of one airline are perfect substitutes, and a frequency raises share only by covering windows that were not covered before.

Figure 5.5 makes Boeing's observations visible: frequencies saturate once windows are covered, short paths fit more windows than long ones, and a departure at a popular time sits in many windows (Boeing Commercial Airplane Group 1993). These are the mechanics behind the frequency-share curve of Chapter 4 and the timetable design of Chapter 11.

The PODS passenger The Passenger Origin-Destination Simulator (PODS) was built by Boeing around the decision window model and developed jointly with MIT from 1994, adding choice among fare products and realistic revenue management systems (Belobaba 2006h). It divides the booking horizon into 16 time frames; at the start of each, every airline's revenue management system sets availability from forecasts built on previously simulated history, and passengers arrive and choose among what is open (Belobaba 2006g). Sell-up, buy-down and recapture then emerge from passenger behaviour rather than being assumed (Chapter 14).

PODS passengers are business or leisure. The two types differ in booking curves and decision windows, and in maximum willingness to pay and the dollar disutility they attach to fare restric-

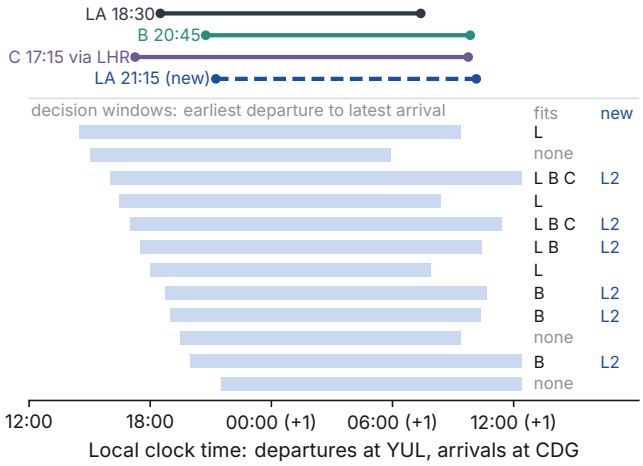


Figure 5.5 Decision windows on YUL–CDG. Paths run from local departure at YUL to local arrival at CDG; each bar is one illustrative traveller's window, from earliest departure to latest arrival. A path is considered only if it lies entirely inside the window. The right-hand columns list the paths that fit today (L: Laurentian 18:30, B: competitor 20:45, C: connection via LHR) and whether Laurentian's proposed 21:15 (L2) would fit.

tions, path quality and re-planning, which each simulated passenger draws from normal distributions (Belobaba 2006g). A passenger first discards every option priced above her willingness to pay, then picks the lowest **total cost**: fare, plus restriction disutilities, plus a path quality cost proportional to the PQI (1 for a nonstop, 3 for a connection), plus a re-planning cost if the path lies outside her window.

Table 5.2 The PODS lecture example (historical simulation settings, USD; adapted from Belobaba, 2006). Mean maximum willingness to pay is 1,200 for business and 300 for leisure passengers. Mean disutilities are 450 and 350 for the Saturday-night stay, 150 and 50 for non-refundability and 150 and 50 for the change fee. Total cost is fare plus the disutilities of the restrictions that apply.

Class	Fare	Restrictions	Business total cost	Leisure total cost
Y	800	none	800	800
M	400	7-day advance purchase, Saturday-night stay	850	750
B	300	14-day advance purchase, Saturday-night stay, non-refundable	900	700
Q	200	21-day advance purchase, Saturday-night stay, non-refundable, change fee	950	650

WORKED EXAMPLE 5.8 Fences at work in PODS

In Table 5.2, the average business passenger's total cost rises as the fare falls, from 800 for Y to 950 for Q: she buys Y even when Q is open, because the restrictions cost her more than the fare differences save. The average leisure passenger, with a willingness to pay of 300, can afford only Q and B, and Q costs her 650 against 700: she buys Q. If revenue management closes Q she sells up to B; if it closes B too, M at 400 exceeds her willingness to pay, and she goes elsewhere or stays home. A business passenger whose Saturday-night cost is 250 rather than 450 sees 650 for M against 800 for Y, and buys down.

These settings describe the restricted fare structures of the early 2000s, which branded fares have since replaced in many markets. The logic of a willingness-to-pay screen plus the dollar cost of each product feature carries over to branded fares directly (Chapter 14).

5.6 Choice models inside planning tools

A network planning system turns a schedule into passengers and revenue in five steps. It sizes each origin-destination market (Chapter 4); a connection builder generates the itineraries that can be sold, within connecting-time and circuity limits; a **share model** allocates each market's travellers to its itineraries; the itinerary demands are

loaded onto legs, where capacity spills some passengers and recapture reassigns some of them (Chapter 7); and revenues are prorated to legs and compared with costs. The share model is where this chapter lives. Its simple ancestor is the **quality of service index** (QSI) of Chapter 4, which scores each itinerary by multiplying factors for its level of service, departure time, aircraft and carrier.

RESULT 5.3 QSI shares are logit shares, and so are QSI recapture rates

If each itinerary's score is a product of factors for its attributes, its QSI share $\text{QSI}_i / \sum_j \text{QSI}_j$ is a logit probability with $V_i = \ln \text{QSI}_i$. The recapture rate of passenger mix and itinerary-based fleet assignment models (Barnhart 2003), $\text{QSI}_r/(\text{QSI}_r + \text{msc})$ with msc the total score of competitors' itineraries, is the logit probability of choosing r when a spilled passenger's choice set is reduced to r and the competitors, with no outside option.

Example 5.1 was a QSI calculation. A QSI model inherits IIA in full, usually with factors set by judgement, and so do its recapture rates. With both Laurentian departures to Paris, the inside scores normalized to one are 0.310 for each Laurentian flight, 0.267 for B and 0.114 for C, so QSI recaptures $0.310/(0.310 + 0.381) = 44.9$ percent of passengers spilled from the 18:30 onto the 21:15. The models of Example 5.5 divert 34.5 percent of the 18:30's passengers to the 21:15 under logit, 61.6 percent under carrier nesting and 21.9 percent under time-of-day nesting when the 18:30 is unavailable. The recapture rates that spill and fleet assignment models take as inputs (Chapters 7 and 12) are outputs of a choice model, only as good as its substitution structure.

Modern systems replace or supplement QSI with estimated logit and nested logit models (Coldren et al. 2003; Lufthansa Systems 2025c), with time-of-day preferences that vary with Delta-T and direction (Koppelman, Coldren, and Parker 2008), and with constants recalibrated so that the model reproduces each market's observed shares. Those constants absorb everything the model leaves out: large adjustments mean the model does not explain the market, and holding them fixed is acceptable for a retiming but risky for a new entrant or a new hub.

IN PRACTICE Itinerary choice inside planning systems in 2026

Lufthansa Systems' 2025 product information for NetLine/Plan describes a market model based on nested logit that predicts choices from itinerary type, connection quality and airline preference, with multinomial and nested logit options trained on PaxIS, DDS or MIDT data, alongside spill and recapture and cost and revenue models (Lufthansa Systems 2025c). The handbook cautions that such public product information does not establish any airline's configuration or calibration (Network Planning Handbook 2026). Data matter as much as the model: booking records such as MIDT and DDS show which itinerary was bought but not what was displayed or who searched without buying, whereas shopping data supply both the choice set and an outside option. On search and booking data from European markets, Lhéritier et al. (2019) report that a machine-learning classifier predicted itinerary choices better than multinomial and latent class logit.

A planning analyst's job is less to build these models than to know what the one in use assumes. Ask what the choice set contains (codeshare duplicates, an outside option); whether the model is QSI, logit or nested logit, and nested how; whether business and leisure are separated; what values of time and connection penalties it implies, and how fare endogeneity was handled (Section 5.7); how much its constants are recalibrated; how well it back-casts past retimings and entries; and whether the recapture rates used for spill come from the same model.

Every network planning forecast contains a theory of substitution. Find out what it is before you trust the answer to a frequency, timing or entry question.

5.7 Why naive fare elasticities are wrong

Sooner or later someone will hand you an elasticity, estimated, if at all, from data in which the airline chose the fares. The handbook is blunt: fares and frequency are endogenous, a historical regression can forecast well while giving misleading causal elasticities, and the first question to ask of any coefficient is how it was identified (Network Planning Handbook 2026).

Fares move with demand for three reasons. The strongest is revenue management: when a departure books ahead of forecast the cheap classes close early and the average fare rises, and when it books behind, classes stay open and a sale may follow, so busy departures sell at high fares and quiet ones at low fares (Williams 2022). Second, a product of better unmeasured quality (departure time, reliability, cabin, loyalty program) costs more and sells more, so a comparison across flights credits its extra traffic to its higher fare (Train 2009, ch. 13). Pricing adds to this: under logit demand with fare coefficient $-\alpha$, a carrier with share s maximizes profit with a markup of $1/(\alpha(1-s))$ over marginal cost, so the more appealing product wins a larger share and charges a larger markup. Third, planners add frequency and capacity where they expect demand to grow, so controlling for them does not help.

IN PRACTICE **Fares that watch demand**

Modern pricing systems make the problem sharper. IATA defines dynamic pricing as setting prices from the shopping context, including the days to departure, competitor activity and the seats remaining, and continuous pricing as allowing an unlimited number of price points so that the price can track demand at each moment (International Air Transport Association 2021). Air Canada relies on PROS's real-time dynamic pricing product, according to the vendor's case study (PROS 2025), and PROS describes its pricing as moving along a continuous curve between filed fares as demand and willingness to pay change (PhocusWire n.d.). In such data the fare is, by design, a function of the demand forecast, and an elasticity read naively from it describes the pricing system as much as the travellers.

Each observation is the intersection of a demand curve and a pricing rule. If demand shifts while the pricing rule stays put, the points trace the pricing rule and a fitted line slopes upward; if costs shift the pricing rule while demand stays put, the points trace the demand curve (Working 1927). Suppose revenue management passes a demand shock into the fare at a rate κ (a day 10 percent busier raises the fare by 10κ percent), and a share R of the variation in log fares comes from such shocks, the rest from costs and noise. The slope of a naive regression of log traffic on log fare then tends to

$$\hat{\eta}_{\text{naive}} \to (1-R)\,\eta + R\left(\eta + \frac{1}{\kappa}\right), \tag{5.4}$$

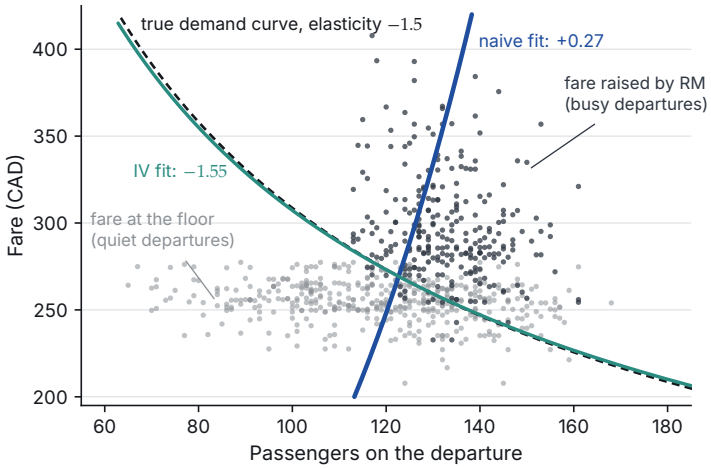


Figure 5.6 Two years of Laurentian's simulated daily YUL–YVR departure (728 departures of a 169-seat 737-8). Grey dots are departures sold at the fuel-indexed floor, dark dots departures where revenue management raised the fare. The dashed curve is the true demand curve for an average day. The naive log-log fit slopes the wrong way; the fit that uses only the fare changes caused by the fuel price (labelled IV, for instrumental variables) recovers the demand curve.

a weighted average of the true elasticity η and the slope of the path that demand shocks trace through the pricing rule. Cost-driven fare variation identifies the demand curve; demand-driven variation identifies the pricing rule. With $\eta = -1.5$, $\kappa = 0.5$ and 80 percent of fare variation due to demand, the naive estimate is $-1.5 + 0.8/0.5 = +0.1$: travellers appear to like paying more.

WORKED EXAMPLE 5.9 The elasticity of a revenue-managed departure

Simulate two years (728 departures) of Laurentian's daily evening flight from Montreal to Vancouver on a 169-seat 737-8, with a true fare elasticity of -1.5. Pricing files a ladder of eight fares whose lowest step is indexed to the weekly jet fuel price; revenue management opens the lowest step whose forecast demand fits into 88 percent of the seats. All numbers are illustrative.

On 63.5 percent of departures demand is weak and the floor is the fare. The busy departures carry more passengers (132.8 on average against 119.2) at higher fares (CAD 297 against 255),

because they are busy. The naive regression gives +0.27: a 10 percent fare increase would raise traffic by 2.7 percent. Using only the fare changes caused by the fuel price, which has nothing to do with this flight's demand, gives −1.55, with a 95 percent interval from −2.17 to −0.92 (Figure 5.6). Fuel moves fares only on quiet departures: when seats are scarce, revenue management sets the fare from demand and capacity, not cost. Read naively, the airline's own fare history says that travellers do not mind paying more.

The same mechanism damages choice models: the fare coefficient is biased toward zero and every dollar value derived from it is inflated. In the full reference edition's simulated panel of Laurentian markets, Laurentian's own-fare elasticity on Montreal–Vancouver comes out at −1.00 against a true −1.95, and the value of a nonstop at CAD 337 against a true CAD 167: the naive model would tell a planner that travellers pay twice as much for a nonstop as they do. Lurkin et al. (2017) find the pattern in airline itinerary choice models estimated on US markets: correcting for fare endogeneity lowers the implied values of time.

The remedies belong to the estimation team. **Instruments** use only the fare variation caused by something unrelated to demand, typically cost: fuel cost per seat scales with distance (with the running case's illustrative burn, about CAD 148 per seat on YUL–SVQ against CAD 6 on YUL–YQB). They fail when the cost shock also changes the product: in the second quarter of 2026 Air Canada paid 51 percent more per litre of fuel than a year earlier and reports responding with fares and fees and with selective capacity adjustments (Air Canada 2026d), and capacity affects demand directly. **Demand models built for endogenous prices**, from Berry's inversion and the random-coefficients model of Berry, Levinsohn and Pakes to control functions, combine choice models with instruments (Berry 1994; Berry, Levinsohn, and Pakes 1995; Petrin and Train 2010). **Natural experiments**, such as a low-cost carrier's entry on some routes and not others, compared by difference-in-differences (Angrist and Pischke 2009; Goolsbee and Syverson 2008), identify the effect of the event, which is what a competitive scenario needs; the Competition Bureau's 2025 market study reports that one new competitor on a route between two cities lowers fares by 9 percent on average (Competition Bureau Canada 2025). They do not identify a fare elasticity, because entry adds a competing product that takes passengers at any fare.

Before using someone else's elasticity, ask:

1. **What moved the fares?** If it was the airline's own pricing and revenue management, the estimate is biased toward zero unless an instrument or an experiment was used.
2. **Which elasticity is it?** In the simulated Montreal–Vancouver market a uniform change in all fares has an elasticity of −1.29, Laurentian's own fare −1.95.
3. **What was held constant, and is the scenario inside the data?** A coefficient estimated with rivals' fares fixed does not describe a fare cut that rivals match, and a 40 percent cut where fares never moved by more than 10 percent is an extrapolation.
4. **How were demand and the market measured?** Loads on full flights are censored (Chapter 7), and a logit market elasticity is proportional to the assumed outside share: rescaled to a 15 percent outside share, the Montreal–Vancouver market elasticity falls from −1.29 to −0.38.
5. **When, and estimated or set by default?** Elasticities from before branded fares or the pandemic describe another market. Price sensitivities inside revenue management methods such as fare adjustment (Fiig et al. 2010) inherit the problem of this section when they come from the airline's own booking history.

Published compilations (Gillen, Morrison, and Stewart 2004; InterVISTAS Consulting 2007) and the market elasticities of −0.8 to −2.0 in the MIT lecture notes, with business demand less elastic than leisure (Belobaba 2006d), are useful as historical ranges and checks on orders of magnitude, not as parameters for a 2026 route.

> **PITFALL**
>
> One elasticity for every market. The handbook warns against importing a universal fare elasticity or frequency-share curve into every market (Network Planning Handbook 2026). The mix of business and leisure travel, distance, competition and the outside option differ from one market to the next, and so do the elasticities.

An elasticity is a property of a market and of the variation that identified it. Ask for both before you use it.

5.8 Summary

- Travellers choose the itinerary with the highest utility, part of which the analyst cannot see. Logit shares are $e^{V_i} / \sum_j e^{V_j}$, an S-curve that responds most where alternatives are close.
- Ratios of coefficients are money values: in the illustrative Paris market a connection 4.4 hours slower than a nonstop must be about CAD 476 cheaper to compete. Own-fare elasticities are $\beta_f f (1 - P)$; market elasticities are much smaller and depend on the outside share. Apply models to samples or segments, never to an average traveller.
- Under IIA a new itinerary draws from all others in proportion to their shares. Nested logit groups close substitutes: calibrated to the same base shares, a second Laurentian frequency to Paris gains 8.2 points with carrier nests and 21.1 with time-of-day nests.
- A single logit fitted to business and leisure travellers undervalues demand at high fares and misattributes the losses from a low-fare entrant.
- Decision windows exclude paths that do not fit, so frequency raises share only by covering new windows. PODS passengers screen fares by willingness to pay, then minimize fare plus the dollar costs of restrictions, path quality and re-planning.
- QSI shares and QSI recapture rates are logit probabilities and inherit IIA: recapture rates are outputs of a choice model, not independent facts.
- Fare elasticities read from airline data are biased toward zero or even positive, because revenue management raises fares when demand is high. Ask what moved the fares.

5.9 Further reading

The Boeing decision window presentation (Boeing Commercial Airplane Group 1993) and Belobaba's PODS and ePODS lectures (Belobaba 2006g, 2006h) are short and repay careful reading, and the handbook (Network Planning Handbook 2026) states the planner's version of the outside-option, IIA and identification warnings. Train (2009, chs. 2, 3, 4, 6 and 13) covers the models and endogeneity; the full reference edition derives them in its Chapters 5 and 6 and develops instruments, BLP, control functions and difference-in-differences in its Chapter 7. For airline applications, see Coldren et al. (2003), Coldren and Koppelman (2005), Garrow's survey (2010) and Lurkin et al. (2017) on fare endogeneity.

CHAPTER

6

Forecasting Demand

IN THIS CHAPTER

- Which decision needs a forecast of which quantity, at which level of detail and how far ahead.
- The seasonal naive benchmark and the main families of forecasting methods, in words.
- Honest evaluation: holdouts, rolling origins, why MAPE misleads, and beating the naive benchmark.
- Booking curves and pickup: how revenue management projects the bookings still to come.
- New routes without history: gravity models, analogues and share models.
- Prediction intervals, fan charts and coherent scenarios for the people who decide.

DRAWS ON Hyndman and Athanasopoulos, Forecasting: Principles and Practice (2021); Belobaba, Introduction to PODS passenger choice model (MIT 16.75J, 2006); Belobaba, Introduction to revenue management (MIT 16.75J, 2006); Belobaba, O&D control: what have we learned? (MIT 16.75J, 2006); handbook

Every number in a network plan rests on a forecast. The route proposal for Seville needs the size of the market and Laurentian's share of it, two years ahead. The winter schedule needs monthly demand by direction and day of week, a season ahead. Fleet assignment needs the mean and variability of demand for each departure. Revenue management needs, every night, the bookings still to come for every fare class. These are different forecasts, at different levels of detail and horizons, and a planner who confuses them builds careful analysis on the wrong quantity.

This chapter is about prediction, not explanation. Chapters 4 and 5 explained why demand responds to price, time and frequency and how travellers choose; a forecast has the narrower goal of being close to what happens. The handbook draws the line sharply: a model can predict well while implying misleading causal elasticities (Network

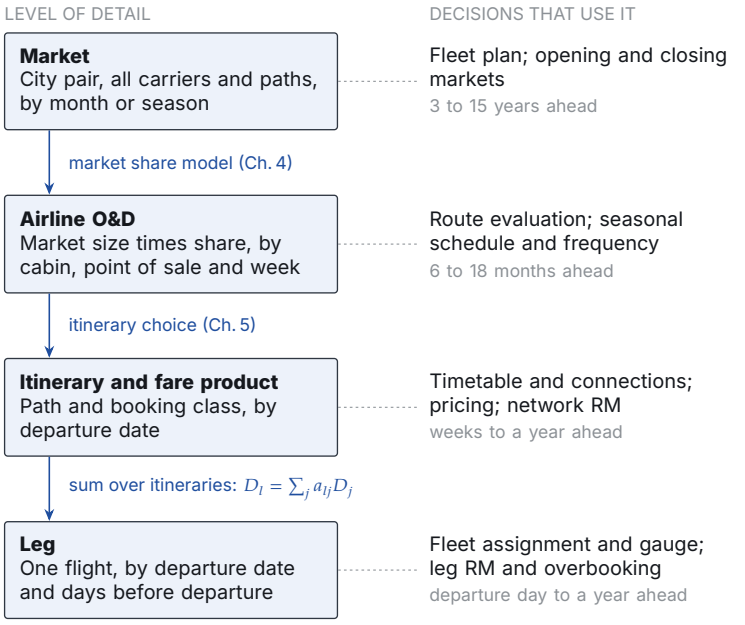


Figure 6.1 The forecasting hierarchy. Each level is derived from the one above by a model (market share, Chapter 4; itinerary choice, Chapter 5) or by a sum over the itineraries that use a leg, and each serves different decisions at different horizons.

Planning Handbook 2026). The methods come from Hyndman and Athanasopoulos (2021), applied here to simulated Laurentian data whose true process is known, so that every claim about accuracy can be checked.

6.1 What to forecast, and at which level

Airline demand is forecast at four levels of detail (Figure 6.1). The **market** is total travel between two cities by all carriers and paths. The **airline O&D** level is the airline's share of it, by cabin, point of sale and week. The **itinerary and fare product** level splits that demand among the airline's paths and booking classes for each departure date. The **leg** level adds up the itineraries that use each flight, by departure date and days before departure.

Each level has its own users. Fleet planners and whoever opens or closes markets work with market forecasts three to fifteen years ahead. The network planner's route evaluations, seasonal schedules and frequencies use airline O&D forecasts six to eighteen months ahead. Timetable construction, pricing and network revenue man-

agement need itinerary and fare-product forecasts, from a year out to the last weeks. Fleet assignment, gauge swaps, leg revenue management and overbooking need leg forecasts, up to the day of departure.

The levels are linked by models, not only by sums: O&D demand is market demand times a share, itinerary demand comes from a choice model, and leg demand is the sum of the itinerary demands that use the leg. The last link has a consequence that is easy to miss. A connecting passenger appears on two legs, so leg forecasts do not add up to O&D totals: travellers counted by O&D and passengers counted by leg measure different things and do not belong in one summing hierarchy (Network Planning Handbook 2026).

Three principles apply at every level.

Forecast demand, not bookings. Bookings and loads are censored by capacity and by revenue management, which closes fare classes before flights fill, so the history must be unconstrained first (Chapter 7).

Forecast what the decision needs, at its horizon. A seasonal schedule decision needs monthly or weekly demand a season ahead, not next week's bookings (Network Planning Handbook 2026). Accuracy one month ahead says little about accuracy at twelve.

Use only what was knowable at the time. A forecast for a decision made in March can use only data available in March: first-release traffic estimates, the competitor schedules published by then, economic data as first reported. A backtest that uses revised data or later schedules flatters the model (Network Planning Handbook 2026).

Forecasts that add up Forecasts made separately at different levels rarely agree. Suppose Laurentian forecasts 31,000 Montreal–Europe O&D passengers for next August, while its forecasts of 19,800 for Paris, 6,900 for Lisbon and 3,500 for Seville sum to 30,200 (illustrative numbers). **Bottom-up** keeps the routes, **top-down** keeps the total, and **reconciliation**, such as the minimum-trace (MinT) method (Wickramasuriya, Athanasopoulos, and Hyndman 2019), shares the gap according to past accuracy. With backtest error standard deviations of 1,500 for the total and 1,100, 600 and 500 for the routes, the total falls by 442 to 30,558, and Paris rises by 238, Lisbon by 71 and Seville by 49: the forecast trusted least moves most.

6.2 Benchmarks and the main method families

Every model is judged against two simple benchmarks.

DEFINITION 6.1 Naive and seasonal naive forecasts

The **naive** forecast repeats the last observation. The **seasonal naive** forecast repeats the observation from the same season of the last observed year: every January is forecast by the most recent January.

They are surprisingly hard to beat on seasonal airline series. The handbook's advice is to begin with the seasonal naive forecast and to accept a more complex model only when it does better out of sample, on a task that matters for the decision (Network Planning Handbook 2026). A model that cannot beat "same month last year" has no place in a route proposal. Three families of models compete with it.

Exponential smoothing forecasts with a weighted average of past observations whose weights decline geometrically, so that recent months count most. The airline workhorse, the **Holt–Winters** method, tracks a level, a trend and a seasonal pattern, each nudged towards the latest evidence as every new month arrives. Because seasonal swings grow with traffic, it is best applied to the logarithm of passengers, which makes the season a set of percentages and growth a constant rate (Hyndman and Athanasopoulos 2021).

ARIMA models describe how each month depends on the months before it and on recent surprises, once trend and season have been removed by differencing. The seasonal naive forecast is the simplest member of the family, and the best-known seasonal model is called the **airline model** because Box and Jenkins fitted it to monthly international airline passengers (Box and Jenkins 1970).

Regression with calendar effects puts known structure into the model directly. Easter moves between March and April; school breaks differ by province; Canadian and US Thanksgiving fall in different months; a month with five weekends carries more traffic than one with four. Smoothing and ARIMA cannot see these effects coming. A **dynamic regression** combines a trend and month and holiday indicators with a model of the leftover errors, so that a recent surprise fades out gradually (Hyndman and Athanasopoulos 2021). Economic drivers such as income or exchange rates help only as much as they can themselves be forecast at the decision's horizon.

WORKED EXAMPLE 6.1 Forecasting Montreal–Paris two years ahead

Data. Laurentian's monthly Montreal–Paris O&D passengers are simulated for ten years: about 12,000 a month at the start, growing 3 percent a year, with a season running from 0.71 of the average in February to 1.315 in July, 7 percent more traffic in the month that contains Easter (March in some years, April in others), and shocks that persist for a few months. Each method is fitted on years 1 to 8 and forecasts years 9 and 10, which are held out.

Results. Over the 24 held-out months the mean absolute error is 922 passengers for the seasonal naive forecast, 681 for Holt–Winters and 518 for the dynamic regression. The largest seasonal naive error is in April: April of year 8 had no Easter, so for April of year 9 it forecasts 13,066 passengers against an outcome of 15,315. Holt–Winters forecasts 15,560, because its April term averages several years, most of which had Easter in April. Only the regression knows when Easter falls.

Uncertainty. The upper end of the Holt–Winters 80 percent interval (Figure 6.2) is 12 percent above its lower end one month ahead, 21 percent at twelve months and 30 percent at 24: uncertainty compounds with the horizon. The regression wins partly because the simulated world was built in its image; on real data the ranking is an empirical question.

6.3 Measuring accuracy honestly

A forecast is useful only if its errors are known, and the way errors are summarized changes which model looks best. The basic test is a **holdout**: fit on the past up to some date, forecast the periods after it, and compare with what happened, as in Example 6.1. One holdout is one draw of luck. **Rolling-origin evaluation**, the time-series version of cross-validation, repeats the test from many forecast origins, each time refitting on the data available at that origin only (Hyndman and Athanasopoulos 2021).

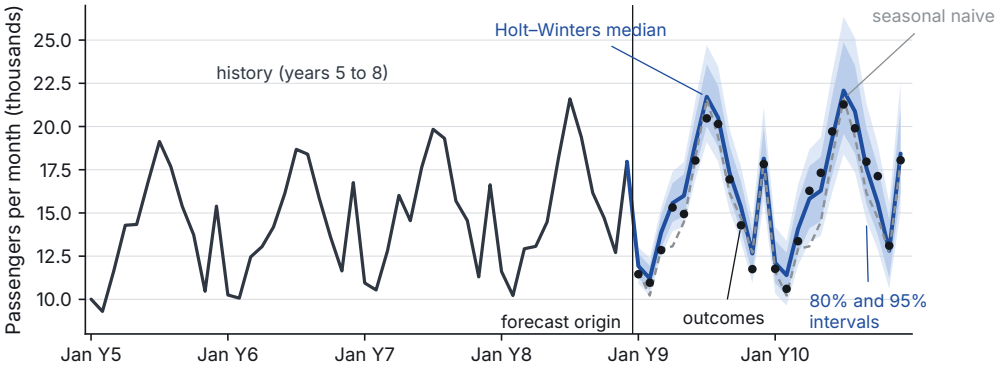


Figure 6.2 Simulated monthly Montreal–Paris O&D passengers for Laurentian. Forecasts made at the end of year 8 for 24 months: the Holt–Winters median with 80 and 95 percent prediction intervals from 5,000 simulated sample paths, the seasonal naive benchmark, and the outcomes that actually followed.

DEFINITION 6.2 Forecast accuracy measures

For the errors (actual minus forecast) over a set of test periods:

- the **mean error**, or **bias**, is their average; positive values mean forecasts too low;
- the **mean absolute error** (MAE) is the average size of the errors, ignoring sign;
- the **mean absolute percentage error** (MAPE) is the average size of each error as a percentage of the actual value;
- the **mean absolute scaled error** (MASE) is the MAE divided by the in-sample MAE of the seasonal naive forecast (Hyndman and Koehler 2006).

MASE below one means the method beats the naive benchmark; it is scale-free and works when some outcomes are zero. MAPE, the most familiar measure, has two defects.

RESULT 6.1 Which forecast each measure rewards

On average, squared error is smallest for a forecast equal to the mean of the outcome, absolute error for its median, and percentage error for a value below the median. MAPE therefore rewards forecasts biased low, and it becomes unstable when outcomes are small.

The reason is that MAPE divides each error by the actual value: an over-forecast of a small outcome is a large percentage, an under-forecast of a large one a small percentage, so shading every forecast down lowers the score (Gneiting (2011) gives the theory). Both defects matter for airlines, which forecast thousands of thin flows (Network Planning Handbook 2026).

WORKED EXAMPLE 6.2 Rolling origins, and why MAPE misleads on thin flows

Rolling origins. On the Montreal–Paris series, place the forecast origin at each month from 60 to 108, refit every method on the data up to that month, and forecast one to twelve months ahead: 49 origins. One month ahead, the mean absolute errors are 2,621 passengers for the naive forecast, 742 for the seasonal naive, 507 for Holt–Winters and 398 for the dynamic regression; twelve months ahead, 857, 700 and 538 for the last three. In MASE the seasonal naive sits at 1.0 to 1.17, Holt–Winters rises from 0.68 to 0.93 and the regression from 0.53 to 0.72. The seasonal naive forecast is biased low by 2 to 3.4 percent because it ignores a year of growth.

A thin flow. Take Laurentian's flexible-fare passengers from Quebec City to Seville, about four per departure, and suppose they are Poisson with mean 4. The perfect forecast, 4, has a MAPE of 56 percent over the departures with at least one passenger (MAPE is undefined on the 1.8 percent with none). A forecast of 3 scores better, 46 percent, and is the best constant forecast by MAPE. A revenue management system tuned on MAPE would protect too few seats for its highest fares. By MASE the perfect forecast scores 0.70 and the forecast of 3 does worse, as it should.

Four habits turn this procedure into a decision tool. **Break errors down** by market size, season, direction, cabin and horizon (Network Planning Handbook 2026): an average can hide a model that is poor on the marginal markets. **Track bias separately**: over-forecasting marginal routes is costlier than errors of the same size on established ones, because it is what gets weak routes approved (Network Planning Handbook 2026). **Weight errors by what they cost**: a new model can lower the network-wide error while doing worse on the few capacity-constrained, high-revenue routes where decisions are made (Network Planning Handbook 2026). And **check the inter-**

vals: across the 49 origins, the Holt–Winters 80 percent intervals contained the outcome 88 percent of the time, so they are somewhat too wide for this series.

> **PITFALL**
>
> Look-ahead in backtests. A backtest that uses the latest, revised data instead of what was available at each origin, or a competitor schedule filed after the forecast date, measures a forecast nobody could have made. Store data by vintage and rebuild each origin's information set exactly (Network Planning Handbook 2026).

6.4 Booking curves and pickup

Revenue management forecasts a different object: for each future departure, the demand still to come, given the bookings already on hand. Bookings accumulate along a **booking curve** whose shape differs by segment. In the PODS simulator's inputs, leisure travellers have made 75 percent of their final bookings 21 days before departure, business travellers only 35 percent (Belobaba 2006g). RM systems revise their forecasts at fixed checkpoints before departure, and the basic RM system that Belobaba describes does so with simple **pickup** methods (Belobaba 2006l).

> **DEFINITION 6.3 Pickup forecasts**
>
> From comparable flown departures, take the average bookings on hand a given number of days out and the average final demand. For a departure with B bookings on hand at that point:
>
> - **additive pickup** adds the average bookings that arrived afterwards: B plus (average final minus average on hand);
> - **multiplicative pickup** scales up by the average growth: B times (average final over average on hand);
> - the **historical** forecast ignores B and uses the average final demand;
> - a **regression** of final demand on bookings on hand contains all three as special cases.

Which is right depends on why departures differ. If they differ only by chance, the bookings already made say nothing about those

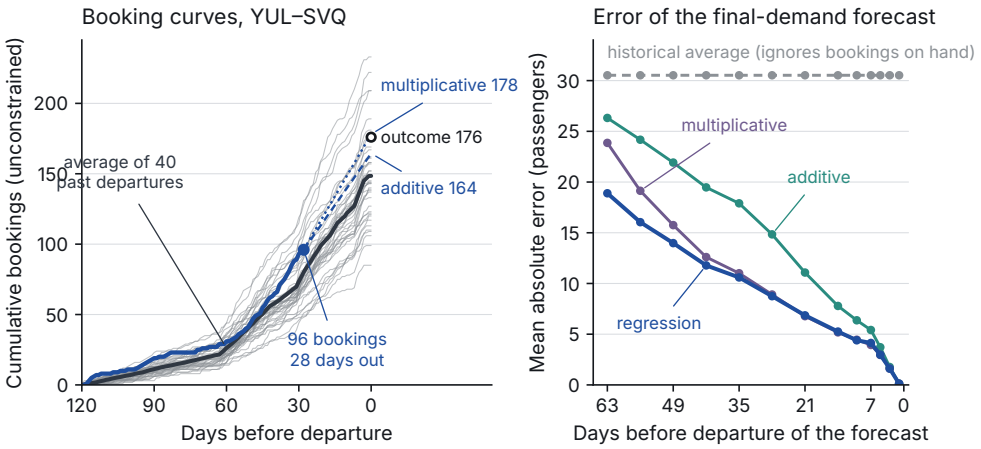


Figure 6.3 Left: booking curves of 40 simulated past YUL–SVQ departures, their average, and a current departure with 96 bookings 28 days before departure, projected to departure by additive and multiplicative pickup. Right: mean absolute error of four forecasts of final demand, by the day on which the forecast is made, over 3,000 simulated departures.

to come, and additive pickup is right. If some departures are simply more popular, a departure ahead of its usual curve will probably stay ahead, and multiplicative pickup is closer. With both features, the best forecast is a straight line in bookings on hand between the two, which is why the fitted regression is a good default. In Belobaba's account of PODS simulations, better forecasts of bookings to come and better unconstraining could matter as much for revenue as the choice of optimization method (Belobaba 2002).

WORKED EXAMPLE 6.3 How many passengers will the Seville flight have?

Data. Simulated YUL–SVQ shoulder-season departures have mean demand 150, and popularity varies by about 25 percent from one departure to the next, so final demand has a standard deviation of about 40, the running case's values. A fifth of the passengers book on the PODS business curve and the rest on the leisure curve, so 54 percent of final demand has booked 28 days out. The history is 40 flown departures; a current departure has 96 bookings 28 days out (Figure 6.3).

Forecasts. The past departures averaged 80.0 bookings at 28 days and 148.5 at departure. Additive pickup adds the average pickup of 68.5: 96 + 68.5 = 164. Multiplicative pickup applies the ratio 148.5/80.0 = 1.855: 96 × 1.855 = 178. The regression

fitted on the history, $20.1 + 1.604 \times 96$, gives 174. The departure eventually carries 176 passengers.

Accuracy. Over 3,000 simulated departures, 28 days out, the mean absolute errors are 30.5 passengers for the historical forecast, 14.8 for additive pickup, 8.9 for multiplicative pickup and 8.7 for the regression. Popularity differences dominate here, so multiplicative beats additive. Sixty-three days out, with few bookings on hand, the multiplicative ratio is 6.9 and amplifies noise: its error is 23.9 against 18.9 for the regression. Had departures differed only by chance, additive pickup would have won, 6.7 against 8.9 at 28 days.

For the planner. Forecasts improve sharply as departure approaches, so capacity decisions made months ahead, such as gauge swaps, rest on far noisier numbers than revenue management's last-week decisions (Chapter 15). And if low fares closed early on the busiest past departures, their bookings understate demand and every pickup estimate inherits the bias (Chapter 7).

6.5 Forecasting a new route

A new route has no history, and a thin market served only by connections has little that can be trusted. Planners combine three kinds of evidence: a cross-sectional model of market size, analogues from comparable launches, and a share model.

The **gravity model** is the standard cross-section. By analogy with Newton's law, traffic between two cities rises with their size and falls with the distance between them, with adjustments for language ties, tourism appeal or a nonstop. It is fitted by regression across many city pairs, in logarithms, so its prediction for a new pair is a median and its honest uncertainty is a factor above and below that median.

WORKED EXAMPLE 6.4 A gravity model for Montreal–Seville

Data. Annual O&D traffic is simulated for the 63 pairs between nine North American and seven European cities, from populations (4.3 million for Montreal, 1.5 million for Seville), distances, a language tie between Paris and three francophone Canadian cities, and a nonstop that raises traffic by 25 percent. The simulated airlines placed their nonstops in the 40 percent

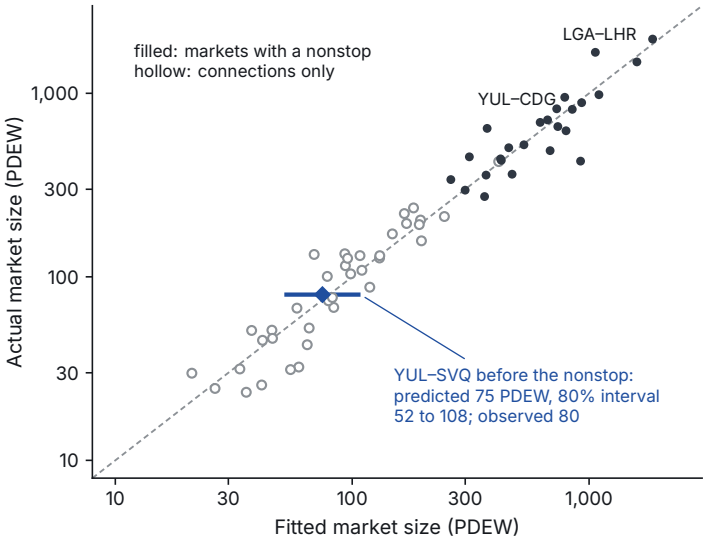


Figure 6.4 A gravity model fitted to 62 simulated city pairs between North America and Europe, actual against fitted market size on logarithmic scales. Pairs with a nonstop (filled) cluster at the top because nonstops were placed in the strongest markets. The bar shows the 80 percent prediction interval for Montreal–Seville before its nonstop, against its true size of 80 PDEW.

of markets they saw as strongest. Montreal–Seville has no nonstop and a true size of 80 PDEW; it is left out of the estimation.

Prediction. Fitted on the other 62 pairs, the model explains 95 percent of the variation in log traffic (Figure 6.4). For Montreal–Seville it predicts a median of 54,650 passengers a year in both directions, 75 PDEW, with an 80 percent interval from 52 to 108 PDEW. The truth, 80, is comfortably inside. That interval, a factor of about 1.45 either side of the median, is a fair statement of how much a cross-section knows about one market, and a useful cross-check on an estimate built from booking data.

The trap. The fit is excellent, yet the nonstop coefficient says that a nonstop multiplies traffic by 2.2, when the truth is 1.25. Markets with nonstops are larger partly because of the nonstop and partly because airlines chose them for being large. A planner who read the coefficient as the stimulation from a new Seville nonstop would forecast 165 PDEW instead of 100.

The second kind of evidence is **analogues**: past launches that resemble the new route in distance, market size, mix of business and

leisure, competition and frequency. For each, measure first-year traffic relative to the pre-launch market, the share the new service won, and the **ramp-up**, the months it took to reach mature traffic. The spread of those ratios is the evidence for stimulation, which the handbook insists must come from comparable launches rather than be assumed (Network Planning Handbook 2026).

The third is the share model of Chapter 4, which turns a market estimate into the new service's traffic and separates its sources, which the handbook asks to be kept apart: stimulated demand, demand diverted from competitors, demand diverted from the airline's own services, and spill (Network Planning Handbook 2026). The three kinds of evidence fail in different ways (booking data miss channels, gravity models miss market idiosyncrasies, analogues are never exact), which is the argument for using all of them.

> **PITFALL**
>
> Reading a cross-sectional coefficient as a causal effect. A nonstop dummy, a frequency variable or a fare in a cross-section of markets measures association, and airlines put nonstops, frequencies and fares where demand already is. Use such models to predict the size of markets like those in the sample, not what a change in service will do (Chapter 5).

6.6 Uncertainty, scenarios and the decision

A forecast without a statement of its uncertainty invites the decision maker to treat it as a fact. A **prediction interval** gives a range that contains the outcome with stated probability. The handbook recommends such distributional forecasts where the cost of a bad outcome dominates the decision (Network Planning Handbook 2026), which describes most capacity decisions: an aircraft committed to a route that underperforms cannot be recovered quickly.

Statistical intervals capture only part of the uncertainty. The handbook lists five sources to keep apart (parameter uncertainty, macroeconomic scenarios, competitor response, launch ramp-up and operational feasibility) (Network Planning Handbook 2026). The practical answer combines intervals for the statistical part with **scenarios** for the structural part. Scenarios must be internally coherent: lower demand tends to come with lower fares, and toggling every input independently produces combinations that cannot occur (Network Planning Handbook 2026).

WORKED EXAMPLE 6.5 Communicating the Seville forecast

The running case gives mature eastbound demand per departure of 175 passengers at the autumn and spring peaks, 150 in the summer shoulder season and 110 in winter, from November to March (the calendar of Chapter 10). Suppose analogues suggest that a new transatlantic route starts at 80 percent of mature demand and ramps up linearly over its first year. Uncertainty about the market as a whole (size, stimulation and share) is one factor common to all months, with a standard deviation of about 20 percent; month-specific noise adds about 6 percent.

Break-even. With the running case's costs, the A321XLR's seven-hour trip costs CAD 80,500 before passenger costs, each passenger contributes CAD 600 (a net fare of 620 plus 30 of ancillary revenue, less 50 of passenger costs), and cargo contributes CAD 3,500, so the leg breaks even at $(80{,}500 - 3{,}500)/600 = 128$ passengers.

Fan chart. Figure 6.5 summarizes 20,000 simulated paths. The median starts at 140 in the launch month. The chance of beating break-even is 66 percent in the first September, 6 percent in the first November, 87 percent in the first April and 77 percent in the first August, then 93 percent in the second autumn and 23 percent in the second winter. The 80 percent band spans a factor of 1.7 in every month, because the common market factor does not average out over time.

Scenarios. An adverse case with demand 20 percent lower and a net fare of CAD 590 raises the break-even load to 135 passengers while mature shoulder demand falls to 120; a favourable case with 15 percent more demand and a fare of CAD 640 lowers it to 124 against 172.

The message is not "the route carries 150 passengers". It is that autumn and spring are very likely profitable at the leg level once the route has ramped up, summer probably so, and winter probably not, unless connecting traffic changes the picture. That points to the decision worth discussing, a winter down-gauge or suspension, and to the assumption that would reverse it: the level of the common market factor.

Three habits make forecasts useful to decision makers: state the decision threshold and the probability of crossing it; name the as-

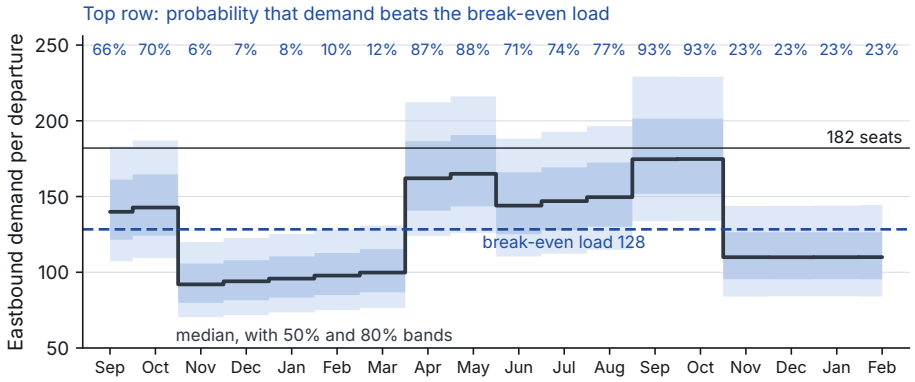


Figure 6.5 Fan chart of eastbound YUL–SVQ demand per departure for the 18 months after a September launch: median with 50 and 80 percent bands, against the leg-level break-even load and the 182 seats of the A321XLR. The top row gives the probability of beating break-even in each month. Illustrative assumptions.

sumption that would reverse the recommendation, which the handbook calls the most useful sensitivity; and after launch, compare outcomes with the forecast as approved, separating a slow ramp-up from a persistent bias (Network Planning Handbook 2026).

IN PRACTICE **Forecasting in 2026**

Revenue management vendors describe a move from pickup and time series methods toward Bayesian and machine-learning models that forecast willingness to pay rather than demand by booking class. PROS reports that adding willingness-to-pay data raises revenue by about 1 to 3 percent, a vendor figure without published methodology (PhocusWire n.d.); Fetcherr markets a generative "large market model" that recommends fares from schedules, inventory and market data (Boynton and Massy-Beresford 2025). A 2025 editorial in the Journal of Revenue and Pricing Management calls for empirical validation of such methods (Yeoman 2025), and which engine a given airline runs is usually not public. The data move too: Cirium releases monthly O&D estimates about 55 days after the end of a month and restates them at 115 days (Cirium 2026a), and the US ticket sample changed from a quarterly 10 percent sample to a monthly 40 percent sample in July 2025, a break that should not be spliced naively (Network Planning Handbook 2026).

6.7 Summary

- Forecast unconstrained demand at the level and horizon the decision needs: markets for fleet and entry, airline O&D for routes and seasonal schedules, itineraries for pricing and network RM, legs for fleet assignment and leg RM.
- Start from the seasonal naive benchmark. Holt–Winters smooths level, trend and season; ARIMA models how months depend on each other; regression with calendar effects sees holidays such as Easter coming.
- Test on held-out data with rolling origins, using only what was knowable at each origin; break errors down, track bias and weight errors by what they cost.
- MAPE rewards forecasts biased low and is unstable on thin flows; MASE compares a method with the naive benchmark at any scale.
- Pickup projects bookings on hand to departure: additive when departures differ by chance, multiplicative when they differ in popularity, a fitted regression in between.
- For new routes, combine a gravity model, analogues and a share model; cross-sectional coefficients on nonstops are not causal.
- Report intervals and coherent scenarios, state the probability of crossing the decision threshold, and name the assumption that would reverse the recommendation.

6.8 Further reading

The handbook's forecasting guidance (Network Planning Handbook 2026) is short and worth rereading after this chapter. Hyndman and Athanasopoulos (2021) is the methods companion, with code for every method named here. For booking curves and RM forecasting, Belobaba's PODS and RM lectures (Belobaba 2006g, 2006l, 2002) give the airline context. The full reference edition, in its Chapter 8, writes out the smoothing, ARIMA, reconciliation and pickup models in full, with proofs.

CHAPTER

7

Spill and Recapture

IN THIS CHAPTER

- Why a flight that is 85 percent full on average can still be turning away one customer in eight.
- The expected-spill formula under normal demand, and how to read it as a picture.
- The k-factor and spill tables: why the same load factor means different things in different markets.
- Why recorded loads understate demand, and how planners recover it.
- What one more seat is worth, recapture, and why revenue management changes the answer.

DRAWS ON Belobaba, Airline demand analysis and spill modeling (MIT 16.75J, 2006); Boeing, Load factor analysis (1978); handbook Exercise B; Barnhart, Kniker and Lohatepanont (2002); Abramovich (2013)

Every network decision is a bet on demand that nobody observes. When a planner assigns a 137-seat A220 instead of a 169-seat 737-8 to the 17:00 departure to Vancouver, the customers who would have flown on the bigger aircraft leave no trace in the airline's records: the flight closes, and they fly another airline, travel another day or stay home. This chapter is about those invisible customers: the vocabulary, the one formula that sits underneath most gauge and fleet decisions, and the economics that turns spill into dollars.

Spill is also where the two halves of this book meet. The planner decides how many seats exist; revenue management decides who gets them. A spill estimate that ignores how revenue management fills the aircraft misprices capacity, a point the chapter closes with and Chapter 15 takes further.

7.1 Demand, load and spill

Fix one flight leg, say Laurentian's Friday 09:00 departure from Montreal, and look at a long series of its departures operated with the same aircraft. For each departure, distinguish three quantities.

DEFINITION 7.1 Demand, load and spill

The **unconstrained demand** D of a departure is the number of customers who would book it at the prevailing fares and schedule if seats were unlimited. The **capacity** C is the number of seats offered for sale. The **load** is the number of passengers carried, $Y = \min(D, C)$, and the **spill** is the demand that could not be accommodated, $S = \max(D - C, 0)$.

Two remarks prevent most early confusion. **Demand here is leg demand**: it adds up every journey that wants a seat on this leg, local and connecting alike (Chapters 4 and 8). And **spill is not a denied boarding**. Spill happens while the flight is on sale: no seat is available at a fare the customer accepts, and the customer goes elsewhere. A denied boarding happens at the gate, when more confirmed passengers show up than there are seats, a consequence of overbooking (Chapter 14).

Demand also varies from one departure to the next. The predictable part (seasons, holidays) is removed by working with homogeneous sets of departures or by forecasting it (Chapter 6); what remains is summarized by the mean μ and standard deviation σ of demand, and by their ratio.

DEFINITION 7.2 k-factor

The **k-factor** of demand is its standard deviation divided by its mean, $k = \sigma / \mu$.

Empirical studies of airline data typically find k between 0.20 and 0.40, with leisure markets at the high end and business or mixed markets at the low end, because business demand is more regular from week to week (Belobaba 2006b). With $k = 0.3$, a flight with mean demand of 150 routinely sees departures with 105 and with 195. The classical spill model assumes that demand is **normally distributed**, a bell curve pinned down by these two numbers. Swan (2002) concludes that the normal shape suits spill modelling; for flights with mean demand above about 50 it is accurate enough for planning.

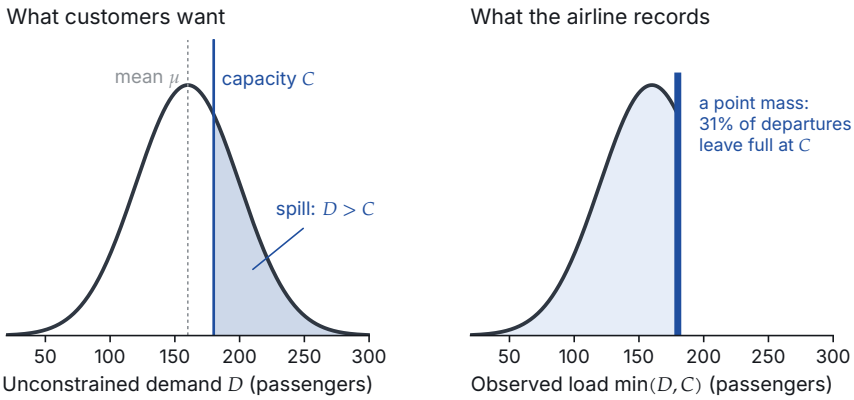


Figure 7.1 Left: the distribution of unconstrained demand for a departure with $\mu = 160$, $\sigma = 40$ and $C = 180$ seats; the shaded tail is the demand that cannot be carried. Right: the distribution of the load the airline actually records. Below capacity it follows demand; at capacity it has a point mass equal to the probability that demand reaches C, here 31 percent.

Figure 7.1 shows what capacity does to the data: the load distribution is the demand distribution with its right tail folded onto a single value. That fold is why statistics on loads mislead. Their mean is too low, and so is their spread, because the departures with the highest demand are exactly the ones that hit the ceiling. The lecture example from the reading library (Belobaba 2006b) makes the point with five Friday departures of a 125-seat flight from Montreal to Los Angeles (Table 7.1).

Table 7.1 Five hypothetical Friday departures of a 125-seat YUL–LAX flight (adapted from Belobaba, 2006). The average load factor is 85.0 percent.

Date	Load	Capacity	Load factor	Spill?
1 Apr	92	125	73.6%	no
8 Apr	125	125	100.0%	likely
15 Apr	108	125	86.4%	no
22 Apr	83	125	66.4%	no
29 Apr	123	125	98.4%	possibly

On 1, 15 and 22 April the flight left with empty seats, so its load is its demand. On 8 April it left full: demand was at least 125, and probably more. On 29 April it left two seats short, which is what a flight that sold out and then lost two no-shows looks like. Statisticians call such observations **censored**: for some departures we see the true value, for others only a lower bound.

Loads are what the airline sold, not what it could have sold. Every estimate of demand on a flight constrained by capacity is a model, never a measurement.

7.2 The expected spill formula

Planning decisions need the expected spill per departure, $\mathbb{E}[S]$, as a function of capacity: the average size of the shaded tail in Figure 7.1, counted beyond the capacity line. For normal demand it has a closed form. Measure capacity in standard deviations above mean demand,

$$z = \frac{C - \mu}{\sigma},$$

so that $z = 0$ means as many seats as mean demand and $z = 1$ one standard deviation of spare room.

RESULT 7.1 Expected spill and load under normal demand

If demand is normal with mean μ and standard deviation σ, expected spill per departure is

$$\mathbb{E}[S] = \sigma\, L(z), \qquad L(z) = \varphi(z) - z(1 - \Phi(z)), \tag{7.1}$$

where φ and Φ are the standard normal density and distribution function. Expected load is $\mathbb{E}[Y] = \mu - \sigma L(z)$, and the expected load factor is $\mathbb{E}[Y]/C$.

The **loss function** $L(z)$ is the average amount by which a standard normal variable exceeds z, counting zero when it does not. The formula therefore reads: find how far into the tail the capacity sits, take the average overflow of a standard bell curve beyond that point, and scale it by the spread of demand. Appendix A.4 derives it and tabulates $L(z)$; in a spreadsheet it is `NORM.S.DIST(z,FALSE) - z*(1-NORM.S.DIST(z,TRUE))`. Three values are worth remembering. At $z = 0$, $L(0) \approx 0.40$: an aircraft sized at mean demand spills on average 40 percent of a standard deviation, although it is "the right size on average". Far below demand, $L(z) \approx -z$, so spill is close to $\mu - C$, the answer that ignores variability. With two standard deviations of spare room, $L(2) \approx 0.008$ and spill is negligible. Variability matters twice: for a given margin of seats over mean demand, a more volatile market has a larger σ and a smaller z, and both raise spill.

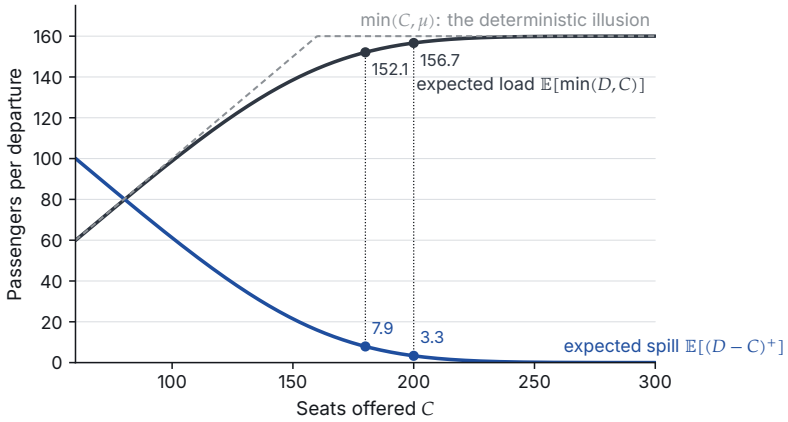


Figure 7.2 Expected load and expected spill as capacity varies, for $\mu = 160$ and $\sigma = 40$. The dashed line $\min(C, \mu)$ is what a planner who ignores variability would expect. The marked points are the two aircraft compared in Example 7.1.

Seat number x is occupied only on departures whose demand exceeds x, so expected load is the sum, over seats, of the probability that each one is used. That gives the most useful single result in capacity planning.

RESULT 7.2 The marginal seat

The C-th seat carries on average $\Pr(D \geq C)$ passengers. One more seat raises expected load, and lowers expected spill, by the probability that demand reaches capacity, $1 - \Phi(z)$ under normal demand.

Each additional seat is used less often than the previous one, so expected load is **concave** in capacity (Figure 7.2). This is the economic heart of every gauge decision, and it needs no assumption of normality.

WORKED EXAMPLE 7.1 Twenty seats are not twenty passengers

This is the handbook's Exercise B. Unconstrained demand on a departure is normal with $\mu = 160$ and $\sigma = 40$. Compare a 180-seat aircraft with a 200-seat aircraft.

180 seats. $z = (180 - 160)/40 = 0.5$. From standard tables $\varphi(0.5) = 0.3521$ and $1 - \Phi(0.5) = 0.3085$, so $L(0.5) = 0.3521 -$

$0.5 \times 0.3085 = 0.1978$. Expected spill is $40 \times 0.1978 = 7.91$ passengers and expected load $160 - 7.91 = 152.09$, a load factor of 84.5 percent.

200 seats. $z = 1$, $\varphi(1) = 0.2420$, $1 - \Phi(1) = 0.1587$ and $L(1) = 0.0833$, so expected spill is 3.33 and expected load 156.67, a load factor of 78.3 percent.

The extra seats. The twenty extra seats carry $156.67 - 152.09 = 4.58$ additional passengers per departure, not twenty. Result 7.2 explains why: the probability that demand reaches the seat falls from 0.31 at seat 180 to 0.16 at seat 200, and averages $4.58/20 = 0.229$ over those twenty seats.

Their value. The handbook assumes that 25 percent of the newly carried passengers would otherwise have been recaptured on another of the airline's flights with the same contribution, and that each truly incremental passenger contributes CAD 180. The upgauge is then worth $4.58 \times 0.75 \times 180 \approx 618$ dollars per departure; if the larger aircraft costs CAD 900 more per trip, it loses about CAD 282. The arithmetic is simple. Its value is that every assumption (normal demand, fixed fares, constant recapture, no stimulation, a single cabin) is visible and can be challenged.

PITFALL

Replacing demand by its mean. A planner who reasons "demand is 160, so 180 seats carry 160 passengers" overstates the load by 7.9 passengers and understates spill by the same amount. The error is largest exactly where decisions are made, when capacity is close to mean demand.

7.3 Load factor, demand factor and spill rate

Airline managers talk in ratios, because ratios compare flights of different sizes.

DEFINITION 7.3 **Spill ratios**

For a series of departures with capacity C: the **demand factor** $\mathrm{DF} = \mu/C$, which can exceed 1; the **average load factor** $\overline{\mathrm{LF}} = \mathbb{E}[Y]/C$, which cannot; the **spill factor** $\mathrm{SF} = \mathbb{E}[S]/C$; and the **spill rate** $\mathrm{SR} = \mathbb{E}[S]/\mu$, the share of all demand that is turned away. The demand factor is the average load factor plus the spill factor.

A less obvious fact makes these ratios practical. Under normal demand, the average load factor, the spill factor and the spill rate depend only on the demand factor and the k-factor, not on the size of the aircraft: dividing (7.1) by capacity gives $\sigma/C = k\,\mathrm{DF}$ and $z = (1 - \mathrm{DF})/(k\,\mathrm{DF})$, so the spill factor is $k\,\mathrm{DF}\,L(z)$ and the spill rate $k\,L(z)$. A single table indexed by demand factor therefore serves every aircraft. Boeing's 1978 spill model packaged it as printed **spill tables** (Boeing Commercial Airplane Company 1978); Table 7.2 reproduces one for $k = 0.35$.

Table 7.2 A spill table for normal demand with $k = 0.35$. Each row satisfies $\mathrm{DF} = \overline{\mathrm{LF}} + \mathrm{SF}$. The last column is $\Pr(D \geq C)$.

Demand factor	Average load factor	Spill factor	Spill rate	Departures full
0.70	0.687	0.013	1.9%	11%
0.75	0.726	0.024	3.2%	17%
0.80	0.761	0.039	4.9%	24%
0.85	0.792	0.058	6.9%	31%
0.90	0.818	0.082	9.1%	38%
0.95	0.841	0.109	11.5%	44%
1.00	0.860	0.140	14.0%	50%
1.05	0.877	0.173	16.5%	55%
1.10	0.891	0.209	19.0%	60%
1.20	0.914	0.286	23.9%	68%

Read left to right, the table predicts the load factor and spill of a flight whose demand you know. Read backwards, it **unconstrains**: from an observed average load factor and an assumed k, it recovers mean demand. Figure 7.3 plots that reading for several k-factors; it is the picture every network planner should carry in their head.

The curves rise slowly and then explode. At an average load factor of 85 percent the spill rate is 2.8 percent if $k = 0.20$, 8.6 percent if $k = 0.30$, 12.6 percent if $k = 0.35$ and 17.4 percent if $k = 0.40$. Belobaba notes the industry rule of thumb that spill rates much above

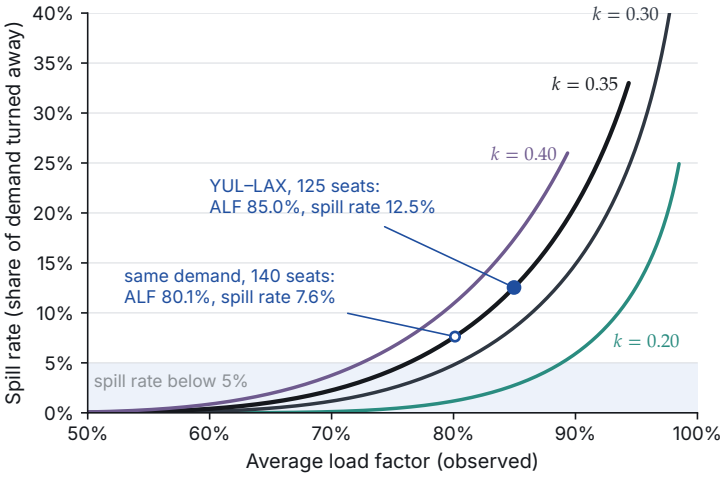


Figure 7.3 Spill rate as a function of the observed average load factor, for k-factors from 0.20 to 0.40. The shaded band marks spill rates below 5 percent. The two marked points are the YUL–LAX flight of Example 7.2 with 125 and with 140 seats.

5 percent are too high, since a lost-revenue share of that size is comparable to a typical airline operating margin (Belobaba 2006b). The 5 percent line corresponds to an average load factor of 88.9 percent when $k = 0.20$ but only 76.3 percent when $k = 0.35$. A high average load factor is therefore a symptom, not a target: the same 85 percent can mean a well-sized flight in a stable business market and a badly undersized one in a volatile leisure market.

WORKED EXAMPLE 7.2 Five Fridays to Los Angeles

Return to Table 7.1. The mean load is 106.2, an average load factor of 85.0 percent. The standard deviation of the loads is 18.6, an apparent k-factor of 0.175, but two of the five observations are capped, which compresses the spread. Following the lecture, assume instead a typical $k = 0.35$.

Unconstrain. The demand factor whose average load factor is $106.2/125 = 0.8496$ (from the formula, or interpolating in Table 7.2) is 0.971; the lecture reads 0.972 from the printed Boeing table. Mean demand is $0.971 \times 125 = 121.4$, so each departure spills on average $121.4 - 106.2 = 15.2$ passengers, a spill rate of 12.5 percent (the 12.6 percent quoted above is for an average load factor of exactly 85.0 percent).

Upgauge. With a 140-seat aircraft demand is unchanged, so the demand factor falls to 121.4/140 = 0.867. The average load factor becomes 0.801, an expected load of 112.2 (six more passengers per departure), with spill of 9.3 and a spill rate of 7.6 percent, as in the lecture's table-based answer.

How much hangs on k. The value 0.35 is an assumption. Fitting the five loads directly, with the departures of 8 and 29 April treated as censored at 125 (Section 7.4), gives mean demand 113.6 and a standard deviation of 26.4: a k-factor of 0.23 and expected spill of 5.8 passengers instead of 15.2. Five departures cannot tell the two answers apart; a season of loads, or k-factors measured on comparable flights, can. Always report a spill estimate with the k-factor it assumes.

The analysis did not conclude "85 percent is high, so upgauge"; it priced the question.

7.4 Unconstraining: recovering demand from censored loads

The spill table needs an assumed k-factor. With a season of departures, both the mean and the spread of demand can be estimated from the loads themselves, provided each full departure is treated as what it is: a lower bound on demand. Averaging the loads understates both, and the bias grows with the share of departures that closed. The standard fix is **censored estimation**: find the bell curve of demand that best explains two facts at once, the loads of the departures that left with empty seats and how often departures sold out. A common way to compute it, the **EM algorithm** (Dempster, Laird, and Rubin 1977), alternates two steps until they agree: replace each full departure by the average demand of a sellout under the current estimate, then re-estimate the mean and spread from the completed data. The full reference edition writes out the model in its Chapter 9.

The censoring is often richer than a flight that closes. Revenue management closes fare classes one by one long before the flight fills, so low-fare demand is censored on many departures that leave with empty seats. RM systems therefore unconstrain class by class along the booking curve, for instance with the **booking profile** method, which extrapolates bookings after a closure from the curves of departures that stayed open (Weatherford and Pölt 2002). Errors in

this step feed into the forecasts that set future availability, a mechanism behind the "spiral down" described in Chapter 14.

WORKED EXAMPLE 7.3 Unconstraining a season to Seville

Laurentian operates YUL–SVQ with a 182-seat A321XLR. A simulated peak season of 26 departures is drawn from normal demand with mean 175 and standard deviation 42, and capped at 182. Eleven departures closed; the other fifteen loads average 141.1.

The loads have a mean of 158.4 and a standard deviation of 30.0. Start EM there: under that estimate, a departure that sells out has an expected demand of 199.1, so each closed departure is counted at 199.1 instead of 182. Refitting raises the estimates to 165.6 and 37.4, and after further rounds the process settles at a mean of 172.7 and a standard deviation of 46.3, close to the truth.

The business consequence is large. Plugged into (7.1), the naive estimates imply 3.7 spilled passengers per departure; the censored estimates imply 14.2, a spill rate of 8.2 percent (the true model has 13.5). A route review based on the naive numbers would conclude that the XLR is comfortably sized for this season. It is not. With 26 departures the estimated mean is still uncertain by roughly 9 passengers, so pool comparable departures and report intervals.

7.5 What is a seat worth?

Suppose each carried passenger contributes r dollars (net fare minus passenger-variable cost) and each seat flown costs c. By Result 7.2, one more seat earns r times the probability that it fills, a probability that falls as seats are added, while it costs c whatever happens.

RESULT 7.3 The newsvendor rule for capacity

Expected contribution is highest when the probability of filling the last seat equals the ratio of its cost to its contribution:

$$\Pr(D > C^{*}) = c/r.$$

The rule explains why airlines deliberately accept spill: unless seats are free, the best capacity always leaves some demand un-

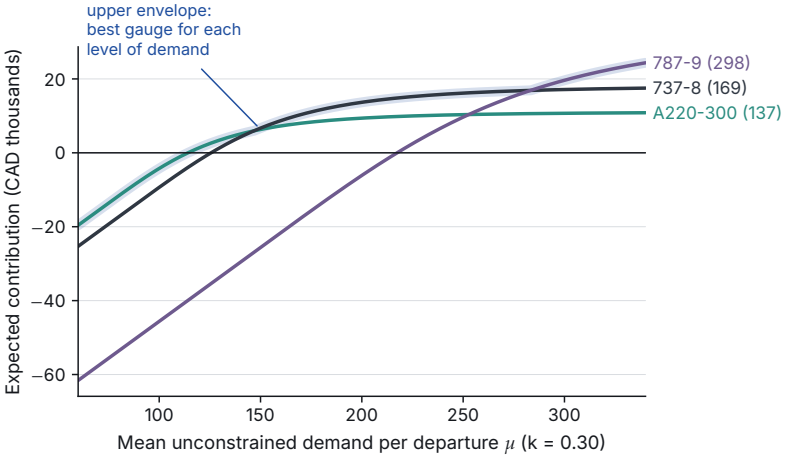


Figure 7.4 Expected contribution per departure on YUL–YVR for three Laurentian aircraft types as mean demand grows ($k = 0.30$, CAD 400 contribution per passenger, 5.2 block hours). The pale band marks the upper envelope: the best type for each level of demand.

served. Real gauge decisions are not continuous, though. Aircraft come in a few sizes and trip costs have large fixed components, so the comparison is between discrete options: for each type, expected contribution is r times the expected load given by (7.1), minus the trip cost, and the best type for each level of demand lies on the upper envelope of these curves (Figure 7.4).

WORKED EXAMPLE 7.4 Choosing the gauge for YUL–YVR

Laurentian's illustrative trip costs for a 5.2-hour YUL–YVR sector, with passenger costs netted out of the contribution, are CAD 43,600 for the A220-300 (137 seats, CAD 318 per seat), CAD 49,300 for the 737-8 (169 seats, CAD 292 per seat) and CAD 85,600 for the 787-9 (298 seats, CAD 287 per seat). Each passenger contributes CAD 400, and $k = 0.30$.

At a mean demand of 150, the A220 carries 124.8 passengers on average (a load factor of 91.1 percent) and contributes CAD 6,321; the 737-8 carries 140.0 (82.8 percent) and contributes CAD 6,688; the 787-9 loses CAD 25,602 at a 50 percent load factor. The 737-8 wins **with the lower load factor**: its extra 15.17 passengers are worth CAD 6,067, more than its CAD 5,700 extra trip cost. At a mean demand of 190 the gap widens to CAD 12,794 against CAD 9,033.

The curves cross at mean demands of about 147 (A220 to 737-8) and 283 (737-8 to 787-9). As a cross-check, the 737-8 costs $5{,}700/32 = 178$ dollars per extra seat, and the newsvendor rule, which adds seats until the last one fills with probability $178/400 = 0.445$, then puts the ideal continuous capacity at about 156 seats for a demand of 150, nearer the 737-8, which wins. Seats come in lumps, which is why fleet planners value families of aircraft with closely spaced sizes.

Aircraft are shared across the network, so putting the 737-8 on YUL–YVR means taking it from somewhere else. Fleet assignment computes these spill-adjusted contributions for every flight and type and chooses the best assignment for the fleet as a whole (Chapter 12).

7.6 Recapture

A spilled customer is not necessarily a lost customer. Some take another Laurentian departure: an earlier flight the same day, the next day, or a connection over Toronto. The airline loses only the spill it fails to recapture.

DEFINITION 7.4 Recapture rate

The **recapture rate** b is the fraction of passengers spilled from an itinerary who travel instead on another itinerary of the same airline (or of a partner whose revenue counts within the analysis's boundary). The contribution lost to spill is then roughly $(1 - b)\, r\, \mathbb{E}[S]$.

The approximation hides a constraint: recaptured passengers need seats, and when they take one they may displace someone else. Recapture is itself a capacity-constrained flow, which is why the itinerary-based fleet assignment model of Barnhart, Kniker and Lohatepanont (2002) treats it explicitly.

WORKED EXAMPLE 7.5 Recapture needs empty seats

Laurentian flies YUL–YVR twice a day with A220s (137 seats). The popular evening departure has mean demand 160 and a standard deviation of 48; the morning departure has mean demand 100 and a standard deviation of 30. Assume, for illustration, that 30 percent of spilled evening customers would accept the next morning's flight and that the two demands are independent.

The evening flight spills $48\,L((137-160)/48) = 32.8$ customers per day. If seats were never a problem, $0.30 \times 32.8 = 9.8$ of them would be recaptured. The morning flight has 38.6 empty seats on average, but not every day: simulating both flights shows that only 7.9 customers are recaptured on average, because some of the evening's worst days coincide with busy mornings. The net loss is $32.8 - 7.9 = 24.9$ customers, about CAD 10,000 of contribution per day at CAD 400 each.

The general lesson is that **spill cost is a network quantity**. Whether a spilled passenger is lost depends on the alternatives the network offers and how full they are, and adding capacity on one flight can pull passengers off another. This is the main reason network planners prefer itinerary-level models to leg-level arithmetic (Chapter 10).

7.7 Spill when revenue management controls the flight

The classical spill model assumes that bookings are accepted first come, first served until the aircraft is full, so that spilled customers are a random sample of all customers, each worth the average fare. No modern airline sells seats that way: revenue management turns away early low-fare requests to protect seats for later, higher-fare customers (Chapter 14). Three consequences matter for planning.

Spill is concentrated at the bottom of the fare ladder. The customers turned away are mostly those asking for the lowest fares, so the revenue lost per spilled passenger is well below the average fare.

The marginal seat sells at the lowest open fare. One more seat raises the booking limit of the cheapest open class by one. Figure 7.5 shows a two-fare flight: a flexible fare of CAD 1,200 with mean demand 40 (standard deviation 12) and a restricted fare of CAD 600 with mean demand 150 (standard deviation 40). The spill model values the 150th seat at CAD 604, the average fare of 726 times the prob-

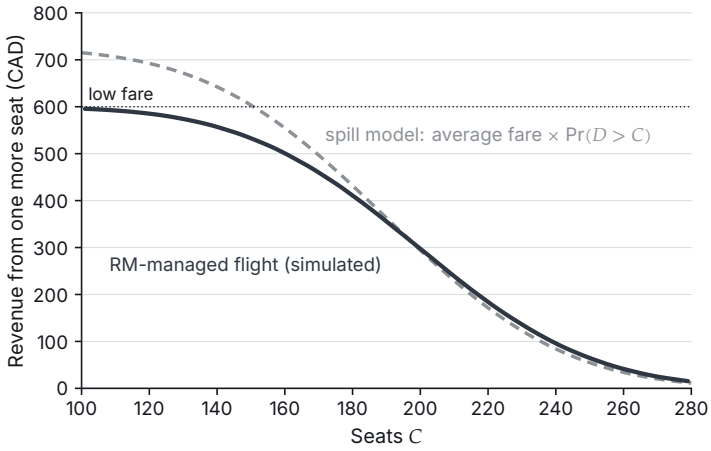


Figure 7.5 The value of one more seat on a two-fare flight. The dashed curve is the classical spill model: average fare times the probability of filling the seat. The solid curve simulates a flight controlled by revenue management, which protects 40 seats for the high fare; its marginal seat is mostly a low-fare seat.

ability of filling it; the managed flight earns about CAD 533 from it, 505 of it from low-fare demand. The spill model overstates the value of the marginal seat wherever capacity is tight, which is where gauge decisions are made.

More capacity can change who buys what. When an aircraft grows, revenue management opens cheaper classes more often, and some customers who would have paid more **buy down**. Simulation studies at MIT (Abramovich 2013) find settings in which the marginal revenue of extra seats is far below the spill model's estimate, even negative in specific scenarios: a warning against mechanical capacity arithmetic, not a claim that capacity normally destroys revenue.

The practical conclusion is to value capacity with the quantities revenue management uses: displacement costs and **bid prices**, the RM system's estimate of the opportunity cost of a seat on each leg, which already reflect fare mix and connecting flows (Chapter 14). Chapter 15 shows how to price a gauge change with them.

IN PRACTICE How spill is estimated today

In 2026 the logic of this chapter is unchanged, but it is embedded in software rather than spill tables. Network planning systems simulate O&D demand, itinerary choice, spill and recapture across the whole schedule; Lufthansa Systems, for example, documents O&D simulation with market-share and spill and recapture functions in its NetLine Plan product (Network Planning Handbook 2026; Lufthansa Systems 2026). Revenue management systems unconstrain demand by fare product and day before departure, increasingly with forecasting engines that vendors describe as Bayesian and AI-based (PROS Holdings 2021). Where airlines price continuously instead of opening and closing fare classes (Szymański, Belobaba, and Papen 2021), demand is censored by price rather than by availability. Your job is less to compute $L(z)$ by hand than to know which assumptions the tool makes, and to sanity-check its output with the formulas above.

7.8 Summary

- Airlines observe **loads**, demand capped at capacity. Averages of loads understate both the level and the spread of demand.
- Under normal demand, expected spill is $\sigma L(z)$ with $z = (C-\mu)/\sigma$, and expected load is $\mu - \sigma L(z)$.
- The C-th seat fills with probability $\Pr(D \geq C)$, so extra seats carry far fewer passengers than their number: in Exercise B, twenty seats carry 4.58.
- Spill rises steeply with the load factor, faster in volatile markets: a high load factor is a symptom, not a target.
- The best capacity always leaves some spill, and the best aircraft often runs at the lower load factor.
- Recapture needs empty seats elsewhere, and on RM-controlled flights the marginal seat sells at the lowest open fare: spill cost is a network quantity, best valued with bid prices.

7.9 Further reading

Belobaba's lecture note (Belobaba 2006b) is short and worth reading in full. Weatherford and Pölt (2002) compare unconstraining methods, Barnhart, Kniker and Lohatepanont (2002) bring recapture into fleet assignment, and Abramovich (2013) shows why spill estimates

must account for revenue management. The full reference edition (Chapter 9) gives the censored likelihood and the EM algorithm.

PART III

Designing the Network

This part is the core of the network planner's job: deciding which markets to serve, with which aircraft, how often and when, and checking that the resulting plan can be flown. It starts with the shape of the network and the aircraft that fly it, values a single route against the rest of the network, designs frequencies and timetables, follows the schedule to the teams that assign aircraft and crews, and ends with how the plan holds up on the day of operation.

CHAPTER

8

Network Structure and Connectivity

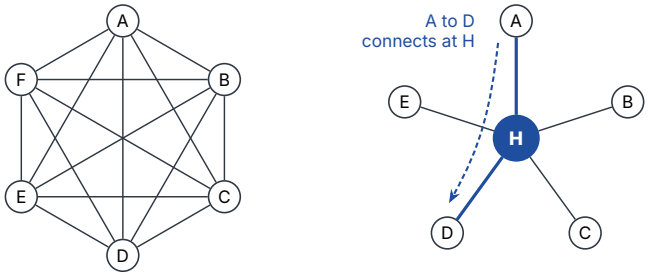
IN THIS CHAPTER

- Why most network airlines route passengers through hubs, and what that choice costs them.
- How n spokes become $n(n+1)/2$ markets, and why many of those markets are not real.
- Economies of density and scope, and when a market deserves its own nonstop.
- Banks, connecting and sixth-freedom traffic, and what Seville adds to Laurentian's Montreal hub and its feeders.
- Interline, codeshare, alliances and joint ventures; freedoms of the air and Canada's bilateral policy.
- Canadian market structure in the Competition Bureau's 2025 study, and why hub carriers earn fare premiums.

DRAWS ON Belobaba, Airline economics review, Airline schedule development, ePODS airline management game and Route planning and evaluation (MIT 16.75J, 2006); handbook; Competition Bureau, Cleared for take-off (2025); Air Canada Q2 2026 MD&A

A network planner rarely decides about one route in isolation. When Laurentian adds a flight from Montreal to Seville, it does not only open the Montreal–Seville market: it opens Quebec City–Seville, Boston–Seville, Toronto–Seville and every other market whose travellers can reach Montreal in time for the evening departure. It also puts more passengers on the feeder flights that bring them, and some of those flights were already full. The value of a route depends on the network around it.

This chapter explains why networks look the way they do: the markets a hub connects, the cost advantage of routing traffic through it, the connecting traffic it creates, the partnerships and traffic rights



Point to point
15 routes serve 15 markets

Hub and spoke
5 routes serve 15 markets:
5 local and 10 connecting

Figure 8.1 Six cities served point to point need fifteen nonstop routes to connect every pair. Served through one of them as a hub, five routes connect the same fifteen markets: five local markets to and from the hub, and ten spoke-to-spoke markets whose travellers connect at H.

that extend or limit a network, and the Canadian market as the Competition Bureau described it in 2025. Chapter 10 turns the network effects counted here into dollars, and Chapter 11 designs the banks that make connections happen.

8.1 Point to point and hub and spoke

Airline demand is generated by **origin-destination (O&D) markets**, while supply comes as **flight legs** (Chapter 2). One leg supplies seats to many markets at once, and one market is served by many itineraries, often of several airlines. Belobaba calls this the dichotomy of airline demand and supply (Belobaba 2006d). How an airline links its cities decides how many markets each leg serves.

> **DEFINITION 8.1 Network structures**
>
> A **point-to-point network** serves each O&D market it sells with a nonstop flight; passengers do not connect. A **hub-and-spoke network** concentrates flights at one or a few **hubs**: most routes link a hub to an outlying **spoke** city, and travellers between two spokes change planes at the hub. A **gateway** is a hub where domestic and regional flights connect to long-haul international flights. Real networks mix the two.

Figure 8.1 shows the trade-off with six cities. Through a hub, five routes cover all fifteen markets, and each spoke flight carries a mix

of travellers: some going to the hub city, others continuing to four other spokes. Belobaba calls this **load consolidation**: one flight serving several markets at once lets the airline offer higher frequency, larger aircraft, or both, and he names it a fundamental reason for the economic success of hubs (Belobaba 2006f).

The price is paid by the passengers who connect: a longer trip, a change of aircraft that can be missed, and competition from any nonstop another airline offers. Belobaba states the economics in one comparison: a hub pays when the cost saved by flying fewer, larger and fuller flights exceeds the revenue lost from travellers who prefer a competitor's nonstop (Belobaba 2006i). The cost side is economies of density (Section 8.3); the revenue side is how travellers value time (Chapters 4 and 5), and frequency softens it: a market with a dozen daily one-stop options can offer a shorter total trip time than one daily nonstop (Belobaba 2006d).

8.2 The combinatorics of connectivity

The arithmetic of Figure 8.1 generalizes, and the general version is the most important number in network planning.

RESULT 8.1 Markets connected by one hub

A hub with n spokes, each linked to the hub by one route, connects

$$M(n) = \underbrace{n}_{\text{local}} + \underbrace{\frac{n(n-1)}{2}}_{\text{connecting}} = \frac{n(n+1)}{2} \tag{8.1}$$

O&D markets with n routes, counting each city pair once: every pair either includes the hub and is flown nonstop, or joins two spokes and connects at the hub. A point-to-point network needs $n(n+1)/2$ routes for the same cities. Adding spoke $n+1$ adds $n+1$ markets: one local and n connecting.

Markets per route grow as $(n+1)/2$, so each new spoke is worth more than the last: a hub with 30 spokes connects 465 markets, and its 31st spoke brings 31 more. This is why hub carriers defend their hubs, and why a new spoke at a large hub is far easier to justify than the same route flown by an airline without one (Belobaba 2006i).

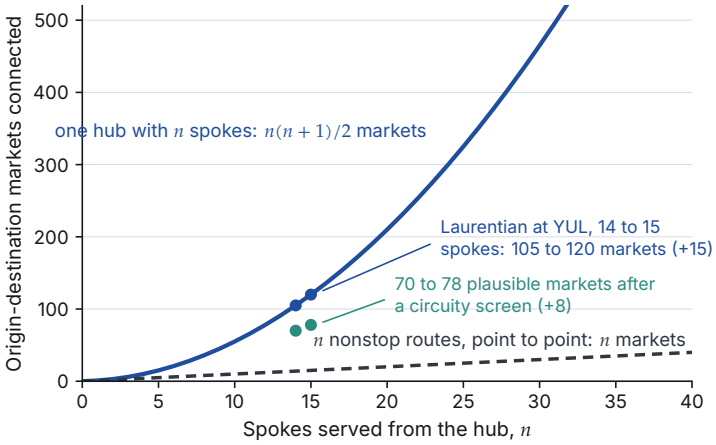


Figure 8.2 O&D markets connected by a single hub as spokes are added, against the markets served by the same number of point-to-point routes. The points mark Laurentian's illustrative YUL network before and after it adds Seville: 105 to 120 markets by the count of Result 8.1, and 70 to 78 after removing connections that backtrack (Example 8.2).

A hub turns n routes into $n(n+1)/2$ markets. That leverage, not the profit of any single route, is what a network planner manages.

Counting pairs gives an upper bound, not a forecast: a connection has commercial value only if a legal, sellable itinerary exists with enough transfer time, a tolerable elapsed time and actual demand (Network Planning Handbook 2026). Two screens remove most of the illusion. The first is timing: an arrival and a departure must fall in the same connecting window (Section 8.4.1). The second is geometry. The **circuity** of a connection from i to j through hub H is the distance flown divided by the direct distance, $(d_{iH}+d_{Hj})/d_{ij}$. Near 1 the path is natural; at 1.8 the traveller flies almost twice the distance and buys only at a deep discount, if at all. The cut-off is an analyst's choice; this book uses 1.25.

> **PITFALL**
>
> Counting geometric markets as commercial ones. A hub with 40 spokes does not serve 820 markets. Between two spokes on the same side of the hub (Ottawa to Toronto through Montreal

has a circuity of 1.8), between spokes whose banks do not meet, and between cities with a better nonstop, the count is zero. Report connectivity after the screens, and say which screens you used.

8.3 Economies of density and scope

Most airline costs are incurred per departure or per block hour whether the seats are full or not (crew, landing and navigation charges, ground handling, most of the fuel), and larger aircraft have a lower cost per seat (Chapter 3). More passengers on the same routes therefore lower the cost per passenger twice: per-departure costs are shared by more travellers, and the airline can fly larger aircraft.

DEFINITION 8.2 Economies of density, scale and scope

- **Economies of density**: unit cost falls when more traffic is carried over an unchanged network. Returns to density are the proportional rise in traffic divided by the proportional rise in cost it causes; above 1 means economies of density.
- **Returns to scale**: the same ratio when traffic and the number of points served grow together, so density is unchanged.
- **Economies of scope** between two outputs, here two O&D markets, exist when producing them together costs less than producing them separately (Panzar and Willig 1981).

Caves, Christensen and Tretheway (1984), for US carriers over 1970–1981, estimated returns to density of about 1.24 but roughly constant returns to scale: carrying more passengers on the same network lowers unit cost, serving more cities at the same density does not. The magnitudes are historical; the mechanism is durable, and it says what a hub is for. A hub is a device for buying density: by consolidating many thin markets onto a few spoke routes, it raises the traffic on each leg without serving fewer markets. Economies of scope follow, because every spoke flight produces a bundle of markets that share its departure cost.

8.3.1 When does a market deserve its own nonstop?

The clearest way to see density at work is to ask when a market should leave the hub. Approximate the trip cost of a flight as a cost

per departure a, independent of aircraft size, plus a cost per seat; dividing the seat cost by the load factor and adding the passenger-related cost gives κ, the cost of each passenger carried. A daily nonstop for Q travellers each way costs $a + \kappa Q$. Routed through the hub, the same travellers ride existing spoke flights and, if those have room, add no departures.

RESULT 8.2 The hub-versus-nonstop threshold

Let a nonstop in market m cost $a_m + \kappa_m Q$ per day and direction, and routing its Q travellers over two hub legs cost $(\kappa_1 + \kappa_2)Q$. The hub routing is cheaper as long as

$$Q < Q^* = \frac{a_m}{\kappa_1 + \kappa_2 - \kappa_m}. \tag{8.2}$$

The numerator is the cost of the extra departure; the denominator is the extra cost per passenger of flying two legs instead of one. Behind a gateway, where the long-haul seat is needed either way, $Q^* = a_m/\kappa_1$: the nonstop must pay a whole long-haul departure to save one short feeder seat. Travellers who refuse to connect lower the threshold.

The result ignores discrete aircraft sizes, minimum commercial frequencies and spill (Chapter 7), and it assumes the feeders have room. Its value is that it names the quantity that matters: not whether a market can fill an aircraft, but whether it can pay for an additional departure.

WORKED EXAMPLE 8.1 Two thresholds on Laurentian's network

Fit the linear trip cost to two Laurentian gauges. On a 1.5-hour transborder sector an A220-300 (137 seats) costs CAD 14,000 per trip before passenger costs and a 737-8 (169 seats) CAD 16,000, so a seat costs 2,000/32 = 62.5 dollars and the departure itself 14,000−62.5×137 = 5,437.5. With an 80 percent load factor and a passenger cost of CAD 30, κ = 62.5/0.80 + 30 = 108.1 dollars.

Between two spokes, such as Quebec City and Boston, the three sectors are similar, so the denominator of (8.2) is one κ and Q^* = 5,437.5/108.1 = 50 travellers a day each way. If 20 percent of the market would refuse a connection and each con-

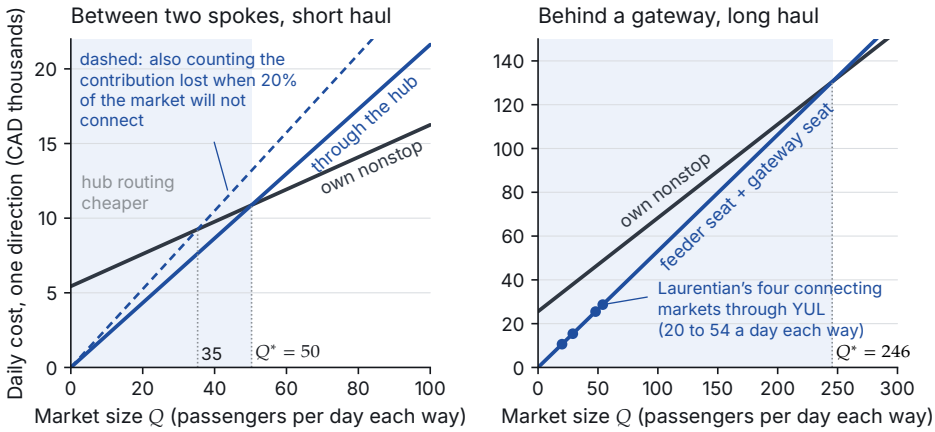


Figure 8.3 The daily cost of serving one market by its own nonstop and through the hub, as the market grows. Left: between two spokes on short sectors, the hub routing is cheaper below about 50 travellers a day each way, or 35 once the contribution lost to travellers who refuse to connect is counted. Right: behind a long-haul gateway the nonstop must pay a whole long-haul departure to save a feeder seat, and the threshold rises to about 246; Laurentian's four connecting markets sit far below it.

tributes CAD 230, the hub routing loses $0.20 \times 230 = 46$ dollars per traveller and the threshold falls to $5{,}437.5/(108.1+46) = 35$.

Behind the gateway, the same fit to the A321XLR (CAD 80,500) and the 787-9 (CAD 115,500) on a 7-hour sector gives a departure cost of CAD 25,586, while the connection adds only a feeder seat on a 1.4-hour A220 sector at $\kappa_1 = 104.2$ dollars. A Boston–Seville nonstop needs $25{,}586/104.2 = 246$ travellers a day each way.

Laurentian's connecting markets through its evening bank (Example 8.2) carry 20 (Quebec City–Seville), 29 (Boston–Seville), 48 (Quebec City–Paris) and 54 (Boston–Paris) travellers per departure: between a twelfth and about a fifth of that threshold. On cost alone none could support a nonstop. The numbers are illustrative; the order of magnitude is the lesson.

Figure 8.3 explains why long-haul nonstops from secondary cities appear only when the market is large or the aircraft small. The A321XLR lowers the long-haul departure cost a_m, which is exactly what moves the threshold; Cirium's 2026 review describes it as opening markets too thin for widebodies, at the price of less belly cargo (Cirium 2026b). Chapter 9 returns to the aircraft.

8.4 Connecting traffic, banks and the sixth freedom

A hub changes who is on board. The Montreal–Seville flight carries travellers from Montreal, but also from Quebec City and Boston, and westbound travellers continuing across North America. Planners describe the mix with standard categories.

> **DEFINITION 8.3 Local, connecting and sixth-freedom traffic**
>
> On a leg from hub H to destination B: **local** traffic travels only between H and B; **behind** (or feed) traffic originates behind the hub and connects at H onto the leg, such as Quebec City to Seville over Montreal; **beyond** traffic continues past B on another flight; **flow** traffic is all traffic that is not local. **Sixth-freedom traffic** is connecting traffic between two foreign countries through the airline's home country, such as Boston to Paris over Montreal on a Canadian airline. Quebec City to Paris over Montreal is connecting but not sixth-freedom traffic.

Connecting travellers have three economic properties that recur throughout this book. When the legs they use have empty seats, they cost almost nothing to carry. They pay prices set in their own O&D market, where they can choose a competitor's nonstop or another hub, so their fares need not exceed the long-haul local fare by much: Laurentian's illustrative restricted Boston–Paris fare is CAD 599 against CAD 549 for Montreal–Paris. And when the legs are full, a connecting passenger takes a seat a local passenger could have bought, so the network gains only the difference. Valuing that displacement is the subject of Chapters 10 and 14.

Belobaba draws the planning consequence: at an established hub, a new route to a small spoke can be justified by one or two connecting passengers per flight to each of 30 or more destinations (Belobaba 2006i). He warns that this incremental logic tends to overlook displacement on the connecting legs, and makes an unprofitable spoke that feeds the network hard to drop. It is also why a hub serves markets of five travellers a day, at the cost of five feeder seats, as long as the feeder has room.

IN PRACTICE **Hubs, sixth freedom and partners at Air Canada in 2026**

Air Canada describes Toronto Pearson, Montreal-Trudeau and Vancouver as its international hubs (Air Canada 2025). Its second-quarter 2026 report attributes 9 percent year-on-year growth in sixth-freedom revenue to the US–Atlantic and US–Pacific corridors, mainly United States to Europe (Air Canada 2026d). In December 2024 it set a goal of 2 percent of the market between the United States and international destinations, 80 percent of which it said could connect efficiently over Toronto, Vancouver or Montreal (Skies Magazine 2024). The Seville service announced on 14 October 2026 is described as timed to connect through Montreal, where US preclearance lets travellers continuing to the United States arrive as domestic passengers (Air Canada 2026f). The partnership with United brings about two million customers a year who connect to or from the United States onto the domestic network (Air Canada 2026g). These are company statements about the network, not evidence about any route's profitability.

PITFALL

Calling every connection sixth-freedom traffic, a common error in the handbook's list (Network Planning Handbook 2026). Domestic-to-international connections (Quebec City–Paris) and international-to-domestic ones (Boston–Vancouver) depend on different traffic rights and competitors from true sixth-freedom flows (Boston–Paris), and respond differently to a change in the long-haul schedule.

8.4.1 How banks make connections

A connection needs an arrival and a departure separated by at least the time to transfer passengers and bags, and by not so much that the trip becomes unattractive. The handbook writes the screen directly (Network Planning Handbook 2026): arrival i connects to departure j if

$$\mathrm{MCT}_{ij} \;\le\; t_j^{\mathrm{dep}} - t_i^{\mathrm{arr}} \;\le\; T_{\max}, \tag{8.3}$$

where MCT_{ij} is the **minimum connecting time** for that pair of flights (it depends on terminals and on whether the passenger

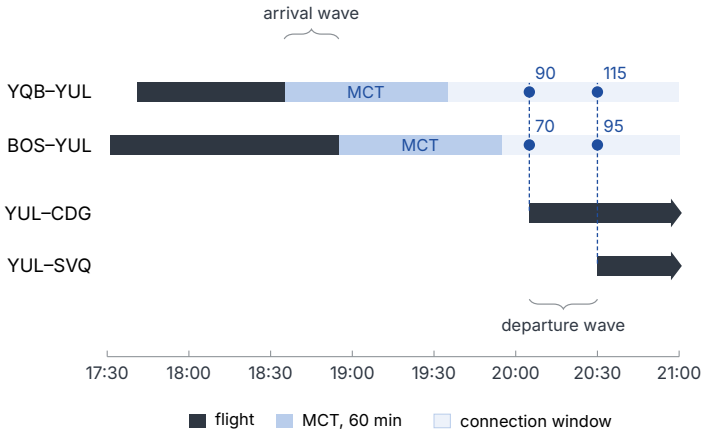


Figure 8.4 Four legs of Laurentian's evening bank at YUL. The Quebec City and Boston flights arrive at 18:35 and 18:55; each opens a connection window once the 60-minute MCT has passed. The Paris departure at 20:05 and the Seville departure at 20:30 fall inside both windows, so the bank creates four connections, with connecting times of 70 to 115 minutes. Times are illustrative except the 20:30 Seville departure, which mirrors the public schedule; the MCT is an assumption, not a YUL rule.

crosses a border) and $T_{\max}$ is the longest connection the analyst considers sellable, a commercial assumption rather than an industry rule.

A **bank** (or connecting complex, or wave) makes this hold for many pairs at once: a group of arrivals close together, followed after at least the MCT by a group of departures. With n_a arrivals and n_d departures a bank creates up to $n_a n_d$ connections, less those that send passengers back where they came from. Banks can be directional: the hubs of the ePODS teaching network have two eastbound and two westbound banks a day (Belobaba 2006h), so that east-west traffic connects without wasting connections between two eastern cities.

Figure 8.4 shows the part of Laurentian's evening bank in the case data: two feeders, two transatlantic departures, four connections. Tight banks shorten and multiply connections but concentrate the demand for gates, security, baggage systems, crews and runways into short peaks; spreading flights out eases the peaks and may lengthen or remove connections (Network Planning Handbook 2026). Chapter 11 designs banks in detail.

8.4.2 What one more spoke adds: Seville at YUL

This worked example puts the chapter's tools together on the case that runs through the book: counting markets, estimating feed, and checking what the feed does to the flights that carry it.

WORKED EXAMPLE 8.2 Adding Seville to Laurentian's Montreal hub

Counting markets. Suppose, for illustration, that Laurentian serves from YUL the fifteen other airports of the running case: Quebec City, Ottawa, Halifax, Toronto, Vancouver, Boston, New York LaGuardia, Chicago, Paris, London, Lisbon, Seville, Madrid, Barcelona and Rome. Without Seville it has 14 spokes and $14 \times 15/2 = 105$ markets; with Seville, 120. The new spoke adds 15 markets, one local and 14 connecting.

Most of the 14 are not markets. The circuity screen removes the six European spokes at once (Lisbon to Seville over Montreal flies 33.6 times the direct distance, London to Seville 6.6 times) and Halifax too, at 1.33, because Halifax lies east of Montreal. Seven North American markets remain, with circuities between 1.00 and 1.10. Seville adds 8 plausible markets, not 15, and the whole network has 70 plausible markets before Seville and 78 after (Figure 8.2).

Estimating feed. The market model of Chapter 4 gives the nonstop about 82 local travellers per departure, so the case's shoulder-season mean demand of 150 includes about 68 connecting travellers: feed is 45 percent of the flight. Only spokes with a flight in the evening bank can supply it. The case's four-leg bank includes two, with mean demand per departure of 20 travellers from Quebec City and 29 from Boston; the leg's own fare products (mean 112) stand for the 82 locals and about 30 connectors from spokes outside the model. In the model the flight's demand is $112 + 20 + 29 = 161$ per departure; feed is $49/161 = 30$ percent of it and CAD 40,321 of the flight's CAD 121,709 of ticket revenue (33 percent): a small model treats as local whatever arrives from outside it.

What the feed does to the feeders. Mean demand on the 137-seat Quebec City A220 rises from 121 to 141, and on the Boston A220 from 124 to 153. By the spill formula of Chapter 7, expected spill on the Quebec City flight rises from 2.0 to 10.0 trav-

ellers per departure and the share of departures that leave full from 20 to 58 percent; on the Boston flight, from 2.8 to 18.6 and from 25 to 78 percent. The feeders turn away about 24 more travellers per bank, half as many as the new feed; which travellers depends on revenue management (Chapter 14). The Seville flight itself, 161 against 182 seats, spills 3.0.

What the planner learns. The network effect of a spoke is plausible markets times their demand, minus what the new traffic displaces on shared legs. Geometric counts nearly double the first term, and the second is not small: Seville turns two comfortable feeders into constrained ones and candidates for an upgauge (Chapters 12 and 15). Chapter 10 values both effects in dollars.

8.5 Alliances, codeshares, interline and joint ventures

An airline can extend its network without flying more aircraft by cooperating with other airlines. The forms differ in what they permit, how revenue is shared and how competition authorities treat them (Table 8.1), and the handbook insists on keeping them apart (Network Planning Handbook 2026).

Table 8.1 Forms of airline cooperation. Summarized from the handbook and the US Department of Transportation (Network Planning Handbook 2026; U.S. Department of Transportation 2024).

Form	What it permits	How revenue is shared	Competition treatment
Interline	One ticket across airlines, bags checked through	Prorate of the through fare	Ordinary commercial agreement
Codeshare	Marketing airline sells a partner's flight under its own number	Settlement by prorate or negotiated rate	Cooperative agreement; no joint pricing
Alliance	Codeshares, loyalty reciprocity, lounges, coordinated connections	As for interline and codeshare	No joint pricing or capacity coordination
Joint venture	Coordinated fares, schedules, capacity and sales within a defined scope	Revenue or profit pooled by formula, whoever operates	Needs antitrust immunity or approval, usually with conditions

The joint venture is the deepest form. Its partners share revenue or profit on the markets in scope whichever partner operates the aircraft, a property called **metal neutrality**, so neither cares whose flight a customer takes and both can plan capacity for the combined network. In the United States, antitrust immunity lets the partners coordinate fares, services and capacity as if they were one carrier in the relevant markets, subject to conditions set after a competitive and a public-benefits analysis (U.S. Department of Transportation 2024).

Why would a competition authority let two airlines set prices together? For interline travellers the partners' flights are complements. If each carrier prices its own segment of a through itinerary, each markup reduces the traffic on which the other earns, and the through fare ends up above what a single owner would charge: Cournot's problem of complements, or **double marginalization**. Joint pricing removes it, lowering the fare and raising both traffic and combined profit. Brueckner (2001) applies it to international alliances: cooperation lowers fares for interline passengers, but on gateway-to-gateway routes where the partners would otherwise compete it can raise them, which is why approvals come with conditions and carve-outs.

WORKED EXAMPLE 8.3 Pricing a beyond itinerary with a partner

Laurentian flies Montreal–Paris and a European partner flies onward from Paris. With an illustrative through demand $Q = 1{,}600 - P$ passengers a month and marginal costs of CAD 300 and 200 on the two segments, separate pricing gives a through fare of CAD 1,233, 367 passengers and CAD 268,889 of combined profit. A metal-neutral joint venture charges CAD 1,050, carries 550 passengers and earns CAD 302,500, and consumer surplus rises from CAD 67,222 to 151,250 a month. For the planner, the revenue a route receives from partner feed depends on the agreement: a prorate and a joint-venture allocation give different answers for the same passenger (Network Planning Handbook 2026).

Air Canada is a founding member of Star Alliance, and its partner page (dated 2022, so a starting map) names joint ventures with United and the Lufthansa Group on transatlantic travel, with United on most flights between Canada and the United States, and with Air China between Canada and China (Air Canada n.d.b; Network Planning Handbook 2026). On 16 October 2026 Air Canada and United



Figure 8.5 Freedoms three to nine for an airline of home country H. Accent arrows carry the traffic each right concerns; grey dashed arrows are legs flown under another right. The examples are for a Canadian airline.

announced that they would renew and expand their transborder agreement, in place since 2022, subject to regulatory approval, and Air Canada applied to Transport Canada for approval (Air Canada 2026g); the expansion would bring in ten city pairs that had been carved out (Travelweek 2026a). For any route that touches a partner, the handbook recommends three views: the operated flight, the airline's incremental network economics and the partnership's economics (Network Planning Handbook 2026).

> **PITFALL**
>
> Assuming that a codeshare or alliance membership permits joint pricing or capacity planning. Neither does (Network Planning Handbook 2026). Coordinating fares and capacity requires an approved joint venture, within its approved scope and outside any carved-out markets. Sharing capacity plans with a partner outside that scope creates a legal problem, not a network benefit.

8.6 Freedoms of the air, bilateral agreements and Blue Sky

A commercially attractive route also needs the right to fly it. International services are governed by air transport agreements between states, which specify which airlines may fly, between which points, how often and with what pricing freedom. The rights they exchange have a standard vocabulary.

DEFINITION 8.4 Freedoms of the air

For an airline of home country H, Transport Canada defines seven freedoms (Transport Canada n.d.):

1. to fly over another country without landing;
2. to land in another country for non-commercial reasons, such as refuelling;
3. to carry traffic from H to another country;
4. to carry traffic from another country to H;
5. to carry traffic between two foreign countries on a service that starts or ends in H;
6. to carry traffic between two foreign countries through H;
7. to carry traffic between two foreign countries on a service that does not touch H.

Cabotage, the carriage of domestic traffic within a country by a foreign airline, completes the list; its two forms are often called the **eighth** and **ninth freedoms**. ICAO calls every freedom beyond the fifth "so-called", because only the first five are recognized by treaty (International Civil Aviation Organization n.d.).

A hub carrier's connecting traffic is built from third and fourth freedoms (Figure 8.5): a Boston–Paris traveller on a Canadian airline uses Canada–United States rights for the first leg and Canada–France rights for the second, and the sixth freedom names the combination. Fifth freedoms are rarer and more valuable, because they let an airline compete in a market between two foreign countries.

Canada negotiates these rights under its **Blue Sky policy** of 2006. Its primary goal is reciprocal Open Skies-type agreements where they are in Canada's overall interest: open third and fourth freedoms, no limits on airlines, frequencies or aircraft types, market-based pricing, flexible codesharing, unrestricted fifth and sixth freedoms, and seventh freedoms for all-cargo services. It excludes cabotage outright, and where a partner will not agree to Open Skies, Canada may negotiate a smaller exchange of rights (Transport Canada n.d.).

For planning, three consequences matter. Traffic rights and airport slots are separate constraints: the right to serve a country does not guarantee usable times (Chapter 11). A route outside an agreement may still be possible through an extra-bilateral authority, a separate application. And licences, designations and permissions are distinct workstreams; the handbook advises asking the responsible team for

the actual treaty and its limits rather than inferring them from the freedom categories (Network Planning Handbook 2026).

8.7 Market structure and hub premiums in Canada

In June 2025 the Competition Bureau published a market study of Canada's domestic airline industry, Cleared for take-off (Competition Bureau Canada 2025). The numbers below are from its executive summary.

Concentration. At major airports Air Canada and WestJet together carried between 56 and 78 percent of domestic passengers in 2023. Concentration fell by one tenth from 2019 as Porter and Flair entered and grew, even though Lynx Air exited, but the Bureau calls average concentration extremely high and new competition fragile (Competition Bureau Canada 2025). Nationally, it estimates 2023 domestic passenger shares of 34.1 percent for Air Canada, 30.1 for WestJet, 9.8 for Flair, 9.0 for Porter and 17 for all others, and the four busiest airports (Toronto Pearson, Vancouver, Calgary and Montreal-Trudeau) handle 55 percent of domestic passengers.

Entry and fares. When one new competitor starts flying a route, average fares fall by 9 percent (Competition Bureau Canada 2025), about CAD 27.80 per base fare in the full report. The share of Air Canada's major routes on which WestJet and at least one smaller carrier also fly rose from 11 to 16 percent after 2019, and the corresponding share for WestJet from 13 to 21 percent; the history of entry and exit, from Canada 3000 to Lynx Air in 2024, is why the Bureau calls these gains fragile.

Barriers and recommendations. New airlines face small scale, high start-up costs, no access to interline agreements and alliances, loyalty programs and slot rules that favour incumbents, user-pay fees of about 30 percent of what passengers pay full-service airlines, and foreign ownership limits (Competition Bureau Canada 2025). Of the Bureau's ten recommendations, two bear directly on network planning: removing the Minister of Transport's power to override the Competition Act's review of airline mergers and competitor collaborations, which include the joint ventures the Minister reviews (Travelweek 2026a); and raising the single-investor foreign ownership limit to 49 percent (Competition Bureau Canada 2025).

WORKED EXAMPLE 8.4 How concentrated is the domestic market?

The Herfindahl-Hirschman index (HHI) sums squared market shares in percent. The four named carriers contribute $34.1^2 + 30.1^2 + 9.8^2 + 9.0^2 = 2{,}246$, and the 17 percent held by others up to $17^2 = 289$, so the national HHI lies between 2,246 and 2,535. That understates concentration where it matters: the relevant antitrust market is the O&D market, and many routes have one or two carriers. The Bureau's full report puts average concentration at more than three times the level at which the Competition Act's merger provisions consider markets concentrated (Competition Bureau Canada 2025). At Montreal-Trudeau, its airport chart shows Air Canada with roughly three fifths of domestic passengers (read off the chart, not published as a number).

8.7.1 Hub premiums

Fares on routes to and from an airline's hub tend to be higher than in comparable markets elsewhere; that difference is the **hub premium**. Borenstein (1989) showed with US data that dominance at a route's endpoints is strongly associated with higher fares, and Berry, Carnall and Spiller (1996) found the premium concentrated on price-insensitive business travellers who prefer the carrier that dominates their home airport. The sources of that preference are familiar: the most departures from the hub, the loyalty program local travellers collect in, corporate contracts with local employers, and control of gates and slots. The estimates are historical and should not be transferred to current markets (Network Planning Handbook 2026).

The mechanism that turns preference into price comes from discrete choice (Chapter 5). In a logit model with price sensitivity α, a carrier with share s of a traveller segment sets its markup over marginal cost at $1/(\alpha(1-s))$. Anything that raises the hub carrier's appeal raises its share and therefore its markup, even with identical costs, and most where price sensitivity is low.

WORKED EXAMPLE 8.5 Price and mix effects in a hub premium

Take two segments with marginal cost CAD 300: business travellers with $1/\alpha = 450$ dollars and leisure travellers with $1/\alpha = 150$. A carrier with 25 percent of each segment charges $300 + 450/0.75 = 900$ dollars and $300 + 150/0.75 = 500$. At its hub its business share rises to 55 percent but its leisure share only to 30 percent, so its fares become $300 + 450/0.45 = 1{,}300$ and $300 + 150/0.70 = 514$: premiums of 44 and 3 percent. With 30 percent business travellers in both markets, the average fare rises from CAD 620 to CAD 750, a premium of 21 percent. If the hub also attracts more business travellers, say 40 percent, the average is CAD 829 and the measured premium 34 percent, more than a third of it from mix rather than price. All values are illustrative.

For a planner, hub premiums have three implications. They are part of the revenue that makes thin spokes viable, so a hub's economics cannot be read from costs alone. They should not be imported into a market where the airline lacks the same presence: Laurentian's Montreal-origin fares say little about what it could earn from Toronto. And they are fragile where they come from the absence of competition: if one more competitor lowers fares by 9 percent (Competition Bureau Canada 2025), part of the premium can disappear with a single entrant.

8.8 Summary

- A hub consolidates many O&D markets onto each leg, so the airline can fly more frequencies and larger aircraft; connecting passengers pay with longer trips.
- A hub with n spokes connects $n(n+1)/2$ markets, and its next spoke adds $n+1$. Timing and circuity screens cut the count: Seville adds 8 plausible markets, not 15.
- Airlines show economies of density but roughly constant returns to scale; a hub buys density. A market deserves a nonstop only when it can pay for an extra departure: behind a long-haul gateway, a few hundred travellers a day.
- Feed is nearly free when feeders have empty seats and displaces other travellers when they do not: Seville's connectors turn two comfortable feeders into constrained ones.

- Only an approved joint venture permits joint pricing, which removes double marginalization on complementary flights.
- Traffic rights come from bilateral agreements; Blue Sky seeks Open Skies but excludes cabotage. Rights, slots and licences are separate constraints.
- The Competition Bureau found the domestic market highly concentrated; one more competitor lowers fares by 9 percent. Hub premiums come from schedule, loyalty and airport presence.

8.9 Further reading

Belobaba's lectures on airline economics, route planning and schedule development (Belobaba 2006d, 2006i, 2006f) set out the hub economics of this chapter. The handbook (Network Planning Handbook 2026) is the best short guide to connections, partnerships and market access, and the Competition Bureau's executive summary (Competition Bureau Canada 2025) is the essential Canadian reference. On the economics, read Caves, Christensen and Tretheway (1984), Borenstein (1989), Berry, Carnall and Spiller (1996) and Brueckner (2001). The full reference edition derives the threshold, the joint-venture prices and the logit markup in its Chapter 10.

CHAPTER

9

Fleet Planning

IN THIS CHAPTER

- Fleet planning as a staged decision under uncertainty, and what the network analyst contributes to it.
- Top-down and bottom-up fleet models: what each sees and what each misses.
- How airlines judge an aircraft: economics, range and payload, commonality, availability and financing.
- Reading a payload-range diagram from the A321 manual, and the A321XLR's mission to Seville against winter winds and over the ocean.
- Owning versus leasing, and why commonality pays.
- A Laurentian fleet decision for thin transatlantic markets, and the parameter that decides it.

DRAWS ON Belobaba, Airline fleet planning models (MIT 16.75J, 2006); Airbus A321 aircraft characteristics, airport and maintenance planning (rev. 33, 2024); handbook; Air Canada Q2 2026 MD&A

A fleet is the most expensive and least reversible commitment an airline makes about its network. Aircraft are ordered years before delivery, financed over a decade or more and flown for two or three decades. The analyst who designs next summer's schedule inherits fleet decisions taken years earlier: which markets can be served at all, at what trip cost and with how many seats. Fleet decisions in turn rest on assumptions about routes, frequencies and demand that come from the planning team.

This chapter follows the fleet decision from the planning process to the aircraft itself, and ends with a fleet decision for Laurentian that combines the trip-cost model of Chapter 3 with the spill model of Chapter 7. Assigning the fleet to individual flights is the subject of Chapter 12.

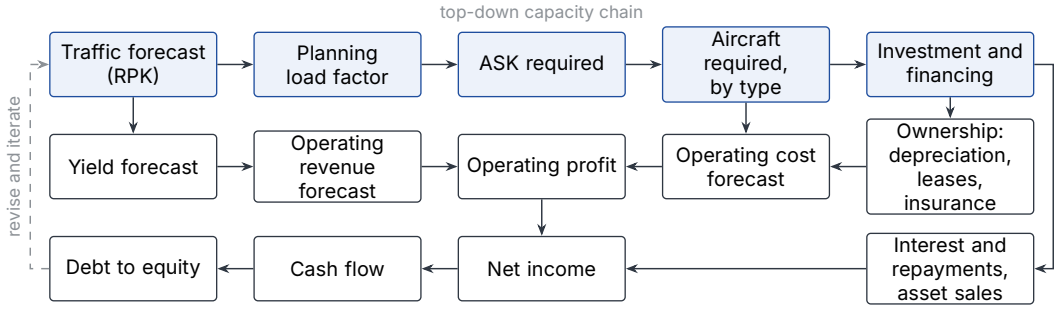


Figure 9.1 The fleet planning evaluation process (adapted from Belobaba, 2006). The top row is the capacity chain used by top-down models; the lower rows carry the plan through costs, revenues and financing to the balance sheet.

9.1 The fleet planning problem

Fleet composition shapes an airline's costs, its finances and, above all, the routes it can serve, because each type can carry only so much payload so far (Belobaba 2006k). In 2006 a 150-seat narrowbody cost USD 40 to 60 million and a 747-400 more than USD 180 million, and some aircraft were flown economically for more than 30 years (Belobaba 2006k). The prices are historical; the proportions are not.

> **DEFINITION 9.1 Fleet plan**
>
> A **fleet plan** specifies, for each future period, the number of aircraft of each type the airline will operate, together with the deliveries, lease returns, retirements and transfers between subfleets that produce those numbers. Choosing it is a **staging problem**: decisions are taken in sequence under uncertainty about demand, costs and competition, and each is constrained by the existing fleet, by the ease of disposing of older aircraft and by the delivery positions manufacturers still have (Belobaba 2006k).

Belobaba describes fleet planning as an evaluation chain (Figure 9.1). Traffic and yield forecasts give revenues; a planning load factor converts the traffic forecast into required seat-kilometres; dividing by what one aircraft produces gives the number of aircraft by type; and the aircraft imply an investment whose costs and financing carry the plan through to profit, cash flow and the balance sheet. Each output feeds back into the assumptions, so the process is iterative (Belobaba 2006k).

At ten years or more, the airline chooses fleet families and places orders. At two to five years, it adjusts delivery timing, extends or returns leases, retires aircraft and moves them between brands or bases. Within a season the fleet is fixed, and the question becomes which type flies which flight (Chapter 12). The handbook asks which regions and aircraft capabilities deserve scarce capital (Network Planning Handbook 2026). Table 9.1 shows such a plan at Air Canada.

Table 9.1 Air Canada's operating fleet at 31 December 2025 and 30 June 2026, and planned at 31 December 2026 and 2027, for selected types (Air Canada 2026d). Planned transfers are subject to the necessary approvals.

Type	Dec 2025	Jun 2026	Plan Dec 2026	Plan Dec 2027
Boeing 787-9	32	32	32	32
Boeing 787-10	0	0	2	8
Airbus A321XLR	0	2	9	19
Airbus A321, mainline	21	21	26	34
Airbus A220-300	41	46	57	65
Boeing 737 MAX 8, mainline	47	31	0	0
Boeing 737 MAX 8, Rouge	0	21	52	52
Mainline and Rouge, all types	256	262	267	288

The table is a staging plan in miniature: narrowbody capacity grows through A220 and A321XLR deliveries, the 737 MAX 8s move to the leisure brand, and the 787-10 joins the widebody fleet. The MD&A also reports eight A350-1000s ordered in February 2026 for delivery from 2030, and CAD 12.7 billion of committed capital expenditure on 30 June 2026 (Air Canada 2026d).

9.2 Top-down and bottom-up models

Belobaba distinguishes two approaches (Belobaba 2006k). A **top-down** (macro) model works with aggregates: total traffic, an average stage length, an average utilization. A **bottom-up** (micro) model builds future networks and schedules route by route and derives the fleet from them.

The top-down capacity gap The top-down model follows the capacity chain of Figure 9.1. Divide the traffic forecast RPK_t for year t by the planning load factor $\mathrm{LF}^{\mathrm{plan}}$, and subtract what the remaining

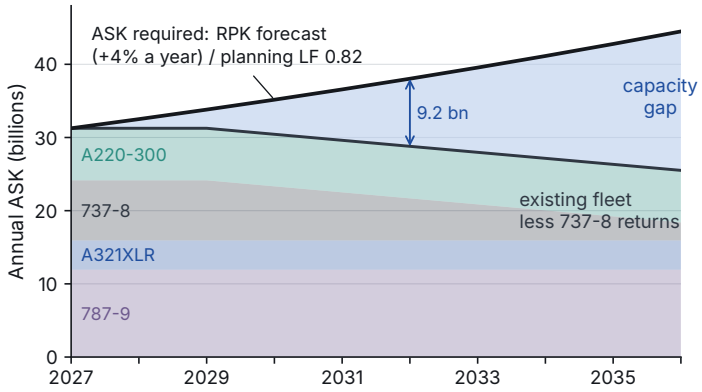


Figure 9.2 A top-down capacity gap for an illustrative Laurentian fleet: 30 A220-300s, 20 737-8s, 6 A321XLRs and 10 787-9s in 2027, with two 737-8 leases returned each year from 2030. Traffic grows 4 percent a year and the planning load factor is 82 percent.

fleet can produce. The remainder is the **capacity gap** G_t that new aircraft must fill:

$$G_t = \frac{\mathrm{RPK}_t}{\mathrm{LF}^{\mathrm{plan}}} - \sum_k N_{kt}\, q_k, \qquad q_k = 365\, U_k\, C_k\, s_{b,k}, \tag{9.1}$$

where N_{kt} aircraft of type k remain in year t, each producing q_k seat-kilometres a year: C_k seats times block speed $s_{b,k}$ times U_k block hours a day (Chapter 3). Candidate types are then compared on the annual cost of producing the gap.

RESULT 9.1 The capacity gap is leveraged on planning assumptions

The gap is a small difference between two large numbers. A 1 percent change in the planning load factor or the traffic forecast moves it by A_t/G_t percent, where A_t is the required capacity: when the gap is a quarter of the requirement, a 1 percent error becomes a 4 percent error in the aircraft to buy.

WORKED EXAMPLE 9.1 Laurentian's capacity gap

Give Laurentian, for illustration, a 2027 fleet of 30 A220-300s, 20 737-8s, 6 A321XLRs and 10 787-9s, flown at 9, 10, 13 and 14 block hours a day on typical stages of 900, 2,000, 5,500 and 6,500 km. One aircraft of each type produces 237, 411, 667 and 1,193 million ASK a year, and the fleet 31.3 billion, enough for 25.6 billion RPK at a planning load factor of 82 percent. Traffic grows 4 percent a year and two 737-8 leases end each year from 2030.

By 2032 the required capacity is 38.0 billion ASK and the remaining fleet supplies 28.8 billion: the gap is 9.2 billion ASK (Figure 9.2). Filled with A321XLRs it needs 14 aircraft; with 787-9s, 8. At typical stages the XLR costs 8.17 cents per ASK to operate plus 1.22 cents of illustrative ownership, against 7.00 and 1.61 cents for the 787-9, so the gap costs CAD 868 million a year with XLRs and 795 million with 787-9s. The top-down answer is the widebody, by CAD 73 million a year.

The answer is fragile. The elasticity of the gap with respect to the planning load factor is $-38.0/9.2 = -4.1$: planning at 80 percent instead of 82 raises the gap to 10.2 billion ASK, planning at 85 percent shrinks it to 7.9 billion, a difference of four XLRs or two 787-9s. Traffic growth of 3 or 5 percent a year moves it to 7.4 or 11.1 billion ASK.

The top-down model is fast and transparent, and Belobaba calls it the more common approach, because detailed scenarios ten or fifteen years ahead are speculative anyway (Belobaba 2006k). Its weakness is structural: it treats seat-kilometres as interchangeable. A 787-9 produces cheap seat-kilometres only if it flies full, and in a market of 150 passengers a day it does not.

The bottom-up model A bottom-up model starts from the network. Future routes and schedules are generated, the airline's share of each O&D market is assumed, and demand and revenue are allocated to flights; the outputs are aircraft assignments, route statistics and a financial projection for each fleet plan (Belobaba 2006k). As an optimization, it chooses how many departures each type flies on each route and how many aircraft of each type to own, to maximize the contribution of all departures minus ownership costs, subject to each route's frequency, the hours an aircraft can fly in a week, delivery positions and reach: a type cannot fly a route beyond its payload-range

(Section 9.4). The contribution of a departure is the planner's familiar quantity: contribution per passenger times the spill-adjusted expected load of Chapter 7, plus cargo, minus the trip cost of Chapter 3 without its passenger term, which the contribution per passenger already nets out.

The bottom-up model gives route-level detail but needs detailed future scenarios and handles competitors poorly. Airlines use both: top-down to size the fleet, bottom-up to check that the aircraft fit the markets, with net present values simulated over fuel prices, traffic growth and yields (Belobaba 2006k).

> **PITFALL**
>
> Treating seat-kilometres as interchangeable. A top-down comparison that picks the type with the lowest cost per ASK assumes every seat can be sold at the planning load factor. On thin routes a large aircraft's cheap seat-kilometres are cheap only on paper: spill, frequency and trip cost decide, and only a route-level evaluation sees them.

9.3 Evaluating an aircraft

Belobaba groups airlines' criteria into technical performance, economics, marketing and environment, and politics (Belobaba 2006k). Table 9.2 arranges them around the questions a planner is asked.

Table 9.2 Criteria for evaluating an aircraft type, adapted from Belobaba (2006).

Criterion	What it measures	Typical evidence
Economics	trip cost, seat cost, CASK; cabin, cargo and revenue	trip-cost model, spill-adjusted contribution
Range and payload	payload-range, runways, seasonal winds	manufacturer charts, operator data
Commonality	type ratings, spares, tooling, crew pools	fleet and crew plans
Availability	delivery positions, lease market, engine reliability	backlog, lessors
Financing	price, residual value, lease rates	net present value, lease versus buy
Regulation	noise, emissions, extended diversion approvals	certification, operator approvals

Size and range used to go together, but by 2006 180-seat 757s already flew thin long-haul routes such as Newark–Lisbon (Belobaba 2006k). The A321XLR extends that logic across the Atlantic, and with it the lesson of Chapter 3: a smaller aircraft has a higher seat cost but a lower trip cost, which wins in thin markets.

Availability is a first-order risk in 2026. IATA estimated in June 2026 that the industry's order backlog had reached 18,100 aircraft, almost 60 percent of the active fleet, with delivery shortfalls absorbed through higher utilization and delayed retirements (IATA 2026d), and Pratt & Whitney engine problems grounded, at the peak, about 650 GTF-powered A320neos (Hamilton 2026). A fleet plan that assumes every aircraft arrives on time and stays in service has no downside case.

Utilization moves in whole departures. An A321 shuttling 1.5-hour sectors in an 18-hour day fits 7 sectors with the manual's 52-minute full turn and 8 with the 41-minute turn shown for the XLR (Airbus 2024): 14.3 percent more utilization. On two-hour sectors six fit either way, and the shorter turn is worth nothing. On long haul the schedule sets the turn: Laurentian's Seville aircraft leaves Montreal at 20:30 and is back 18.2 hours later (the handbook works through the same arithmetic for the published schedule (Network Planning Handbook 2026)), so at four flights a week it flies only 8.5 block hours a day unless paired with another rotation. Weekly patterns decide long-haul utilization, and the fleet count with it.

9.4 Payload-range diagrams

The payload-range diagram is the most important technical description of an aircraft for a network planner (Belobaba 2006k). It answers one question: how much can this aircraft carry over a given distance?

DEFINITION 9.2 Design weights

The **operating empty weight** (OEW) is the aircraft ready for service without payload or usable fuel. **Payload** is passengers, baggage and cargo. The **zero-fuel weight**, OEW plus payload, must not exceed the **maximum zero-fuel weight** (MZFW), a structural limit; zero-fuel weight plus fuel must not exceed the **maximum take-off weight** (MTOW); the fuel must fit in the tanks; and the landing weight must not exceed the **maximum landing weight** (MLW).

Fuel follows the physics of cruise. An aircraft burns fuel in proportion to its weight, so carrying fuel costs fuel, and for a given payload the fuel needed grows faster than distance. The **Breguet range equation**, $R = \Lambda \ln(W_0/W_1)$, says range grows with the logarithm of the ratio of start to end weight, where the range factor Λ sums up aerodynamics and engines. Reserves for diversion, holding and contingencies come on top.

RESULT 9.2 Reading a payload-range diagram

The maximum payload at a given range is the smallest of three limits, which give the diagram its three segments:

1. a **structural plateau** at MZFW minus OEW;
2. a **weight-limited segment**, where take-off weight is at MTOW and every extra mile of fuel displaces payload;
3. a **fuel-limited segment**, where the tanks are full and payload is removed only to make the aircraft lighter so the fuel goes further.

Past the knee between the last two segments, payload falls four to six times faster per mile on the A321neo curves.

The A321 manual lists the design weights for every variant (Airbus 2024). The long-range A321neo-ACF reaches an MTOW of 97 tonnes; the A321neo-XLR reaches 101 tonnes, with a MZFW of up

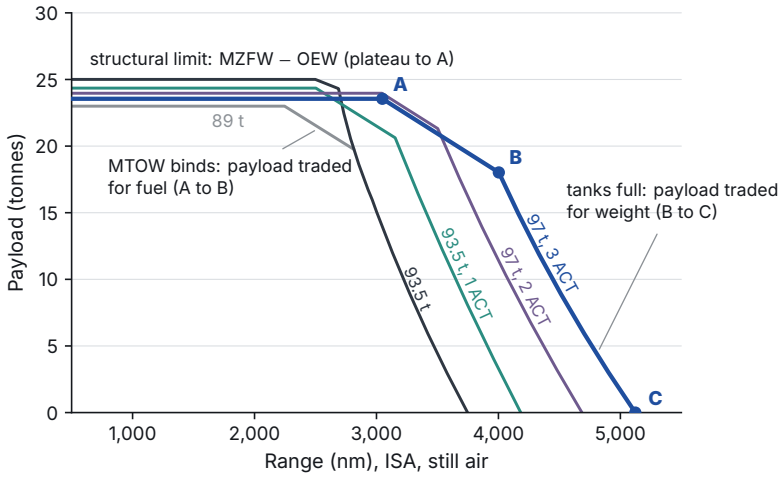


Figure 9.3 Payload-range of the A321neo at ISA in still air, redrawn from the Airbus A321 airport and maintenance planning manual (revision 33, June 2024, figure 3-2-1-991-021). Manufacturer data, given for information only. The 97-tonne, three-ACT curve is marked with its corners A, B and C.

to 75.6 tonnes. Usable fuel is 18.4 tonnes in a standard A321neo, up to 25.9 tonnes with three additional centre tanks (ACTs), and 31.2 tonnes in the XLR with its permanent rear centre tank and optional forward tank.

Figure 9.3 redraws the manual's A321neo chart (Airbus 2024). The 97-tonne, three-ACT aircraft carries its structural maximum of about 23.5 tonnes to about 3,050 nm (corner A), trades payload for fuel at about 0.58 tonnes per 100 nm down to about 18.0 tonnes at 4,000 nm (B), and reaches a ferry range of about 5,120 nm (C). The practical fact is the knee: before it, each extra 100 nm costs about half a tonne of payload; beyond it, several tonnes. A two-parameter Breguet model, with a range factor of 15,800 nm and fixed reserves of 3.2 tonnes, reproduces the five curves with a root-mean-square error of 25 nm; the full reference edition fits it in its Chapter 11.

Headline ranges deserve suspicion. The manual describes the A321LR as flying up to 4,000 nm with 206 passengers (Airbus 2024); the curve offers about 18 tonnes there, or 87 kg per passenger with baggage, a light assumption. Planners need the range with their own cabin, weights, reserves and winds.

9.5 The A321XLR on Montreal–Seville

The A321 manual has no payload-range chart for the XLR (Airbus 2024), but the Breguet model fitted on the A321neo turns the XLR's weights and fuel capacity into an illustrative curve. It is this book's construction, not manufacturer or Air Canada data.

A payload-range chart is drawn for still air. Over the North Atlantic the westerlies, strongest in winter, help eastbound flights and hinder westbound ones. A flight that covers a ground distance D at true airspeed V with an average wind component w (positive for a tailwind) burns as much fuel as a still-air flight of

$$\mathrm{ESAD} = D\,\frac{V}{V+w}, \tag{9.2}$$

the **equivalent still-air distance**, which is where the payload-range curve must be read.

WORKED EXAMPLE 9.2 An illustrative A321XLR mission to Seville

Distance and winds. The great-circle distance from Montreal to Seville is 3,000 nm (5,556 km); allowing 2 percent for routing, the track flown is about 3,060 nm. At 450 knots, Laurentian's block times of 7.0 hours eastbound and 7.9 hours westbound, less the 0.65 hours of taxi, climb and approach in the block-time line of Chapter 3, imply average winds of roughly 30 knots each way.

Aircraft. We take the manual's variant with MTOW 101 tonnes, MZFW 75.6 tonnes and 31.2 tonnes of fuel, and assume an OEW of 52.0 tonnes for a 182-seat cabin, since the manual quotes none. At 95 kg per passenger with baggage (the mission rule in Belobaba's lecture (Belobaba 2006k)), 182 passengers weigh 17.3 tonnes.

The curve. The model puts corner A at 3,709 nm with a structural payload of 23.6 tonnes, corner B at 4,841 nm with 17.8 tonnes, and the ferry range at 6,100 nm. With 182 passengers and no cargo, the still-air range is about 4,870 nm, close to the manual's headline of up to 4,700 nm (Airbus 2024): a sanity check, not a validation. The landing-weight limit would trim another half tonne or so from the plateau, so the cargo room below is slightly optimistic.

The missions (Figure 9.4). Eastbound with a 30-knot tailwind, the ESAD is $3{,}060 \times 450/480 = 2{,}869$ nm; westbound against 30

knots it is 3,060 × 450/420 = 3,279 nm. Both lie on the plateau: the aircraft carries 182 passengers and 6.3 tonnes of cargo. On a strong winter day with a 90-knot headwind, the westbound ESAD is 3,060 × 450/360 = 3,825 nm, past corner A, and the payload limit falls to 23.0 tonnes; with 2 tonnes of extra winter fuel (holding, a more distant alternate), to 21.0 tonnes, leaving 3.7 tonnes for cargo. Cargo starts to suffer above a headwind of about 79 knots, or 37 knots with the extra fuel; passengers would be restricted only beyond 140 knots, which does not happen on this route.

A longer route. Montreal–Rome is 3,555 nm. Against a 90-knot headwind with 2 tonnes of extra fuel its ESAD is about 4,530 nm, beyond the knee, and the payload limit is 16.0 tonnes: about 13 passengers with their bags below a full cabin.

What the planner learns. Range is not a circle on a map. A route well inside the headline range can face payload limits westbound in winter, and because the curve steepens sharply past the knee, a route whose winter westbound ESAD sits near corner B is fragile. The handbook asks for separate eastbound and westbound assessments by season, covering payload, cargo, winds, reserves, runways, alternates and diversion requirements (Network Planning Handbook 2026).

PITFALL

Reading range from a circle. A route inside the headline range is not necessarily feasible with a full cabin in every season. Ask for the range with the airline's cabin, passenger weights and reserve policy, read at the westbound winter ESAD, and check how far that point sits from the knee.

Weight is not the only limit on cargo. The XLR's rear centre tank occupies the front of the aft hold, leaving eight container positions, or six with the optional forward tank, against ten on a standard A321neo (Airbus 2024); with 182 passengers' bags on board, cargo is often limited by volume first. Industry analysis lists the loss of belly cargo among the type's main trade-offs (Cirium 2026b).

Extended diversion operations A twin-engine aircraft crossing the North Atlantic spends part of its flight far from any airport where it

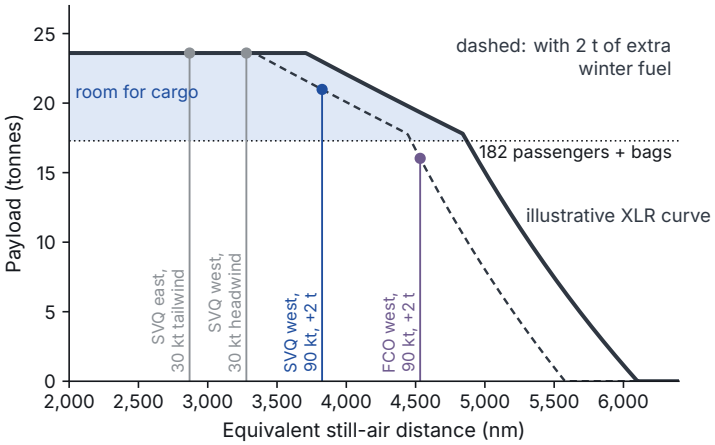


Figure 9.4 An illustrative A321XLR payload-range curve, this book's construction and not manufacturer data, since the manual has no XLR chart (manual weights and fuel capacity, Breguet model fitted to the A321neo charts, OEW assumed at 52.0 tonnes), with Montreal–Seville and Montreal–Rome missions at their equivalent still-air distances. The dashed curve carries 2 tonnes of extra winter fuel; the shaded band is the payload available for cargo above 182 passengers and their bags (17.3 tonnes).

could land after an engine failure; between Newfoundland and the Azores lie about 1,200 nm of ocean. Such flights fall under **extended diversion time operations** (EDTO), long called ETOPS. The rules apply when a flight is planned beyond a threshold flying time from an adequate en-route alternate airport, and they add three layers of approval: the aircraft and engine combination for the diversion time, the operator by its regulator, and each operation, planned with suitable alternates within the approved diversion time and the fuel to reach them. The handbook notes that Canadian guidance now uses EDTO terminology (Transport Canada's TP 6327) and warns against equating type certification with the approval of an operator and route (Network Planning Handbook 2026). EDTO shapes routings and fuel, and so payload; the feasibility check belongs to flight operations.

9.6 Owning, leasing and commonality

An airline can buy aircraft with cash, debt or equity, or lease them (Belobaba 2006k). Leasing can cost more but buys flexibility: an operating lease lets an airline introduce or exit a type without buying or selling aircraft, and needs less capital up front. Air Canada combines the two: of its 30 A321XLRs, 15 are leased and 15 bought

from Airbus, with purchase rights for 10 more, and it reports CAD 501 million of sale-and-leaseback proceeds in the first half of 2026, a third route by which owned aircraft become leased ones (Air Canada 2026d).

The comparison is simple. Buying costs the price less the present value of the resale value at the end; leasing costs the present value of the rents plus the cost of meeting the return conditions. Leasing wins when the rent is below the **break-even rent** that equalizes the two, and two inputs dominate: the discount rate and the expected residual value.

WORKED EXAMPLE 9.3 Leasing or buying an A321XLR

Take, for illustration, a price of CAD 90 million, a 12-year holding period, a residual value of 45 percent (CAD 40.5 million), return conditions costing CAD 3 million and a discount rate of 8 percent. A dollar in 12 years is worth 0.3971 today, and 144 monthly rents paid in advance are worth 94.31 times one rent. Buying costs $90 - 40.5 \times 0.3971 = 73.92$ million in present value. A lease at CAD 0.68 million a month costs $0.68 \times 94.31 + 3 \times 0.3971 = 65.32$ million, so leasing is cheaper by CAD 8.6 million. The break-even rent is $(73.92 - 1.19)/94.31 = 0.771$ million a month, 0.86 percent of the price per month.

At a 6 percent discount rate (Table 9.3) the same rent is above the break-even of 0.658 million, and buying is cheaper by CAD 2.2 million. A higher residual value favours buying; a higher discount rate favours leasing. Leasing pays when the lessor's cost of capital is lower or its residual-value view more optimistic, and it moves residual-value risk off the airline.

Table 9.3 Break-even monthly lease rent (CAD millions) for the illustrative A321XLR of Example 9.3. Leasing is cheaper when the offered rent is below the entry.

Discount rate	Residual 35%	Residual 45%	Residual 55%
6%	0.702	0.658	0.615
8%	0.809	0.771	0.733
10%	0.917	0.884	0.851

Fleet commonality is the sharing of type ratings, spare parts, tooling, maintenance programs and crew pools across aircraft. It reduces the training, equipment and spares every new type brings (Belobaba 2006k), and it pools risk. If each aircraft is unavailable on

a given day with probability 5 percent, covering the schedule with 95 percent confidence takes 1 spare for 4 aircraft (25 percent of the subfleet), 2 for 8 (25 percent), 4 for 30 (13 percent) and 6 for 60 (10 percent). A small subfleet of a unique type carries a heavy spare burden; a type that shares spares and crews with a larger family draws on a bigger pool. The same logic applies to reserve crews and spare engines.

9.7 A Laurentian fleet decision

Laurentian's fleet team must equip a program of thin transatlantic routes from Montreal: Seville, Lisbon, Madrid and Barcelona, between 2,827 and 3,192 nm away and well inside the XLR's reach (Section 9.5). Two proposals are on the table, with roughly equal capital: four A321XLRs flying each market daily, or two 787-9s flying each market three or four times a week.

WORKED EXAMPLE 9.4 Four A321XLRs or two 787-9s?

Inputs. Demand per daily departure in each direction follows the case's Seville numbers by season: a mean of 175 (standard deviation 42) for 16 peak weeks, 150 (40) for 14 shoulder weeks and 110 (35) for 22 winter weeks, a generic calendar (Chapter 10 builds Seville's own). Lisbon, Madrid and Barcelona are assumed 10, 15 and 25 percent larger, with net fares of CAD 600, 630 and 640 against Seville's 620. Each passenger brings CAD 30 of ancillary revenue and costs CAD 50. Trip costs come from Chapter 3; cargo contributes CAD 3,500 per XLR departure and an assumed CAD 12,000 per 787-9 departure; ownership costs CAD 8.16 million a year per XLR and 19.2 million per 787-9. All inputs are illustrative.

Frequency and demand. Each aircraft flies one rotation a day, so option A (four XLRs) serves each market daily and option B (two 787-9s) 3.5 times a week. Fewer flights lose customers who cannot travel on the days offered. We assume weekly demand varies with weekly frequency f as $(f/7)^{\theta}$, with a **frequency elasticity** $\theta = 0.4$: halving frequency loses 24 percent of demand, and each 787-9 departure carries the remaining demand of two days.

One market, one season. Take Seville in the shoulder season, eastbound. Under A, mean demand per departure is 150, the expected load 145.2 (79.8 percent) and the contribution

$$600 \times 145.2 + 3{,}500 - (11{,}000 \times 7.0 + 3{,}500) = 10{,}120$$

dollars per departure, CAD 70,800 a week. Under B, mean demand per departure is $2 \times 150 \times 0.758 = 227.4$; the 298-seat aircraft carries 223.7 on average (75.1 percent) and contributes

$$600 \times 223.7 + 12{,}000 - (15{,}500 \times 7.0 + 7{,}000) = 30{,}720$$

dollars per departure, CAD 107,500 a week from 3.5 departures.

The program. Summing over markets, seasons and directions gives Table 9.4. Option A contributes CAD 16.3 million a year before ownership and −16.4 million after; option B, 37.9 million before and −0.5 million after. The 787-9s win by CAD 15.8 million a year.

Table 9.4 Annual contribution of three fleet options for Laurentian's thin transatlantic program (both directions, frequency elasticity 0.4; illustrative inputs). A: four A321XLRs, each market daily; B: two 787-9s, each market 3.5 times a week; C: two A321XLRs, each market 3.5 times a week.

CAD millions a year	Option A	Option B	Option C
Seville	−0.69	5.27	6.47
Lisbon	4.30	9.30	7.69
Madrid	6.07	11.11	8.34
Barcelona	6.59	12.19	7.74
Contribution before ownership	16.27	37.86	30.24
Ownership	32.64	38.40	16.32
Contribution after ownership	−16.37	−0.54	13.92
Departures a year	2,912	1,456	1,456
Passengers a year (thousands)	424	330	249
Load factor	80.0%	76.1%	93.9%

Where the difference comes from. Comparing A with B mixes two changes: fewer departures and bigger aircraft. Separate them with option C: two XLRs flying B's 3.5 weekly pattern. Going from C to A doubles frequency with the same gauge; it gains demand ($2^{0.4} = 1.32$ times as many customers) but doubles trip costs and ownership, and costs CAD 30.3 million

a year. Going from B to C swaps the gauge at equal frequency; the XLRs spill more and lose the widebody's cargo, but their trip cost is CAD 35,000 lower and two XLRs cost CAD 22 million a year less to own, so C beats B by CAD 14.5 million. Option C, which nobody proposed, is best: CAD 13.9 million after ownership.

Sensitivity. Figure 9.5 varies the parameter that matters most. C beats B for any frequency elasticity above 0.19 and beats A for any value below 1.15. With demand 30 percent higher in every market, B becomes best for elasticities below 0.57, because the XLRs then spill too much. Cheaper 787-9 cargo, dearer fuel or a premium-cabin uplift on the 787-9 all leave C first, and a Monte Carlo simulation over elasticity, demand, fuel and cargo ranks C first in 86 percent of draws, B in 10 percent and A in 4 percent.

What the planner learns. The question as posed mixed a frequency decision with a gauge decision, and the best answer was a plan nobody had proposed. At equal frequency, the XLR's lower trip cost beats the 787-9's lower seat cost in markets of this size. Whether to fly them daily depends on how much demand daily service earns, a parameter planners rarely know well (Chapter 11). Feed at Montreal is not counted yet (Chapter 10), and the recommendation states what would reverse it: demand 30 percent above the base case with a frequency elasticity below about 0.57.

IN PRACTICE Long-range narrowbodies in 2026

Cirium counted 50 A321XLRs delivered by the third quarter of 2026, with 28 airlines committed to 440 aircraft and cabins from 138 to 239 seats; Air Canada and Iberia fit 182 (Cirium 2026b). It lists up to 30 percent lower fuel burn per seat than the 757 and access to thin markets as advantages, lost belly cargo and passengers' preference for widebodies as trade-offs, and reports that United avoided some New York–Italy routes because of range (Cirium 2026b), the payload problem of Example 9.2. Air Canada's first XLR entered service on 9 June 2026 with 14 business and 168 economy seats (Network Planning Handbook 2026), and its announced uses favour moderate frequencies: Montreal–Seville from 8 September 2027 at three, then four, flights a week (Net-

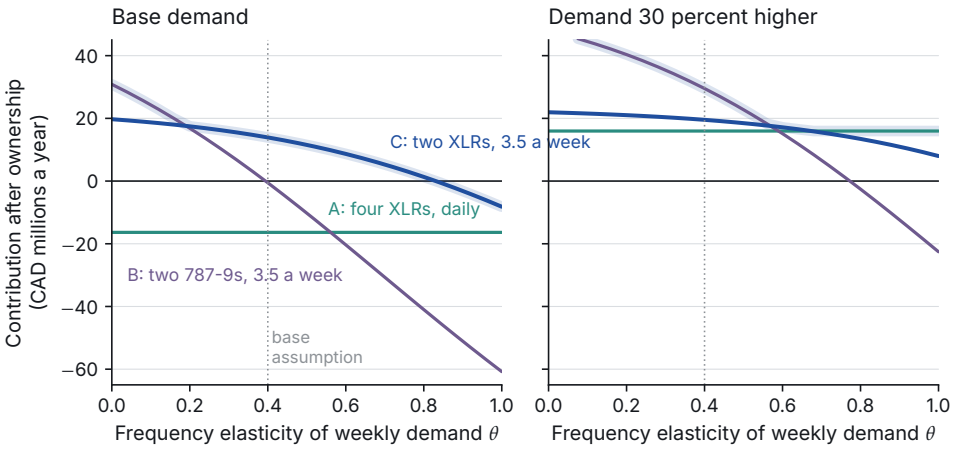


Figure 9.5 Annual contribution after ownership of the three fleet options as the frequency elasticity of demand varies, with base demand (left) and demand 30 percent higher in every market (right). The pale band marks the best option; the dotted line is the base assumption $\theta = 0.4$.

work Planning Handbook 2026), and for summer 2027 Toronto–Shannon three times a week and Montreal–Basel-Mulhouse four times a week (PAX News 2026). These resemble option C more than option A.

9.8 Summary

- A fleet plan stages deliveries, leases, retirements and transfers under uncertainty.
- The top-down capacity gap is fast but highly leveraged on its assumptions and treats seat-kilometres as interchangeable; the bottom-up model checks, route by route, that aircraft fit markets.
- Aircraft are judged on economics, range and payload, commonality, availability and financing; in 2026 availability is a first-order risk. Utilization moves in whole departures.
- A payload-range diagram is the minimum of a structural, a weight and a fuel limit; past the knee, payload falls four to six times faster per mile. Read it at the equivalent still-air distance: winter westbound winds eat the XLR's cargo margin to Seville first.
- Over the ocean, EDTO requires approval of the type, the operator and each operation.
- Leasing wins below the break-even rent, set by the discount rate and residual value. Commonality pools spares and crews.

- In Laurentian's decision, two XLRs at 3.5 flights a week beat four XLRs daily and two 787-9s under most assumptions; the frequency elasticity decides.

9.9 Further reading

Belobaba's short lecture on fleet planning models (Belobaba 2006k) sets out the process, the models and the criteria. The Airbus A321 manual (Airbus 2024) gives the weights, payload-range curves and turn times; operators' approved data govern. The handbook (Network Planning Handbook 2026) frames the XLR and EDTO for a planner, and Air Canada's MD&A (Air Canada 2026d) gives the fleet plan. The full reference edition, in its Chapter 11, writes out the bottom-up optimization, the payload-range limits and the lease formula.

CHAPTER

10

Route Evaluation

IN THIS CHAPTER

- Allocated against incremental profitability: three views of a route, and why a route can pass one and fail another.
- Prorate, and why the revenue allocated to a leg is not what the leg is worth.
- Feed, network contribution and displacement: why a route can lose money on its own report and still be worth flying, and the reverse.
- Avoidable costs and the opportunity cost of aircraft time.
- The handbook's Exercise A and Montreal–Seville case, then Laurentian's YUL–SVQ season by season.
- Scenarios, sensitivity, the thresholds that reverse the decision, and the decision memo.

DRAWS ON Belobaba, Route planning and evaluation (MIT 16.75J, 2006); handbook (route economics, Exercise A, the Montreal–Seville case, the decision memo); the spill model; Air Canada, Seville announcement (2026)

Sooner or later every network analyst is asked whether to open a route, keep it, change it or drop it. Every airline has a report that prints a profit for each flight, and that number cannot answer the question on its own: a flight's revenue depends on how connecting fares are split among legs, its costs on the horizon of the decision, its passengers may come from other flights or push other passengers off, and its aircraft could be doing something else. Belobaba states the economic criterion in one sentence: the potential for incremental profit in the short run, given the opportunity cost of taking the aircraft from another route (Belobaba 2006i). The handbook states the planner's core decision in another: choose the feasible schedule and aircraft allocation that creates the greatest incremental network value, compared with the best realistic alternative (Network Planning Handbook 2026).

This chapter turns those sentences into a method, using the costs of Chapter 3, the demand of Chapter 4, the spill model of Chapter 7

and the network effects of Chapter 8. It ends with a full evaluation of Laurentian's Montreal–Seville route and the memo that carries it.

10.1 Three views of route economics

Costs are incurred per flight, driven by the fleet and the schedule; revenues are earned per itinerary, driven by O&D markets (Belobaba 2006i). A flight's profit can therefore be measured in two ways. The **incremental** way adds or removes the flight, re-optimizes the rest of the network and measures the difference; it captures every interaction but needs a model of the whole operation. The **allocated** way divides all revenues and costs among flights and treats each as independent; it is easy to explain, but the allocation is subjective and the network effects disappear (Belobaba 2006i). The handbook asks for both, and one more.

DEFINITION 10.1 Three views of route economics

1. **Flight contribution**: the revenue allocated to the flight under a stated convention, minus its costs under a stated definition (avoidable, or including ownership, or fully allocated).
2. **Incremental network contribution**: the change in the whole network's revenue minus avoidable cost when the route is added or removed, relative to a dated baseline, counting feed, displacement and traffic taken from the airline's other services.
3. **The capital and strategic case**: the longer-term view that adds aircraft ownership, launch and station costs, fleet commitments and strategic effects such as competitive presence.

A route can pass one view and fail another (Network Planning Handbook 2026).

The lecture's sample network, taken from an article by Baldanza, makes the point with one flight (Belobaba 2006i). A Syracuse–Omaha leg carries local passengers and feeds onward flights to San Francisco and Los Angeles. Its local O&D revenue is USD 6,000; connecting revenue prorated to the leg is USD 1,500, and the share prorated to the onward legs USD 4,000. Variable operating costs are USD 4,500, aircraft ownership USD 2,000 and allocated overhead USD 1,500. The network bears USD 700 of extra variable costs for the

connecting passengers and USD 500 of opportunity cost for seats beyond Omaha that other passengers could have used. Table 10.1 combines these into nine measures.

Table 10.1 Nine profitability measures for one Syracuse–Omaha flight (adapted from Belobaba, 2006, who draws on Baldanza). Historical teaching numbers.

Measure	Calculation	USD
Variable leg profitability	6,000 + 1,500 – 4,500	3,000
Plus aircraft ownership	3,000 – 2,000	1,000
Fully allocated leg profitability	1,000 – 1,500	(500)
Variable, with network contribution	3,000 + 4,000 – 700	6,300
Variable, network contribution and opportunity cost	6,300 – 500	5,800
Ownership and network contribution	6,300 – 2,000	4,300
Ownership, network contribution and opportunity cost	4,300 – 500	3,800
Fully allocated, with network contribution	4,300 – 1,500	2,800
Fully allocated, network contribution and opportunity cost	2,800 – 500	2,300

The same flight earns anywhere from a loss of USD 500 to a profit of USD 6,300, and each number answers a different question (Belobaba 2006i). For short-term schedule changes, ownership and overhead are fixed, so the variable measure with network contribution applies. In the middle term aircraft can be moved between routes, so ownership enters. A hub's monthly profitability is measured without network contribution, because summing every flight's network contribution counts the same connecting revenue twice. Whether a hub should exist at all is judged fully allocated, because then everything is variable.

The table also shows the mechanism behind two surprises every planner meets. A route can lose money on its own report and still be worth flying. Fully allocated, the Syracuse flight loses USD 500; but cancelling it in the short run saves none of the ownership and overhead behind that loss, while giving up its USD 3,000 variable margin and the USD 4,000 earned on the Omaha connections, net of USD 1,200 of connecting costs and displacement: the network loses USD 5,800. The reverse also happens: a flight can show a profit on its own report and still be the wrong choice, because its connecting passengers push others off full feeders, its passengers were already flying with the airline, or its aircraft could earn more elsewhere.

A route's profit is not a property of the route. It is the answer to a question about a decision, and the decision fixes the revenue convention, the cost horizon and the alternative.

10.1.1 Revenue allocation and prorate

The handbook's rule is to follow the journey first and allocate second: forecast the net revenue the airline keeps from each O&D journey, after partner settlements and without government taxes, and only then, if a flight view is needed, split connecting revenue among legs (Network Planning Handbook 2026).

DEFINITION 10.2 Prorate

A **prorate** splits the fare of a connecting itinerary among the legs it uses, with shares that add up to the fare. A **mileage prorate** makes each leg's share proportional to its distance; a **straight-rate prorate** makes it proportional to a reference local fare on the leg (here the full unrestricted fare). Inside one airline a prorate is a reporting convention agreed with Finance; between airlines it is set by prorate agreements.

WORKED EXAMPLE 10.1 Three ways to split a Boston–Paris fare

In Laurentian's case data a restricted Boston–Paris ticket over Montreal costs CAD 599 and a flexible one CAD 1,349. The legs are 254 miles (Boston–Montreal) and 3,432 miles (Montreal–Paris); the full local fares are CAD 389 and CAD 1,149.

Mileage prorate gives Boston–Montreal 254/3,686 = 6.9 percent of the fare: CAD 41 of the restricted fare and CAD 93 of the flexible one. **Straight-rate prorate** gives it 389/1,538 = 25.3 percent: CAD 152 and CAD 341. On the mileage convention the feeder earns less from a restricted Boston–Paris passenger than a quarter of its own cheapest local fare (CAD 189), so a mileage-based report shows Boston–Montreal as a weak flight however much feed it supplies.

Network value. Revenue management prices a seat on each leg with a **bid price** π_l, the opportunity cost of selling it (Chap-

ter 14); a passenger is worth to the network the fare minus the bid prices of the legs used. At mean demand, Laurentian's four-leg bank has bid prices of CAD 109 on Boston–Montreal and CAD 490 on Montreal–Paris. The restricted Boston–Paris fare is exactly marginal ($599 - 109 - 490 = 0$), so the network carries only part of that demand; the flexible fare leaves $1{,}349 - 599 = 750$ dollars. Viewed from one leg, the flexible passenger is worth the fare minus the other leg's bid price: $1{,}349 - 490 = 859$ to Boston–Montreal and $1{,}349 - 109 = 1{,}240$ to Montreal–Paris. These add up to 2,099, far more than the fare: not an allocation, but the numbers that matter for decisions about one leg at a time.

RESULT 10.1 Marginal leg values do not add up

The marginal value of a leg is the network's best contribution with the leg minus its best contribution without it, with passenger flows re-optimized. When legs are complements, as a feeder and the long-haul flight it feeds are, their marginal values add up to more than the network's total contribution; when they are substitutes, as parallel frequencies in one market are, they add up to less.

If one itinerary with fare f uses two legs and nothing else, removing either leg loses the whole fare, so the two marginal values sum to $2f$. If two parallel flights can carry the same passengers, removing one loses little because the other recaptures them. Hub networks are full of both relationships, so revenue allocated to a leg is not the marginal value of that leg (Network Planning Handbook 2026).

10.2 Feed, network contribution and displacement

Belobaba lists what a leg adds to the rest of the network, the revenue earned on other legs by passengers who travel only because it exists, and what it takes: the cost of handling its connecting passengers, and the opportunity cost of seats beyond the hub that travellers from other markets could have used, which he calls **network displacement cost** (Belobaba 2006i).

DEFINITION 10.3 Network contribution and displacement

The **network contribution** of a leg is the contribution it generates on other legs: revenue from connecting passengers allocated elsewhere, net of their passenger-related costs. Its **displacement cost** is the contribution lost on other legs because its passengers take seats that other travellers would have bought. Both are measured against a baseline network without the leg.

Displacement depends on load. A connecting passenger on a feeder with empty seats costs only the passenger-related costs; on a feeder that would otherwise be full, the passenger takes a seat someone else would have bought, and the network loses that traveller's contribution. The bid price measures this loss per seat: near zero with spare seats, large on a feeder full every night. The same new route can bring real feed in one season and mostly reshuffle passengers in another.

The handbook turns all of this into one comparison across every affected flight and journey (Network Planning Handbook 2026). For a scenario s and baseline b, with retained passenger revenue R, other incremental revenue O (cargo, ancillaries) and avoidable operating cost AC,

$$\Delta\mathrm{NC} = \sum_{\text{flights, journeys}} \Big[(R+O-\mathrm{AC})_s - (R+O-\mathrm{AC})_b \Big] - F_{\text{launch}}, \quad (10.1)$$

with launch costs F_{launch} and longer-horizon capital added as the decision requires. Two habits keep (10.1) honest: count every passenger once, at the revenue the network keeps, never prorating a connecting ticket and also counting its full value elsewhere; and do not add a generic feed uplift to the output of a network simulator that already models connections and recapture (Network Planning Handbook 2026).

WORKED EXAMPLE 10.2 What the Seville leg is worth to Laurentian's bank

In Laurentian's four-leg evening bank (Chapter 8), Quebec City and Boston feed Montreal, which connects them to Paris and Seville. A deterministic network program, which chooses how many passengers of each fare product to carry at mean demand within every leg's capacity, earns CAD 365,949 per bank with

the Seville flight and CAD 245,499 without it. The Seville leg is worth CAD 120,450 of revenue to the bank.

The Seville passengers pay CAD 121,709 in O&D fares: 81,388 on the leg's own fare products and 40,321 from Quebec City and Boston. But their arrival changes who else flies. Both feeders become full, and the program drops 16 restricted Boston–Paris passengers (CAD 9,584), 3 restricted Quebec City–Paris passengers (CAD 1,947) and one Quebec City local (CAD 159). The Paris seats they free are refilled by 19 restricted Montreal–Paris locals who were spilled before (CAD 10,431). Net displacement is CAD 1,259, and 121,709 − 1,259 = 120,450.

Allocations get close to this number only by accident. Mileage prorate credits the Seville leg with CAD 119,421, because it keeps 93 to 96 percent of every connecting fare; straight-rate prorate credits it with CAD 112,620. The differences, CAD 1,029 and CAD 7,830, are the network contribution net of displacement that each flight report leaves out. At mean demand the program also understates displacement: with random demand the two feeders turn away about 24 more travellers per bank once Seville is added (Chapter 8). And these are fares before costs.

10.3 Avoidable costs and the opportunity cost of aircraft time

On the cost side the rule is the one of Chapter 3: count the costs that differ between the scenario and its baseline over the horizon of the decision. Fuel, crew, airport, navigation, maintenance and passenger costs move with the flight; launch costs, station costs and conditional airport incentives are shown separately; aircraft ownership is often unavoidable in the short run, though the aircraft time still has an opportunity cost; overhead is never avoidable (Network Planning Handbook 2026). Hence the handbook's two symmetric conclusions: an accounting loss is not a reason to cancel a flight whose costs are mostly unavoidable and which contributes positively, and a positive contribution is not a reason to launch one if a better use of the same aircraft exists or the longer-run costs are not covered (Network Planning Handbook 2026).

Laurentian's trip cost model (Chapter 3) is of the avoidable kind. For the A321XLR on Montreal–Seville it charges CAD 11,000 per block hour, CAD 3,500 per departure and CAD 50 per passenger.

The eastbound flight blocks 7 hours and the westbound 7 hours 55 minutes, the times implied by the published schedule when Seville is six hours ahead of Montreal (Network Planning Handbook 2026), so with Y passengers the westbound trip costs CAD 10,083 more:

$$\mathrm{TC}_{\mathrm{E}} = 80{,}500 + 50\,Y, \qquad \mathrm{TC}_{\mathrm{W}} = 90{,}583 + 50\,Y.$$

With CAD 620 of allocated net fare, CAD 30 of ancillary revenue and CAD 50 of passenger cost, each passenger contributes $r = 600$ dollars, and cargo's CAD 3,500 per departure happens to equal the departure cost. A flight therefore covers its avoidable cost at $77{,}000/600 = 128.3$ passengers eastbound and $87{,}083/600 = 145.1$ westbound: break-even load factors of 70.5 and 79.7 percent.

10.3.1 The opportunity cost of aircraft time

The aircraft a new route uses is rarely idle in the baseline: if the route flies, something else does not. That forgone contribution is the opportunity cost in Belobaba's criterion, and it often decides.

DEFINITION 10.4 Shadow price of aircraft time

The **shadow price** ω of a fleet's aircraft time is the increase in the network's best achievable contribution from one more hour of that fleet's aircraft time, in CAD per aircraft-hour. Fleet assignment models produce it as the value of the fleet's availability constraint (Chapter 12).

The simplest model is a merit order of the candidate uses m of a fleet's weekly hours, each with contribution V_m and aircraft time h_m.

RESULT 10.2 The ratio rule for scarce aircraft time

Rank the candidate uses by contribution per aircraft-hour, V_m/h_m, and fly them in that order until the hours run out. The shadow price of aircraft time is the ratio of the use that straddles the limit, $\omega = V_k/h_k$, and a use is worth flying only if its contribution covers the value of the hours it takes:

$$V_m - \omega\, h_m \;\geq\; 0.$$

Figure 10.1 draws the rule as a supply curve. The shadow price is set by the least valuable use that still flies, not by the average use, and it rises in seasons when more valuable uses compete for the air-

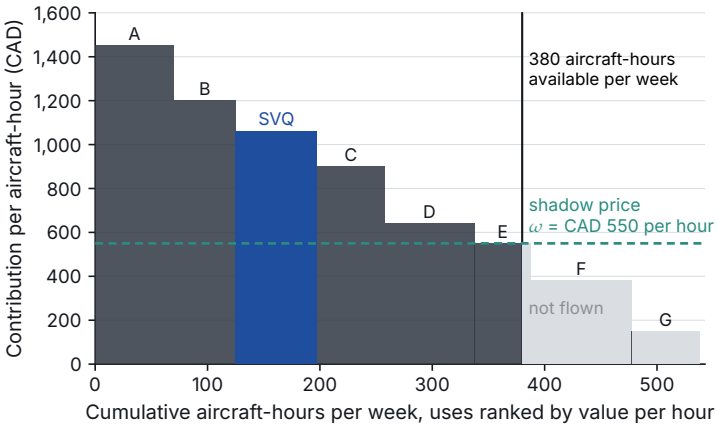


Figure 10.1 A merit order of candidate uses of Laurentian's A321XLR aircraft time in a spring or autumn week, ranked by contribution per aircraft-hour (uses A to G are illustrative). With 380 hours available, use E straddles the limit and sets the shadow price at CAD 550 per hour. Seville, at CAD 1,061 per hour, is comfortably inside the plan.

craft. It is also a marginal value. Seville's four weekly rotations use 72.7 of the 380 hours: removing them would release hours that go first to the rest of use E at CAD 550 and then to use F at CAD 380. Re-solving without Seville gives an exact opportunity cost of CAD 28,917 a week, or CAD 7,229 per round trip, against the screening charge $\omega h = 550 \times 18.17 = 9{,}992$. For large blocks of hours, re-solve the fleet model with and without the route.

The handbook allows two ways to account for aircraft scarcity: compare explicitly modelled, mutually exclusive alternatives, or charge a shadow price for aircraft time in a screening model (Network Planning Handbook 2026). Mixing them counts the opportunity cost twice.

PITFALL

Double counting the opportunity cost. In the handbook's Seville case the candidate contributes CAD 8,000 and the best alternative for the same aircraft CAD 11,000, so the alternative is better by CAD 3,000. An analyst who subtracts the alternative's 11,000 from the candidate as an opportunity cost (leaving minus 3,000) and then also compares the result with the alternative's 11,000 reports a gap of 14,000; one who nets each option against the other reports a gap of 6,000. Both rank

the options correctly here but overstate what is at stake, and with more than two options either can distort the ranking. Use one method per analysis and say which.

10.4 The handbook's exercises

Besides the spill exercise of Chapter 7, the handbook contains two worked exercises on route economics, both with invented numbers (Network Planning Handbook 2026).

WORKED EXAMPLE 10.3 Exercise A: a full-looking route that is not the best use

One departure of a candidate flight carries 110 local passengers paying CAD 220 and 40 connecting passengers paying CAD 450 for their whole itineraries. Cargo and ancillaries contribute CAD 3,000 net. The flight's avoidable operating cost is CAD 32,000. Carrying the connecting passengers on existing feeder flights adds CAD 2,000 of operating cost there and displaces CAD 4,000 of other contribution, and passengers switching to the new flight from the airline's other services take CAD 3,500 of revenue with them. All figures are incremental except ticket revenue, whose baseline losses are deducted explicitly. Then

$$\begin{aligned} \text{gross ticket revenue} &= 110 \times 220 + 40 \times 450 = 42{,}200, \\ \Delta\text{NC} &= 42{,}200 + 3{,}000 - 32{,}000 \\ &\quad - 2{,}000 - 4{,}000 - 3{,}500 = 3{,}700. \end{aligned}$$

An independently evaluated use of the same aircraft earns CAD 4,500, so the alternative is better by CAD 800 (Network Planning Handbook 2026).

A full flight of 150 passengers does not settle the decision; the next question is whether another departure time could raise the candidate's contribution without another aircraft. And the CAD 450 connecting fares enter once, in full, never prorated to the flight and counted again elsewhere.

The second exercise is the handbook's Montreal–Seville case: the schedule of the announced service with invented economics for one departure, in a leg-allocation view in which connecting revenue is

split consistently across legs and later lines capture only changes elsewhere (Network Planning Handbook 2026) (Table 10.2).

Table 10.2 The handbook's invented economics for one Montreal–Seville departure (adapted from the handbook). The numbers are illustrative and are not estimates for any airline.

Item	Assumption or calculation	CAD
Passengers	150 on a 182-seat aircraft, 82.4% load factor	
Net passenger revenue allocated to this leg	150×620	93,000
Ancillary revenue	150×30, not already in passenger revenue	4,500
Cargo contribution	Net of its handling costs	3,500
Avoidable flight cost	Includes passenger-related costs	(88,000)
Initial flight contribution	Sum of the above	13,000
Additional contribution on affected other flights	Net of their costs and any displacement	6,000
Lost contribution on own alternative services	Cannibalization	(9,000)
Expected incremental disruption cost	Not already included	(2,000)
Incremental network contribution versus no new flight	Under this convention	8,000

A feasible alternative using the same aircraft and time windows contributes CAD 11,000, so the alternative is better by CAD 3,000: neither the positive contribution nor the 82.4 percent load factor establishes that the candidate should be chosen (Network Planning Handbook 2026). The bridge also shows the reverse of the Syracuse surprise: feed adds CAD 6,000, but cannibalization of the airline's own services takes back CAD 9,000, so the route is worth less to the network than on its own report. Carrying 170 passengers at CAD 530 instead of 150 at CAD 620 would lower passenger revenue from CAD 93,000 to 90,100. And the ranking reverses at CAD 20 more retained fare per passenger (CAD 3,000 per departure) with demand and network effects held fixed, a threshold for discussion rather than a pricing recommendation, because a higher fare would change demand and mix (Network Planning Handbook 2026).

The case assumes 150 passengers. Chapter 7 explains why that cannot be a forecast of mean demand: a flight whose demand averages 150 carries fewer than 150 on average, because some departures spill.

WORKED EXAMPLE 10.4 The Seville case with spill-based loads

Laurentian forecasts unconstrained demand per eastbound departure of $\mathcal{N}(150, 40^2)$ in its shoulder season, $\mathcal{N}(175, 42^2)$ at its peak and $\mathcal{N}(110, 35^2)$ in winter, for 182 seats; its calendar (Section 10.5) applies the peak level to autumn and spring and the shoulder level to summer. By the spill formula of Chapter 7, expected load is mean demand less expected spill, $\mu - \sigma L(z)$, where $z = (182 - \mu)/\sigma$ and L is the normal loss function (Appendix A.4).

Shoulder. $z = 32/40 = 0.80$, $L(0.80) = 0.1202$, so expected spill is $40 \times 0.1202 = 4.8$ and expected load 145.2 (load factor 79.8 percent); 21 percent of departures leave full. **Peak.** $z = 0.167$, $L(0.167) = 0.321$, expected spill 13.5 and expected load 161.5 (88.7 percent); 43 percent of departures leave full. **Winter.** $z = 2.06$, expected spill 0.25 and expected load 109.7 (60.3 percent).

Keep the handbook's fares and cost model, and scale its network effects with the load: CAD 40 of feed contribution and CAD 60 of cannibalization per passenger (6,000 and 9,000 at 150 passengers), plus CAD 2,000 of disruption cost per departure. Each passenger contributes $620 + 30 - 50 = 600$ dollars and cargo offsets the departure cost, so the flight contribution is $600\,\mathbb{E}[Y] - 77{,}000$:

- shoulder: $600 \times 145.2 - 77{,}000 = 10{,}115$; network effects $-20 \times 145.2 - 2{,}000 = -4{,}904$; incremental network contribution CAD 5,211;
- peak: flight contribution 19,907; incremental 14,677;
- winter: flight contribution −11,153; incremental −15,348.

Against the CAD 11,000 alternative, the shoulder departure falls short by CAD 5,789 instead of 3,000, the peak departure beats it by CAD 3,677, and the winter departure falls short by CAD 26,348. Treating mean demand as the load overstated the shoulder flight's contribution by 4.8 passengers, or CAD 2,885 per departure, nearly doubling the margin by which it loses. The season matters far more: the same route is a clear winner at its peak and a heavy loser in winter.

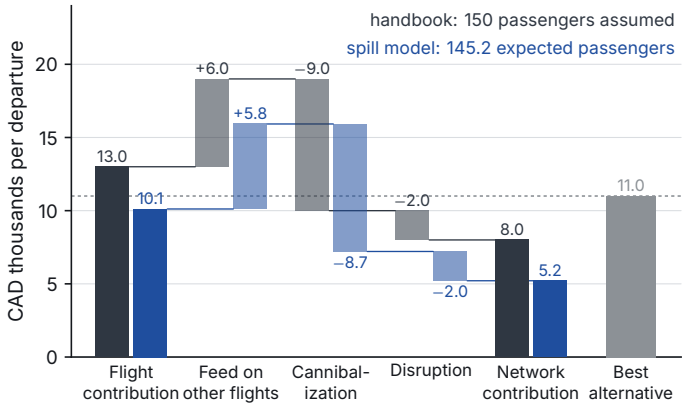


Figure 10.2 The handbook's Montreal–Seville bridge from flight contribution to incremental network contribution for one eastbound departure, at the handbook's assumed 150 passengers and at the spill-based expected load of 145.2 for the same demand forecast. Both fall short of the CAD 11,000 alternative; the spill correction widens the gap from CAD 3,000 to CAD 5,789.

10.5 Laurentian YUL–SVQ: a full seasonal evaluation

A launch decision is about a schedule for a year, in both directions, against the best use of the aircraft in each period.

The real schedule. Air Canada announced Montreal–Seville on 14 October 2026: flight AC958 leaves Montreal at 20:30 and arrives in Seville at 09:30 the next day, AC959 leaves Seville at 12:45 and arrives in Montreal at 14:40, with an A321XLR of 14 business and 168 economy seats. Service starts on 8 September 2027 at three flights a week and rises to four from 1 October 2027. The published operating periods run from 8 September 2027 to 8 January 2028 and from 18 February to 3 June 2028 (Air Canada 2026f): an autumn-to-spring service with a mid-winter break, for reasons the release does not give. Laurentian's case borrows the schedule and nothing else.

Laurentian's assumptions are in Table 10.3; the calendar, the westbound demand, the scaling of network effects and the value of aircraft time are illustrative. Demand is strongest in autumn and spring and weak in winter. The fleet model values XLR time most in the northern summer, when every long-haul aircraft has strong uses, and least in winter.

Table 10.3 Assumptions of the Laurentian YUL–SVQ evaluation. Westbound demand is assumed equal to eastbound demand in each period.

Input	Value	Source
Aircraft, seats	A321XLR, 182	case
Block time, cycle	7 h 00 east, 7 h 55 west; 18 h 10 from departure to return	public schedule
Demand per departure (each direction)	$\mathcal{N}(175, 42^2)$ autumn and spring; $\mathcal{N}(150, 40^2)$ summer; $\mathcal{N}(110, 35^2)$ winter	case; calendar assumed
Calendar	Autumn 9 weeks (Sep–Oct), winter 21 (Nov–Mar), spring 9 (Apr–May), summer 13 (Jun–Aug)	assumed
Frequency	4 round trips a week	public schedule
Revenue	CAD 620 net fare, CAD 30 ancillary per passenger; CAD 3,500 cargo per departure	case
Avoidable trip cost	CAD 11,000 per block hour, 3,500 per departure, 50 per passenger	case
Network effects	feed CAD 40 and cannibalization CAD 60 per passenger; disruption CAD 2,000 per departure	handbook, scaled
Value of aircraft time ω	CAD 550 per hour autumn and spring; 1,000 summer; 150 winter	assumed

The mean of 150 is about 82 local travellers (Chapter 4) and 68 connecting travellers from seven markets behind the gateway (Chapter 8). The cannibalization loss covers the roughly 11 travellers per departure (43 a week, Chapter 4) whom the nonstop moves from Laurentian's connection over Paris: Laurentian loses their Paris contribution less the value of the Paris seats they free, which is not zero on an over-demanded flight.

Per round trip. Writing r for contribution per passenger, e_{f} and e_{c} for the feed and cannibalization effects per passenger, δ for the disruption cost, K for cargo and TC_d^0 for the trip cost of direction d before passenger costs, the value of one round trip against the best alternative use of the aircraft is

$$V = \sum_{d \in \{\mathrm{E},\mathrm{W}\}} \left[(r + e_{\mathrm{f}} - e_{\mathrm{c}})\, \mathbb{E}[Y_d] + K - \mathrm{TC}_d^0 - \delta \right] - \omega h, \qquad (10.2)$$

where $\mathbb{E}[Y_d]$ comes from the spill model and $h = 18.17$ hours is the aircraft time the rotation occupies. In words: the net contribution of the expected passengers and cargo, less fixed trip cost and

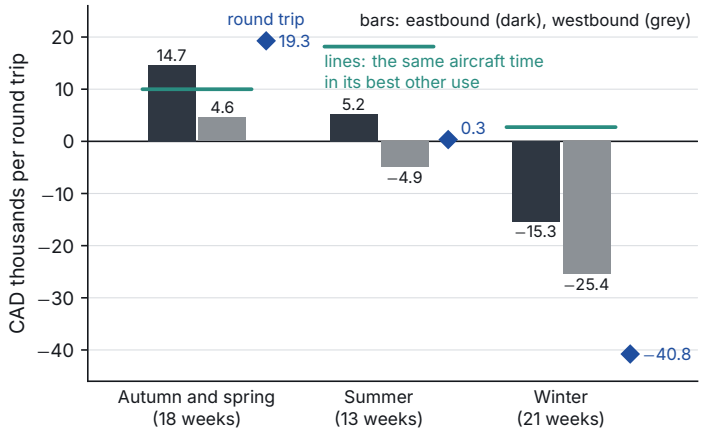


Figure 10.3 Incremental network contribution per round trip of Laurentian's YUL–SVQ by period and direction, against the value of the same aircraft time in its best alternative use (lines). Autumn and spring beat the alternative; summer covers its avoidable costs but not the value of the aircraft; winter loses money even before the aircraft's opportunity cost.

disruption, in both directions, less what the aircraft hours would earn elsewhere. Table 10.4 applies it period by period.

Table 10.4 Laurentian YUL–SVQ by period and direction, in CAD thousands. Load is expected passengers per departure; East and West are incremental network contributions per departure; ωh is the value of the aircraft time per round trip in its best alternative use; Net deducts it from the round trip. A mature year has 36 round trips in autumn, 84 in winter, 36 in spring and 52 in summer.

Period	Load	East	West	Round trip	ωh	Net	Net per year
Autumn	161.5	14.7	4.6	19.3	10.0	9.3	334
Winter	109.7	(15.3)	(25.4)	(40.8)	2.7	(43.5)	(3,654)
Spring	161.5	14.7	4.6	19.3	10.0	9.3	334
Summer	145.2	5.2	(4.9)	0.3	18.2	(17.8)	(927)

Figure 10.3 reads the table as a picture, and three conclusions stand out.

Direction. With identical demand in both directions, the westbound flight earns CAD 10,083 less than the eastbound one, the cost of 55 extra minutes of block time against the prevailing winds. A route judged on its eastbound departure, as the handbook's case is, looks better than its round trip. The block times also assume Seville is six hours ahead, which fails for a few weeks around the clock changes (Chapter 11).

Season. Summer covers its avoidable costs (CAD 339 per round trip) but the aircraft is worth CAD 18,167 elsewhere: the route is profitable and still wrong. Winter loses CAD 40,779 per round trip before any opportunity cost.

> **PITFALL**
>
> Judging a route by its load factor. The handbook's case runs at 82.4 percent and is not the best use of the aircraft; 170 passengers at CAD 530 earn less passenger revenue than 150 at CAD 620; and Laurentian's summer flights, at 79.8 percent, cover their costs while losing CAD 17,828 per round trip against the aircraft's alternative. Load factor is a symptom of demand, price and gauge, not a measure of value (Network Planning Handbook 2026).

Pattern. Totals over the year depend on which periods are flown (Table 10.5). The year-round average round trip loses CAD 9,713 against not flying at all; an average can hide an unprofitable season, and here it is not even attractive. Flying only in autumn and spring earns CAD 1.39 million against not flying and CAD 668,103 against the best alternative use of the aircraft.

The recommended pattern is two seasons a year. Laurentian's weak winter explains the difference with the real service's longer periods: an airline with stronger winter demand, holiday traffic or idle winter aircraft could fly through most of it. A monthly evaluation, as the handbook recommends (Network Planning Handbook 2026), and the restart costs of suspending service twice a year would refine the pattern.

Table 10.5 Annual value of four operating patterns for Laurentian YUL–SVQ, in CAD, in a mature year.

Operating pattern	Round trips	Against not flying	Against the best alternative
Year-round	208	(2,020,279)	(3,913,246)
Autumn, winter and spring (September to May)	156	(2,037,910)	(2,986,210)
Autumn, spring and summer	124	1,405,133	(258,933)
Autumn and spring	72	1,387,503	668,103

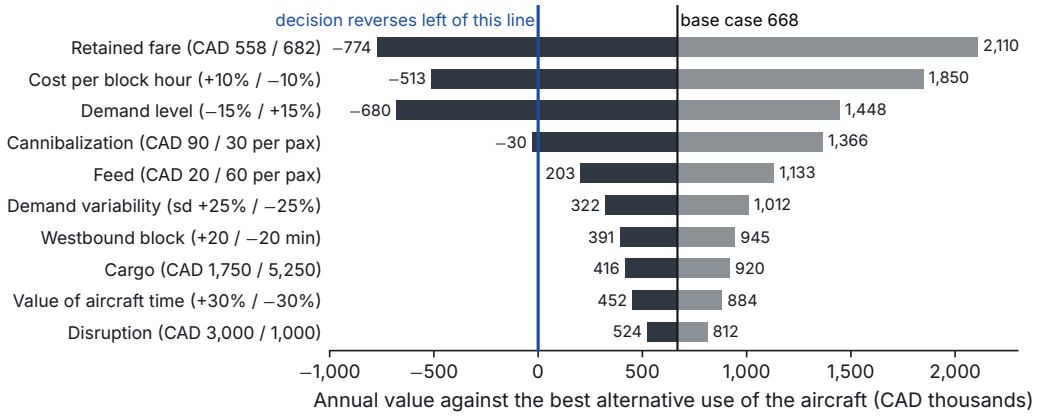


Figure 10.4 Tornado chart for the recommended autumn-and-spring pattern: its annual value against the best alternative use of the aircraft when one input moves across the range shown and the others stay at base. Bars that cross the accent line reverse the recommendation.

10.6 Scenarios, sensitivity and the threshold that reverses the decision

The handbook asks for base, adverse and favourable cases with coherent assumptions, because lower demand tends to come with lower fares and higher fuel prices with other costs; for structural uncertainty, forecast error and execution risk to be kept apart; and for the sensitivity that matters most, demand capture against retained fare, with the thresholds that reverse the recommendation (Network Planning Handbook 2026).

Coherent scenarios. In Laurentian's adverse case demand is 10 percent below forecast, the retained fare CAD 600, the cost per block hour 8 percent higher, and the value of aircraft time 15 percent lower, because the same conditions depress the aircraft's other uses. In the favourable case demand is 8 percent higher, the fare CAD 640, block-hour cost 5 percent lower and aircraft time 10 percent more valuable. The autumn-and-spring pattern, worth CAD 668,103 a year against the alternative in the base case, is worth minus CAD 1.43 million in the adverse case (minus CAD 0.82 million even against not flying) and CAD 2.14 million in the favourable case, in which adding summer would raise it to CAD 2.33 million. The downside is larger than the base-case gain: the route is a bet on fares and demand.

One input at a time. A tornado chart moves each input across a plausible range with the others at base (Figure 10.4): a guide to which inputs deserve attention, not a scenario analysis.

Retained fare, the cost per block hour and the demand level dominate, and each can reverse the decision within its range; a 10 percent

fare shortfall alone turns CAD 668,103 into a loss of CAD 774,000. The demand bar is asymmetric (minus 1.35 million, plus 0.78 million) because spill caps the upside: a 15 percent shortfall removes 16 expected passengers per departure, a 15 percent excess adds only 9, because 182 seats cannot carry the extra demand on the busiest days. Cannibalization comes next, and at CAD 90 per passenger it too reverses the decision, if only just: network effects are not a rounding error. The value of aircraft time matters less, because in autumn and spring the aircraft's alternatives are only moderately valuable.

The threshold. Setting (10.2) to zero and solving for the fare gives the fare below which the recommendation reverses: fixed costs net of cargo, plus disruption and the aircraft's alternative value, spread over the expected passengers,

$$f^*(\theta) = \frac{\sum_d (\mathrm{TC}_d^0 + \delta - K) + \omega h}{\sum_d \mathbb{E}[Y_d \mid \theta]} - f_{\mathrm{anc}} + c_{\mathrm{pax}} + e_{\mathrm{c}} - e_{\mathrm{f}}, \qquad (10.3)$$

where θ is **demand capture**, the realized share of forecast demand (which scales both μ and σ), f_{anc} is ancillary revenue and c_{pax} the passenger cost. At the forecast ($\theta = 1$) in autumn and spring, the numerator is $171{,}083 + 4{,}000 - 7{,}000 + 9{,}992 = 178{,}075$ and the denominator $2 \times 161.5 = 323.0$, so $f^* = 551.3 - 30 + 50 + 60 - 40 = 591$ dollars: CAD 29, or 4.6 percent, below the planned fare. Because the fixed costs are spread over fewer passengers, the threshold rises at an increasing rate as demand falls: CAD 607 at 95 percent capture, 627 at 90 percent and 653 at 85 percent. At the planned fare, demand capture below 91.7 percent reverses the decision.

Figure 10.5 is the handbook's matrix. The same algebra gives the cost-side thresholds: the decision reverses at a block-hour cost of CAD 11,622, 5.7 percent above plan, the kind of increase a fuel price movement can produce; at a westbound block of 8 hours 43 minutes, 48 minutes more than planned; and at 29 westbound seats blocked by hot-day or winter-wind performance. Each is a threshold for the people who own the input (revenue management, fuel and Finance, flight operations), not a forecast (Network Planning Handbook 2026).

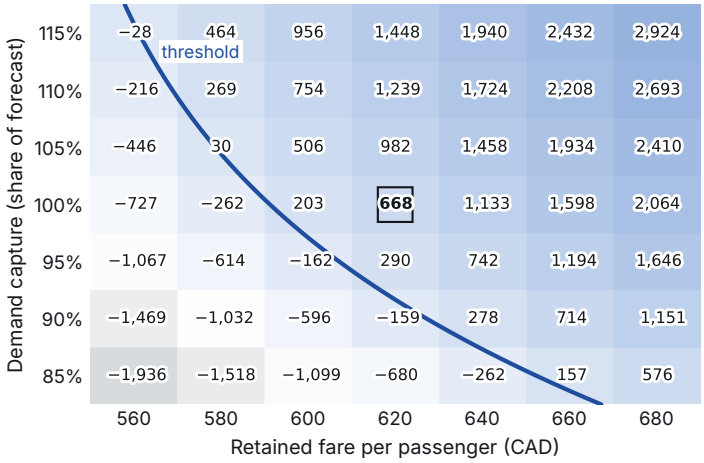


Figure 10.5 Two-way sensitivity of the autumn-and-spring pattern's annual value against the best alternative (CAD thousands) to demand capture and retained fare. The line is the threshold of (10.3); the base case is boxed.

IN PRACTICE **Route evaluation tools in 2026**

Lufthansa Systems describes its NetLine/Plan as producing O&D-based flight and network profitability by chaining market-share, spill and recapture, and cost and revenue models, and claims full scenario evaluations in under 30 seconds for up to 250,000 markets (Lufthansa Systems 2026); these are vendor claims. Such a tool computes something between views 1 and 2 of Definition 10.1, depending on its revenue allocation, recapture and cost settings, and the analyst must know which. The XLR is also widening the set of routes evaluated: Cirium's October 2026 review counts 50 delivered and describes them as opening long-haul markets too thin for widebodies, at the price of reduced cargo capacity (Cirium 2026b).

10.6.1 Ramp-up after launch

A new route does not start mature: awareness, distribution, corporate contracts and partner feed take time to build, and the handbook asks for the ramp-up and any time-limited airport incentives to be explicit (Network Planning Handbook 2026). Laurentian assumes, for illustration, that demand starts at 80 percent of the forecast and closes the gap exponentially with a 12-week time constant, and a one-off launch cost of CAD 600,000. The first autumn's 28 round trips, worth CAD 259,818 against the alternative at mature demand,

then lose CAD 262,697, because load falls with demand while costs and the aircraft's alternatives do not. With the launch cost, the mature path pays back in the 53rd week (September 2028), the ramp-up path in the 90th (late May 2029); after three years the ramp-up path has accumulated CAD 0.72 million against CAD 1.29 million. A first season is a poor guide to a route's long-run economics in either direction (Network Planning Handbook 2026).

10.7 The one-page decision memo

The output of all this work is a document a decision-maker can read in five minutes. The handbook's template covers the decision and deadline, baseline and alternatives, schedule and fleet, market evidence, revenue, cost definitions, network effects, feasibility, downside cases, the recommendation, and open conditions with owners; a recommendation should state what would make it wrong, and can be that the evidence is insufficient pending named inputs (Network Planning Handbook 2026). Table 10.6 fills it in for Laurentian, with the inputs of Table 10.3.

Table 10.6 A one-page decision memo for Laurentian YUL–SVQ, following the handbook's template. All figures are illustrative.

Memo element	Laurentian YUL–SVQ
Decision and deadline	Approve YUL–SVQ for 2027–28 with the A321XLR before the schedule is published
Baseline and alternatives	Baseline: no Seville service, XLR time used by the fleet plan. Alternatives: year-round; autumn, spring and summer; autumn and spring
Result	Autumn and spring: CAD 9,279 per round trip better than the alternative, CAD 668,103 a year. Summer: CAD 17,828 worse per round trip. Winter: CAD 43,504 worse
Downside	Adverse case CAD 1.43 million a year worse than the alternative; first season CAD 262,697 worse with ramp-up; payback in the 90th week
Recommendation	Launch for autumn and spring only; do not schedule summer or winter
What would make it wrong	Retained fare below CAD 591; demand capture below 91.7 percent; block-hour cost above CAD 11,622; 29 or more westbound seats blocked
Conditions and owners	Fare and mix, feeder displacement (revenue management); westbound performance (flight operations); crew pairing and station costs (crew planning, airports); contribution definitions (Finance)

10.8 Summary

- Route economics has three views: flight contribution, incremental network contribution against a baseline, and the capital case. The decision horizon fixes which applies.
- Prorates make allocations add up; marginal leg values do not, because feeders and long-haul legs are complements.
- A leg's network value is the change in the network's best contribution, net of the passengers it displaces from full feeders: Seville is worth CAD 120,450 of fares per bank after CAD 1,259 of displacement.
- Count only costs that change with the decision, and charge aircraft time at its shadow price or compare explicit alternatives, never both.
- Spill-based loads lower the handbook's Seville contribution by CAD 2,885 per departure; seasons matter more, from a CAD 3,677 gain at the peak to a CAD 26,348 loss in winter against the alternative.
- Laurentian's YUL–SVQ is worth flying in autumn and spring only (CAD 668,103 a year against the alternative); a fare 4.6 percent below plan or demand capture below 91.7 percent reverses the decision.
- The memo states the recommendation, the downside and the thresholds that would make it wrong, with an owner for each open input.

10.9 Further reading

Belobaba's eighteen-slide lecture on route planning and evaluation (Belobaba 2006i) is worth reading in full, and the handbook's route economics sections and Montreal–Seville case (Network Planning Handbook 2026) before every business case. Barnhart, Kniker and Lohatepanont (2002) show why passenger flows and recapture belong inside fleet decisions, and The Global Airline Industry (Belobaba, Odoni, and Barnhart 2015) covers route evaluation at textbook length. The full reference edition proves that marginal leg values do not add up and derives the ratio rule from a linear program in its Chapter 12.

CHAPTER

11

Frequency and Schedule Design

IN THIS CHAPTER

- Why frequency is part of the product: schedule delay, total trip time and the S-curve of market share.
- Frequency against aircraft size, priced with the spill arithmetic of Chapter 7.
- How a timetable is built: four linked decisions, time-of-day demand and the transatlantic window.
- Hub banks: minimum connecting time, why a bank multiplies connections and what its peaks cost.
- Rotations in UTC and the fleet a timetable needs, on Laurentian's Montreal–Seville pattern.
- Seasons, SSIM, slots under the WASG, Montreal's night restrictions, and what schedule design tools do.

DRAWS ON Belobaba, Airline schedule development and Airline economics review (MIT 16.75J, 2006); Barnhart, The schedule design problem (MIT 1.206J, 2003); IATA, ACI and WWACG, Worldwide Airport Slot Guidelines, edition 4 (2025); handbook; Lohatepanont and Barnhart (2004)

A route is not yet a product. What a customer buys is an itinerary on a date, at particular times, with a particular frequency, aircraft and connection; the same city pair can perform very differently at 07:00 and at 14:00 (Network Planning Handbook 2026). Much of a network planning analyst's week is spent on these attributes: adding a second daily frequency, swapping aircraft between routes, moving a departure 20 minutes to catch a bank, deciding whether a seasonal route runs three or four times a week. Each change looks small on the departure board and ripples through connections, rotations, crews, slots and the demand of other flights.

This chapter starts with why frequency creates demand and share, and prices the choice between more flights and bigger aircraft. It then builds a timetable: departure times, hub banks, rotations computed in UTC, and the slots that decide whether a time can be flown at all. It closes with what schedule design tools do and a worked 20-minute retiming. Route profitability is the subject of Chapter 10, fleet assignment and crews of Chapter 12, and schedule slack of Chapter 13.

11.1 Frequency, schedule delay and the S-curve

A traveller has a time at which she would like to leave. Unless a flight departs then, she must leave earlier or later, and the gap is a cost even though it is spent on the ground. Frequency shrinks it.

> **DEFINITION 11.1 Schedule delay**
>
> A traveller's **schedule delay** (or schedule displacement) is the gap between the time she would like to leave and the departure she takes. The **expected schedule delay** of a timetable averages it over travellers, each taking the departure closest to her preferred time. If that flight is full and she must move further, the extra displacement is **stochastic delay** (Douglas and Miller 1974).

The arithmetic is simple when preferred times are spread evenly over an operating day of H hours (06:00 to 22:00 gives $H = 16$). With f departures at the centres of equal slices of the day, the average traveller is a quarter of a slice from her flight: expected schedule delay is $H/(4f)$ hours, and no other placement does better (Appendix A.2). Belobaba writes total trip time as fixed time (access, egress and airport processing) plus flight time plus a schedule displacement of K/f hours (Belobaba 2006d); the even spread gives $K = H/4$:

$$T(f) = t_0 + t_b + \frac{K}{f}, \qquad K = \frac{H}{4}, \tag{11.1}$$

with t_0 the fixed time and t_b the block time. The saving from each extra departure shrinks fast: with $H = 16$, the second daily frequency saves two hours of expected displacement, the third 40 minutes, the fourth 20 and the eighth 4.3.

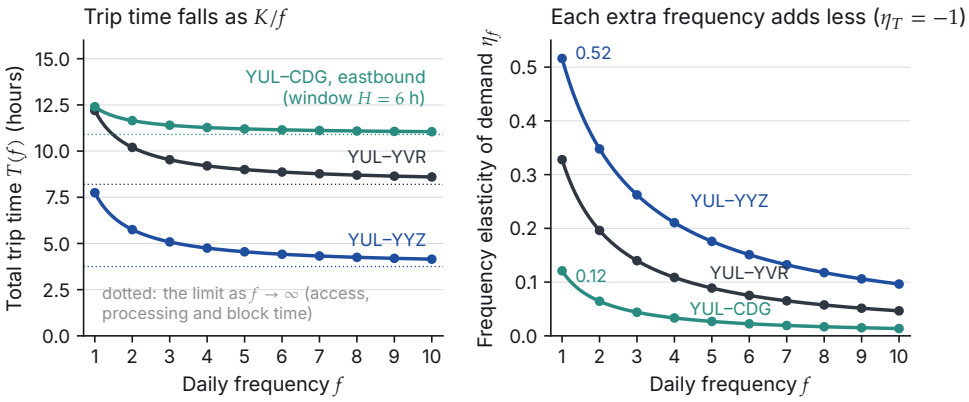


Figure 11.1 Left: total trip time as daily frequency grows for three Laurentian routes; the dotted lines are the limits as frequency grows without bound. Right: the frequency elasticity of demand (11.2) with a time elasticity of -1. Access, processing and block times are illustrative; the transatlantic route uses a six-hour eastbound departure window instead of a full day.

Trip time matters because it enters demand. If market demand has a constant elasticity $\eta_T < 0$ with respect to total trip time (Chapter 4), frequency acts on demand only through T.

RESULT 11.1 The frequency elasticity of demand

The percentage change in market demand for a one percent increase in daily frequency is the time elasticity times the share of schedule delay in total trip time:

$$\eta_f = |\eta_T| \, \frac{K/f}{T(f)}. \tag{11.2}$$

It falls towards zero as frequency grows. Adding one frequency multiplies demand by $(T(f+1)/T(f))^{\eta_T}$.

Hence three pieces of airline lore: frequency matters more in short-haul markets, where schedule delay is a large share of trip time (Belobaba 2006d); each market has a **saturation frequency** beyond which departures add almost no demand; and frequency matters more to business travellers, whose time elasticity is larger.

WORKED EXAMPLE 11.1 How much demand does a second frequency create?

Take three illustrative Laurentian routes with $\eta_T = -1$: YUL–YYZ (2.5 hours of access and processing, 1.25 hours of block), YUL–YVR (3.0 and 5.2) and eastbound YUL–CDG (4.0 and 6.9). The domestic routes have $H = 16$ and $K = 4$; eastbound transatlantic departures fit an evening window of about six hours (Section 11.3), so YUL–CDG has $H = 6$ and $K = 1.5$.

With one daily flight, total trip time is $2.5+1.25+4 = 7.75$ hours to Toronto, 12.2 hours to Vancouver and $4.0 + 6.9 + 1.5 = 12.4$ hours to Paris; schedule delay is 52, 33 and 12 percent of it. A second daily flight cuts trip time to 5.75, 10.2 and 11.65 hours, which raises market demand by 34.8, 19.6 and 6.4 percent. A third adds 13.1, 7.0 and 2.2 percent. Chapter 4, with a time elasticity of -1.2 and a full-day window, finds 46 percent for Toronto and 21 for Paris; the gap at Paris comes mainly from the narrow window, which alone cuts 21 percent to 8.

On short haul the first few frequencies create travel. On long haul a second daily frequency must be justified by capacity, connections or competition rather than by stimulated demand.

These numbers describe the whole market. When several airlines serve it, frequency also decides how it is shared, and more than proportionally: the airline with more frequency alone serves the travellers who want to leave when only it flies (Belobaba 2006d). Belobaba's **S-curve** writes airline i's market share as

$$\mathrm{MS}_i = \frac{f_i^{\alpha}}{\sum_j f_j^{\alpha}}, \qquad \alpha > 1, \tag{11.3}$$

where f_j is airline j's daily frequency and historical estimates of α lie between 1.3 and 1.7 (Belobaba 2006d); Chapter 4 discusses the curve and Appendix A.2 derives it. Against a competitor with three daily nonstops and $\alpha = 1.5$, Laurentian's share is 16.1 percent with one daily flight, 35.2 with two, 50.0 with three and 60.6 with four: the second frequency adds 19.1 points, against 15.0 under proportional shares.

That amplification drives frequency competition. Each carrier adds flights until the share it takes from the other just pays for the extra trip; in a stylized duopoly with equal costs and fixed fares, the two together spend a fraction $\alpha/2$ of the market's revenue on

frequency, three quarters with $\alpha = 1.5$ (Appendix A.2). Steep share effects push competitors towards many frequencies with small aircraft.

11.2 Frequency versus gauge

Suppose a route needs about 280 seats a day, offered as one large aircraft or two smaller ones. The large aircraft wins on cost per seat and pools demand into one cabin; the smaller ones win on frequency, which creates demand and share. The contribution arithmetic of Chapter 7 settles the question, applied to each complete plan with the demand that plan generates.

A plan flies f daily departures of an aircraft with C seats and trip cost TC, each passenger contributing r. Departure j faces mean demand μ_j with standard deviation $\sigma_j = k\mu_j$ (k is the k-factor). Expected daily contribution is

$$\pi(f, C) = r \sum_{j=1}^{f} \Big[\mu_j - \sigma_j L\Big(\frac{C - \mu_j}{\sigma_j}\Big) \Big] - f\,\mathrm{TC}, \tag{11.4}$$

where the bracket is passengers carried, mean demand minus expected spill, with L the normal loss function of Chapter 7. The step that matters is the demand: the μ_j add up to the one-flight demand times the multiplier frequency creates, from Result 11.1 where the airline serves the market alone or from the S-curve where it competes.

WORKED EXAMPLE 11.2 Two A220s or one 787-9 to Vancouver?

Laurentian flies YUL–YVR with two daily A220-300s of 137 seats, the service used in Chapter 7. Demand is $\mathcal{N}(160, 48^2)$ for the evening flight and $\mathcal{N}(100, 30^2)$ for the morning flight. The alternative is one evening 787-9 (298 seats). Two A220 trips cost CAD 87,200 (CAD 318 per seat) and one 787-9 trip CAD 85,600 (CAD 287 per seat): 24 more seats for CAD 1,600 less. Each passenger contributes CAD 400, and $k = 0.30$.

Demand. Laurentian is the only nonstop carrier, so frequency acts through trip time alone. Example 11.1 gives a multiplier of $12.2/10.2 = 1.196$, so the 260 daily passengers of the two A220s correspond to 217.4 with one daily flight.

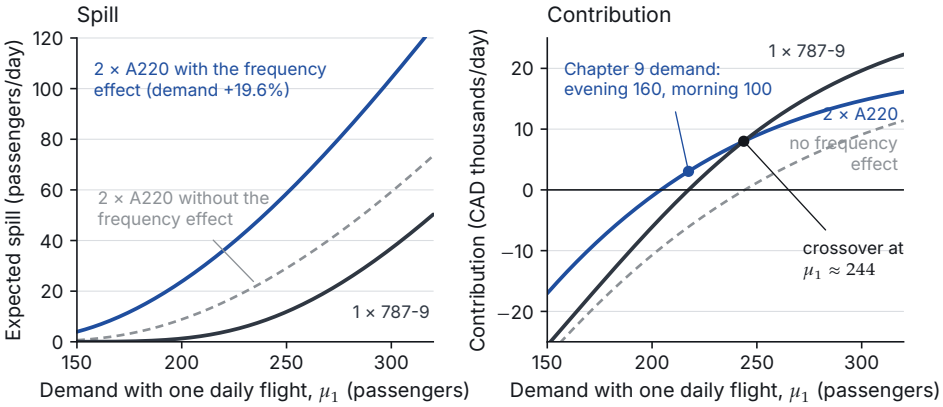


Figure 11.2 Two daily A220-300s against one daily 787-9 on YUL–YVR as Laurentian's one-frequency demand grows: expected spill (left) and contribution (right) per day. The blue curve applies the 19.6 percent demand gain of a second frequency (Example 11.1); the dashed grey curve omits it. Laurentian trip costs, CAD 400 per passenger, $k = 0.30$.

Two A220s. The evening flight spills 32.8 passengers a day and the morning flight 1.6, so 225.6 are carried and contribution is CAD 3,049 a day.

One 787-9. With $\sigma = 65.2$, $z = (298 - 217.4)/65.2 = 1.24$. Spill is 3.4, 214.0 are carried and contribution is $400 \times 214.0 - 85{,}600 \approx 0$.

The A220s win by about CAD 3,000 a day, CAD 1.1 million a year, while spilling ten times as many passengers at a higher seat cost. Without the frequency effect they would carry 202.8 passengers and lose CAD 6,088 a day: the frequency effect is worth about CAD 9,100 a day.

Figure 11.2 lets demand grow. The 787-9 overtakes only when one-flight demand reaches about 244, where the A220s face 292 passengers and spill more than 50 a day; by then the better plan is usually to keep both frequencies and upgauge them. In a competitive market the frequency effect is far larger. Against a rival with three daily nonstops and $\alpha = 1.5$, the second frequency multiplies Laurentian's demand by $0.352/0.161 \times 1.022 = 2.23$ (share gain times market stimulation): the 260 passengers of the two A220s would shrink to about 116 on a single 787-9, which would fly 61 percent empty.

> **PITFALL**
>
> Choosing the aircraft with the lowest cost per seat. Seat cost compares capacity, not plans: it ignores the demand frequency creates, the share it wins and the spill capacity causes. Compare the contributions of complete plans.

At a hub, two daily flights can also sit in two banks (Section 11.4), which is why hub carriers prize small aircraft with competitive seat costs on thin spokes. As a route grows, airlines usually add frequencies until the useful times are filled, then upgauge, unless slots (Section 11.6) block the extra frequency.

Frequency is a demand decision and gauge is a cost decision. Price both with the same contribution arithmetic, letting demand respond to frequency before spill is computed.

11.3 Building the timetable

Belobaba splits **schedule development** into four linked decisions: **frequency planning** (how often), **timetable development** (at what times), **fleet assignment** (with which aircraft type) and **aircraft rotation planning** (which sequence of flights each aircraft flies) (Belobaba 2006f). Frequency plans come a year or more before departure; timetables and rotations are set two to six months ahead; revisions continue until the flight operates. The steps are iterated, because a departure time fixes which connections exist, which aircraft and crews can fly the flight and which slot it needs (Barnhart and Cohn 2004). Most changes are **incremental** what-ifs against a baseline, recorded on a **schedule card**: both directions, dates, local and UTC times, block times, turns, aircraft, feasible connections, the implied rotation, and the owner of every unverified assumption (Network Planning Handbook 2026). Table 11.2 shows one for Seville.

Evenly spread preferences are a benchmark, not a description. Business demand in North America peaks early morning and late afternoon, around 09:00 and 17:00 by Belobaba's rule of thumb (Belobaba 2006f); leisure demand is flatter. With peaked demand the best time for a departure is the **median** of the travellers who would choose it, half wanting to leave earlier and half later. A second cri-

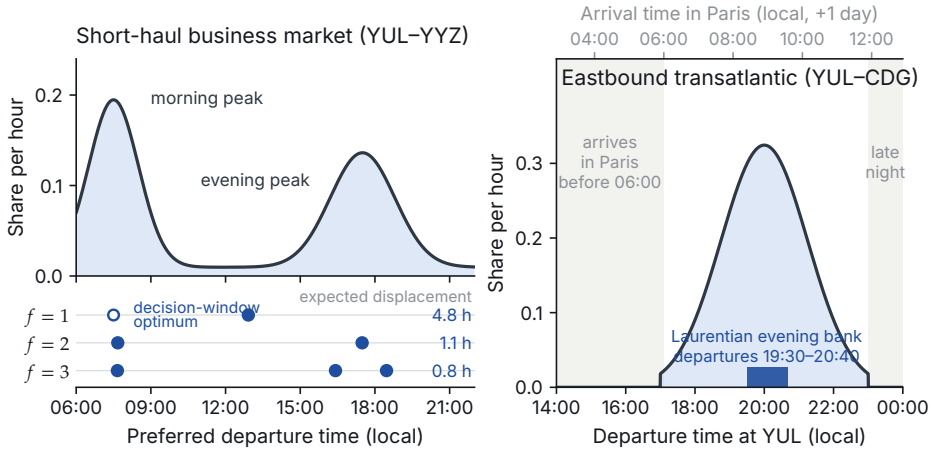


Figure 11.3 Time-of-day demand profiles (illustrative). Left: a business-heavy short-haul market; the rows show the departure times minimizing expected schedule delay for one, two and three daily flights, and the hollow marker the single flight maximizing demand within one hour. Right: the eastbound transatlantic window; departures before about 17:00 reach Paris before 06:00.

terion is often closer to behaviour: a traveller offered a flight five hours from her preferred time picks another airline, another day or no trip, so placing a flight to catch the most travellers within an hour of it can beat minimizing average delay.

Figure 11.3 applies both criteria to an illustrative YUL–YYZ profile with 45 percent of travellers in a morning peak and 40 percent in a broader evening peak. A single flight minimizing average delay lands at 12:55, between the peaks: expected delay is 4.8 hours (against 4.0 in an evenly spread market) and only 2.1 percent of travellers are within an hour of it. Catching the most travellers within an hour moves it to 07:30, within an hour of 18.1 percent. Two flights go to the peaks, 07:40 and 17:29, cutting expected delay to 1.1 hours (2.0 when evenly spread) with 58 percent of travellers within an hour. A third lowers delay only to 0.84 hours. In a peaked market the first two frequencies are worth more, and the third much less, than the even-spread model suggests.

Time zones compress the window. Flights from the United States to Europe do not leave before about 17:00, because travellers do not want to arrive before 06:00 (Belobaba 2006f); from YUL, with 6.9 block hours and six hours of time difference, a 17:06 departure reaches Paris at 06:00, and late departures meet YUL's night restrictions (Section 11.6). Within this six-hour window, transatlantic times are set by the bank they connect with.

Other constraints push departures off the peaks (Belobaba 2006f). An aircraft arriving at 11:00 can leave again at 12:00, off-peak; hold-

ing it for the 17:00 peak costs utilization, so airlines keep turns short and fly some off-peak departures. Banks, slots, curfews, crews and maintenance restrict times further.

11.4 Hub banks and connections

A hub multiplies markets by timing: if flights from many cities arrive together and flights to many cities leave together shortly afterwards, every arriving passenger can reach every departing city with one stop.

DEFINITION 11.2 Bank and connection window

A **bank** (or wave) is a group of arrivals at a hub followed by a group of departures, timed so that passengers can connect (Network Planning Handbook 2026). Its **connection window** runs from the **minimum connecting time** (MCT), the shortest interval the applicable airline, airport, terminal and connection-type rules allow for transferring a passenger and baggage, to $\bar{\tau}$, the longest connection the analyst treats as sellable.

DEFINITION 11.3 Feasible connection

An arrival i at time a_i and a departure j at time d_j form a **feasible connection** when

$$\mathrm{MCT}(i,j) \;\le\; d_j - a_i \;\le\; \bar{\tau}, \tag{11.5}$$

with both times as full timestamps (date, time and zone) rather than clock faces (Network Planning Handbook 2026).

A retiming moves pairs across both limits at once: in the handbook's screen, moving a 17:00 departure to 17:15 gains a 16:05 arrival that missed a 60-minute MCT but, with a 120-minute commercial limit, loses a 15:10 one, so only the value of the two itineraries decides (Network Planning Handbook 2026). Laurentian's evening transatlantic bank shows the screen at hub scale (Table 11.1 and Figure 11.4; times illustrative, and set independently of the four-leg bank drawn in Chapter 8). Of the 56 pairs formed by eight North American arrivals between 17:55 and 18:45 and seven European de-

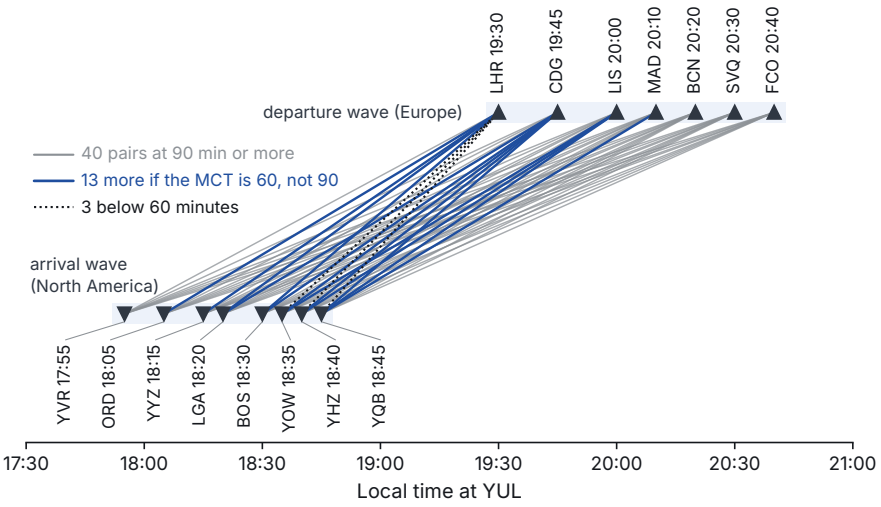


Figure 11.4 Laurentian's illustrative evening bank at YUL. Each line joins an arrival to a departure; grey lines are connections that survive a 90-minute MCT, blue lines are the 13 additional connections available with a 60-minute MCT, and dotted lines the three pairs below 60 minutes.

partures between 19:30 and 20:40, 53 are feasible with a 60-minute MCT and 40 with a 90-minute MCT; the 53 average 108 minutes and the longest, Vancouver to Rome, takes 165. Connections are lost first in the corner where the last arrivals meet the first departures, which makes the MCT a planning variable, not an operational detail.

Table 11.1 Connecting times in minutes in Laurentian's illustrative evening bank at YUL. Values in parentheses fail a 60-minute MCT; values marked † pass a 60-minute but fail a 90-minute MCT.

Arrival	LHR 19:30	CDG 19:45	LIS 20:00	MAD 20:10	BCN 20:20	SVQ 20:30	FCO 20:40
YVR 17:55	95	110	125	135	145	155	165
ORD 18:05	85†	100	115	125	135	145	155
YYZ 18:15	75†	90	105	115	125	135	145
LGA 18:20	70†	85†	100	110	120	130	140
BOS 18:30	60†	75†	90	100	110	120	130
YOW 18:35	(55)	70†	85†	95	105	115	125
YHZ 18:40	(50)	65†	80†	90	100	110	120
YQB 18:45	(45)	60†	75†	85†	95	105	115

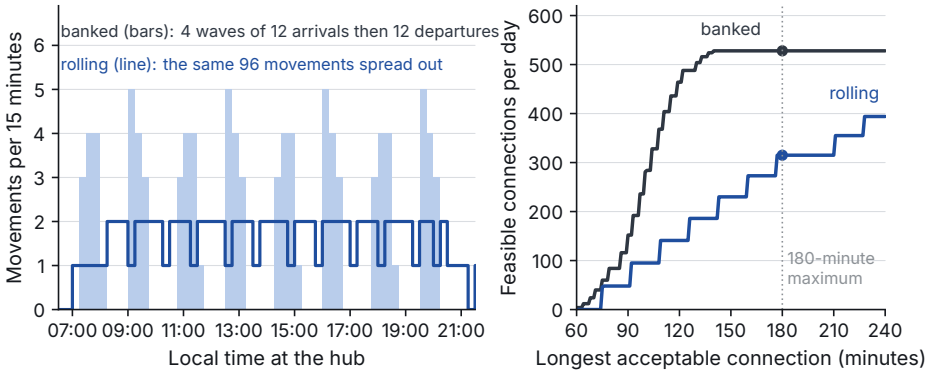


Figure 11.5 The price of connectivity in an illustrative hub with 48 arrivals and 48 departures a day. Left: runway movements per 15 minutes in a day of four banks (bars) and in a rolling version of the same flights (line). Right: feasible connections as the longest acceptable connection grows, with a 60-minute MCT; the dots mark a 180-minute limit.

RESULT 11.2 Why banks create connections

If all of a bank's arrivals fit the connection window of all its departures, a bank with n_b spokes each way offers $n_b(n_b - 1)$ useful connections (the other pairs return passengers to their origin): connections grow with the square of the bank. If the same flights are spread evenly over the day (a **rolling** hub), each arrival finds only the departures that fall inside its window. Banking multiplies connections by roughly the length of the operating day divided by the combined width of the banks' connection windows.

Doubling a bank from 10 to 20 spokes each way raises its connections from 90 to 380. A rolling hub still connects, but less often and with longer waits.

Figure 11.5 prices the trade-off in an illustrative hub with four banks, each of twelve arrivals within 40 minutes followed an hour later by twelve departures. With a 60-minute MCT and a 180-minute limit, the banked day offers $4 \times 12 \times 11 = 528$ connections, with a median wait of 100 minutes. Spreading the arrivals evenly from 07:00 to 20:15, each aircraft turning in 75 minutes, leaves 315 with a median wait of 126. The ratio, 1.68, is close to the rule's 1.66 (a 795-minute day over four windows of 120 minutes). The bank's price is its peak: five movements per 15 minutes against two, twelve aircraft on the ground at once against five, and 100 minutes of ground time per aircraft against 75, utilization spent waiting for the bank.

Those peaks load gates, ramp crews, customs, baggage and the runway: a historical profile of a U.S. banked hub in the MIT course shows operations per 15 minutes exceeding instrument-weather capacity at every bank (Barnhart 2003). Banks also make delays contagious, since a late inbound flight either holds its connecting departures or strands its passengers (Chapter 13). A rolling (de-peaked) hub flattens the peaks at the cost of fewer and longer connections (Jiang 2006), so hubs bank the flows that matter most (for YUL, North America to Europe in the evening and Europe to North America in the afternoon) and spread the rest.

IN PRACTICE **Banks at Air Canada's hubs in 2026**

Air Canada develops Toronto Pearson, Montreal-Trudeau and Vancouver as its international hubs (Air Canada 2025). At an investor conference in September 2026, its chief financial officer described the banks as synchronized so that U.S. inbound traffic meets international departures "within a couple of hours", called the A220 (about 130 seats at the seat cost of a Boeing 737 or Airbus A320) the key enabler for serving smaller U.S. cities, and noted the appeal of arriving in Europe around 07:00 to 08:00 (Investing.com 2026). These are this chapter's levers: bank timing, small-gauge feeders and time-zone windows. Treat the transcript's figures as management statements.

PITFALL

Inflating connectivity. A feasible pair is not a market: each candidate must still be scored by demand, share, elapsed time and contribution, without counting mutually exclusive itineraries as separate passengers. And treating the MCT as a target produces connections that misconnect whenever the inbound is late: MCT compliance makes a connection schedule-valid, not reliable (Network Planning Handbook 2026).

11.5 Rotations, clocks and the fleet a timetable needs

A timetable is operable only when every departure has an aircraft. The unit is the **rotation**, the sequence of flights an aircraft flies before returning to its start. For an out-and-back route the handbook's cycle is outbound block plus away turn plus return block plus base turn,

$7.0 + 1.5 + 8.0 + 2.0 = 18.5$ hours in its example (Network Planning Handbook 2026). Aircraft **utilization**, block hours per aircraft per day, falls whenever aircraft wait for peaks or banks or routes run less than daily.

RESULT 11.3 How many aircraft a timetable needs

A flight occupies an aircraft from departure until it is ready again (arrival plus minimum turn). With BH block hours a day, h the block hours one aircraft can sustain a day after maintenance, and $A(t)$ the number of flights occupying aircraft at time t, the fleet N must satisfy

$$N \geq \frac{\mathrm{BH}}{h} \qquad \text{and} \qquad N \geq \max_t A(t). \tag{11.6}$$

The exact minimum is counted at any one instant: the aircraft occupied by flights, plus, for each station, the largest shortfall of aircraft that station would run into over the day if it started with none on the ground.

The exact count is the **count line** that fleet assignment uses to cap each type's aircraft (Chapter 12) (Hane et al. 1995). The pooling bound's denominator must reflect realistic availability, not 24 hours a day (Network Planning Handbook 2026). Flown daily, the handbook's 18.5-hour cycle fits one aircraft by the count, but at an illustrative 14 sustainable block hours a day its 15 daily block hours need $15/14 = 1.07$ aircraft, so such a route is flown from a pool rather than by one tail.

Schedules are sold in local time but computed in UTC, and offsets change on different dates: Quebec observes daylight saving time from the second Sunday of March to the first Sunday of November (Québec 2007), the European Union from the last Sunday of March to the last Sunday of October (European Parliament and Council of the European Union 2001), and IATA seasons start on the European dates (IATA, ACI, and WWACG 2025). These rules can change: Quebec consulted on ending clock changes in 2024 (Ministère de la Justice du Québec 2024) and British Columbia stopped them in 2026 (Government of British Columbia 2026), so the dates below assume the law in force in October 2026. Around both season boundaries Montreal and Seville are five hours apart instead of six: from 31 October to 6 November 2027 and from 12 to 25 March 2028.

Laurentian's Seville service mirrors a publicly announced pattern, subject to change: YUL 20:30, Seville 09:30 the next day; Seville 12:45, YUL 14:40; A321XLR; three weekly from 8 September 2027 and four weekly from 1 October, with no departures from 9 January to 17 February 2028 and a last departure on 3 June 2028 (Network Planning Handbook 2026; Air Canada 2026f). Its days, and its operation through the whole winter, are illustrative: Wednesday, Friday and Sunday, adding Monday from October.

WORKED EXAMPLE 11.3 The Seville rotation in UTC

Summer time (YUL at UTC−4, Seville at UTC+2). The 20:30 departure is 00:30 UTC the next day and the 09:30 arrival is 07:30 UTC, a 7-hour block. The 12:45 departure is 10:45 UTC, a turn of 3 hours 15 minutes, and the 14:40 arrival is 18:40 UTC, a block of 7 hours 55 minutes. The aircraft is back 18 hours 10 minutes after leaving, 5 hours 50 minutes before another 20:30 departure (Network Planning Handbook 2026).

Five-hour weeks and winter. Holding Montreal times and block times, Seville's local times move an hour earlier in the five-hour weeks (arrival 08:30, departure 11:45); keeping the published 12:45 departure instead would stretch the turn to 4 hours 15 minutes. In winter every UTC time is an hour later than in summer (Table 11.2).

Dates. A Sunday 20:30 departure is a Monday flight in UTC, so the commercial Wednesday, Friday and Sunday flights are Thursday, Saturday and Monday operations in Seville (Network Planning Handbook 2026).

Aircraft time. A round trip flies 14.92 block hours: 44.75 a week at three weekly, with the aircraft away 32 percent of the week, and 59.67 at four weekly, away 43 percent. At 14 sustainable block hours a day, Seville uses 0.46 and 0.61 of an aircraft-week; the rest must be sold to other routes for the aircraft to earn its ownership cost. The October pattern leaves one Montreal turn of only 5 hours 50 minutes, between the Sunday flight's return and the Monday departure.

Table 11.2 Schedule card for Laurentian's YUL–SVQ rotation (A321XLR, 182 seats), holding Montreal times and block times fixed. Summer time: 8 September to 30 October 2027 and from 26 March 2028; five-hour weeks: 31 October to 6 November 2027 and 12 to 25 March 2028; winter: 7 November 2027 to 11 March 2028. Seville times are on the day after the Montreal departure.

Item	Summer time	Five-hour weeks	Winter
Clocks, YUL and SVQ	UTC−4, UTC+2	UTC−4, UTC+1	UTC−5, UTC+1
Depart YUL, local (UTC)	20:30 (00:30)	20:30 (00:30)	20:30 (01:30)
Arrive SVQ, local (UTC)	09:30 (07:30)	08:30 (07:30)	09:30 (08:30)
Depart SVQ, local (UTC)	12:45 (10:45)	11:45 (10:45)	12:45 (11:45)
Arrive YUL, local (UTC)	14:40 (18:40)	14:40 (18:40)	14:40 (19:40)
Block east, west	7:00, 7:55	7:00, 7:55	7:00, 7:55
SVQ turn; time away	3:15; 18:10	3:15; 18:10	3:15; 18:10

PITFALL

Clock-face arithmetic. Subtracting local times across time zones, midnight or a daylight-saving change gives wrong block times, turns and connection windows. Compute in UTC, keep local times for commercial meaning, and never hard-code a single year-round UTC offset (Network Planning Handbook 2026).

11.6 Seasons, slots and airport restrictions

Schedules are planned by **IATA scheduling season**: summer begins on the last Sunday of March and winter on the last Sunday of October (IATA, ACI, and WWACG 2025), so S27 runs from 28 March to 30 October 2027, W27 from 31 October 2027 to 25 March 2028 and S28 from 26 March to 28 October 2028. Schedule data move in the formats of IATA's Standard Schedules Information Manual (**SSIM**), which also covers coordination messages, MCT data and time zones; edition 36 is effective from 1 March 2026 to 28 February 2027 (Network Planning Handbook 2026). Where airport capacity is short, schedules must also fit the Worldwide Airport Slot Guidelines (WASG), the industry standard wherever national rules do not

override it; slot messages use SSIM formats and UTC (IATA, ACI, and WWACG 2025).

DEFINITION 11.4 Coordination levels, slots and historic precedence

Under the WASG (IATA, ACI, and WWACG 2025):

- a **Level 1** airport has infrastructure generally adequate for demand at all times;
- a **Level 2** airport may be congested at some periods, resolved by schedule adjustments agreed with a **facilitator**; no slots are allocated;
- at a **Level 3** airport a **coordinator** allocates slots, and an airline must hold one before operating;
- an **airport slot** is a coordinator's permission to arrive or depart at a Level 3 airport on a specific date and time;
- a **series of slots** is at least five slots at about the same time on the same day of the week in a season; from Summer 2027 a series needs at least seven slots in a Northern Summer season and still five in winter;
- **historic precedence** entitles an airline to keep a series in the next equivalent season if it operated the series at least 80 percent of the time.

Use it or lose it. The 80/20 rule makes slots valuable and sticky: an airline that flies its series keeps it season after season, one that does not loses it to the pool. Use is measured against the series held on the **Historics Baseline Date**, 31 January for summer and 31 August for winter, and later cancellations count as non-use unless caused by events outside the airline's control, such as airport closures, severe weather or strikes (IATA, ACI, and WWACG 2025). Each day of the week is a separate series, and where daylight-saving dates differ at a season boundary, as between Canada and Europe, airline and coordinator agree in advance how to file those weeks (IATA, ACI, and WWACG 2025).

New entrants and the calendar. Unchanged historic slots are allocated first; half of the remaining **pool** goes to **new entrants**, airlines that would hold fewer than seven slots at the airport that day (an arrival and a departure per rotation, so up to three daily rotations), and historic slots cannot be withdrawn to make room for them (IATA, ACI, and WWACG 2025). The Slot Conference meets in November for the next summer and in June for the next winter,

with discussion of fares or market entry prohibited (IATA, ACI, and WWACG 2025): a new summer frequency at a coordinated airport must be in the plan by the autumn of the year before.

WORKED EXAMPLE 11.4 Counting slots

The 80 percent test. S28 has 31 weeks, so a weekly series held all season has 31 slots and must operate at least $\lceil 0.8 \times 31 \rceil = 25$ times, tolerating six unjustified non-uses; a W27 series of 21 slots needs 17.

Seville at a Level 3 airport. Suppose, for illustration, that Seville were coordinated. Its slots fall on the UTC day after the Montreal departure, so the Wednesday flights form a Thursday series: eight Thursdays between the 8 September launch and the end of S27, enough even under the new seven-slot summer minimum. The Monday flights added in October would be Tuesday slots, only four before the season ends: not a series, so no historic precedence. A six-week peak-summer program no longer qualifies from S27.

Two distinctions keep slot analysis honest. A slot is not a traffic right (Chapter 8): allocation is independent of the rights granted under bilateral air service agreements (IATA, ACI, and WWACG 2025). And coordination is the exception: IATA counted 395 airports under coordination in 2026, 216 of them fully slot-coordinated, from which about 43 percent of the world's passengers depart (IATA 2026a). In 2024 the Canadian Airports Council told a House of Commons committee that only Toronto Pearson, Vancouver and Billy Bishop Toronto City were Level 3 in Canada, citing Montreal-Trudeau, Calgary and Quebec City as Level 2 (Canadian Airports Council 2024); designations change, so check the current list.

Slots are one line of the constraint sheet the handbook recommends for every airport, with runway performance, gates, customs, de-icing and noise (Network Planning Handbook 2026). At home, it warns against saying that YUL closes at night: Aéroports de Montréal describes 24-hour operation with restrictions and exceptions by aircraft category (Network Planning Handbook 2026). A summary last updated by the airport in March 2011 bars Chapter 3 jets above 45,000 kg maximum take-off weight from departing between 00:00 and 07:00 or arriving between 01:00 and 07:00 without an exemption (Boeing 2011), and in December 2025 Aéroports de Montréal proposed barring take-offs and landings from 01:00 to

06:00 (CBC News 2025). The sources differ in date and status, so obtain the rule in force for the aircraft and season planned. Either version caps how late the evening transatlantic bank can leave and how early overnight flights from the west can land.

11.7 Schedule design tools and the retiming decision

Schedule design chooses the flights and aircraft types that maximize contribution, given origin-destination demands and fares, the fleet and its costs (Barnhart 2003). Belobaba's 2006 view was that no model could find an optimal timetable among so many combinations (Belobaba 2006f), so research optimized changes to an existing schedule instead. **Fleet assignment with time windows** lets each departure move by a few minutes (plus or minus 10 in the lecture) when that improves utilization or the match of capacity to demand (Rexing et al. 2000; Barnhart 2003). **Incremental schedule design** lets planners mark flights as mandatory or optional and add candidate flights and times to a **master flight list**, from which the model selects legs while assigning fleet types (Barnhart 2003; Lohatepanont and Barnhart 2004).

The hard part is demand. Removing a flight does not simply delete its passengers: some move to the airline's other flights (recapture), some to competitors, in amounts that depend on what else is removed. In Barnhart's example, a market served by two mandatory and two optional flights carries 450 passengers; removing one optional flight leaves 410, removing both leaves 300, while adding up the effects of each removal predicts 330 (Barnhart 2003). The models therefore iterate: solve, recompute demand for the chosen schedule with a richer model, adjust recapture rates, and solve again. The lecture's caveats are a checklist for any optimized schedule: plans are disrupted, competitors respond, and the models assume deterministic demand, optimal revenue management, accurate forecasts and known recapture rates (Barnhart 2003).

IN PRACTICE **Schedule tools in 2026**

Commercial planning systems evaluate schedules by simulating origin-destination demand. Lufthansa Systems describes NetLine/Plan as chaining demand and market-share, spill and recapture, and cost and revenue models into flight and network profitability, with scenario evaluation in under 30 seconds (a vendor claim) (Lufthansa Systems 2026); Chapter 16 gathers what

is known of Air Canada's own tools. The 2026 AGIFORS Schedule and Strategic Planning program includes planning with bank structures, robust scheduling and, from Air Canada, aircraft routing under uncertainty (AGIFORS 2026). The planner's job is to know which demand and recapture assumptions a tool uses and to test its answers against this chapter's arithmetic.

11.7.1 When is a retiming worth it?

A schedule change is worth making when it raises the network's contribution (Chapter 10) across all affected flights and journeys, after the expected cost of the delays and misconnections it invites. The handbook's checklist for a retiming covers connections, demand, competitiveness, arrival restrictions, aircraft return timing and crew feasibility, under low, base and high delay cases, and ends with the binding constraint and the assumption most likely to reverse the decision (Network Planning Handbook 2026).

WORKED EXAMPLE 11.5 18:00 or 18:20 to Vancouver?

Laurentian's evening YUL–YVR flight, now a 737-8 with 169 seats, departs at 18:00; the proposal is 18:20. All inputs are illustrative.

Connections. Feeders arrive from Quebec City at 17:05, Boston at 17:10, Halifax at 17:15 and Ottawa at 17:20, with MCTs of 45 minutes for domestic connections and 60 for the transborder one from Boston. At 18:00 only Quebec City and Halifax (exactly at the minimum) connect; at 18:20 all four do.

Demand and spill. Local demand falls from 140 to 136 a day, because a competitor leaves at 18:30. The existing connections carry 12 a day (7 from Quebec City, 5 from Halifax) and the new ones add 8, so mean demand rises from 152 to 156. With $k = 0.30$, carried passengers rise only from 141.06 to 143.11 (2.055 more before rounding), because the flight is often full: CAD 822 a day at CAD 400 each.

Reliability. If a feeder is late with probability 0.30 and its lateness is exponential with a mean of 20 minutes, expected misconnections fall from 2.77 a day, mostly on the Halifax connection, to 2.40, because the existing connections gain slack: CAD 95 a day at CAD 250 per misconnection.

The bank peak. At 18:20 the departure falls inside the arrival wave of the evening bank (Figure 11.4). Operations estimates 2, 4 or 7 extra minutes of taxi-out in the low, base and high cases, at the 737-8's CAD 150 per block minute.

Aircraft and crew. The aircraft reaches Vancouver at 20:32 instead of 20:12 local time in summer; British Columbia has kept UTC−7 all year since 2026, so Vancouver times are an hour later in winter (Government of British Columbia 2026). Its 23:55 overnight return, reaching Montreal at about 07:30, after YUL's restricted window, keeps a turn of 3 hours 23 minutes instead of 3 hours 43, and crew planning confirms, by assumption, that the shorter layover remains legal.

Table 11.3 Evaluating the 18:20 retiming of Example 11.5 against 18:00. The base case is worth about CAD 116,000 a year.

Daily effect (CAD)	Low delay	Base	High delay
Extra passengers carried, at CAD 400	822	822	822
Fewer misconnections, at CAD 250	95	95	95
Taxi-out in the bank peak (2, 4, 7 min at 150)	−300	−600	−1,050
Net change	617	317	−133

Table 11.3 gives the recommendation: retime to 18:20 unless operations expects the bank peak to add more than about 6.1 minutes of taxi-out. The binding constraint is the evening bank at YUL, not the aircraft or the crew. The assumption most likely to reverse the decision is the local share lost to the 18:30 competitor: losing 5.6 local passengers a day instead of 4 wipes out the base-case gain. A good memo states both thresholds, so the forecasting and operations teams know which of their numbers the decision rests on.

> **PITFALL**
>
> Evaluating a retiming on its own flight. The flight's revenue may rise while the network's falls: new connections can displace higher-value local passengers on a full flight, a later departure can strand passengers on the day's last connection, and a departure moved into a bank peak can delay other flights (Network Planning Handbook 2026).

11.8 Summary

- The product is an itinerary at a time, not a route. Schedule development sets frequencies, times, aircraft types and rotations, iteratively and mostly by incremental what-ifs.
- Frequency creates demand by cutting schedule delay, most on short haul and less with each addition; across competitors the S-curve gives the frequency leader a more than proportional share.
- Compare frequency and gauge by the contribution of complete plans, letting demand respond to frequency before computing spill: two A220s beat one 787-9 to Vancouver despite a higher seat cost.
- Departure times follow peaked demand and time zones (a six-hour eastbound transatlantic window), then bend to rotations, banks, slots and curfews.
- A bank's connections grow with the square of its size, at the price of peaks and of aircraft waiting on the ground. A timetable needs at least its block hours over sustainable utilization; compute rotations in UTC.
- WASG slots are held by weekday series, kept with 80 percent use and separate from traffic rights. A retiming is worth making when the network's contribution, net of disruption cost, beats the alternative.

11.9 Further reading

Belobaba's schedule development lecture (Belobaba 2006f) introduces frequency planning and timetable constraints, and his economics review (Belobaba 2006d) supplies total trip time and the S-curve, derived in Appendix A.2. Barnhart's lecture (Barnhart 2003) and Lohatepanont and Barnhart (2004) develop incremental schedule design. The handbook (Network Planning Handbook 2026) contains the screening rule, rotation arithmetic, the Seville case and the retiming exercise, and the Worldwide Airport Slot Guidelines (IATA, ACI, and WWACG 2025) are short enough to read before a first slot conference. The full reference edition's Chapter 13 proves the bank-counting and minimum-fleet results and writes out the schedule design models in full.

CHAPTER

12

From Schedule to Aircraft and Crews

IN THIS CHAPTER

- What happens to the timetable after it leaves the network planner, and why it comes back with questions.
- The time-space network: the picture that turns aircraft into flows and counts the aircraft a timetable needs.
- Fleet assignment: what it decides, the spill cost of each aircraft type, its three constraints, and a sixteen-flight Laurentian bank solved.
- Passengers buy itineraries, not flights: the passenger mix model and recapture.
- Maintenance routing, lines of flying, crew pairings, crew pay, fatigue rules and rostering.
- What a twenty-minute schedule change breaks, and how to find out before it is published.

DRAWS ON Barnhart, Airline fleet assignment (MIT 16.75J, 2006); Barnhart, Airline schedule planning: fleet assignment, maintenance routing and crew scheduling, with Problem Sets 2 and 3 (MIT 1.206J, 2003); Belobaba, Airline schedule development (2006); Boeing, Demand driven dispatch; handbook; Transport Canada, AC 700-047

The network planner publishes a timetable: which flights operate and when. Three more steps turn it into something an airline can fly. Fleet assignment decides which aircraft type flies each departure, aircraft routing which individual aircraft, and crew scheduling which crews, at the lowest cost the rules allow. Each step takes the previous one as fixed, and each can find that the timetable does not work: the type the planner had in mind cannot be in two places at once, an aircraft never sleeps at the base, a crew would rest five minutes too little in Boston. That is when the schedule comes back to the planner's desk with questions.

This chapter follows the timetable through those steps on two small Laurentian examples, a sixteen-flight bank through Montreal

and a day of 32 A220 flights; Chapter 13 takes the same A220 day into operations. The models belong to other teams and are described here in words. Appendix A.5 writes the fleet assignment model in symbols, and the full reference edition derives the routing and crew models in its Chapters 14 and 15.

12.1 From timetable to tails and crews

Airline schedule planning is usually split into steps solved in sequence: schedule design selects the flights, fleet assignment chooses an aircraft type for each, aircraft routing strings them into maintenance-feasible sequences for individual aircraft, and crew scheduling builds legal work schedules (Barnhart 2006, 2003). In Belobaba's description, timetables and aircraft rotations are defined two to six months before departure, and revisions continue until the day of operation (Belobaba 2006f). Table 12.1 lists the problems downstream of fleet assignment. The order follows what each step needs: routing needs the type of each flight, because aircraft of different types cannot replace one another, and crew scheduling needs both, because pilots are qualified by type and the routing decides when a crew can stay with its aircraft.

Table 12.1 The planning problems downstream of fleet assignment (adapted from Barnhart, 2003).

Problem	Decides	Horizon	Objective	Main constraints
Aircraft (maintenance) routing	which aircraft flies each flight	days to weeks	feasibility, through value	maintenance, turns, fleet size
Crew pairing	multi-day work sequences from a crew base	a season, by day of week	crew cost	duty, rest and pay rules
Rostering or bidlines	which person flies which pairings	a month	preferences, fairness	days off, monthly limits, training
Recovery	repairs to all of the above	hours	passenger and cost impact	everything above, in real time

The handbook puts the planner's risk in one phrase: fleet assignment is not tail assignment (Network Planning Handbook 2026).

A feasible fleet assignment guarantees that aircraft rotations exist, but not rotations that bring each aircraft to a maintenance base in time, nor cheap or even legal crew pairings (Barnhart 2006, 2003). A sequence that fits the aircraft turns may still create an expensive overnight, need another crew, or fail the duty and rest rules (Network Planning Handbook 2026). A proposal therefore faces three tests: does it fleet, does it route, does it crew. Chapter 13 adds a fourth: does it survive a bad day.

12.2 The time-space network

An aircraft is a unit of flow that moves through stations and time. Fleet assignment, routing and recovery models all draw it the same way.

> **DEFINITION 12.1 Time-space network**
>
> For each aircraft type, the **time-space network** has one time line per station. Each flight is an arc from its departure at one station to its **ready time** at the other: the arrival time plus the type's minimum turn time there. **Ground arcs** along a station's line hold aircraft that wait, and a **wrap-around arc** joins the last event of the day to the first, so aircraft on the ground at night start the next day. A vertical **count line** at any moment crosses ground and flight arcs; the aircraft on them are all the aircraft of that type in service.

Ending a flight arc at the ready time builds the turn into the picture: every path through the network is a sequence the type can fly (Barnhart 2003). Because turn times differ by type, so do the networks. Figure 12.1 draws Cheapo Airlines, a six-flight teaching example from the MIT course, for a Boeing 737-500 that turns in 25 minutes (Barnhart 2003). At airport B its two arrivals are ready at 10:40 and 10:55, before departures at 11:00 and 14:00, so either aircraft can take either departure. For the A320, which needs 35 minutes, the second arrival is ready only at 11:05, after the 11:00 has left. Five minutes of turn time change which aircraft can feed which flight.

If aircraft flow in balance at every station, the number crossing a vertical line is the same wherever it is drawn, so it is the fleet the plan uses. A spreadsheet computes it. At each station, starting from the count time, keep a running total of departures minus ready arrivals;

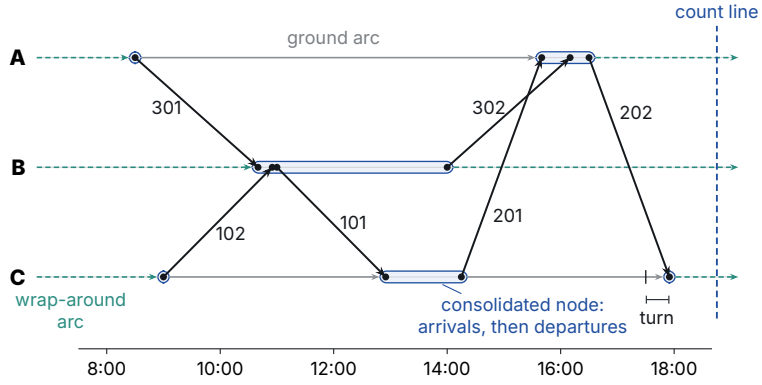


Figure 12.1 The time-space network of the Boeing 737-500 in Cheapo Airlines, the six-flight teaching example of MIT 1.206J (Barnhart, 2003). Flight arcs end at the ready time, arrival plus the 25-minute turn. Shaded boxes group arrivals followed by departures, where any arriving aircraft can take any departing flight. Wrap-around arcs carry the aircraft that stay overnight; the count line crosses them.

its largest value is the number of aircraft that must be on the ground there at the count time. A type needs the sum of these maxima over stations plus the aircraft airborne at the count time (Barnhart 2003). If that exceeds the fleet, no assignment can fix it, and the timetable goes back to schedule design (Chapter 11).

12.3 Fleet assignment

On one flight, choosing a type is the gauge choice of Chapter 7: a 137-seat A220-300 is cheaper to fly, a 169-seat 737-8 spills fewer passengers. On a schedule, every aircraft that lands must take off again, each type exists in a fixed number, and an aircraft on the morning Vancouver flight is not available for the Toronto peak. The best type for each flight in isolation is almost never a feasible plan.

DEFINITION 12.2 Fleet assignment

Given a daily timetable, the number of aircraft of each type, minimum turn times, the operating cost of each flight with each type, and demand and fare data, the **fleet assignment problem** chooses one type for every flight to maximize total contribution (revenue minus the costs that depend on the choice), while aircraft of each type flow in balance through every station and no type is used more than it exists (Barnhart 2006).

The decision is a type, not a tail: the 09:00 to Vancouver is a 737-8, not a particular 737-8 (Network Planning Handbook 2026). Only what changes with the type counts: block-hour and departure costs, and revenue lost to spill. Ownership costs are sunk, because the fleet is given.

Inputs: operating cost and spill cost by type. The operating cost OC_{ik} of flight i with type k is block hours times the type's hourly cost plus a departure cost (Chapter 3). The revenue side needs, for every flight and type, the revenue lost to spill. Deterministic demand understates it, because a flight whose mean demand fits still spills on its busy days (Chapter 7). With demand of mean μ_i and standard deviation σ_i, the **assignment cost** uses expected spill net of recapture:

$$c_{ik} = \mathrm{OC}_{ik} + (1 - b_i)\, r_i\, \sigma_i\, L\left(\frac{C_k - \mu_i}{\sigma_i}\right), \tag{12.1}$$

where C_k is the seats of type k, r_i the contribution per passenger (leg fare net of passenger costs), L the loss function of Chapter 7, and b_i the **recapture rate**, the share of spilled passengers who take another of the airline's flights. The contribution of the pair is $r_i\mu_i - c_{ik}$.

Two inputs are modelling choices rather than data. Recapture can be set from the quality of service index of the alternative against the competitors' share (Chapters 4 and 5): a same-day flight with a QSI of 0.18, against competitors holding 42 percent of the market, recaptures $0.18/(0.18 + 0.42) = 0.30$ of spilled passengers (Barnhart 2003). The leg fare is an allocation, because a connecting passenger pays one fare for two flights, and the MIT lecture shows that the allocation rule changes the assignment (Barnhart 2003).

WORKED EXAMPLE 12.1 Pricing the Vancouver departure

Laurentian's 09:00 YUL–YVR has mean demand $\mu = 172$ with $k = 0.30$, so $\sigma = 51.6$. Prorating the fares of its 48 connecting passengers by distance and averaging with the local fare of CAD 430 gives a leg fare of CAD 455.65, so $r = 435.65$ net of the CAD 20 passenger cost. Spill is assumed lost ($b = 0$).

A220-300. $z = (137 - 172)/51.6 = -0.678$ and $L(z) = 0.8265$, so expected spill is 42.6 passengers, costing CAD 18,579. The operating cost for 5.2 block hours is \$8{,}000 ×5.2 + 2{,}000 = \$ CAD 43,600. From an unconstrained contribution of \$435.65 ×172 = \$ CAD 74,932, the A220 keeps CAD 12,753.

737-8. $z = -0.058$, $L = 0.4287$, expected spill 22.1 passengers, spill cost CAD 9,637; operating cost \$9{,}000 ×5.2 + 2{,}500 = \$ CAD 49,300; contribution CAD 15,995. The 737-8 saves CAD 8,943 of spill for CAD 5,700 of extra operating cost: it is worth CAD 3,243 more here.

Recapture. On the 16:20 YYZ–YUL, 144 of 178 passengers are locals who could take the 10:50; at a 30 percent rate for them, $b = 0.30 \times 144/178 = 0.243$, and the A220's spill cost falls from CAD 8,520 to CAD 6,452.

A flight that looks right-sized at its mean (172 passengers for 169 seats) still spills 22 passengers a day on average, and each coefficient is only as good as the leg fare and recapture rate behind it.

Figure 12.2 shows the comparison for every flight of the bank of Section 12.4. The 737-8 earns its extra seats on the Vancouver round trip and, more modestly, on the Toronto peaks; elsewhere its extra operating cost exceeds the spill it saves.

Constraints in words. The model has three families of constraints (Barnhart 2003, 2006):

- **Cover**: every flight in the timetable is flown by exactly one type. Whether a flight should operate is a schedule design question. Combinations that are not allowed, such as a type without the range, are removed.
- **Aircraft balance**: at every point of every type's network, the aircraft that arrive equal the aircraft that leave. This is the constraint that rules out the best type for each flight in isolation.

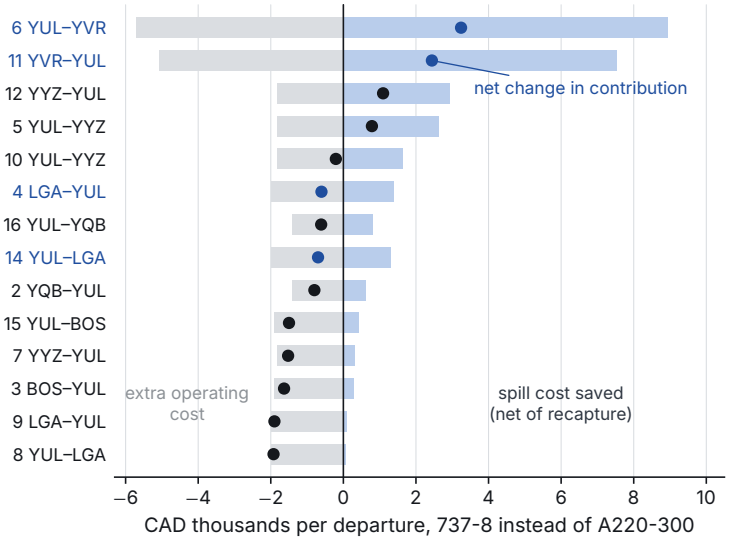


Figure 12.2 Where the larger aircraft pays. For each flight either narrowbody can fly, the spill cost (net of recapture) the 737-8 saves over the A220-300 and its extra operating cost; the dot is the net change in contribution. Blue labels mark the flights that the optimal assignment of Section 12.4 gives to the single 737-8.

- **Fleet count**: the aircraft of each type crossing the count line may not exceed the number the airline has.

Real models add side rules, such as one type for both legs of a through flight or enough overnights at maintenance stations (Barnhart 2003; Clarke et al. 1996). They interact: in the bank of Section 12.4, two one-stop through services requested by sales would together cost CAD 5,640 a day, although one alone costs CAD 1,831 and the other nothing.

Output. The output is a type for every flight, and with it the seats each flight sells, the operating cost of the day and the pilot group that flies each flight. Each type's flights can be strung into rotations, but the model does not choose them; routing does (Section 12.5). The model solves fast even for thousands of flights (Barnhart 2006; Hane et al. 1995), and can be re-solved as bookings come in: **demand driven dispatch**, Boeing's name for swapping aircraft of different sizes close to departure, is attractive only between types the same pilots can fly, because any other swap breaks a crew pairing (Boeing Commercial Airplane Group, n.d.; Berge and Hopperstad 1993).

12.4 A Laurentian bank solved

Take, as an illustration, 16 daily Laurentian flights through Montreal, with times chosen for this example rather than taken from the banks of Chapters 8 and 11. Narrowbodies serve Quebec City, Boston and New York LaGuardia, spending the night there, plus Toronto twice a day and Vancouver once; a 787-9 flies a daily Paris rotation, leaving at 19:00 and returning at 12:25, and may also fly the Toronto trunk during its afternoon at Montreal. The fleet is two A220-300s (137 seats, 35-minute turn), one 737-8 (169 seats, 40 minutes) and one 787-9 (298 seats, 75 minutes). Demand comes from 16 local and 8 connecting itineraries with $k = 0.30$; locals on the Toronto and New York flights are recaptured on the other same-day frequency at rate 0.30. All data are illustrative.

WORKED EXAMPLE 12.2 Sixteen flights, four aircraft

The planner's plan. The planner built four lines of flying, one per aircraft: a Quebec City line (two Toronto round trips between nights in Quebec City), a Boston line that includes the Vancouver round trip, a New York line and the Paris line. The best type for each whole line puts the 737-8 on the Boston line. This plan earns CAD 211,972 a day.

The model's plan earns CAD 213,803 (Figure 12.3). The 787-9 flies only Paris: the Toronto round trip in its afternoon would contribute CAD 4,783 with it against CAD 21,923 with A220s. The 737-8 sleeps in New York and flies LaGuardia–Montreal, the Vancouver round trip and Montreal–LaGuardia; the A220s fly the rest. The count line at 01:00 crosses four aircraft: two A220s at Quebec City and Boston, the 737-8 at LaGuardia and the 787-9 on its way to Paris.

Why the model does better. The Boston and New York aircraft are both on the ground at Montreal in the morning and in the evening, so they can exchange their work. The 737-8 still flies to Vancouver but sleeps in New York: on the Boston flights it would lose CAD 1,634 and CAD 1,498 against an A220, on the LaGuardia flights only CAD 603 and CAD 698, because New York demand (128 and 126 passengers) is closer to its size. The exchange gains CAD 1,831 a day, about CAD 668,000 a year, 0.9 percent.

The price of real aircraft. Giving every flight its best type regardless of balance earns CAD 216,985 but needs six narrow-

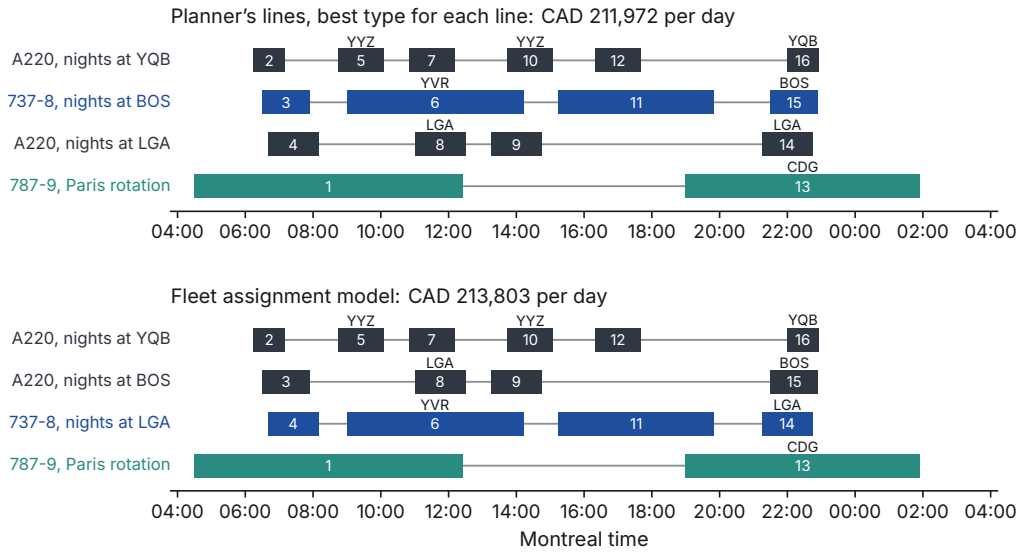


Figure 12.3 The planner's lines of flying with the best type for each line (top) and the rotations implied by the optimal fleet assignment (bottom). The model exchanges the daytime work of the Boston and New York aircraft at Montreal, so that the 737-8 spends its nights where its extra seats are worth most while still flying to Vancouver.

bodies instead of three: flying real aircraft costs CAD 3,182 a day. Re-solving with one more aircraft values each fleet count constraint: a third A220 would add CAD 1,301 a day and a second 737-8 CAD 887; a second 787-9 adds nothing.

Each choice in the optimum is easy to defend. What the planner missed is the combination, because it breaks lines of flying that were built by hand and look natural.

The value of one more aircraft is its value to this timetable, which was built for four; a fleet decision changes the schedule too (Chapters 9 and 10).

Passengers buy itineraries. The model values each flight as if its passengers flew only that flight. A connecting passenger needs a seat on every leg, which creates two errors no choice of leg fare removes (Barnhart 2003, 2006). With **network effects**, a passenger spilled from one leg also frees a seat on the other leg of the trip, so one leg's spill depends on the other's capacity. With **recapture**, a spilled passenger may take another of the airline's flights, but only if it has a seat. In general no leg-by-leg spill cost can reproduce the

true network spill of every assignment, as Barnhart's two-leg example shows (Appendix A.5) (Barnhart 2006).

DEFINITION 12.3 Passenger mix model

Given the seats on every flight, the mean demand and fare of every itinerary, and recapture rates between itineraries, the **passenger mix model** chooses how many passengers of each itinerary to carry, to redirect to another itinerary or to lose, so as to maximize revenue (Barnhart, Kniker, and Lohatepanont 2002).

Placed inside fleet assignment it gives the **itinerary-based fleet assignment model**, which computes spill and recapture consistently with the seats each type provides (Barnhart, Kniker, and Lohatepanont 2002). In the Cheapo example, the leg data credit the full connecting fare to each leg, so the leg-based model values spilled passengers at about USD 143 and puts the larger A320 on the four flights through the connecting airport. The itinerary model sees that the cheapest passengers to spill are connecting ones paying USD 130 for seats on two flights; refusing 32 of them in each direction relieves both legs at once, the smaller 737-500 flies everything, and daily cost falls 8.6 percent (historical teaching numbers) (Barnhart 2003).

On the Laurentian bank the itinerary model moves the 737-8's nights from New York to Quebec City. The cause is averages, not network effects: at their means the New York flights fit in an A220, yet they exceed 137 passengers on about 40 percent of days. Over 1,000 simulated days, with passengers allocated optimally each day (Barnhart 2003), the leg-based assignment beats the planner's lines by CAD 1,967 a day and the itinerary-based one by CAD 1,540. Barnhart's simulations of hub networks found the opposite ranking (historical) (Barnhart 2003); which effect dominates depends on the network.

PITFALL

Treating a deterministic network model's revenue as a forecast. Valued at mean demand, the passenger mix model credits Laurentian's optimal plan with CAD 261,296 a day; the simulated average is CAD 225,641, 14 percent less. Use determinis-

tic models to compare plans, and expected spill or simulation for the level.

IN PRACTICE Fleet assignment in 2026

At 30 June 2026 Air Canada's operating fleet counted 357 aircraft: 211 mainline (205 passenger aircraft of ten types and six 767 freighters), 51 at Air Canada Rouge and 95 operated by regional partners as Air Canada Express (Air Canada 2026d). Each subfleet with its own cabin, crew group or operator is a separate type in a fleet assignment model, and the set changes as 787-10s and A321XLRs arrive and 737 MAX 8s move to Rouge (Air Canada 2026d). Lufthansa Systems describes a module that reassigns types using origin-destination or revenue management forecasts, up to close to the day of operation (Lufthansa Systems 2025b, 2025d); which planning modules Air Canada uses is not public (Network Planning Handbook 2026). Open-source solvers solve models like Example 12.2 in milliseconds (Huangfu and Hall 2018); the hard part is the inputs, above all spill that reflects how revenue management sells the seats (Chapter 15).

12.5 Maintenance routing and lines of flying

Routing decides which aircraft, which tail, flies each flight. From here on the example is a day of Laurentian A220 flying: 32 flights between Montreal and five stations in its time zone, Quebec City (YQB), Ottawa (YOW), Toronto (YYZ), Boston (BOS) and New York LaGuardia (LGA). The schedulers built it as four **lines of flying**, one per aircraft (Table 12.2). Every night one aircraft sleeps at Quebec City, one at Boston and two at Montreal. The minimum turn is 35 minutes.

Aircraft must visit a maintenance base regularly: in the United States practice of 2003, an A check every 40 to 45 flying hours and no more than three or four calendar days apart (historical) (Barnhart 2003). Today's intervals come from each operator's approved maintenance program, and the handbook's advice is to ask maintenance planning whether a rotation reaches the right station in time, with a long enough ground window and hangar capacity that night (Network Planning Handbook 2026). An aircraft that sleeps at an outstation every night never gets its overnight check.

Fleet assignment cannot see this: it can require enough overnights at maintenance stations in aggregate, but one aircraft could be serviced every night and another never (Barnhart 2003). Routing follows individual aircraft. It chooses **strings**, sequences of flights from one maintenance check to the next, that cover each flight once and fit the fleet (Barnhart 2003; Barnhart et al. 1998).

Table 12.2 Laurentian's illustrative A220 day, as built (Montreal time). Each line starts and ends where its aircraft spends the night. Block times to or from Montreal: YQB 55 minutes, YOW 45, YYZ 1 h 15, BOS 1 h 25, LGA 1 h 30. The 32 flights total 37.5 block hours.

Line	Starts at	Flights: number, destination, departure	Ends at
A	YQB	101 YUL 06:00, 102 YYZ 07:45, 103 YUL 09:40, 104 LGA 11:40, 105 YUL 13:50, 106 YYZ 16:00, 107 YUL 17:55, 108 YQB 20:00	YQB
B	BOS	201 YUL 06:15, 202 BOS 08:20, 203 YUL 10:25, 204 YQB 12:30, 205 YUL 14:05, 206 LGA 15:45, 207 YUL 17:55, 208 YOW 20:10, 209 YUL 21:30	YUL
W	YUL	301 LGA 07:00, 302 YUL 09:10, 303 YYZ 11:20, 304 YUL 13:15, 305 YOW 15:05, 306 YUL 16:25, 307 BOS 17:45	BOS
X	YUL	401 YOW 07:30, 402 YUL 08:50, 403 YQB 10:15, 404 YUL 11:50, 405 BOS 13:30, 406 YUL 15:35, 407 YYZ 17:50, 408 YUL 19:45	YUL

Before any model, a count tells the planner whether an overnight pattern can work at all.

RESULT 12.1 Base overnights limit maintenance frequency

If a daily schedule leaves B aircraft overnight at the maintenance base, and each of the N aircraft must spend a night there at least once every k nights, a maintenance-feasible routing can exist only if $B \geq N/k$: the base supplies B aircraft-nights per night and the fleet needs N/k. The condition is necessary, not sufficient; the aircraft must also be able to swap through the base nights.

WORKED EXAMPLE 12.3 Routing four A220s through Montreal

Take two illustrative maintenance rules: every aircraft sleeps at Montreal at least every second night ($k = 2$), or every third night ($k = 3$).

The lines as built fail. Line A starts and ends at Quebec City; an aircraft that flies it every day never sees Montreal at night. Lines of flying are a scheduling convenience, not a routing.

The count passes. With $N = 4$ and $B = 2$, Result 12.1 allows $k = 2$, just: each night the two base nights must go to the two aircraft that slept away the night before.

The swap points decide. Only three places let two aircraft exchange their onward flights at Montreal: the morning bank (the 06:55 arrival from Quebec City can continue on 401 at 07:30 or on 102 at 07:45), the 17:00 bank and the 19:00 bank. Every other connection is forced, so the day can be routed in $2 \times 2 \times 2 = 8$ ways. One satisfies $k = 2$, four more only $k = 3$, and three, including the lines as built, leave some aircraft away from Montreal forever.

The model finds it. With $k = 2$ there are 11,817 legal strings of up to two days, and the routing model picks two (Figure 12.4). They differ from the lines as built only at the morning bank, where the aircraft from Quebec City turns onto the Ottawa round trip.

The overnight pattern, a network planning choice, sets the maintenance frequency the fleet can achieve; the swap opportunities at the banks decide whether the aircraft can rotate through it.

12.6 Crew pairings, pay and rostering

Barnhart's course presents crew cost as the second-largest operating expense after fuel, a 2003 ranking that moves with fuel prices (Barnhart 2003). Crews are scheduled in layers: flights into duties, duties into pairings, pairings into monthly lines for individual crew members (Figure 12.5).

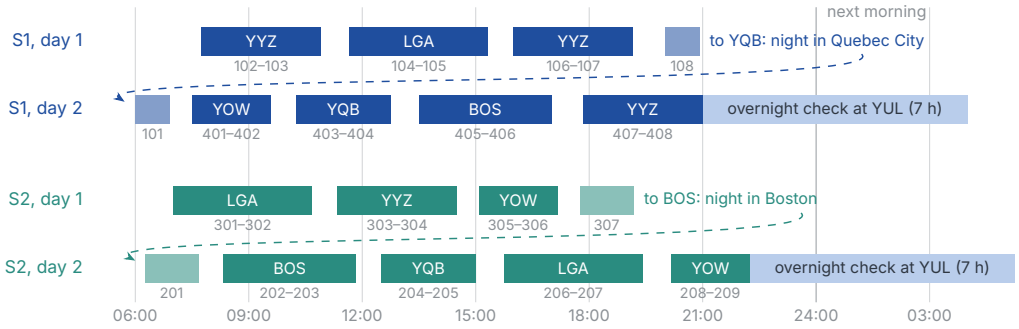


Figure 12.4 The two strings that route Laurentian's four A220s under an every-second-night maintenance rule. Each string covers two consecutive days and ends with an overnight check at Montreal; boxes are round trips from Montreal, labelled with the outstation, and the pale boxes are the one-way legs to and from an outstation night. String S1 needs the aircraft from Quebec City to take the 07:30 Ottawa turn (401) instead of the 07:45 Toronto flight it was built for.

> **DEFINITION 12.4 Duty period, pairing and time away from base**
>
> A **duty period** (duty) is a sequence of flights, roughly a working day, that one crew operates before it must rest. It runs from **report**, before the first departure, to **release**, after the last arrival; the gaps between flights are **sits**. A **pairing** is a sequence of duties separated by rests that starts and ends at the crew's base; a rest away from base is a **layover**. The **time away from base** (TAFB) runs from the pairing's first report to its last release (Barnhart 2003).

Legality rules come from regulation and collective agreements (Barnhart 2003). Duties cap flying and duty time and bound each sit; pairings bound rests and their number of days. Cockpit crews stay together and are qualified on one type or family; cabin crews are scheduled as individuals. Crews travelling as passengers to reposition are **deadheads**. Table 12.3 gives the simplified rules of the Laurentian example, adapted from the course's third problem set (Barnhart 2003); they are not Canada's regulations.

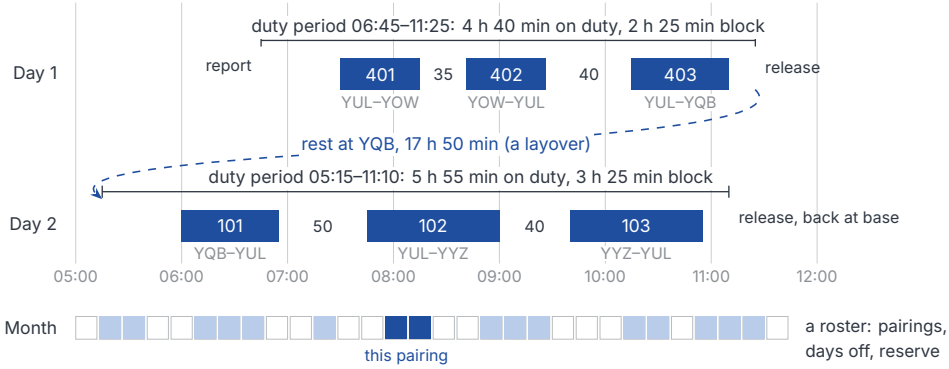


Figure 12.5 A two-day pairing from the Laurentian example. Day 1 is a 4 h 40 min duty with 2 h 25 min of block time ending in Quebec City; the crew rests 17 h 50 min there and flies a 5 h 55 min duty home the next morning. Numbers between flights are sit times in minutes. A roster strings pairings like this one, days off and reserve days into a month.

Table 12.3 Illustrative crew rules for the Laurentian A220 example.

Rule	Laurentian example
Report before first departure; release after last arrival	45 minutes; 15 minutes
Maximum duty, report to release	13 h up to four flights, 30 min less per extra flight
Maximum block time in a duty	8 h
Sit between flights	at least 30 min on the same aircraft, at least 50 min when changing aircraft, at most 90 min
Rest between duties (layover)	10 h to 18 h
Duties per pairing	at most 3

Why crew cost depends on how flights chain. Crews are not paid for block time alone. The lecture's pay function credits the largest of several measures at two levels (Barnhart 2003):

$$
\begin{aligned}
b_d &= \max\{\,\mathrm{BLK}_d,\; f_d\,\mathrm{DUT}_d,\; g_d\,\},\\
\gamma_p &= \max\Big\{\,\sum_{d\in p} b_d,\; f_p\,\mathrm{TAFB}_p,\; g_p\Big\},\\
c_p &= w\,\gamma_p + h\,n_p.
\end{aligned}
\tag{12.2}
$$

The **credit** b_d of a duty is the largest of its block time, a fraction f_d of its duty time and a guarantee g_d; a pairing's credit γ_p is the largest of its summed duty credits, a fraction f_p of its TAFB and a guarantee g_p. The cost adds w per credit hour and h per layover night n_p. Each term pays for a kind of unproductive time: the guarantee for short

days, the duty fraction for long sits, the TAFB fraction for long layovers. The ratio of credit to block time measures crew efficiency; 1.00 means crews are paid only for flying. For Laurentian, illustratively, $w =$ CAD 400 for two pilots, $h =$ CAD 300, $f_d = 1/2$, $f_p = 2/7$, $g_d = 4$ h and no pairing guarantee.

WORKED EXAMPLE 12.4 Pricing one pairing

Price the pairing of Figure 12.5. Day 1 flies 401 YUL–YOW at 07:30, 402 back and 403 to Quebec City, arriving 11:10. Day 2 flies 101 YQB–YUL at 06:00, 102 to Toronto and 103 back, arriving 10:55.

Duty 1. Report 06:45, release 11:25: 4.67 h on duty, 2.42 h of block. Credit $\max\{2.42,\ 0.5\times4.67,\ 4.00\} = 4.00$ h: the guarantee binds.

Duty 2. After 17 h 50 min of rest, report 05:15, release 11:10: 5.92 h on duty, 3.42 h of block. Credit $\max\{3.42,\ 2.96,\ 4.00\} = 4.00$ h.

Pairing. TAFB is 28 h 25 min, so the trip term is $\frac{2}{7}\times28.42 = 8.12$ h, above the 8.00 h of duty credit. The crew is paid 8.12 h for 5 h 50 min of flying, a ratio of 1.39, and the pairing costs CAD $400 \times 8.119 + 300 = 3{,}548$.

The crew sits in a Quebec City hotel from late morning to dawn because the 06:00 departure must be flown by someone already there and rested. A schedule that puts a midday arrival and an early departure at the same outstation buys that layover every day.

From pairings to a crew plan. Crew planning chooses the cheapest set of legal pairings that covers every flight exactly once (Barnhart 2003). The difficulty is their number: a typical problem has hundreds of millions of legal pairings (Barnhart 2003). The Laurentian day's 32 flights form 652, and 44,778 if the 90-minute cap on sits is dropped: rules protect crews and also keep the problem finite. Optimizers generate promising pairings as they go rather than listing them (the full reference edition, Chapter 15).

The Laurentian optimum uses five pairings and costs CAD 16,533 a day: 39.08 credit hours for 37.50 block hours, a ratio of 1.042, plus three hotel nights. Read it as a planner. The 06:00 from Quebec City and the 06:15 from Boston must be flown by crews who slept there, which forces layovers. The evening arrival at Quebec City (108, in

at 20:55) cannot feed the 06:00, because its crew would rest only 8 h 05 min. The Boston night works only because 307 arrives at 19:10, leaving 10 h 05 min of rest before 201: five minutes to spare. The crew-expensive flights are the evening departures to outstations and the edges of the day.

Routing and crews meet at short connects. A **short connect** is a connection legal for a crew only if the same aircraft flies both flights, because a crew changing aircraft needs time to cross the terminal (Barnhart 2003); in the Laurentian rules, sits of 30 to 49 minutes. Each routing provides different short connects, so each gives a different crew cost. Under the every-third-night rule the five feasible routings cost CAD 16,181 to CAD 16,533 a day in crews: choosing the routing with crews in mind is worth up to CAD 352 a day. The cheapest crew plan of all, CAD 16,014, needs a routing that strands aircraft away from Montreal. Integrated models offer the crew problem a menu of routings for this reason (Cohn and Barnhart 2003).

Fatigue rules in Canada. Flight crew hours of work and rest in Canada are governed by Division III of Part VII of the Canadian Aviation Regulations, published in 2018 and amended in 2022, and explained in Transport Canada's AC 700-047 (Transport Canada 2024). The framework sets flight duty limits that depend on the duty's start time and number of flights, so an early start or many short sectors shorten the allowed duty; cumulative flight time limits, for example 112 hours in any 28 days and 1,000 in any 365; minimum rests, such as 10 hours in suitable accommodation after a duty ending away from base; time free from duty; and limited extensions on the day, with a schedule that needs them on more than 10 percent of occasions in 90 days to be re-planned (Transport Canada 2024). Legality depends on the clock time and sector count of each duty, which is why the handbook tells planners to obtain a crew feasibility review rather than apply a remembered limit (Network Planning Handbook 2026).

Rostering. Pairings are built for anonymous crews. Rostering assigns them to people for a month, with days off, reserve days, vacations and training (Barnhart 2003). In the **bidline** approach, generic monthly lines are posted and crew members bid for them by seniority; in **rostering**, personalized lines are built directly. For the planner, rostering matters through its limits: cumulative flight time caps how much flying crews can absorb in a peak month, and reserve crews are the buffer for disruptions (Chapter 13).

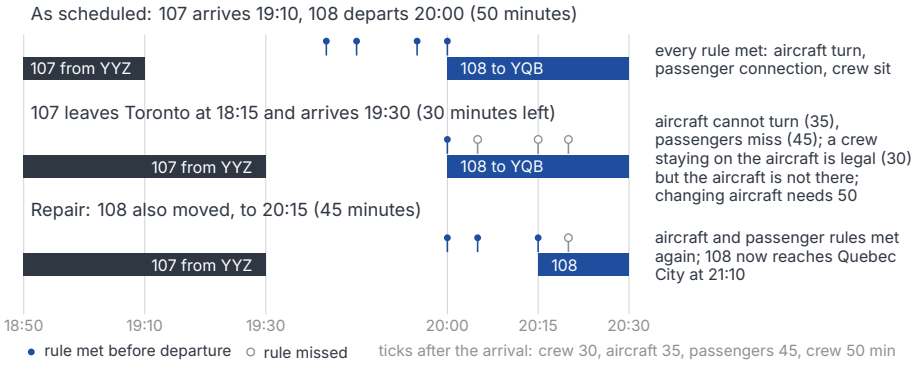


Figure 12.6 What a twenty-minute change does at Montreal's 19:00 bank. Ticks mark the earliest time each rule allows the onward departure after flight 107 arrives: a crew staying with its aircraft (30 minutes), the aircraft turn (35), a passenger connection (45) and a crew changing aircraft (50). Moving 107 breaks three of the four; moving 108 fifteen minutes later restores the aircraft turn and the passenger connection, so the crew can again stay with its aircraft.

IN PRACTICE **Crew and maintenance as commercial constraints in 2026**

The handbook treats crew and maintenance as commercial constraints: a small schedule change can create a discrete extra crew or hotel requirement (Network Planning Handbook 2026). Air Canada's second-quarter 2026 MD&A reports "crew cycle" as a separate line of other operating expenses, CAD 98 million in the quarter against CAD 86 million a year earlier, and records charges related to collective agreements, including the pilots' agreement with ALPA ratified in 2024 (Air Canada 2026d). A network analyst should ask crew planning three questions about any proposal: which pairings and layovers it changes, which rule binds, and what it does to credit per block hour.

12.7 What breaks when a planner moves a flight

The handbook warns that a 20-minute departure change may alter a connection, a duty period and a maintenance window at once (Network Planning Handbook 2026). The Laurentian day shows each failure.

WORKED EXAMPLE 12.5 Twenty minutes later from Toronto

The commercial team asks to move 107, the evening Toronto–Montreal flight, from 17:55 to 18:15 for Toronto business travellers. It now arrives at 19:30 instead of 19:10. Check the four layers (Figure 12.6).

Passengers. At an illustrative minimum connecting time of 45 minutes, the connection onto 108 to Quebec City at 20:00 shrinks from 50 to 30 minutes and is lost; so is the connection onto 208 to Ottawa, from 60 to 40 minutes.

Aircraft. The aircraft from 107 can no longer turn onto 108 (30 minutes against 35), so 107's aircraft must take 208 and 207's must take 108.

Maintenance. The only routing that met the every-second-night rule used the 107 to 108 turn. Now no routing does: the aircraft that sleeps in Boston would sleep in Quebec City the next night.

Crews. A crew flew 106, 107 and 108 on one aircraft. With 30 minutes between 107 and 108 it could stay on board, but the aircraft no longer continues, and changing aircraft needs 50 minutes. The duty breaks.

Repair. Moving 108 to 20:15 restores the aircraft turn (45 minutes), the Quebec City connection (exactly 45; the Ottawa one stays lost) and the every-second-night routing. The crew plan and its cost, CAD 16,533 a day, come back unchanged. The price is commercial: 108 now reaches Quebec City at 21:10. It also opens a LaGuardia–Quebec City connection that Chapter 13 examines.

A change that cannot be repaired so easily. Moving 307, the evening Boston departure, 20 minutes later breaks nothing for aircraft or passengers. But the crew arriving in Boston would rest only 9 h 45 min before the 06:15 departure, below the 10-hour minimum, and no legal crew plan exists under any of the eight routings. The fix is another retiming, an extra crew positioned to Boston, or an extra hotel night.

The example generalizes into a checklist for any schedule change. List every connection the flight is part of, for passengers, aircraft and crews, against the minimum for each; trace the aircraft rotation to its next maintenance night; ask crew planning which duties, rests and

layovers the change touches; and look for the second change that repairs the first. The planner is not expected to solve these problems, but to recognize them early and to price them with the teams that do (Network Planning Handbook 2026).

> **PITFALL**
>
> Treating legality as a property of a flight. Whether a flight can be crewed, routed or connected depends on its neighbours: the turn before it, the rest after it, the bank around it. A change that is legal for the flight in isolation can make the whole day infeasible, as the 20-minute move of 307 does.

12.8 Summary

- After the timetable come fleet assignment, aircraft routing, crew pairing and rostering, each with the earlier steps fixed. A timetable must fleet, route and crew; passing one test does not guarantee the next.
- The time-space network draws aircraft as flows; the aircraft crossing any count line are the fleet a plan needs.
- Fleet assignment chooses a type for every flight to maximize contribution, using operating cost and expected spill net of recapture, subject to cover, aircraft balance and fleet counts. On the Laurentian bank it beats the planner's best lines by CAD 1,831 a day by letting two aircraft exchange work at the hub.
- Leg-based spill misprices connecting passengers; the passenger mix model values them by itinerary. Deterministic models ignore demand variability, so compare plans by simulation.
- Routing follows tails: base overnights must satisfy $B \geq N/k$, and the swap points at banks decide whether aircraft can rotate through the base.
- Crew pay is the largest of block time, a share of duty time or time away, and guarantees, so how flights chain (early outstation departures, late arrivals, short connects) drives crew cost.
- A 20-minute change can break a passenger connection, an aircraft turn, a maintenance night and a crew rest at once; check all four and look for the repairing second change.

12.9 Further reading

Barnhart's guest lecture (Barnhart 2006) is the shortest complete account of fleet assignment; her schedule planning course (Barnhart 2003) adds the passenger mix model, maintenance routing, crew pairing and their integration. Barnhart, Kniker, and Lohatepanont (2002) introduce the itinerary-based model, and the Boeing brochure (Boeing Commercial Airplane Group, n.d.) explains demand driven dispatch. Read Transport Canada's AC 700-047 (Transport Canada 2024) with the crew planning team. The full reference edition derives fleet assignment in its Chapter 14 and routing and crew pairing in its Chapter 15.

CHAPTER

13

Robustness and Operations Recovery

IN THIS CHAPTER

- Where delays come from, and how one late arrival travels through aircraft, crews and passengers.
- Slack and propagated delay: why delay builds up through the day like a queue, and what a minute of slack is worth.
- Buffers against utilization: punctuality bought with aircraft time.
- Robust routing and retiming, which move existing slack to where delay arrives.
- Operations control on the day: delays, swaps and cancellations.
- Passenger delay, Canada's passenger protection rules, and disruption cost in the business case.

DRAWS ON Barnhart, Airline operations lectures 1 to 3 and Barnhart and Clarke, New approaches to add robustness into airline schedules (MIT 1.206J, 2003); Belobaba, Airline operations control (MIT 16.75J, 2006); EUROCONTROL, Arriving on time (2020); Lufthansa Systems press release on Air Canada (2025); handbook; Canadian Transportation Agency

Every plan in the previous chapters assumes that the airline will fly it as built, and the plans were built to be efficient. Chapter 12 packed a day of Laurentian flying into four aircraft and five crew pairings with tight turns and short connects; each saving removed idle time that could have absorbed a shock. Then the day happens: a thunderstorm at LaGuardia, a de-icing queue in Montreal, a hydraulic leak. Optimizers squeeze slack out of aircraft and crew plans, so an optimized plan is easier to disrupt (Barnhart 2003).

Before the day, the planner decides how much slack the schedule carries and where: block times, turn times, connection times, which aircraft follows which flight. On the day, operations control repairs what went wrong with the options the schedule left it. The hand-

book asks the network analyst to put reliability into the business case (Network Planning Handbook 2026). This chapter simulates the Laurentian A220 day of Chapter 12, with its every-second-night routing and its CAD 16,533 crew plan, over thousands of days; Appendix A.6 derives the formulas stated here.

13.1 Where delays come from

A flight needs an aircraft, a legal crew, and often a slot, a gate and an open runway; anything that takes one away creates a disruption. The causes are shortages of airline resources (a mechanical problem, a crew that is missing or would break a rule) and of airport and airspace resources (weather that cuts runway capacity, a security event) (Barnhart 2003; Belobaba 2006j). Banked hubs are especially exposed, because airlines schedule more movements into a bank than the airport can handle in poor visibility (Barnhart 2003).

Delays then travel. A late arrival delays the next departure of the same aircraft, but its cockpit crew and its cabin crew may go to different next flights, so one late arrival can make three later flights late through three different resources (Barnhart 2003). Delays also accumulate through the day: in US data for August 2000, flights arriving in the morning averaged 5 to 7 minutes late and those arriving after 19:00 21 to 26 minutes (historical) (Barnhart 2003). The same lectures put the cost of disruptions at 2 to 3 percent of annual revenue, an estimate from around 2000 (Barnhart 2003).

> **DEFINITION 13.1 Robustness**
>
> A schedule is **robust** in one or more of four senses: it has the lowest expected cost once disruptions are counted; it suffers the fewest delays and disruptions of aircraft, crews and passengers; it is easy to recover; and it contains disruptions so that their effect downstream stays small (Barnhart and Clarke 2003).

The senses conflict: spare aircraft make recovery easy and cost money. This chapter measures the second sense and prices it in the first.

In the Laurentian simulation, each departure is delayed with a probability of 15 to 45 percent depending on its station and time of day, by a lognormal amount with a median of 10 minutes, and with probability 0.008 suffers a disruption of 60 to 180 minutes. All values

are illustrative; together they put about 78 percent of arrivals within 15 minutes of schedule, close to the 77.7 percent of the US routings in the robustness lecture (historical) (Barnhart and Clarke 2003).

13.2 Slack and propagated delay

Propagation happens at the turn. A flight cannot leave before its aircraft has arrived and been turned, and the minimum turn time (MTT) of the Laurentian A220 is 35 minutes. Planned ground time beyond that is slack, and an inbound delay larger than the slack passes to the next departure.

DEFINITION 13.2 Slack, propagated delay and independent delay

Let flight j follow flight i on the same aircraft. The **slack** s_{ij} is the planned departure of j minus the planned arrival of i, minus the MTT. If i arrives TAD_i minutes late, the **propagated delay** PD_{ij} is the part of j's delay caused by the late aircraft. The **independent** departure and arrival delays IDD_j and IAD_j are what j would have suffered with an on-time aircraft: late passengers, an air traffic control hold, a headwind (Lan, Clarke, and Barnhart 2006; Barnhart and Clarke 2003).

$$\begin{aligned} \mathrm{PD}_{ij} &= \max(\mathrm{TAD}_i - s_{ij},\ 0), \\ \mathrm{TDD}_j &= \mathrm{PD}_{ij} + \mathrm{IDD}_j, \qquad \mathrm{TAD}_j = \mathrm{PD}_{ij} + \mathrm{IAD}_j, \end{aligned} \tag{13.1}$$

where TDD_j and TAD_j are j's total departure and arrival delays (Figure 13.1). The same construction applies to crews, with the minimum sit in place of the MTT, and to passengers, with the minimum connecting time. Independent delay does not depend on the routing, so it can be measured on the rotations actually flown and reused to rebuild the delays of any other routing (Barnhart and Clarke 2003).

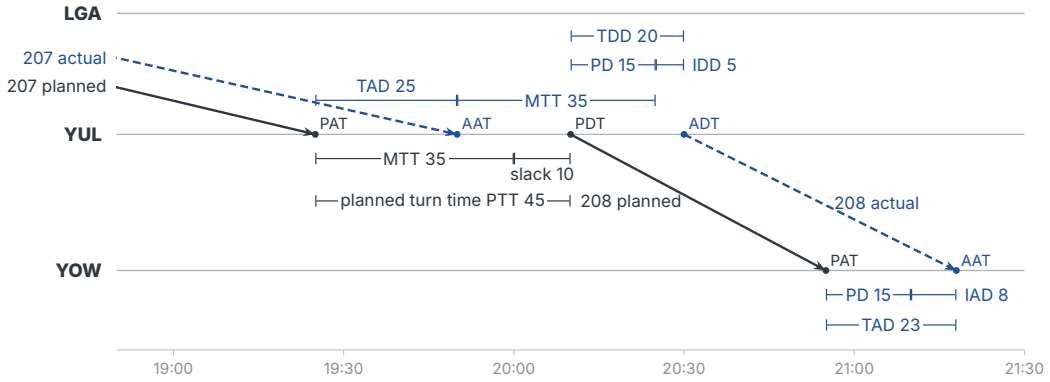


Figure 13.1 Slack and propagated delay at Montreal for the LaGuardia inbound 207 and the Ottawa departure 208, with illustrative times. The planned turn of 45 minutes is the minimum turn time of 35 plus 10 minutes of slack. Flight 207 arrives 25 minutes late, the aircraft is ready 35 minutes later at 20:25, and so 208 inherits 15 minutes of propagated delay; 5 minutes of independent delay make it leave at 20:30, and it arrives 23 minutes late, of which 8 are its own.

RESULT 13.1 Propagation along a rotation is a queue

Let W_k be the propagated delay into the k-th flight of an aircraft's day and s_k the slack after that flight. Then

$$W_{k+1} = \max(W_k + \mathrm{IAD}_k - s_k,\ 0),$$

Lindley's recursion for the wait in a single-server queue (Lindley 1952). The delay inherited by a late-evening flight is the largest excess of independent delay over slack on any run of consecutive earlier flights: the worst stretch of the day (Appendix A.6).

The aircraft is the server, each flight a customer, and propagated delay the time a customer waits because the one before it ran long. A morning problem that the day's slack never absorbs is still there at night, which is why delays grow through the day.

WORKED EXAMPLE 13.1 A late start on line B

Line B of the Laurentian day starts in Boston with 201 to Montreal at 06:15 and flies out and back from Montreal until 209 lands there at 22:15. Its eight turns have slacks of 5, 5, 5, 5, 10, 5, 10 and 0 minutes. Suppose 201 arrives 60 minutes late and nothing else goes wrong.

As planned. The propagated delays into 202 to 209 are 55, 50, 45, 40, 30, 25, 15 and 15 minutes: every later flight leaves late, 275 minutes in total.

With 10 more minutes on every turn. The delays become 45, 30, 15 and then zero: three late flights and 90 minutes.

Eighty extra minutes of ground time removed 185 minutes of propagated delay on this day, and would have removed nothing on a day when 201 was on time. Slack is insurance, valued over the days that go wrong.

13.3 Buffers versus utilization

Take a single turn with inbound delay X and slack s, so that propagated delay is $\max(X - s, 0)$. The first minute of slack saves on average about $\Pr(X > 0)$ minutes of it, the probability that the inbound is late at all; the thirtieth saves only the probability that it is more than 30 minutes late. Slack has diminishing returns, which gives a rule.

RESULT 13.2 How much slack to plan

If a minute of slack costs c_s and a minute of propagated delay costs $c_d > c_s$, add slack until the probability that the inbound delay exceeds it equals the cost ratio: $\Pr(X > s^*) = c_s/c_d$ (Appendix A.6).

This is a newsvendor rule, of the same kind as Littlewood's rule for protection levels (Chapter 14). Suppose Laurentian's LaGuardia inbounds arrive 6 minutes late on average (early arrivals counting as negative delay) and 47 percent of them late at all, with a long right tail (a shifted lognormal, which fitted best in the robustness work (Barnhart and Clarke 2003); parameters illustrative). Expected propagated delay falls from 9.6 minutes with no slack to 6.1 with 10 and 2.9 with 30. At an illustrative CAD 20 per minute of slack and

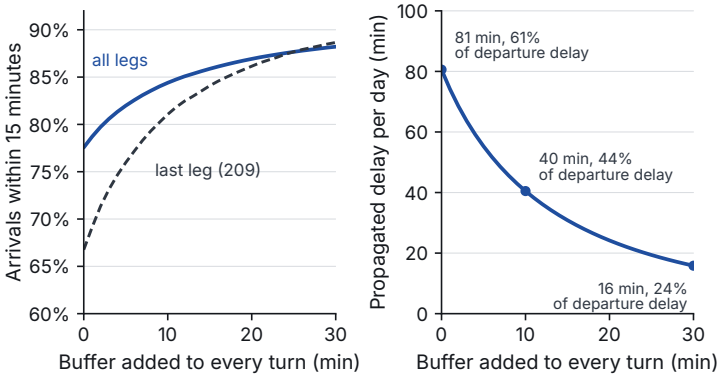


Figure 13.2 Buffer study on line B (nine A220 legs, 40,000 simulated days). Left: share of arrivals within 15 minutes of schedule, for all legs and for the last leg of the day. Right: propagated delay per day and its share of all departure delay. The first minutes of buffer buy the most.

CAD 80 per minute of delay, the critical ratio is 0.25 and the best slack is 10.7 minutes, at an expected cost of CAD 689 a day against CAD 771 with none. The cost curve is flat near its minimum, so precision about the slack matters little; the delay cost matters a great deal (at CAD 50 a minute the best slack is 2.6 minutes, at CAD 120 it is 18.4), and it depends on what follows the turn. Appendix A.6 gives the closed form for the lognormal case; the one printed in the robustness lecture is wrong (Barnhart and Clarke 2003).

Slack on one turn also shields every later flight. Figure 13.2 simulates line B over 40,000 days with the same buffer added to each of its eight turns. As built, 77.6 percent of its arrivals are within 15 minutes, but only 66.8 percent of its last flight's, and propagated delay is 61 percent of all its departure delay, against 20 to 30 percent reported for Continental Airlines at the time (Barnhart and Clarke 2003): line B's turns are all within 10 minutes of the minimum. Ten minutes of buffer per turn lift on-time arrivals to 84.4 percent and halve propagated delay; twenty more add only 3.8 points.

The buffer is not free: ten minutes on eight turns is 80 minutes of aircraft time a day, which would move line B's last arrival from 22:15 to 23:35 or cost a flight. The right panel of Figure 13.3 shows the trade for an A220 shuttling between Montreal and Toronto inside a 06:00 to 23:00 day. With up to 3 minutes of buffer per turn it fits nine legs (74.4 to 78.1 percent on time); from 4 to 20 minutes, eight. Within a number of legs buffer is free, so an eight-leg day should use all 20 minutes (85.9 percent). Between leg counts it is expensive: dropping from nine legs to eight gives up 11 percent of the aircraft's flying for

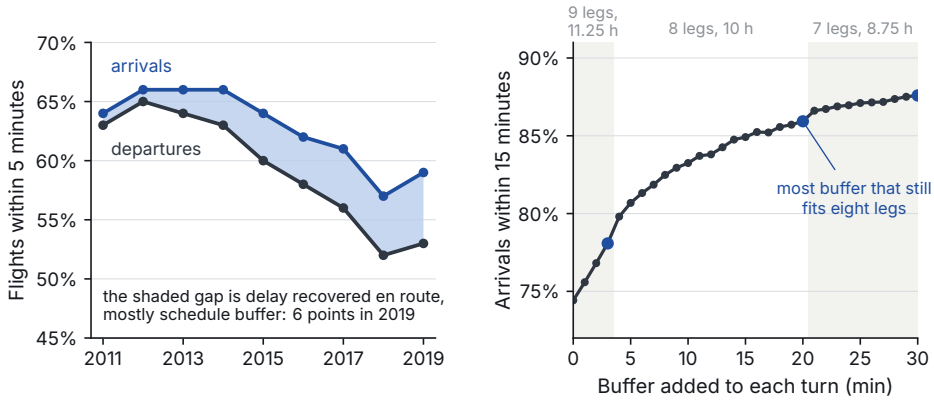


Figure 13.3 Buffers buy arrival punctuality and cost utilization. Left: share of flights within 5 minutes of schedule in the EUROCONTROL network, 2011 to 2019, redrawn from EUROCONTROL (2020); the shaded gap between arrivals and departures is delay recovered en route. Right: an illustrative Laurentian A220 shuttling between Montreal and Toronto (75-minute blocks, 40-minute base turns) inside a 06:00 to 23:00 window; legs and block hours fall in steps as the buffer per turn grows.

7.8 points of on-time performance. Whether the ninth leg is worth the reliability it destroys is a route evaluation question (Chapter 10).

EUROCONTROL's 2020 think paper shows the same trade across Europe (left panel). As air traffic flow management delays grew after 2013, airlines added schedule buffer: flights departing within 5 minutes of schedule fell from 63 percent in 2011 to 53 percent in 2019, those arriving within 5 minutes only from 64 to 59 percent, at the price of resources planned but not used and lower utilization (EUROCONTROL 2020). These pre-2020 figures use a 5-minute threshold and are not comparable with North American 15-minute statistics; the handbook treats EUROCONTROL's delay costs as context, not Canadian parameters (Network Planning Handbook 2026).

13.4 Robust routing and retiming

Buffers buy reliability with aircraft time. Lan, Clarke and Barnhart showed two ways to get it from the slack a schedule already has (Lan, Clarke, and Barnhart 2006; Network Planning Handbook 2026). **Robust routing** uses the fact that routings satisfying the maintenance rules hardly differ in direct cost: among them, it chooses the one that puts long turns behind delay-prone inbounds, scoring each aircraft sequence by its expected propagated delay (Barnhart and Clarke 2003). **Retiming** moves departures by 5 to 15 minutes after fleeting and routing are fixed, shifting slack from where it is idle to

where passengers need it. A connection of c_{ij} passengers disrupts on average c_{ij} times the probability that the actual connection time falls below the minimum connecting time (MCT), and the model picks one departure time per flight to minimize the total without breaking any aircraft turn (Lan, Clarke, and Barnhart 2006; Barnhart and Clarke 2003).

On four US sub-networks of 20 to 278 flights, built on July 2000 data and tested on August data (historical), re-routing alone cut propagated delay by 44 to 47 percent (37 to 38 percent by another, inconsistent column of the source) and disrupted passengers by 10.8 percent, at almost no cost. With a 30-minute MCT, retiming within 15 minutes cut expected disrupted passengers by 40 percent, and within 5 minutes by 20 percent (Barnhart and Clarke 2003).

WORKED EXAMPLE 13.2 Which Laurentian routing is robust?

Chapter 12 found eight ways to route the four A220s through the day's three swap points. Here all eight are simulated over the same 20,000 days (Table 13.1), with illustrative passengers: 80 locals per flight, 12 on each plausible connection at Montreal, and 79 a day connecting to two assumed evening departures to Seville and Paris. One MCT of 45 minutes applies everywhere, a simplification, since transborder arrivals clear customs at Montreal and need longer (Chapter 11).

Propagated delay barely moves. It ranges from 269 to 282 minutes a day, because every routing inherits the same thin turns.

Passengers rank the routings differently. Keeping 107 and 108 on one aircraft at the 19:00 bank protects the Toronto to Quebec City connection: each routing that does so disrupts 6.5 fewer passengers a day than the same routing with the 19:00 swap.

Crew cost pulls the other way. Under the every-third-night rule, the routing with the cheapest crews saves CAD 352 a day but disrupts 4.3 more passengers a day than the every-second-night routing, so the saving pays only if a disrupted passenger costs Laurentian less than CAD 82. With an average delay of 17 hours per disrupted passenger, it almost certainly costs more.

A routing chosen for one objective is rarely the best for the others.

Table 13.1 The eight Laurentian routings of Chapter 12 simulated over the same 20,000 days. Arrivals within 15 minutes range only from 77.5% to 78.1%.

Routing (morning, 17:00, 19:00 bank)	Maintenance rule met	Crew cost (CAD per day)	Propagated delay (min per day)	Disrupted passengers per day
as built: 101→102, 406→407, 107→108	none	16,633	270.1	22.8
as built, 107→208 at 19:00	every third night	16,467	268.9	29.2
406→307 at 17:00	none	16,400	275.0	27.0
406→307 and 107→208	every third night	16,233	273.9	33.5
101→401 in the morning	every second night	16,533	278.3	24.5
101→401 and 107→208	every third night	16,367	276.7	31.0
101→401 and 406→307	every third night	16,181	282.3	28.8
101→401, 406→307 and 107→208	none	16,014	280.7	35.3

13.5 Operations control and recovery

On the day, the plan belongs to operations control. Belobaba describes three levels: system operations control, which manages the day in real time and reassigns aircraft and crews; dispatch, which plans each flight's route and fuel; and station control, which runs the turnarounds (Belobaba 2006j). The aim is to fly as close to plan as possible; in the recovery considerations Belobaba lists from a US carrier, customers inconvenienced come first and route profitability is only a tie-breaker (Belobaba 2006j).

The controllers' levers are few (Barnhart 2003). They can delay a flight until its aircraft and crew are ready; swap aircraft between flights, turning one long delay into several short ones; cancel, usually a round trip from the hub to a spoke, which leaves every aircraft where it should be; and call reserve or deadhead crews. In the practice the lectures describe, decisions were taken in sequence: aircraft, then crews, then passengers. Aircraft recovery models use the time-space network of Chapter 12, with copies of each flight at candi-

date times and fixed costs per minute of delay and per cancellation (Barnhart 2003). Those costs are their weakness, because the true cost of a delay depends on the passengers it strands and the crews it breaks. Passenger-centred models, which minimize disrupted passengers, reduced them by 10 to 20 percent on bad days in simulation (historical) (Bratu and Barnhart 2006; Barnhart 2003).

WORKED EXAMPLE 13.3 An aircraft goes technical at noon

Tail 1 lands in Montreal from Toronto at 10:55 and needs a two-hour repair. It is due to fly 104 to LaGuardia at 11:40, then 105, 106, 107 and 108 to Quebec City. Tail 2, flying line B, lands from Boston at 11:50 and is due out on 204 to Quebec City at 12:30. Figure 13.4 and Table 13.2 compare four options.

(**a**) **Wait for the repair.** 104 leaves 75 minutes late and the delay rolls through tail 1's afternoon. 107 lands at 20:10, too late for the 20:30 to Seville (10 passengers) and the 20:10 to Ottawa (12 passengers, stranded until morning).

(**b**) **Swap at noon.** Tail 2 takes 104 at 12:25 and the rest of tail 1's day; repaired tail 1 takes 204 at 12:55 and line B. Nine flights are late instead of five, but with 225 delay minutes instead of 315, and every transatlantic connection holds. The crew plan survives; maintenance does not, because tail 2 slept in Boston and now ends the day in Quebec City, a second night away from Montreal that the every-second-night rule forbids.

(**c**) **Swap, then swap back at the 19:00 bank.** This sends tail 2 home for the night, but leaves two crews too little time to change aircraft; crew control re-pairs them so that each stays with its aircraft, and both new duties are legal.

(**d**) **Cancel the LaGuardia round trip.** Tail 1 is back on schedule and no flight leaves late, but 208 passengers lose their flights; the later LaGuardia flights have seats for 48 of them, and 160 wait until the next day.

A model that prices only delay minutes and cancellations could choose (d). Passengers rank the options (c), (b), (a), (d), and (d) costs them more than five times the passenger-hours of (b). The menu was written months earlier: the noon swap exists because the schedule brings two A220s to Montreal within an hour of each other.

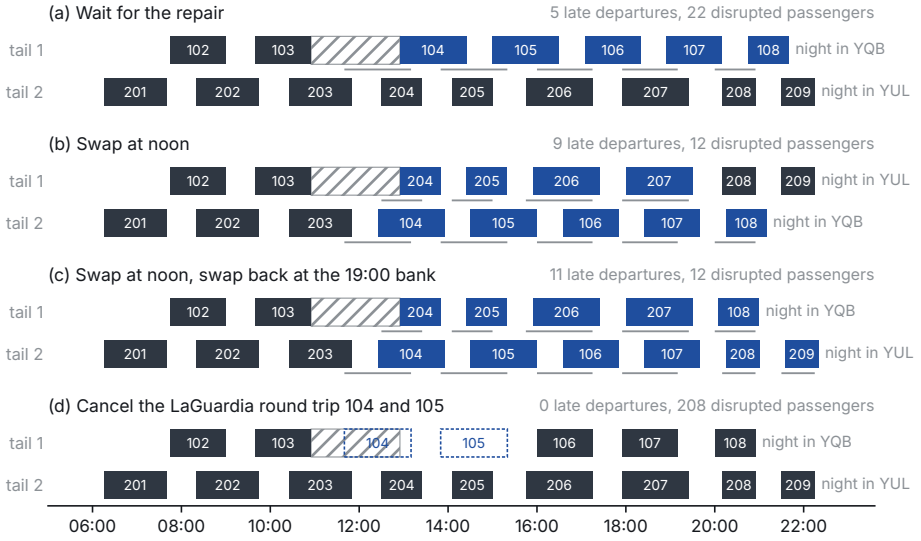


Figure 13.4 Four ways to recover from tail 1's two-hour repair at Montreal (hatched), drawn for the two tails involved. Bars are flights as operated, late ones in the accent colour; grey lines under late flights mark their planned times; dashed outlines are cancelled flights. The note beside each panel title counts late departures and disrupted passengers.

Table 13.2 Recovery options for the noon repair. Departure delay is summed over flights; passengers stranded overnight are counted as 18 hours late and missed transatlantic connections as 24 hours. Rebooking uses Laurentian's own flights only.

Outcome	(a) Wait	(b) Swap	(c) Swap and back	(d) Cancel
Late departures	5	9	11	0
Departure delay (min)	315	225	225	0
Disrupted passengers	22	12	12	208
Stranded overnight	22	12	12	160
Passenger delay (hours)	946	571	566	3,076
Crew plan	as planned	as planned	two duties re-paired	as planned
Maintenance plan	as planned	broken for tail 2	as planned	as planned

IN PRACTICE Operations control at Air Canada in 2026

Air Canada's operations control runs on Lufthansa Systems' NetLine/Ops ++, which the vendor announced as live at the airline in January 2025; the release describes the benefits in qualitative terms only and says nothing about which recovery models the system runs (Lufthansa Systems 2025a). Disruption shows in the airline's reporting: in the second quarter of 2026 operated capacity grew 0.3 percent year over year, below guidance of 0.5 to 1 percent, mainly because weather disruptions reduced completion (Air Canada 2026d). Every swap opportunity and two-flight cancellation in a published schedule is a recovery option the controllers may need.

13.6 Measuring what passengers experience

Punctuality is usually stated per flight, for example as the share of flights arriving within 15 minutes of schedule (A15) (Network Planning Handbook 2026). Passengers care about arrival at their final destination, and the ones who suffer most are those whose trip breaks.

DEFINITION 13.3 Disrupted passenger

A passenger is **disrupted** when the planned itinerary becomes infeasible, because a flight in it is cancelled or a connection becomes shorter than the minimum connecting time. A passenger's **delay** is the actual minus the scheduled arrival time at the final destination (Barnhart and Clarke 2003; Bratu and Barnhart 2006).

If a share p_d of passengers is disrupted, with average delay D_d, and the others have average delay D_{nd}, the average passenger delay and the disrupted passengers' share of it are

$$\bar{D} = p_d D_d + (1 - p_d) D_{nd}, \qquad \text{share of delay} = \frac{p_d D_d}{\bar{D}}. \tag{13.2}$$

Because D_d is measured in hours and D_{nd} in minutes, a small p_d can dominate. In data for one US carrier in August 2000, 3.2 percent of passengers were disrupted, waited 320 minutes on average against 16 for the others, and accounted for 40 percent of all passenger delay (historical) (Barnhart 2003). On the Laurentian day only

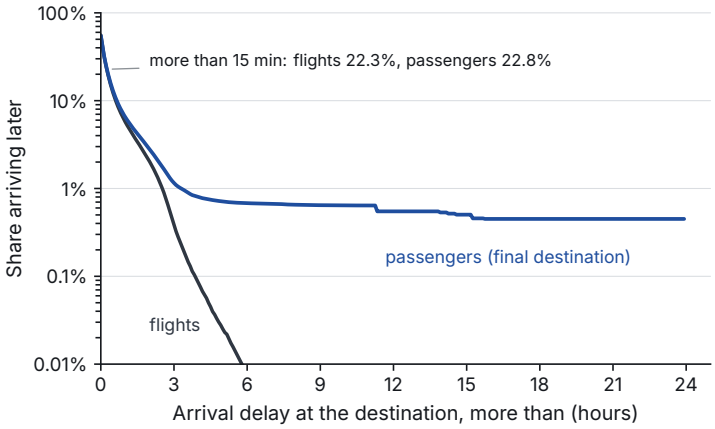


Figure 13.5 Share of flights and of passengers arriving more than a given delay late on the Laurentian day (every-second-night routing, 20,000 simulated days, logarithmic scale). The two curves agree up to about an hour; beyond that, missed connections, the end of the day and transatlantic misconnects give passengers a long tail.

24.5 passengers a day, 0.83 percent, are disrupted, but they wait 17 hours on average and account for 39 percent of all passenger delay (Figure 13.5). At 15 minutes flights and passengers look alike; at six hours almost no flight is that late, while 0.7 percent of passengers are.

> **PITFALL**
>
> Reading flight punctuality as passenger experience. On-time performance can be improved by padding blocks or by cancelling the latest flights, and both can make passengers worse off: in Example 13.3, cancelling the LaGuardia round trip produces no late departure and by far the most disrupted passengers. As the handbook advises, report passenger arrival at the final destination next to A15, because a flight-level improvement can hide worse outcomes for connecting itineraries (Network Planning Handbook 2026).

13.7 Disruption in the business case

A schedule change that adds a connection, tightens a turn or moves the last departure of the day changes expected disruption, and the handbook's first-pass framework puts that change into the decision (Network Planning Handbook 2026):

$$\text{expected net contribution} = \text{planned contribution} - \mathbb{E}[\text{incremental disruption cost}]. \tag{13.3}$$

The expected cost should come from delay scenarios, missed connections, cancellations, re-accommodation and rotation recovery, without double counting disruption costs already in the base cost forecast (Network Planning Handbook 2026). The handbook's invented Montreal–Seville case carries an expected incremental disruption cost of CAD 2,000 per departure (Chapter 10) (Network Planning Handbook 2026).

Canadian passenger protection. Part of the disruption cost is set by regulation. Under the Air Passenger Protection Regulations, the Canadian Transportation Agency distinguishes three categories of disruption with different obligations (Table 13.3) (Canadian Transportation Agency 2023; Government of Canada 2019). In all three the airline must keep passengers informed and rebook them, or in some cases refund, after a cancellation or a delay of three hours or more. Standards of treatment (food, drink and, overnight, a hotel) apply when a passenger told of the disruption less than 12 hours before departure has waited two hours, if the cause is within the airline's control, safety causes included. Compensation is owed only for disruptions within its control, not required for safety and announced 14 days or less before departure; it grows with the arrival delay, with thresholds at three, six and nine hours (Canadian Transportation Agency 2023). Denied boarding has its own rules, which a downgauge on a sold flight can trigger (Chapter 14).

Table 13.3 The three disruption categories of the Air Passenger Protection Regulations as described by the Canadian Transportation Agency's guide, which is not a legal text; the regulations prevail, and the compensation amounts they set are not reproduced here.

Category	Examples given by the regulator	Standards of treatment	Compensation
Within the airline's control	commercial decisions, staffing, scheduled maintenance	yes, after a two-hour wait	yes, by arrival delay
Within its control, required for safety	safety problems found in checks	yes, after a two-hour wait	no
Outside the airline's control	weather, security, air traffic control instructions	no	no

The handbook's planning advice: rebooking, assistance and compensation exposures are not interchangeable, so a model should neither treat every delay as compensable nor assume that weather costs nothing commercially (Network Planning Handbook 2026).

IN PRACTICE **Passenger protection rules in October 2026**

The rules are in flux. In June 2023 Parliament amended the Canada Transportation Act to eliminate the three categories, and proposed regulations published in the Canada Gazette on 21 December 2024 would make compensation owed unless the disruption falls in an exhaustive list of exceptional circumstances (Canadian Transportation Agency 2024). In September 2026 the Agency still described the amendments as proposed and reported a backlog of more than 95,000 complaints (Canadian Transportation Agency 2026b). Bill C-31, tabled in May 2026, would transfer the making of these regulations to the Minister of Transport (Canadian Transportation Agency 2026b; McCarthy Tétrault 2026); in late October 2026 it was still at committee stage (Parliament of Canada 2026). A business case should state which version of the rules it assumes.

WORKED EXAMPLE 13.4 Selling a connection onto the last flight to Quebec City

In Chapter 12 the commercial team moved 107 from Toronto to 18:15, and the planner repaired the day by moving 108, the last departure to Quebec City, to 20:15. That opened a 50-minute connection from 207, the LaGuardia inbound due at 19:25, which clears the 45-minute MCT used here. Suppose 12 passengers a day would buy it at an illustrative CAD 150 of contribution each, CAD 1,800 a day. A passenger who misconnects is stranded until morning, at a cost K to Laurentian (hotel, meals, rebooking, goodwill and any compensation owed), illustratively CAD 400.

Flight 207 comes late in line B's day, when its aircraft carries the day's accumulated delay. With 108 at 20:15 the connection fails 36 percent of the time, stranding 4.3 passengers a day (Table 13.4): an expected cost of CAD 1,709 a day, so the expected net contribution (13.3) is only CAD 91, and negative at any K above CAD 421. Moving 108 to 20:30 cuts failures to 20 percent and earns CAD 838 a day; 20:45 earns CAD 1,177. The cost is a later arrival for every local passenger to Quebec City (Chapter 11).

A legal connection is not a reliable one. The MCT is a minimum, not a target, and a connection onto the last flight of the day, from an inbound at the end of a tight rotation, needs more time or a decision not to sell it.

Table 13.4 Selling the LaGuardia to Quebec City connection onto the last departure (20,000 simulated days). In the original plan 107 leaves Toronto at 17:55; in the other columns at 18:15. A contribution of CAD 150 per connecting passenger and K = CAD 400 per stranded passenger are illustrative; breakeven K is the stranding cost at which the connection stops paying.

Departure of 108	20:00, original	20:15, repair	20:30	20:45
Connection from 207 (min)	35, not sold	50	65	80
Misconnect probability		36%	20%	13%
Passengers stranded per day	0	4.3	2.4	1.6
Expected cost (CAD/day)	0	1,709	962	623
Net contribution (CAD/day)	0	91	838	1,177
Breakeven K (CAD)		421	748	1,157
108 within 15 minutes	74%	77%	82%	84%

Slack is worth most where delay arrives and where a failure is expensive: behind late-prone inbounds, in front of connections onto the last flight of the day, and at the turns that feed a bank. The same minutes spread evenly buy much less.

13.8 Summary

- Delays travel through aircraft, crews and passengers along different paths.
- Propagated delay is the part of an inbound delay that exceeds the slack. Along a rotation it behaves like a queue, so an evening flight inherits the worst stretch of the day.
- Slack has diminishing returns: plan it until the chance that the inbound delay exceeds it equals the cost ratio of a minute of slack to a minute of delay. Buffer is free until it costs a leg.
- Robust routing and retiming move the slack a schedule already has; historical tests removed over a third of propagated delay.
- Operations control recovers from the options the schedule provides, and should judge them by what passengers lose.

- A few disrupted passengers carry a large share of passenger delay; report them alongside A15.
- Expected net contribution subtracts expected disruption cost from planned contribution; Canada's passenger protection rules decide which costs apply, and they were under revision in 2026.

13.9 Further reading

Barnhart and Clarke's robustness lecture (Barnhart and Clarke 2003), the operations lectures of Barnhart's course (Barnhart 2003) and Belobaba's lecture on operations control (Belobaba 2006j) are the core of this chapter. The published versions are Lan, Clarke, and Barnhart (2006) on robust routing and retiming and Bratu and Barnhart (2006) on passenger-centred recovery; Clausen et al. (2010) survey disruption management. For Canadian obligations, start from the Agency's guide (Canadian Transportation Agency 2023). The full reference edition derives the delay formulas and the recovery and retiming models in its Chapter 16.

PART IV

Revenue

Part III decided how many seats to fly, between which cities and at what times. Part IV is about what those seats earn. Chapter 14 explains how airlines build fare structures and how revenue management decides, flight by flight and day by day, which fares are on sale, how many reservations to accept and which connecting passengers to carry. Chapter 15 turns its outputs into the planner's tools for valuing schedule and capacity changes.

CHAPTER

14

Revenue Management for Planners

IN THIS CHAPTER

- Why one economy cabin is sold at many prices, how fences make travellers sort themselves, and how fare structures moved from restricted to simplified to branded fares.
- Booking classes, nesting, protection levels and booking limits, with Littlewood's rule worked on Laurentian's Montreal–Seville fare ladder.
- What seat control does to the loads, the spill and the value of a seat that a planner reads.
- Overbooking: how many reservations to accept for 182 seats, and what Canada's denied-boarding rules add to the cost.
- Network control: why leg-by-leg control turns away the wrong passengers, and what the bid prices of Laurentian's Montreal bank tell a planner.
- Willingness to pay, sell-up, buy-down and spiral down, and dynamic pricing as it stands in 2026.

DRAWS ON Belobaba, Airline economics review; Introduction to revenue management; Pricing challenges: ePODS and reality; Flight overbooking: models and practice; Network revenue management: origin-destination control; Willingness to pay and competitive revenue management (MIT 16.75J, 2006); Belobaba, O&D control: what have we learned? (2002); Littlewood (1972); Air Passenger Protection Regulations (SOR/2019-150); handbook; 2026 research digest

When Laurentian opens sales for a Montreal–Seville departure, it posts five economy fares, from CAD 419 to CAD 1,249, and from then until the doors close a revenue management system decides, many times a day, which of them a traveller can still buy. The planner who put the A321XLR on the route never sets one of those prices or limits, but every number the planner reads about the flight passes through them: its load factor, its average fare, the passengers it spills and the value of one more seat.

This chapter explains what revenue management does to the flights a planner schedules and what its numbers mean. It follows a booking through the fare structure (Sections 14.1 and 14.2), the booking limits of one flight (Sections 14.3 and 14.4), overbooking (Section 14.5), the bid prices that decide which connecting passengers to carry (Section 14.6) and the willingness to pay on which all of it rests (Section 14.7). The derivations are in Appendix A.

14.1 One cabin, many prices

Two features of the airline business make differential pricing natural. The first is the cost structure. Once a flight is scheduled, almost all of its cost is committed: Laurentian's avoidable cost of a YUL–SVQ departure is about CAD 88,000 with 150 passengers on board, of which only CAD 50 per passenger varies with the number carried (Chapter 3). An empty seat earns nothing and cannot be stored. A single fare set at the average cost, about CAD 587 per passenger, would turn away travellers willing to pay CAD 450 for a seat that will otherwise leave empty, and would charge CAD 587 to travellers willing to pay CAD 1,500.

The second is heterogeneity. Business travellers decide late, change plans and value flexibility, and their demand responds little to price; leisure travellers plan ahead, accept conditions and shop on price. The reading library quotes typical market price elasticities of about −0.8 for business and −1.6 for leisure travel (historical estimates) (Belobaba 2006d).

Charging different prices for the same product according to willingness to pay is **price discrimination**; charging different prices for products that differ in characteristics and cost is **product differentiation** (Belobaba 2006d). Airline fare structures do both, but mostly the first: a flexible and a restricted economy ticket differ in cost to the airline by a few dollars of expected refunds and changes, and in price by hundreds. Differential pricing raises revenue from both ends of the market. Discount travellers who would not fly at a single fare fill seats that would otherwise leave empty, and high-fare travellers pay more than a single fare would extract. Belobaba reports studies showing that most traditional high-cost airlines could not cover their total operating costs with any single fare (Belobaba 2006d). Revenue per passenger is very unequal: Air Canada's premium cabins produced about 30 percent of its passenger revenue in the second quarter of 2026 (Air Canada 2026d).

Figure 14.1 shows what is at stake on the Seville flight. Laurentian's five economy fares, stacked in order with their mean demands,

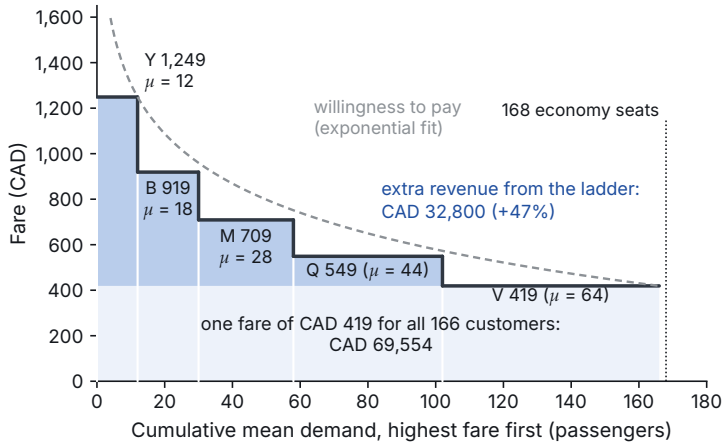


Figure 14.1 Laurentian's YUL–SVQ fare ladder drawn as a staircase: each block is a class's fare times its mean demand, and the upper corners trace an exponential willingness-to-pay curve. A single fare of CAD 419 accepted by all 166 customers would earn CAD 69,554; the ladder, with every customer buying the product meant for that customer, would earn CAD 102,354.

trace a willingness-to-pay curve: 166 customers per departure would pay at least CAD 419, and 12 of them would pay CAD 1,249. A single fare of CAD 419 earns CAD 69,554. If every customer bought the product meant for that customer, the ladder would earn CAD 102,354, 47 percent more. That difference is the prize the fare structure tries to capture and revenue management must defend.

The difficulty is that willingness to pay is invisible: a traveller asking for the lowest fare does not reveal that her employer would have paid four times as much. Airlines therefore offer a menu designed so that each type of traveller prefers the option meant for it. The device is the **fence**, a condition attached to a cheap fare, such as a 21-day advance purchase or no refunds, that costs the airline little but costs travellers with a high willingness to pay a lot. A fence works only if it hurts the high-paying traveller more than the low-paying one, and pays only if what it gains from the first group exceeds what it gives up on the second.

WORKED EXAMPLE 14.1 Designing a fence for YUL–SVQ

Suppose a departure attracts 40 time-sensitive travellers who value a flexible ticket at CAD 1,400 and 120 leisure travellers who value it at CAD 550. A 21-day advance purchase with no refunds costs the first group CAD 600 of inconvenience and the second CAD 100 (all values illustrative).

Without a fence, the best single price is CAD 550, and all 160 fly for CAD 88,000. With the fence, the restricted fare is set where leisure travellers just accept it, $550 - 100 = 450$, and the flexible fare as high as it can go before time-sensitive travellers prefer the restricted ticket, $450 + 600 = 1{,}050$. Revenue is $40 \times 1{,}050 + 120 \times 450 =$ CAD 96,000, CAD 8,000 more: the fence gains CAD 500 on each of 40 time-sensitive travellers and gives up CAD 100 on each of 120 leisure travellers.

If the restriction is weakened so that it costs a time-sensitive traveller only CAD 250, the flexible fare can be at most CAD 700, and the menu earns CAD 82,000, less than the single price. The revenue a route can earn depends on restrictions that never appear in the schedule.

14.2 From restricted to branded fares

DEFINITION 14.1 Fare product and booking class

A **fare product** is a fare level for travel in an origin-destination market together with its rules: when it must be bought, whether it can be changed or refunded and, in branded structures, which services it includes. Each fare product is assigned to a **booking class**, a one-letter inventory bucket on each flight. Revenue management controls booking classes, not fares: when a class closes on a flight, every fare product mapped to it becomes unavailable on that flight.

Through the 1990s, network carriers in North America sold economy through heavily restricted structures: a full fare with no conditions, then a cascade of discounts, each with a longer advance purchase, a Saturday-night stay and stiffer penalties (Belobaba 2006d). Advance purchase separates trips planned early from trips decided late, and makes cheap fares book first. The Saturday-night stay ex-

cludes the business round trip that returns home for the weekend, historically the most effective fence (Belobaba 2006d).

Low-fare airlines attacked this structure with simple one-way fares, few restrictions and modest ratios between their highest and lowest fares. In the early 2000s, as business travel fell and internet shopping made every fare visible, network carriers followed: by 2005 Delta sold simplified fares, without the Saturday-night stay and with a ratio of highest to lowest fare of about 4 to 1 or less, in all its US domestic markets (Belobaba 2006o). As the fence arithmetic predicts, more travellers bought lower fares, and simplified structures produced higher load factors but revenues 10 to 15 percent lower (Belobaba 2006d). Competition also sets fare levels: the Competition Bureau found in 2025 that on Canadian routes one additional competitor reduces average fares by about 9 percent (Competition Bureau Canada 2025).

MIT's ePODS simulations measured the value of each fence in a market with two competing airlines (Belobaba 2006o). Starting from a fully restricted structure, removing advance purchase alone cost 0.5 percent of revenue, because the Saturday stay still kept business travellers out of the discount classes; removing the Saturday stay cost 16.8 percent, and removing every restriction 45 percent. Revenue management earned 8.2 percent more than selling first come, first served in the restricted structure, and 0.1 percent more with all fences gone, a collapse that Section 14.7 explains (historical simulation results).

> **PITFALL**
>
> Reading a higher load factor as a better fare structure. In the ePODS experiments the structure with the highest load factor, 88.1 percent, earned 45 percent less than the fully restricted structure, whose load factor was 81.6 percent. Fences are worth more than fullness.

Simplification did not end differential pricing; it moved the fences. Carriers rebuilt product differentiation around **branded fares**, named bundles of rules and services sold under the same names across markets, and **ancillaries**, services sold separately such as a checked bag. A basic fare is what Deneckere and McAfee call damaged goods: the seat and the flight are those of the standard fare, minus services that some travellers value much more than others (Deneckere and McAfee 1996). Each brand is sold at one or more price points, each mapped to a booking class, so revenue manage-

ment controls the price within a brand while the traveller chooses among brands.

IN PRACTICE Fare structures in 2026

IdeaWorksCompany estimates global airline ancillary revenue at USD 157 billion in 2025, 15.7 percent of airline revenue against 9.1 percent in 2016 (IdeaWorksCompany 2025). Air Canada sells economy under five brands, Basic, Standard, Flex, Comfort and Latitude, which differ in baggage, seat selection, change and refund conditions and loyalty earning (Prince of Travel 2026). On 28 July 2026 it added Premium Economy Basic and Business Class Basic fares in selected international markets: non-refundable and with paid seat selection, but with the same seat, meals and priority services (Open Jaw 2026).

Laurentian's YUL–SVQ economy structure (Table 14.1) follows the branded pattern, with three brands, the first two sold at two price points each. Within a brand the two classes differ in advance purchase and refund conditions, so every class keeps its own fences. The highest fare is 2.98 times the lowest, a compressed ratio typical of branded structures.

Table 14.1 Laurentian's YUL–SVQ economy fare ladder for the 168 economy seats of the A321XLR. Fare: one-way net fare in CAD. AP: advance purchase in days. Credit: a travel credit instead of a refund. Mean and SD: unconstrained demand per departure, normal and independent by class. All values are illustrative.

Class	Brand	Fare	AP	Changes	Refund	Mean	SD
Y	Flex	1,249	0	free	full	12	5
B	Flex	919	14	free	fee	18	7
M	Standard	709	14	fee	credit	28	11
Q	Standard	549	21	fee	none	44	17
V	Basic	419	21	none	none	64	26
Total						166	34

Pricing designs this menu. Revenue management takes the menu and the capacity as given (Belobaba 2006l) and decides which items are on sale on each flight at each moment by opening and closing booking classes; a traveller sees the lowest fare whose rules she meets among the open classes. The planner supplies the capacity both of them sell. A fare change alters the demand for each class and a capacity change alters the best booking limits, so the two must be evaluated together.

14.3 Booking limits and Littlewood's rule

Revenue management decides how many seats each fare may sell, and its objective is revenue, not yield or load factor: maximizing yield leads to severe limits on cheap fares and empty seats, maximizing load factor to cheap seats (Belobaba 2006l). The problem is solved millions of times: a large carrier with 2,000 flight legs a day, 10 booking classes and 300 days of sales holds 6 million booking limits at any moment (Belobaba 2006l). A revenue management system keeps them current in a loop. It forecasts the demand still to come for each future departure and class from booking history, after unconstraining the history of classes that were closed (Chapters 6 and 7); computes protection levels and an overbooking level; and loads the resulting booking limits into the inventory, where every sales channel sees them as open or closed classes. It repeats the cycle at checkpoints until departure, so a class closed today can reopen tomorrow. Today's limits also shape the bookings from which tomorrow's forecasts are built, a point that returns in Section 14.7.

DEFINITION 14.2 Nesting, protection levels and booking limits

Booking classes $j = 1, \dots, n$ share a cabin of C seats and are numbered from the highest fare, $f_1 > f_2 > \cdots > f_n$. In a **serially nested** inventory, every seat available to a class is also available to all higher classes, so a high-fare request is never refused while a seat remains for sale at a lower fare. The **protection level** y_j is the number of seats reserved for classes 1 to j together. The **booking limit** b_j is the number of seats that may be sold before class j closes. They are linked by $b_1 = C$ and $b_{j+1} = C - y_j$.

Every modern system nests, because a partitioned inventory, in which each class has its own seats, refuses a high-fare request once its allocation is sold even while cheaper seats are unsold (Belobaba 2006l). The question becomes how many seats to keep away from each cheaper class.

Start with two classes, a high fare f_1 and a low fare f_2, and assume that low-fare demand books first, as advance-purchase rules encourage. Consider the last seat the airline might protect for the high fare. Sold now at the low fare, it earns f_2 for certain. Protected, it earns f_1 only if high-fare demand D_1 turns out large enough to reach it. Since that chance falls as more seats are protected, there is a last seat worth protecting.

RESULT 14.1 Littlewood's rule

With two classes, the low fare booking first and independent demands, protect a seat for the high fare as long as the chance of selling it at the high fare, times the high fare, is at least the low fare. The optimal protection level is

$$y^* = \max\{y : f_1 \Pr(D_1 \geq y) \geq f_2\}, \tag{14.1}$$

and the low fare's booking limit is $C - y^*$. If high-fare demand is normal with mean μ_1 and standard deviation σ_1, then $y^* \approx \mu_1 + \sigma_1 \Phi^{-1}(1 - f_2/f_1)$, where Φ^{-1} is the inverse of the standard normal distribution function. The protection level depends neither on low-fare demand nor on capacity (Littlewood 1972).

The rule weighs two errors, like the newsvendor rule that sized aircraft in Chapter 7. Protecting one seat too few costs $f_1 - f_2$ when a high-fare traveller then finds the seat sold cheaply; protecting one too many costs f_2 when it leaves empty. One consequence is worth remembering: when the low fare is more than half the high fare, the airline protects fewer seats than mean high-fare demand, and when it is less than half, more. Appendix A.7 derives the rule.

WORKED EXAMPLE 14.2 Protecting Flex seats on YUL–SVQ

Class Y sells at CAD 1,249 with demand of mean 12 and standard deviation 5, and the next class, B, at CAD 919 (Table 14.1). Protecting a seat for Y pays while $\Pr(D_Y \geq y) \geq 919/1{,}249 = 0.736$. With $\Phi^{-1}(1 - 0.736) = -0.630$,

$$y^* = 12 - 0.630 \times 5 = 8.85.$$

In whole seats, the ninth protected seat is expected to earn $1{,}249 \times \Pr(D_Y \geq 9) = 947$, more than the B fare, and the tenth only 864, less. Laurentian protects 9 seats for Y, and B's booking limit is $168 - 9 = 159$.

The airline protects fewer seats than mean Y demand because B pays 74 percent of the Y fare: the tenth seat would sell at the Y fare only with probability 0.69, and an expected CAD 864 is worth less than a sure CAD 919. Against cheaper fares the same rule protects more: 11.15 seats against M, 12.76 against Q and 14.12 against V (Figure 14.2). Capacity does not enter. A larger

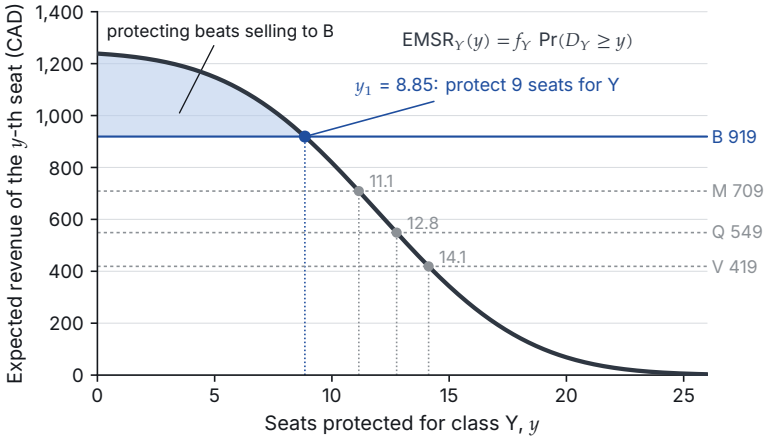


Figure 14.2 Littlewood's rule for class Y on Laurentian's YUL–SVQ flight. The curve is the expected revenue of protecting the y-th seat for Y. Where it crosses the B fare, protecting another seat stops paying. Crossings with the cheaper fares (grey) give the protection Y would need against each of them.

aircraft would still protect 9 seats for Y and give the extra seats to the cheaper classes.

With more than two classes, Belobaba's expected marginal seat revenue (EMSR) heuristics apply Littlewood's rule at every cut between higher and lower classes (Belobaba 1987, 1989). The standard version, **EMSR-b**, merges the classes above a cut into one artificial class, with their summed demand and their demand-weighted average fare, and protects it against the next fare down (Talluri and van Ryzin 2004b; Belobaba 2006l). To set V's limit on YUL–SVQ, classes Y to Q have mean demand 102, a standard deviation of 22.0 and an average fare of CAD 740.6; Littlewood's rule against the V fare of 419 protects 98 seats for them, so V may sell at most 70, a limit that binds on about 40 percent of departures. The four cuts protect 9, 26, 54 and 98 seats, and the nested booking limits are 168, 159, 142, 114 and 70 (Figure 14.3). On this flight they coincide with the exactly optimal protection levels. Appendix A.7 gives the EMSR-b formula; the full reference edition derives the dynamic program that defines optimal control in its Chapter 18.

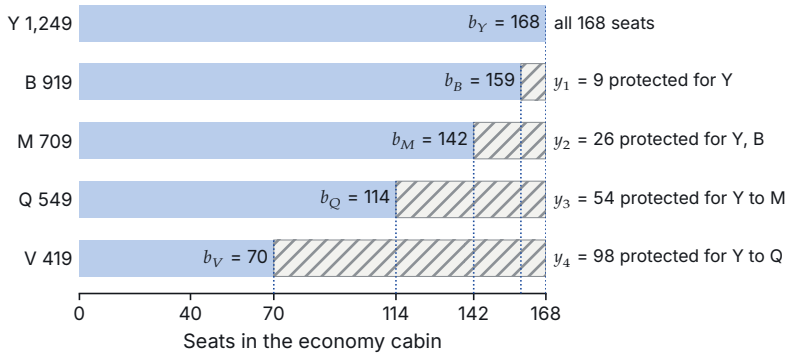


Figure 14.3 Serially nested booking limits and protection levels on Laurentian's YUL–SVQ economy cabin, from EMSR-b. Each bar shows the seats a class may sell (its booking limit b) and the seats protected for the classes above it.

14.4 What seat control does to a flight

A planner rarely sets a booking limit, but revenue management changes four numbers the planner reads.

It lowers loads and raises fares. Under EMSR-b control Laurentian's Seville flight carries 150.6 economy passengers per departure at an average fare of CAD 620, the value the running case uses for the route (Chapter 10). Sold first come, first served, it would carry 153.5 passengers at CAD 585 and earn CAD 89,831 instead of CAD 93,465, 3.9 percent less. A lower load factor on a well-managed flight is a sign that seats were held for high fares, not that demand is weak.

It moves spill to the bottom of the ladder. Under EMSR-b, 7.6 of the 15.4 passengers spilled per departure are V travellers and 3.8 are Q travellers. Under first come, first served, 5.0 Y and 4.7 B travellers are among the 12.5 turned away. This is why Chapter 7 warned against valuing spilled passengers at the average fare.

It values the last seat far below the average fare. On a controlled flight, one more seat is worth its marginal value, the expected revenue of the last seat. For Laurentian's flight that is about CAD 268 to 295, depending on how the booking process is modelled: below even the lowest fare, because the last seat fills only on some departures. Value is concave in capacity: in the dynamic booking model, ten more seats would add CAD 1,896 per departure and ten fewer would cost CAD 3,229. The protection levels for the upper classes do not change with capacity (9, 26, 54 and 98 seats with 158, 168 or 178 seats), so extra seats go to the cheapest fares.

Its value grows steeply with demand. Scaling all class demands together, EMSR-b earns 0.4 percent more than first come, first served at a demand factor (mean demand over seats) of 0.79, 4.0 percent at 0.99 and 12.6 percent at 1.19. Historically, Belobaba put the gain from effective leg-based fare class control and overbooking at 4 to 6 percent of total revenue (Belobaba 2002).

On a flight controlled by revenue management, one more seat is worth the marginal value of the last seat, not the average fare.

PITFALL

Trusting limits that are computed once. In a dynamic booking model of YUL–SVQ in which all classes book during the same months, EMSR-b re-solved at 16 checkpoints came within 0.43 percent of the optimum, while the same limits computed once, 210 days out, lost 8.2 percent, more than having no control at all. Ask how often a system re-optimizes before trusting its limits.

14.5 Overbooking

A planner counts seats; revenue management sells reservations, and some booked passengers never turn up. If the airline stopped selling at the physical capacity, it would fly those seats empty on exactly the flights where other customers had been turned away. **Overbooking** authorizes more bookings than seats so that the expected no-shows are replaced in advance, at the price of the occasional flight on which more passengers show up than the cabin holds. Belobaba's lecture records average US no-show rates of 10 to 15 percent of final bookings (historical figures), and notes that effective overbooking can earn as much as fare-class seat allocation (Belobaba 2006m).

DEFINITION 14.3 Overbooking vocabulary

For one departure with C seats in the cabin:

- the **authorization level** AU is the maximum number of confirmed bookings the airline accepts, and AU/C is the **overbooking factor**;
- the **show-up rate** q is the share of bookings held at departure whose passengers present for the flight; the others are **no-shows**;
- a **denied boarding** is a passenger with a confirmed booking who presents and cannot be carried;
- a **spoiled seat** is a seat that departs empty on a flight that was closed for sale.

Overbooking and spill are different (Network Planning Handbook 2026): spill is demand turned away while the flight is on sale, whereas a denied boarding happens at the gate, to a passenger holding a confirmed booking. No-show rates vary by fare product, connection status and day (at a hub, a missed connection appears as a no-show on the next leg), so they are estimated segment by segment. The simplest model treats each booking as an independent coin flip that shows up with probability q, so that the show-ups among b bookings, $N(b)$, average bq. Real show-ups vary more, because weather, a late inbound bank or a group move many passengers at once.

Two errors are possible. Too low an authorization level spoils seats, a loss nobody sees except as a lower load factor; too high a level denies boardings at the gate. The deterministic rule AU $= C/q$ fills the cabin on average, and risk rules cap the probability of any denied boarding, but neither asks whether the empty seats it accepts are a fair price for the denials it avoids. That needs a cost c_{SP} for a spoiled seat, the contribution of the passenger who could have occupied it, and a cost c_{DB} for a denied boarding: compensation, care, rebooking and goodwill.

RESULT 14.2 The optimal authorization level

Accept one more booking as long as the chance that the passengers already booked fill the cabin is below the critical ratio ρ. The cost-minimizing authorization level is the smallest b such that

$$\Pr(N(b) \geq C) \;\geq\; \rho = \frac{c_{\text{SP}}}{c_{\text{SP}} + c_{\text{DB}}}. \tag{14.2}$$

When a denied boarding costs more than a spoiled seat, ρ is below one half and the optimal level sits below C/q.

The logic is marginal. The extra booking matters only if its passenger shows up. Then either the cabin still has a free seat, which the booking fills, or it is already full, and the booking adds a denied boarding. Accepting pays while the second case is unlikely enough. The rule is the newsvendor rule of Chapter 7 seen from the other side, and Appendix A.8 derives it. Demand uncertainty does not move it: demand decides how often the limit binds, not where it should be.

WORKED EXAMPLE 14.3 How many bookings for 182 seats?

Treat the 182 seats of Laurentian's A321XLR as one cabin and assume, for illustration, that 93 percent of bookings held at departure show up, independently. A spoiled seat costs CAD 550, the restricted (L) Seville fare of CAD 599 in the network model of Section 14.6, net of CAD 50 of passenger-variable cost and rounded, because on a controlled flight the seat an extra booking fills is a low-fare seat. A denied boarding costs CAD 1,650 on average, a blend of volunteers who accept a travel credit and a later flight and the occasional involuntary denial with regulated compensation, rebooking and care. The critical ratio is $550/(550 + 1{,}650) = 0.25$.

Table 14.2 Show-ups and expected overbooking cost per departure on YUL–SVQ ($C = 182$, $q = 0.93$, $c_{\mathrm{SP}} = 550$, $c_{\mathrm{DB}} = 1{,}650$).

Bookings b	$\mathbb{E}[N(b)]$	$\Pr(N(b) \geq 182)$	Expected cost (CAD)
190	176.7	0.079	3,068
191	177.6	0.134	2,719
192	178.6	0.206	2,480
193	179.5	0.295	2,391
194	180.4	0.395	2,483
195	181.4	0.499	2,779
196	182.3	0.601	3,290

With 192 bookings the probability of a full cabin is 0.206, below 0.25, so the 193rd booking is worth accepting; with 193 it is 0.295, so the 194th is not. The optimal authorization level is 193, an overbooking factor of 1.06; the deterministic limit, $182/0.93 = 195.7$, would accept two or three more. At 193 the flight boards 179.0 passengers on average, denies boarding to 0.46 and spoils 2.97 seats, for an expected cost of CAD 2,391. Without overbooking it would spoil 12.7 seats, CAD 7,007, so overbooking saves CAD 4,616 on every departure that books to the limit.

The answer is robust to the cost estimate: 193 stays optimal for any denied-boarding cost between CAD 1,314 and CAD 2,117. It is less robust to the show-up rate: with $q = 0.90$ the optimal limit is 199, with $q = 0.95$ it is 189.

Independence matters: if the show-up rate varies by 2.5 percentage points between departures, expected denied boardings at 193 bookings rise from 0.46 to 1.21. Service targets imply costs too: capping the probability of any denial at 5 percent allows 190 bookings, which is optimal only if a denied boarding costs CAD 6,396 to 12,423, four to eight times the blended cost. Making that implied cost explicit is the most useful conversation an analyst can have about overbooking policy.

A denied boarding is **voluntary** when a passenger agrees to give up the seat, usually for compensation and a later flight, and **involuntary** otherwise. Volunteers are much cheaper: in 2004, 95 percent of US denied boardings were voluntary (historical) (Belobaba 2006m). With illustrative reservation values for the Seville flight, an ascending offer at the gate buys three volunteers for about CAD 726 each,

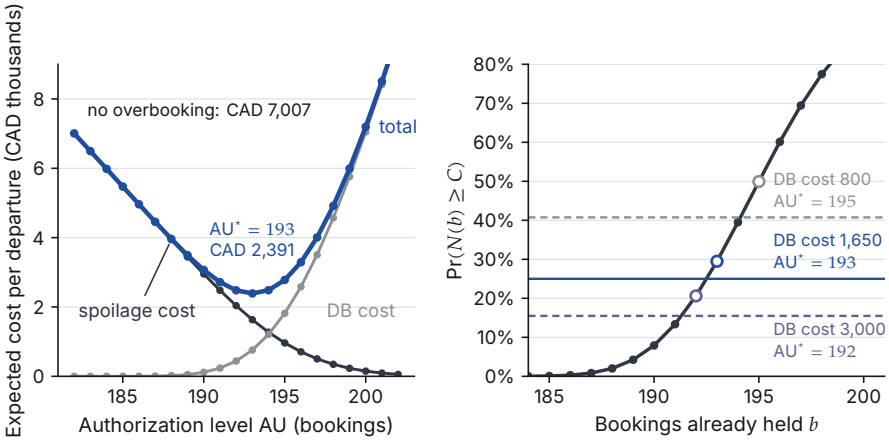


Figure 14.4 Overbooking YUL–SVQ. Left: expected spoilage, denied-boarding and total cost per departure as the authorization level varies; the minimum is at 193 bookings. Right: the marginal rule. The probability that show-ups among b bookings already fill the 182 seats rises steeply with b; the optimal limit is the first b at which it reaches the critical ratio, shown for three denied-boarding costs.

while three involuntary denials arriving nine hours or more late would cost at least CAD 7,200 in regulated compensation alone.

IN PRACTICE **Denied boarding rules for a Canadian carrier in 2026**

Canada's Air Passenger Protection Regulations govern denied boarding; the consolidated text is current to 30 September 2026 (Government of Canada 2019). When a denial is within the carrier's control and not required for safety, which covers commercial overbooking, the carrier must provide care, offer alternate travel or a refund and pay minimum compensation (s. 12(4)). It must first ask all passengers whether they are willing to give up their seat (s. 15(1)), and it may not remove a passenger already on board unless safety requires it (s. 15(2)). Minimum compensation depends on the delay in arrival at the destination: CAD 900 for less than 6 hours, CAD 1,800 for 6 to 9 hours and CAD 2,400 for 9 hours or more (s. 20(1)) (Government of Canada 2019; Canadian Transportation Agency 2026a). The regime is in flux: amendments published for consultation in December 2024 were still described as proposed in September 2026 (Canadian Transportation Agency 2024, 2026b), and Bill C-31, introduced in May 2026, would move regulation-making and complaint handling to the Minister of Transport (McCarthy Tétrault 2026). The return leg, SVQ–YUL,

falls under European Union rules, which owe a denied passenger EUR 600 on a flight over 3,500 km (European Union 2026).

The compensation bands tie the cost of a denial to the rebooking options the network offers, which is the planner's business. On a route with several daily frequencies a denied passenger arrives within hours and the CAD 900 band applies; on a route served three times a week, like Seville, the CAD 2,400 band is the realistic one, which argues for a lower limit on thin long-haul routes. Removing a frequency raises the cost of every denied boarding on the remaining ones. And the seats revenue management sells on the Seville flight number about 193, not 182, so a spill model of booking demand should cap bookings at the authorization level and passengers at the cabin (Chapter 7).

> **PITFALL**
>
> Forgetting that a gauge change is an oversale. When a planner replaces a 169-seat 737-8 by a 137-seat A220 on a departure that is already sold, every booking beyond the new authorization level is a potential denied boarding that no overbooking model chose. Its expected cost at regulated rates belongs in the business case for the change (Chapters 13 and 15).

14.6 Network control and bid prices

At a hub, managing each flight on its own is not enough. A passenger from Quebec City to Paris occupies seats on two aircraft, and whether to sell her a ticket depends on what both seats could otherwise earn. Laurentian's evening bank at Montreal (Table 14.3) connects two feeders to two transatlantic flights and sells sixteen fare products: eight origin-destination (O&D) markets, four local and four connecting, each with a flexible (H) and a restricted (L) fare. A quarter of all O&D demand connects, and three of the four legs are oversubscribed on average.

Table 14.3 Laurentian's bank at Montreal. Mean demand on a leg is the sum of the mean demands of all products using it; DF, the demand factor, is mean demand divided by seats. Bid prices (CAD) are those of Example 14.4. All values are illustrative.

Leg	Aircraft	Seats	Mean demand	DF	Bid price
YQB–YUL	A220-300	137	141	1.03	159
BOS–YUL	A220-300	137	153	1.12	109
YUL–CDG	787-9	298	317	1.06	490
YUL–SVQ	A321XLR	182	161	0.88	0

Under **leg/class control** each leg runs its own booking limits, and a connection can be sold in a class only if that class is open on every leg it uses (Belobaba 2006n). On YQB–YUL, class L then holds the local passenger at CAD 159, the YQB–CDG passenger at CAD 649 and the YQB–SVQ passenger at CAD 699, and the leg cannot tell them apart. Two failures follow. **Short haul blocks long haul**: when YQB–YUL closes L to protect seats for its CAD 329 flexible locals, it also refuses the YQB–SVQ passenger at CAD 699, although the Seville leg usually has empty seats. **Connections displace locals**: when BOS–YUL and YUL–CDG are both full, a BOS–CDG L passenger at CAD 599 occupies two seats that local restricted passengers would buy for CAD 189 + 549 = 738. The first failure matters more, because both legs of a connection are seldom full at the same time (Belobaba 2006n).

Origin-destination control gives different itineraries different availability on the same flight according to their value to the network (Belobaba 2006n). Early systems, developed at American Airlines (Smith, Leimkuhler, and Darrow 1992), mapped itineraries to hidden virtual classes on each leg by total fare, which favours connections even when their second leg is full; displacement-adjusted virtual nesting (DAVN) also charges each itinerary for the value of the seats it uses on its other legs. Accepting an itinerary on a leg when its fare, net of those charges, exceeds the value of the leg's own seat reduces to one rule.

DEFINITION 14.4 Bid price and bid-price control

The **bid price** π_l of leg l is the minimum revenue the airline requires for one more seat on that leg: an estimate of the value of its last seat in its best alternative use. Under **bid-price control**, a request for O&D fare product j with fare f_j is accepted if seats are available and the fare covers the sum of the bid prices of the legs it uses,

$$f_j \geq \sum_l a_{lj}\,\pi_l, \tag{14.3}$$

where a_{lj} is 1 if product j uses leg l and 0 otherwise.

Bid prices usually come from a network optimization: replace each product's random demand by its mean, allocate seats to products to maximize revenue within the leg capacities, and read from the solution what one more seat on each leg would add. They are the shadow prices of the capacity constraints, and the acceptance rule reproduces the optimal allocation for every product whose fare differs from its path's bid price; Appendix A.9 shows why.

WORKED EXAMPLE 14.4 The bid prices of the YUL bank

The deterministic program for Laurentian's bank earns CAD 365,949 per bank, with bid prices of CAD 159 on YQB–YUL, 109 on BOS–YUL, 490 on YUL–CDG and 0 on YUL–SVQ. The three oversubscribed legs are exactly full; Seville carries its 161 expected passengers with seats to spare.

The bid prices form a chain read off the marginal passengers. On YQB–YUL the marginal passenger is the local restricted fare, so a seat is worth CAD 159. On YUL–CDG it is the YQB–CDG L connection, which pays 649 and uses a feeder seat worth 159, so a Paris seat is worth $649 - 159 = 490$. On BOS–YUL it is the BOS–CDG L connection, which pays 599 and uses a Paris seat worth 490, so a Boston seat is worth only 109, less than the Boston local's restricted fare of 189: every Boston local is already accepted, and the best use of one more Boston seat is one more restricted Boston–Paris connection.

Now apply the rule (Figure 14.5). YUL–CDG L at 549 clears the Paris bid price of 490 and is accepted in full, although it

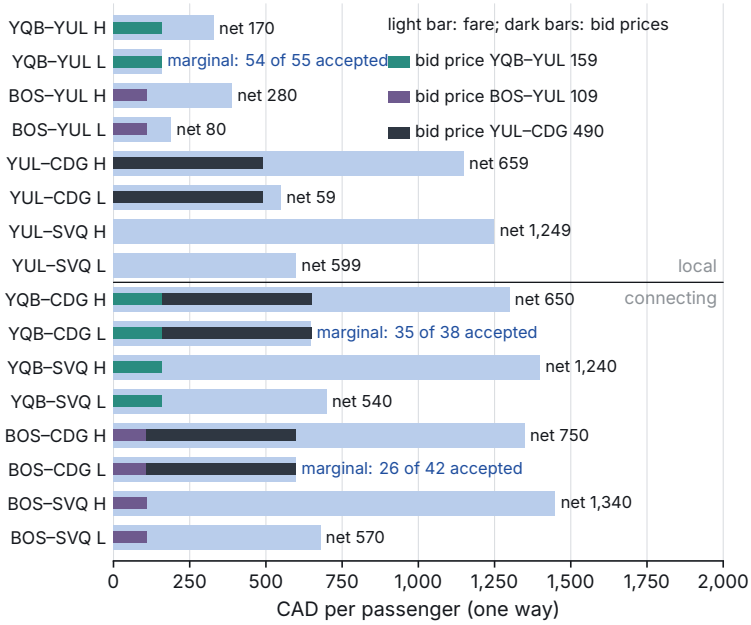


Figure 14.5 The bid-price test on Laurentian's network. Each light bar is a product's fare; the dark bars inside it are the bid prices of the legs it uses, stacked. A product is worth accepting when its fare exceeds the stack; three restricted products sit exactly at the margin, and the deterministic program accepts only part of their demand.

pays less than either restricted Paris connection. YQB–SVQ L at 699 clears 159 + 0 with CAD 540 to spare. Three restricted products sit exactly at their path bid prices, and the program accepts only part of their demand: 54 of 55 YQB–YUL L, 35 of 38 YQB–CDG L and 26 of 42 BOS–CDG L. The network wants all its Seville connections, because that leg has room, and refuses some restricted Paris connections, because they earn no more than the two seats they occupy are worth. Both failures of leg control are corrected by the same four numbers.

A bid price says whether a product is worth selling, not how many units to sell, so it must be recomputed often. In a simulation of 500 departures of the bank, bid prices computed once earned exactly the first come, first served revenue; re-solved 32 times during the booking period, they earned 0.6 percent more than leg/class control, and DAVN 1.07 percent more. MIT's PODS simulations found the same order: effective O&D control added 1 to 2 percent of network rev-

enue to the 4 to 6 percent earned by leg-based control and overbooking, mostly at high load factors and only with substantial connecting traffic (historical results) (Belobaba 2006n, 2002).

The handbook gives the planner's reading: a bid price estimates what scarce capacity is worth in its best alternative use, moves with remaining capacity, time, demand and passengers' alternatives, and is neither a fare nor an accounting cost (Network Planning Handbook 2026). Read that way, Laurentian's bid prices answer three questions.

Which legs are scarce? The Paris leg, by a wide margin. The feeders' bid prices are low not because the feeders are empty but because the passengers at their margin pay little.

What is one more seat worth? On the margin, the bid price. A new connection to Paris over Montreal must earn at least the sum of its feeder's and the Paris leg's bid prices to be worth carrying.

What is a larger aircraft worth? Here the bid price is the wrong tool: it is a derivative, valid for a few seats, while capacity decisions are large. Replacing the A220 from Quebec City by a 169-seat 737-8 is worth 159 × 32 = CAD 5,088 per bank by the bid price, but CAD 459 when the network program is re-solved, because after four seats the feeder's bid price falls to zero, and CAD 1,257 under random demand. Chapter 15 works through such valuations.

> **PITFALL**
>
> Reading a zero bid price as a free seat. The deterministic program prices YUL–SVQ at zero because 161 expected passengers fit into 182 seats. But with a standard deviation of 26 passengers, the Seville flight fills on about one departure in five, and on those its last seats are valuable: averaged over random demand, a Seville seat is worth about CAD 112 (Chapter 15).

14.7 Willingness to pay, sell-up and dynamic pricing

Everything so far assumed that each booking class has its own demand, independent of what else is for sale. Fences make that roughly true. When they weaken, a customer who finds Q closed may pay for M, and one who finds M open may never look at B.

DEFINITION 14.5 Willingness to pay, sell-up and buy-down

A customer's **willingness to pay** is the highest fare at which she buys rather than take her best alternative: another airline, another day, another mode or no trip. **Sell-up** occurs when a customer whose preferred fare is closed buys a more expensive product of the same airline instead. **Buy-down** occurs when a customer who would pay a higher fare buys a cheaper one because it is open.

A standard model in the PODS research describes willingness to pay above the lowest fare f_n with one parameter (Cléaz-Savoyen 2005; Fiig et al. 2010). The share of customers willing to pay f_n who would also pay a fare f is

$$p(f) = e^{-\gamma(f/f_n - 1)}, \qquad \gamma = \frac{\ln 2}{\text{FRAT5} - 1}, \tag{14.4}$$

where FRAT5 is the fare ratio at which half of them sell up. Laurentian's ladder implies a FRAT5 of 1.52 on average, and willingness to pay rises over the booking period because late bookers are different people (Belobaba 2006a).

The consequences for availability are large. Take the Seville ladder with no fences, an early booking period dominated by leisure customers (FRAT5 of 1.3) and a late one (FRAT5 of 2.5). Early, only 49 percent of the customers willing to pay the V fare would pay Q, so opening V earns the most per customer. Late, 87 percent would pay Q and 58 percent B; the best lowest open class is then B, at CAD 529.5 per customer against CAD 419 with V open, and opening Q or V loses revenue even on an empty flight. A system that opens V late because V has the most bookings in its history sells at 419 to customers who would pay 919.

A system that forecasts each class from its own bookings makes exactly that mistake, and it compounds. If protection for the high fare is too low, the cheap fare stays open longer and some customers who would have paid the high fare buy the cheap one. The system records fewer high-fare bookings, forecasts less high-fare demand and protects even fewer seats. Revenue falls, the load factor rises, and the history confirms the forecast at every step. This is the **spiral-down effect**, formalized by Cooper, Homem-de-Mello and Kleywegt (Cooper, Homem-de-Mello, and Kleywegt 2006), and it is why leg revenue management lost almost all its value in the fenceless ePODS structure of Section 14.2.

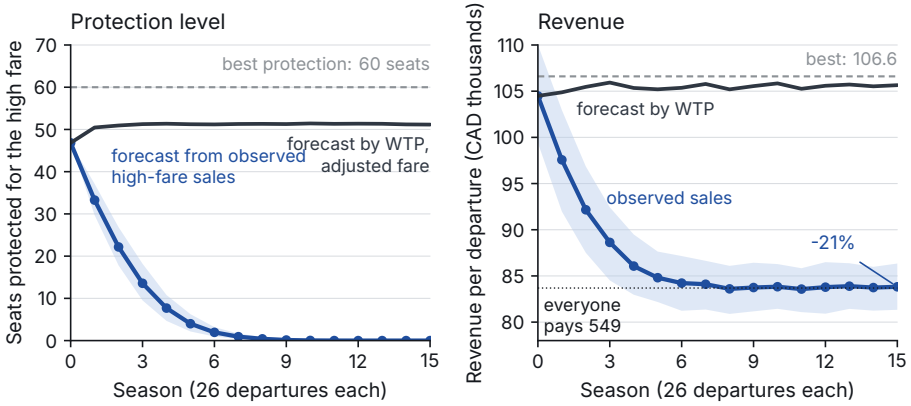


Figure 14.6 The spiral-down effect on the two-class flight of Example 14.5, season by season (26 departures per season, 300 simulated histories). Left: seats protected for the high fare. Right: revenue per departure. A system that forecasts high-fare demand from observed high-fare sales (blue, with the 10th to 90th percentile band) spirals to zero protection; one that forecasts by willingness to pay and uses the adjusted low fare (black) stays close to the best protection level.

WORKED EXAMPLE 14.5 Spiral down on a two-class flight

Take a stylized Seville cabin: 168 seats, a Y fare of CAD 1,249 and a Q fare of CAD 549 with no fences between them, on average 115 customers willing to pay only Q and 45 willing to pay Y, with leisure customers booking first (illustrative). Starting from a forecast inherited from a fenced past, Littlewood's rule protects 46.8 seats for Y. Business customers who arrive while Q is still open buy it, on average 12.2 of them, so observed Y sales average only 30.8. Read as Y demand, they cut the protection to 33.2 seats at the next revision, then to 22.1, 13.4, 7.2 and 3.6.

In the simulation of Figure 14.6, every history protects less than one seat by season 15, and revenue per departure falls from CAD 104,500 to CAD 83,800, the revenue when every customer pays CAD 549. The best protection level is 60 seats, earning CAD 106,603. A system that forecasts by willingness to pay, and charges the cheap fare for the buy-down it causes, settles near 51 seats and earns CAD 105,500.

The repairs work on both links of the loop. **Q-forecasting** estimates demand by willingness to pay: it divides each past departure's bookings by the share of lowest-fare customers who would have bought at the lowest fare that was open, then splits the total

by fare level using sell-up estimates (Belobaba 2006a). **Fare adjustment** charges each cheaper class for the revenue it loses through buy-down by replacing its fare with the marginal revenue of opening it (Fiig et al. 2010). In the two-class example the adjusted Q fare is CAD 275, half the published fare, and Littlewood's rule applied to it protects 54 seats instead of spiralling to none. **Choice-based** revenue management extends the idea to any model of how customers choose among the products offered (Talluri and van Ryzin 2004a).

A rival's open fare caps sell-up, so willingness to pay must be estimated conditional on competitors' availability (Belobaba 2006a). In PODS, an airline that alone adopted forecasting by willingness to pay gained 3.4 percent of revenue while its rival lost only 0.2 percent (historical simulation results) (Belobaba 2006a).

With **continuous pricing**, the airline can quote any price, not just the rungs of a ladder. For exponential willingness to pay, the optimal price is the opportunity cost of a seat, its bid price, plus a constant markup set by willingness to pay (Gallego and van Ryzin 1994). Thirty days before a Seville departure with 20 seats left and late-period customers, optimal dynamic pricing earns an expected CAD 18,376, against CAD 18,056 for the best fixed price and CAD 18,203 for the best use of the five ladder fares (illustrative). Continuous pricing adds 1.8 percent to the best fixed price, and the ladder already captures 0.8 points of it: a booking-class system is dynamic pricing with a coarse menu (Wittman and Belobaba 2019).

IN PRACTICE **Dynamic and continuous pricing in 2026**

The legacy system limits a carrier to 26 booking classes, one letter each, on a flight. IATA defines the newer layers: dynamic pricing (prices set from the shopping context without filing them in advance), continuous pricing (an unlimited number of price points) and dynamic bundling (offers assembled for each request) (International Air Transport Association 2021). ATPCO committed in 2022 to help airlines reach 80 percent of offers created dynamically by 2026, up from 6 percent, and reported 23 percent by mid-2023 (ATPCO 2022, 2023); no published figure for 2026 was found. In PODS simulations, continuous pricing raised revenue by up to 2.0 percent for a first mover (Szymański, Belobaba, and Papen 2021). For Air Canada, a vendor case study states that it uses a PROS real-time dynamic pricing product (PROS 2025); which forecasting and network optimization engines sit behind it is not public. A planner

should ask any system what opportunity cost it puts on a seat, at what level, and how often it is recomputed.

> **PITFALL**
>
> Reading class bookings as class demand. When fences are weak, the absence of Y bookings says nothing about how many customers would pay the Y fare if nothing cheaper were open; it says only that something cheaper was open. Demand data drawn from revenue management's history understate high-fare demand unless they were forecast by willingness to pay.

14.8 Summary

- Airlines sell one cabin at many prices because seats perish, one more passenger costs little, and travellers differ in willingness to pay. Fences make travellers sort themselves, and pay only if they hurt high-paying travellers much more than others.
- Under low-cost competition, fare structures moved from restricted to simplified fares, which raised load factors and cut revenue, and then to branded fares and ancillaries, which rebuilt the fences around services.
- Booking classes are nested; protection levels reserve seats for higher classes and set the booking limits of lower ones. Littlewood's rule protects a seat for the high fare while the chance of selling it high, times the high fare, exceeds the low fare: 9 seats for Y on YUL–SVQ.
- Seat control lowers loads, raises average fares, moves spill to the cheapest classes and values the last seat at CAD 268 to 295, well below the CAD 620 average fare.
- Overbooking accepts bookings while the chance that the cabin is already full stays below $c_{SP}/(c_{SP} + c_{DB})$: 193 bookings for 182 seats to Seville. Canadian compensation for an involuntary denial, CAD 900 to 2,400, depends on a rebooking delay that the schedule sets.
- Network control accepts an itinerary when its fare covers the sum of the bid prices on its legs: CAD 159, 109, 490 and 0 on Laurentian's bank, where the Paris leg is scarce. Bid prices value small changes; large ones need the network model re-solved.

- When fences weaken, customers buy down and class forecasts spiral down; forecasting by willingness to pay with fare adjustment repairs the loop, and continuous pricing sets each price at the opportunity cost of a seat plus a markup.

14.9 Further reading

Belobaba's 2006 lectures are short and readable: on airline economics and pricing (Belobaba 2006d, 2006o), flight leg revenue management (Belobaba 2006l), overbooking (Belobaba 2006m), origin-destination control (Belobaba 2006n, 2002) and willingness to pay (Belobaba 2006a). Talluri and van Ryzin (2004b) is the standard reference for the field. The full reference edition gives the derivations: price discrimination and fences in its Chapter 17, Littlewood's rule, EMSR and the dynamic program in its Chapter 18, the overbooking limit in its Chapter 19, the network linear program in its Chapter 20, and Q-forecasting, fare adjustment and dynamic pricing in its Chapter 21.

CHAPTER

15

Where Planning Meets Revenue Management

IN THIS CHAPTER

- What planning and revenue management each control, on which horizons, and what passes between them.
- Bid prices and displacement costs as the value of a seat, and what their spread across departures shows.
- An upgauge and a new leg valued several ways, and why the answer depends on the control that sells the seats.
- RM-aware spill, and when one more seat loses money.
- Consistent forecasts, how revenue management reacts to a schedule change, and why a first season misleads.
- The questions to ask the revenue management team.

DRAWS ON Belobaba, Introduction to the airline planning process, Airline demand analysis and spill modeling, and O&D control: what have we learned? (MIT 16.75J, 2006); handbook; Abramovich (2013); Talluri and van Ryzin (2004)

A network planner never sells a seat. Every number a planner uses to judge capacity, from a flight's load factor and average fare to its spill and the revenue of a new route, comes out of a revenue management system that decides, request by request, who gets the seats the plan creates. The plan draws the box; revenue management fills it.

This chapter joins the two halves of the book. Chapter 7 valued a seat with an average fare and the probability of filling it, Chapter 10 measured network contribution with prorates and displacement, and Chapter 14 computed bid prices for Laurentian's evening bank at Montreal. Here the same capacity changes are valued all these ways at once. One claim runs through the chapter: the value of capacity is not a property of a flight. It depends on the demand the flight faces, the network around it and the control that sells it.

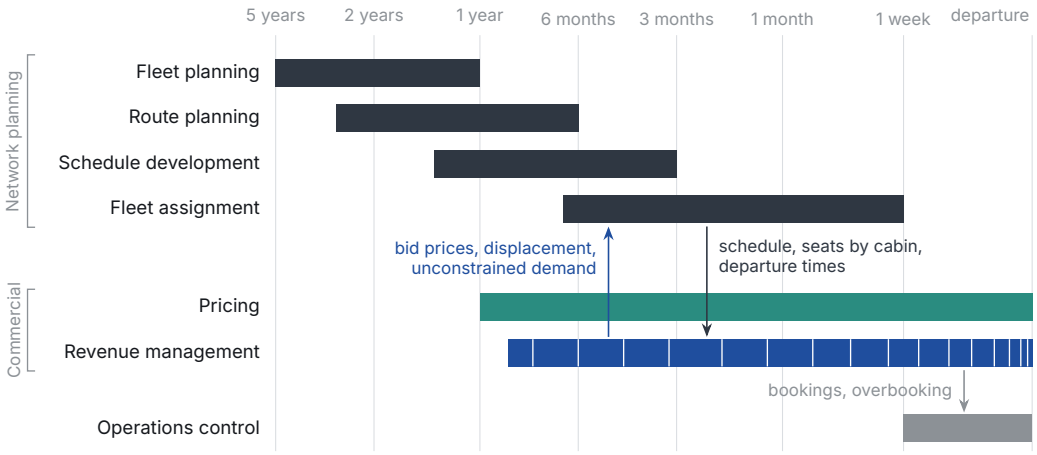


Figure 15.1 What network planning and the commercial functions control, and when. Bars are typical horizons before departure on a roughly logarithmic scale; white ticks mark revenue management's re-optimizations. Arrows are the main flows of information between the functions.

15.1 Two functions, two horizons

Belobaba describes revenue management as the last decision in the planning chain: given a scheduled flight, its capacity and its prices, how many bookings to accept by fare type so as to fill each seat with the highest possible revenue (Belobaba 2006c). Everything before it creates or reshapes capacity (Chapter 1). Figure 15.1 sets the two families of decisions side by side.

Planning controls what exists: the fleet, routes, frequencies, departure times, connection banks and, through fleet assignment, the seats in each cabin of each departure. Most of these decisions are fixed before the first booking arrives. Revenue management controls who gets what exists: which fare products and itineraries are open at each moment, the protection levels or bid prices that decide this, and the overbooking authorization (Chapter 14). It works over a booking period of about 300 days and revises its controls continuously (Belobaba 2006l). Pricing, the third partner, sets the fare products and their restrictions.

Three asymmetries shape the interface. **Information**: the planner decides early, on market forecasts that revenue management later replaces with bookings in hand. **Objective**: for revenue management the cost of a seat is sunk, while the planner must weigh revenue against avoidable cost. **Granularity**: revenue management sees each departure, fare product and day before departure; plan-

ning works with seasons, markets and averages. Each asymmetry produces numbers that disagree for good reasons.

Planning hands revenue management a schedule, seats by cabin and departure times, and with them the itineraries that can connect. Revenue management hands back what planning cannot see: demand by origin-destination and fare product with the censoring removed, the fare mix it can achieve, and the value of a seat on each leg. The handbook describes the division in the same terms: planning sets much of the capacity and schedule, revenue management controls how the product is sold over time, and a bid price estimates the opportunity value of scarce capacity (Network Planning Handbook 2026).

DEFINITION 15.1 Value of capacity

Fix a demand forecast and a booking control u: first come, first served, a revenue management method, or the ex post optimum, which knows each departure's demand in advance. Let $R_u(C)$ be expected network revenue per departure when the legs have capacities C. The **revenue value** of a change from C to C' under u is $R_u(C') - R_u(C)$. Its **contribution value** subtracts the change in avoidable cost.

A seat has no value of its own. Its value is a change in what the network earns, and that change depends on the demand, the rest of the network and the control that sells the seat.

15.2 Bid prices: the value of a seat, departure by departure

Revenue management already computes a value of capacity. Chapter 14 found it for Laurentian's bank: two A220-300 feeders of 137 seats from Quebec City and Boston, a 298-seat 787-9 to Paris and a 182-seat A321XLR to Seville, with a flexible (H) and a restricted (L) fare in each of eight O&D markets. The **deterministic linear program** (DLP) allocates seats to these sixteen products at mean demand and earns CAD 365,949 per bank. The value of one more seat, the **bid price**, is CAD 159 on YQB–YUL, 109 on BOS–YUL, 490 on

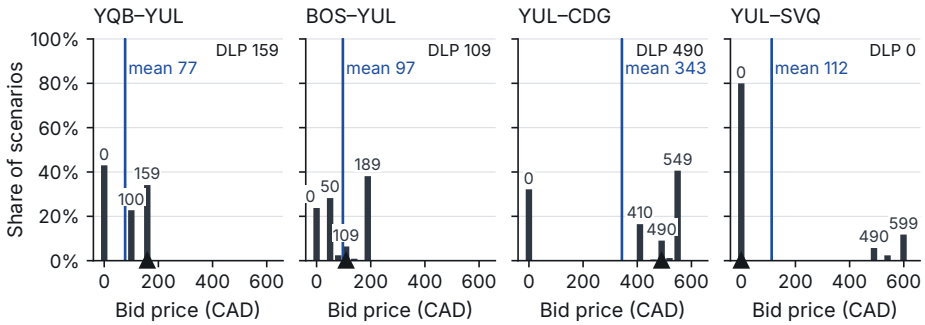


Figure 15.2 Ex post bid prices on Laurentian's bank: for each of 4,000 demand scenarios, the value of one more seat on each leg when seats are allocated knowing that departure's demand. The vertical line is the mean, the triangle the DLP bid price at mean demand.

YUL–CDG and 0 on YUL–SVQ. A request is worth accepting when its fare covers the sum of the bid prices along its path, its **displacement cost**: the revenue the network gives up elsewhere to carry it.

Each bid price is a fare difference, read off the passenger at the margin. A Quebec City seat is worth the CAD 159 of the restricted local who would take it. A Paris seat is worth 649 − 159 = 490: the restricted Quebec City–Paris fare minus the value of its feeder seat. A Boston seat is worth only 599−490 = 109, because every Boston local is already carried and the best use of one more seat is a restricted Boston–Paris connection, which costs a Paris seat. These numbers say where capacity is scarce (Paris, by a factor of three or more) and screen proposals: a new connection to Paris must earn at least the Paris bid price plus its feeder's.

Two limits matter. A bid price is a derivative, valid for a few seats (Appendix A.9). And the DLP replaces random demand by its mean, so it describes an average departure that never occurs. Solving the allocation again for each of many demand scenarios, with realized demand in place of the mean, gives an **ex post bid price** for each departure: the value of one more seat to a controller who knew that departure's demand in advance. Over 4,000 scenarios these average CAD 77, 97, 343 and 112 on the four legs. Figure 15.2 shows their spread.

The distributions are lumpy, because each departure's bid price is set by whichever passenger is marginal on it. On YUL–CDG it is zero on 32 percent of departures (the flight has room), CAD 549 on 41 percent (a restricted local is marginal), CAD 410 on 17 percent (599 − 189: a restricted Boston–Paris connection whose Boston seat could go to a local) and CAD 490 on 9 percent. YUL–SVQ is worth nothing on 80 percent of departures and CAD 599 on 12 per-

cent. Even the mean overstates large additions: for 32 extra Quebec City seats it suggests about CAD 2,470, twice the CAD 1,257 they are worth to a clairvoyant controller (Example 15.2).

For planning, the distribution says more than the mean. A leg whose bid price is zero on most departures and high on a few, like Seville, needs seats on its peak departures, which argues for a seasonal or day-of-week gauge pattern rather than a permanent upgauge. A leg whose bid price is high on most departures, like Paris, is short of seats almost always. Ask revenue management for bid prices by departure and by days before departure, not for a seasonal average.

15.3 Valuing an upgauge and a new leg

The revenue value is half of a capacity case; the other half is avoidable cost. With ticket revenue and trip costs only, the incremental network contribution per bank of a change from C to C' under control u is

$$\Delta\mathrm{NC} = [R_u(C') - R_u(C)] - \sum_l \left[\Delta\mathrm{TC}_l + c_l^{\mathrm{pax}}\,\Delta\mathbb{E}[Y_l]\right], \qquad (15.1)$$

where $\Delta\mathrm{TC}_l$ is the change in the trip cost of leg l (the whole cost of a new flight, the difference between types for a gauge change), c_l^{pax} the cost per passenger on leg l, and $\Delta\mathbb{E}[Y_l]$ the change in its expected load. Ownership is compared separately (Chapter 10).

The revenue bracket is the hard part, and there are eight ways to estimate it. **Bid price times seats** multiplies the bid price by the seats added. The **spill model** of Chapter 7 multiplies the extra expected load by an average fare of the leg's passengers, either their full O&D fares or their fares prorated to the leg by straight-rate prorate (Chapter 10). The **DLP re-solved** with the new capacity and the **ex post optimum**, the average over scenarios of the allocation that knows each departure's demand, are network optimizations; the second bounds what any control can earn. The last three are **booking simulations** of 2,000 departures, in which requests arrive one at a time, restricted ones early and flexible ones late, under bid prices recomputed 32 times or 8 times during the booking period, or under first come, first served.

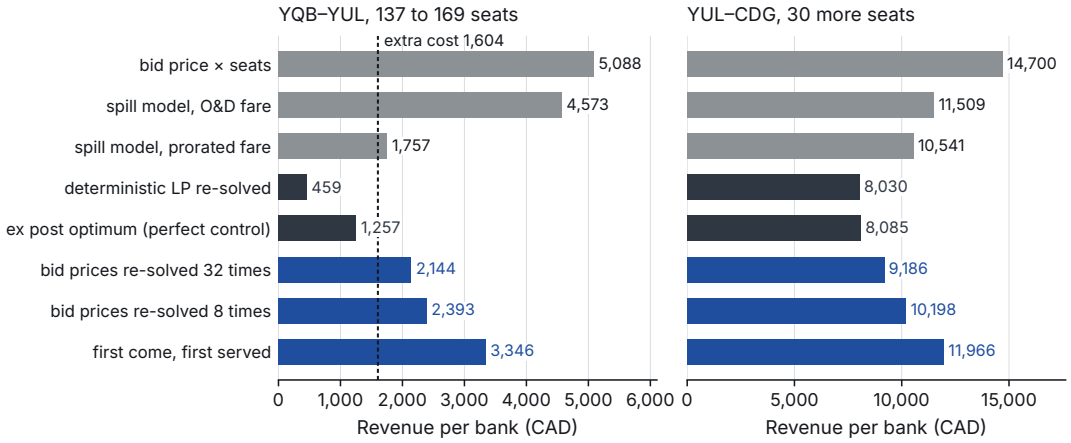


Figure 15.3 The revenue value of the same extra seats under eight methods, for a 737-8 instead of an A220-300 on YQB–YUL (left) and thirty more seats on YUL–CDG (right). Grey: leg arithmetic. Dark: network optimizations. Blue: booking simulations under three controls. The dashed line is the extra cost of the larger aircraft.

WORKED EXAMPLE 15.1 Thirty more seats to Paris

Give YUL–CDG 328 seats instead of 298. Its demand has mean 317 and standard deviation 36.1.

Leg arithmetic. The bid price gives $490 \times 30 = 14{,}700$ dollars per bank. The spill model adds 16.3 passengers, worth CAD 11,509 at the leg's average O&D fare and 10,541 at its average prorated fare.

The network. Re-solving the DLP gives CAD 8,030. The extra seats admit 16 restricted Boston–Paris and 3 restricted Quebec City–Paris connections, but both feeders are full, so as many restricted locals lose their feeder seats. Each extra Paris passenger earns $599 - 189 = 410$ or $649 - 159 = 490$, and the last 11 seats stay empty. The ex post optimum gives 8,085.

The controls. Bid prices recomputed 32 times value the seats at CAD 9,186, recomputed 8 times at 10,198, and first come, first served at 11,966, mostly from late-booking flexible passengers who used to find the flight full.

The three tools a planner reaches for first, the bid price and the two spill models, give three of the four largest values; the prorated spill model lands near the eight-solve simulation only because its errors offset. The control-based numbers are ordered

by the quality of the control: the better the control, the less the extra seats are worth.

That ordering is not an accident. A control that loses revenue on the existing seats, by selling them to the wrong passengers or leaving them empty, gains more from extra seats, because the extra seats repair part of its mistakes. In these simulations capacity and control are substitutes, so capacity must be valued under the control that will sell it.

WORKED EXAMPLE 15.2 Upgauging the Quebec City feeder

Replace the A220-300 (137 seats) by a 737-8 (169 seats) on YQB–YUL. The extra cost is CAD 1,400 per trip with Laurentian's cost model plus about CAD 204 of passenger costs on the legs whose loads change: CAD 1,604 per bank.

Bid price and DLP. The bid price times the seats says $159 \times 32 = 5{,}088$ dollars. Re-solving the DLP says CAD 459: the first extra seat admits the one refused restricted local (159), the next three admit three restricted Quebec City–Paris connections, each worth only $649 - 549 = 100$ because it now displaces a local Paris passenger, and the rest are worthless at mean demand. With random demand the extra seats fill on busy departures, and the ex post optimum gives CAD 1,257.

Spill model. Demand has mean 141 and standard deviation 19.7, and expected load rises by 9.33 passengers: CAD 1,757 at the prorated average fare of CAD 188.3, but 4,573 at full O&D fares, because nearly half the leg's passengers connect to Europe.

Simulated controls. Bid prices recomputed 32 times give CAD 2,144, recomputed 8 times 2,393, and first come, first served 3,346. Under eight recomputations the extra seats sell, per bank, 5.0 more local Quebec City tickets, 2.9 more Quebec City–Paris and 0.5 more Quebec City–Seville connections, and 2.1 fewer local Paris tickets, displaced by those connections. By straight-rate prorate, CAD 1,729 of the gain (72 percent) appears on YQB–YUL; the rest is on Paris (354), Seville (298) and Boston (12).

The decision. Against CAD 1,604, the upgauge loses CAD 347 per bank under perfect control but gains 540, 789 and 1,742 un-

der the three simulated controls; the bid price times the seats would have approved it by a wide margin. The business case for a gauge change contains a hidden assumption about revenue management: if the team is about to improve its control, the case weakens.

PITFALL

Valuing capacity under the wrong control. The ex post optimum assumes a controller who knows demand in advance and understates what extra seats are worth to a real one; first come, first served overstates it. Ask revenue management to simulate the control it actually runs, with the forecasts it will actually have.

A new leg is the largest capacity change, and part of its cost falls where no report of the new flight looks. Suppose Laurentian adds a daily 182-seat A321XLR from Montreal to Lisbon to the evening bank, with six products scaled from Seville's (a market 10 percent larger, fares in the ratio of 600 to 620): mean demand of 177 passengers per bank, of whom 54 connect from Quebec City and Boston.

WORKED EXAMPLE 15.3 Adding Lisbon to the evening bank

The network at mean demand. Re-solving the DLP with five legs gains CAD 119,937 per bank. Every Lisbon product clears its path bid price by a wide margin (the narrowest, the restricted Quebec City–Lisbon fare, by $679 - 159 = 520$), so all Lisbon passengers are accepted, paying CAD 129,243. But the 54 connecting ones need seats on two full feeders, and the program refuses 25 restricted Quebec City locals (CAD 3,975) and 29 restricted Boston locals (CAD 5,481); a reshuffle of Paris connections gives back CAD 150. Displacement costs CAD 9,306, and the Boston bid price rises from 109 to 189.

Random demand and revenue management. The ex post optimum gains CAD 117,607, and bid prices recomputed 8 times gain 112,813 after CAD 7,563 of displacement. First come, first served gains only 100,082, because the Lisbon connections, which book early in their restricted fares, also take feeder seats from Paris and Seville connections.

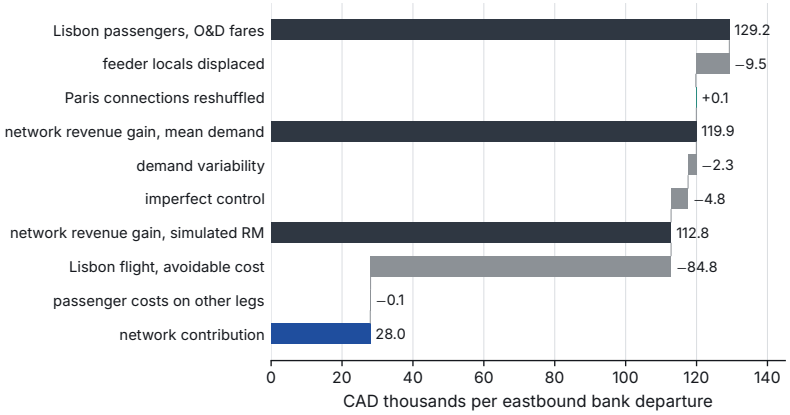


Figure 15.4 From the O&D fares of the Lisbon passengers to the network contribution of the new flight, per bank: the DLP at mean demand, then the ex post optimum and simulated bid-price control, then the avoidable cost of the Lisbon flight and the passenger costs on the other legs.

Contribution. The Lisbon trip cost at the simulated load is CAD 84,761 and passenger costs on the other legs rise by CAD 89, so by (15.1) the network contribution is 112,813 − 84,761 − 89 = 27,963 dollars per bank, before ownership (Figure 15.4).

The leg view. The flight's own report, at a straight-rate prorated average fare of CAD 673.4, shows a contribution of CAD 28,484, within 521 of the network answer, but only because two errors cancel: it charges nothing for the displaced locals and credits the feeders with a large share of each connecting fare. With mileage prorate it shows CAD 35,565.

The displacement falls on the feeders, and their reports hide it too: at mean demand the Quebec City and Boston flights carry as many passengers as before and, with straight-rate prorate, even report slightly more revenue (CAD 252 and 318 per bank), because the Lisbon connections that take the locals' seats are credited with prorated shares close to or above the local fares. Only the network view shows the CAD 9,456 of local fares given up. And the new leg is worth more under better control, the opposite of the upgauges: added demand competes for scarce feeder seats, and allocating scarce seats is what revenue management does. Extra capacity substitutes for control; extra demand calls for more of it.

15.4 RM-aware spill

The classical spill model of Chapter 7 values a seat at the average fare times the probability of filling it, as if seats were sold first come, first served and the spilled customers were a random sample. Revenue management breaks that assumption three ways. It spills from the bottom of the fare ladder, so a spilled passenger is worth less than the average fare. It turns away low-fare requests even on departures that leave with empty seats. And customers respond to what is open: when more seats keep cheap fares open longer, some customers who would have paid more **buy down**.

Abramovich studied these effects with MIT's Passenger Origin-Destination Simulator (Abramovich 2013). He argues that revenue management changes both the number and the value of spilled passengers, that in less restricted fare structures under traditional revenue management adding capacity could reduce revenue because passengers bought down, and that forecasting and optimization that account for willingness to pay reduced these losses (Abramovich 2013).

A two-fare flight shows the mechanism. The flexible fare is CAD 1,200, with demand normal with mean 40 and standard deviation 12; the restricted fare is CAD 600, with demand normal with mean 150 and standard deviation 40; restricted customers book first. Littlewood's rule (Chapter 14 and Appendix A.7) protects 40 seats for the flexible fare: at a fare ratio of one half it protects the median of flexible demand. With **fenced** fares each customer buys the product meant for them. **Without fences** every flexible customer buys the restricted fare while it is open.

Follow one extra seat, which raises the restricted booking limit by one. With fences it earns CAD 600 when restricted demand would have exceeded the limit, or CAD 1,200 when it saves a flexible customer from being spilled. Without fences it can also do harm. When the restricted fare closes only because flexible customers have bought it, and nobody is spilled, the extra seat keeps it open for one more flexible customer, who pays CAD 600 instead of 1,200: a loss of CAD 600 on a seat that carries no extra passenger.

At 150 seats (Figure 15.5) the spill model says CAD 604, the fenced flight earns about 530, and the unfenced flight 448 with fixed protection or 454 with protection chosen for buy-down. At 180 seats the four values are 432, 406, 232 and 257.

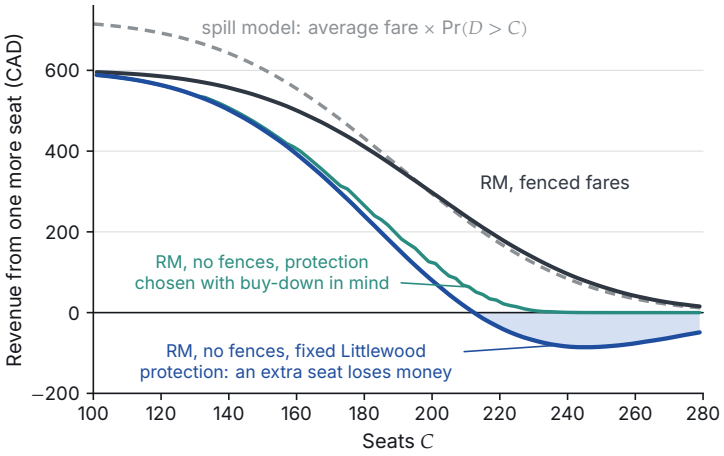


Figure 15.5 The revenue from one more seat on the two-fare flight. Dashed: the classical spill model. Dark: revenue management with fenced fares. Without fences, with protection held at 40 seats (cobalt), extra seats lose money from about 213 seats; with protection chosen with buy-down in mind (teal) they are worth little but never negative. Simulated with 400,000 demand draws.

RESULT 15.1 Re-optimized protection never makes capacity harmful

If protection is re-optimized for each capacity, with buy-down anticipated, expected revenue never falls when capacity grows. The airline can always add the extra seat to the protected seats: restricted sales and buy-down stay as they were, and flexible customers have one more seat.

WORKED EXAMPLE 15.4 When one more seat loses money

Take the unfenced flight with 220 seats and protection held at 40, so the restricted fare closes after 180 sales. In 400,000 simulated departures the restricted fare reaches its limit on restricted demand alone on 22.6 percent of departures; flexible customers are spilled on another 4.0 percent; and on 32.8 percent the restricted fare closes after some flexible customers have bought it, with nobody spilled.

One more seat earns the restricted fare in the first two cases and loses the fare difference in the third:

$$600 \times (0.226 + 0.040) - 600 \times 0.328 = -37 \text{ dollars},$$

and a direct simulation gives −39. The spill model values the same seat at CAD 172, and the fenced flight earns 182 from it. With protection re-optimized for buy-down, rising from 60 to 61 seats, it is worth CAD 24.

On flights with weak fences the classical spill value can overstate the value of capacity several times over, and in the wrong range it gets even the sign wrong. The remedy for buy-down belongs to revenue management and pricing, through fences or through forecasts and optimizers that anticipate buy-down (Chapter 14), not to the fleet plan.

PITFALL

Concluding that capacity destroys revenue. Negative marginal revenue comes from weak fences combined with a control that ignores buy-down. The handbook presents Abramovich's result as a model-specific warning against mechanical capacity arithmetic, not as a claim that capacity normally destroys revenue (Network Planning Handbook 2026).

This is why the planner's spill and the revenue management system's view of the same flight must agree: otherwise the aircraft is sized for one airline and its seats are sold by another. RM-aware spill adjusts the formulas of Chapter 7 in three ways. Value spilled passengers at the fares revenue management would actually refuse, close to the lowest open fares. Count separately the demand rejected on departures with empty seats: it is lost by choice, and capacity does not recover it. And include buy-down and sell-up, whose size depends on the fences and on whether forecasts and optimizer account for willingness to pay. The booking simulations of Section 15.3 make all three adjustments at the network level, which is why they are the numbers to ask for.

15.5 Consistent forecasts

Planning and revenue management forecast the same travellers for different decisions. Many disputes about "the demand" are differences of definition.

> **DEFINITION 15.2 Constrained and unconstrained demand**
>
> The **unconstrained demand** for a product is the number of requests it would receive if it were open throughout the booking period. **Constrained demand**, or bookings, is what was sold given capacity and the availability that revenue management allowed. The difference is **spill**, refused because seats ran out, plus **RM rejections**, refused while seats remained to protect them for others.

With buy-down and sell-up, unconstrained demand for a product depends on which other products are open, so it is defined relative to an availability policy (Chapter 14). Planning needs demand that capacity has not censored; revenue management's bookings are censored by both mechanisms and understate the demand a larger aircraft would face (Chapter 7). Table 15.1 lists what to reconcile.

Table 15.1 Demand quantities that planning and revenue management share, and how to keep them consistent.

Quantity	Planning uses it for	Revenue management uses it for	Consistency check
Unconstrained O&D demand	market size, spill	product forecasts	same definition, same vintage
Leg demand	gauge, fleet assignment	booking limits, overbooking	build from O&D demand, never by adding legs
Fare mix	revenue per passenger	protection, bid prices	the next passenger's fare is not the average
Bookings and loads	post-launch review	forecast history	unconstrain before comparing
Recapture, sell-up, buy-down	spill cost	choice-based forecasts	same customer-choice assumptions

Two rows deserve numbers. **O&D and leg.** At mean demand Laurentian's bank faces 621 O&D travellers, who make 772 leg boardings, because 151 of them connect; the handbook warns that the

two measures cannot simply be summed into one hierarchy (Network Planning Handbook 2026). **Horizon.** A gauge decision taken six months ahead faces both day-to-day variation and the error of the forecast itself, which revenue management's forecasts close to departure barely carry. If the YQB–YUL forecast is uncertain by 15 percent of its mean, the standard deviation of demand rises from 19.7 to 28.9 passengers, and the 32 extra seats of the upgauge carry 11.1 passengers on average instead of 9.3. The handbook adds the fare row: do not assume that the observed average fare is the next passenger's (Network Planning Handbook 2026).

> **PITFALL**
>
> Summing leg forecasts. Adding the forecasts of a bank's legs counts every connecting traveller twice. Build leg demand from O&D demand (Chapter 6).

15.6 How revenue management reacts to a schedule change

A schedule change reaches revenue management as new capacities, and the system responds at once. Bid prices move: thirty more Paris seats send the Paris bid price from 490 to zero and raise Boston's from 109 to 189 at mean demand. Availability follows, opening restricted fares more often on the enlarged leg and its connections and sometimes closing them elsewhere, which is why the gain from the Quebec City upgauge is split across four legs.

Changes to flights already on sale add a constraint. A smaller aircraft turns every booking beyond the new authorization level into a potential denied boarding, a cost that belongs in the case (Chapter 14). A larger one gives revenue management seats to sell late, mostly to customers who pay more.

Forecasts lag the change. They are built from bookings censored by the old capacity, so the first season of a larger aircraft starts with forecasts that describe the smaller one. If unconstraining is weak, lower forecasts reduce protection, more cheap seats are sold, and the history drifts down further: the **spiral-down** effect (Chapter 14). A new route has no history at all and is sold on forecasts borrowed from similar markets (Chapter 6) while its demand ramps up (Chapter 10).

The first season is therefore a poor measure of a change. Revenue management is still learning, demand is ramping up, markets and

competitors move, and the leg's report misses the network effects. Noise adds to this: in the simulation the bank's revenue has a standard deviation of CAD 18,597 per departure, so the difference between two seasons of 180 departures has a standard error of about CAD 1,960, comparable to the CAD 2,393 the Quebec City upgauge is worth.

WORKED EXAMPLE 15.5 Reading the first season

Suppose the Quebec City upgauge has flown for a season. The leg's prorated revenue rose from CAD 25,000 to 28,230 per bank (12.9 percent), and a comparable feeder outside the bank rose from 21,000 to 22,260 (6.0 percent). The numbers are illustrative, built from 6 percent market growth and the simulated leg effect of CAD 1,729.

Before and after. The leg gained CAD 3,230, twice the cost of the upgauge, but CAD 1,500 of it is market growth.

With a control group. Without the upgauge the leg would have grown like the control, to $25{,}000 \times 1.06 = 26{,}500$ dollars, so the effect is $28{,}230 - 26{,}500 = 1{,}730$. This assumes the two legs would have moved alike without the change.

The network. The simulated network gain is CAD 2,393, of which 664 appears on other legs. The leg's report understates the gain by 28 percent; the naive comparison overstates it by 35 percent.

Read a first season with a control group, at the network level, with its uncertainty attached. Such designs are used in practice: an Air Canada presentation at the 2026 AGIFORS scheduling and strategic planning meeting used difference in differences to measure the effect of the closure of Russian airspace on flights between North America and India (AGIFORS 2026).

15.7 Questions for the revenue management team

The handbook lists the questions a planner should put to revenue management: which cabins and departures constrain demand, what incremental capacity is worth, where rejected demand would be recaptured, which fare and cabin assumptions planning should use, and how fares, availability and displacement would change after a schedule adjustment (Network Planning Handbook 2026). Table 15.2 expands them into a checklist.

Table 15.2 Questions for the revenue management team, expanded from the handbook's list.

Question	Why it matters	See
Which legs and departures constrain demand, and how often?	capacity may be scarce only on peaks	Section 15.2
What are the bid prices, at what level, and how often recomputed?	an average hides the departures that matter	Section 15.2
What is the change worth under random demand and the control actually used?	bid price times seats overstates	Section 15.3
How will availability and displacement change on other legs?	the effects fall on other flights	Example 15.3
Where would rejected demand be recaptured?	recapture needs seats elsewhere	Chapter 7
Which forecast is unconstrained, how, and of which vintage?	bookings understate demand	Section 15.5
How much buy-down and sell-up does the system assume?	weak fences devalue extra seats	Section 15.4
How long will forecasts take to adapt?	the first season is sold on old forecasts	Section 15.6
For flights on sale, how many bookings are affected?	downgauges create oversales	Section 15.6

The horizons of Figure 15.1 imply four meeting points: the seasonal schedule build, changes to schedules already on sale, late changes of aircraft type (which revenue management values best, because its forecasts are then sharpest) and the post-implementation review. The handbook's plan for a new analyst's second month asks for this collaboration explicitly: work with forecasting and revenue management on demand and displacement, and with Finance on the contribution bridge (Network Planning Handbook 2026). For Air Canada, public sources say that the Chief Commercial Officer's portfolio includes both the network and the airline's yield management and pricing strategies (Air Canada 2025), and a vendor case study names a Director of Revenue Optimization (PROS 2025).

IN PRACTICE **The interface in 2026**

Commercial planning systems chain demand, market-share, spill and recapture, and cost and revenue models into O&D profitability; Lufthansa Systems describes its NetLine/Plan product in these terms (Lufthansa Systems 2026). At the 2026 AGIFORS scheduling and strategic planning meeting, Stanford University presented a research prototype linking forecasting, scheduling, fleet assignment, crew and revenue management through one shared demand model (AGIFORS 2026). On the selling side, IATA describes continuous pricing with unlimited price points instead of at most 26 booking classes (International Air Transport Association 2021), and a vendor case study reports that Air Canada uses a PROS real-time dynamic pricing product (PROS 2025). The planner's question does not change: what does a seat on this departure displace?

15.8 Summary

- Planning creates capacity over years to weeks; revenue management decides who gets it over about 300 days of bookings. The value of capacity depends on the demand, the network and the control that sells it.
- Bid prices identify scarce legs and value small changes. They are fare differences that jump from departure to departure; ask for them by departure, not as a seasonal average.
- The same extra seats are worth CAD 8,030 to 14,700 per bank on the Paris leg and 459 to 5,088 on the Quebec City feeder, depending on the method; bid price times seats is the most optimistic. The better the control, the less extra seats are worth.
- A new long-haul leg displaces locals on full feeders, and neither its report nor theirs shows it: Lisbon contributes about CAD 28,000 per bank after CAD 7,563 of displacement.
- On flights that revenue management controls, the classical spill model misprices capacity; with weak fences and protection that ignores buy-down, extra seats can lose money.
- Forecasts must agree on what is unconstrained, on O&D versus leg, on vintage and on fares. A first season misleads; read it with a control group, at the network level.

15.9 Further reading

Belobaba's introduction to the planning process (Belobaba 2006c), his spill lecture (Belobaba 2006b) and his review of PODS research on O&D control (Belobaba 2002) are the library's sources for the two halves joined here; the handbook (Network Planning Handbook 2026) is the source of Table 15.2. Abramovich (2013) is the essential study of spill under revenue management, and Talluri and van Ryzin (2004b, ch. 3) give the theory of network bid prices. The full reference edition derives the bounds on bid-price valuations and the marginal seat with and without fences in its Chapter 22, and solves and simulates the bank in its Chapter 20.

PART V

Practice

The last part turns from models to the conditions in which they are used. Chapter 16 describes the industry as it stands in October 2026: pricing technology, planning systems, aircraft supply, carbon rules, the Canadian market and Air Canada's public position. Chapter 17 assembles the analyst's working toolkit: the data, the workflow, the checks and the one-page memo that carries a recommendation.

CHAPTER

16

The Industry in 2026

IN THIS CHAPTER

- Planning systems in 2026, and how a fuel shock re-cuts a plan.
- Engines and delivery backlogs: why aircraft time is scarce and what a late aircraft costs.
- The A321XLR and the thin long-haul markets a long-range narrowbody opens.
- The carbon rules that enter route economics, and why their cost depends on the route and the direction of flight.
- Continuous pricing, offers and NDC, and how to read a vendor's claim.
- The Canadian market, its airports and Air Canada's public position in the autumn of 2026.

DRAWS ON Air Canada, Second quarter 2026 MD&A (11 August 2026); Competition Bureau, Cleared for take-off (2025); Lufthansa Systems, NetLine/Ops ++ release (2025); handbook; public sources dated up to 28 October 2026

The methods in this book come mostly from teaching material of 2003 and 2006. Since then the industry has changed which aircraft it can buy, which regulations price its emissions, how it sells seats and who it competes with. None of these changes repeals the economics of the earlier chapters, but each changes an input, a constraint or a tool. This chapter is a dated snapshot taken at the end of October 2026, just before Air Canada was due to report its third-quarter results on 29 October (Network Planning Handbook 2026). Facts of this kind age quickly: recheck them before you rely on one at work.

Industry news arrives as press releases, vendor brochures and regulatory filings, each written by someone with an interest in how you read it. Separate what is measured from what is claimed, and translate each announcement into the question it raises for a schedule: which input moves, which constraint binds, which routes change rank.

16.1 Network planning practice and tools

The planning systems of 2026 run the chain of models this book describes. Lufthansa Systems describes its NetLine/Plan product as an engine of demand and market share, spill and recapture, and cost and revenue models that evaluates a scenario in under 30 seconds across up to 250,000 markets (Lufthansa Systems 2026). That is the chain of Chapters 4, 5, 7 and 10; what changed is speed, scale and calibration data, not structure.

At Air Canada, Lufthansa Systems confirms only the operations side: the airline went live with NetLine/Ops ++, its operations control system, by January 2025 (Lufthansa Systems 2025a). A June 2026 posting for a schedule optimization analyst, reproduced by a job aggregator, places a Scheduling and Network Planning team in the Network Planning branch, describes schedule adjustments from 3 to more than 300 days before departure using profitability and yield measures, fleet assignment, the integration of regional flying and slot portfolio management, and lists Sabre Schedule Manager as an asset (ResumeGeni 2026). Such traces are clues, not an organization chart; ask which system is authoritative for published schedules and who owns each model input (Network Planning Handbook 2026). The 2026 AGIFORS Schedule and Strategic Planning program, with an Air Canada presentation on aircraft routing under uncertainty (AGIFORS 2026), extends the models of Chapters 11 to 13 rather than replacing them.

IN PRACTICE **Re-cutting a plan in a fuel shock**

After conflict in the Middle East disrupted shipping through the Strait of Hormuz in the first half of 2026 (Air Canada 2026d), jet fuel prices surged; IATA's June outlook assumed jet fuel at US$152 per barrel for the year, almost 70 percent above 2025 (IATA 2026d). Air Canada suspended its full-year guidance on 30 April 2026 (The Canadian Press 2026) and reinstated it on 11 August with capacity growth cut from 3.5 to 5.5 percent to 2.25 to 3.25 percent, on jet fuel assumed at about CAD 1.38 per litre in the third quarter against CAD 0.90 originally (Air Canada 2026c). In planning terms, a fuel shock raises the avoidable cost of every flight in proportion to its fuel burn, so it lowers the contribution of long, thin and low-yield flying first. The re-cut is a fresh ranking of flying by contribution per aircraft-hour, against a shadow price of aircraft time (Section 16.2) that the cuts themselves lower.

16.2 Supply constraints: engines, deliveries and the long-range narrowbody

For most of the history of network planning, a planner could assume that the aircraft in the fleet plan would exist. In 2026 that assumption fails in two ways.

The first is engines. Durability problems with Pratt & Whitney's geared turbofan (GTF) engines grounded about 650 A320neo-family aircraft at the peak; in July 2026 Pratt said the issues should be behind it in 2027 or 2028 (Hamilton 2026). On the A220, whose PW1500G engines belong to the same family, Airbus and Pratt said in June 2026 that no aircraft would remain grounded for engine reasons by the end of the year (Leeham News 2026). Air Canada's A220-300s fly PW1500G engines (Air Canada n.d.a) and it chose GTF engines for its A321XLRs (Aviacionline 2022); Air Transat still cited GTF problems in the quarter to 31 July 2026 (Transat A.T. 2026) and pushed its first A321XLR deliveries back to 2027 (FlightGlobal 2025).

The second is deliveries. IATA counted an order backlog of 18,100 aircraft in May 2026, almost 60 percent of the active fleet, and estimated that deliveries had fallen about 5,600 aircraft short of the pre-pandemic trend; airlines absorbed the shortage through higher utilization, delayed retirements and higher load factors, and the average fleet age rose above 15 years (IATA 2026d). Air Canada reduced its 787-10 order from 18 to 14 in November 2025 (Air Canada 2026d).

The value of a late aircraft Scarcity has a price, and a planner should know how to compute it.

> **DEFINITION 16.1 Shadow price of aircraft time**
>
> The **shadow price** of an aircraft over a period is the change in the network's best achievable contribution when one more aircraft of that type is available for the period, all other plans being re-optimized. It is the opportunity cost of using the aircraft for any particular flying, and the hurdle every use of it must clear.

RESULT 16.1 A missing aircraft costs the marginal line

Rank the candidate lines of flying for a fleet by their contribution over a period, each line needing one aircraft. With K aircraft the best plan flies the K most valuable lines. Losing one aircraft costs exactly the contribution of the K-th line, the marginal line; gaining one adds the next line down; the shadow price lies between the two. A delivery that slips by d periods therefore costs d times the marginal line, not d times the average line.

The rule follows from the ranking: the airline always drops its least valuable flying first (the full reference edition proves it in its Chapter 23).

WORKED EXAMPLE 16.1 An A321XLR delivered eight weeks late

Laurentian plans six A321XLRs for the summer and has seven candidate lines of flying for them, with illustrative contributions of CAD 70,000, 58,000, 51,000, 44,000, 37,000, 29,000 and 22,000 per week. With six aircraft it flies the first six lines and earns CAD 289,000 a week.

An engine shortage delays the sixth aircraft by eight weeks. By Result 16.1 the cost is $8 \times 29{,}000 = $ CAD 232,000: the airline drops its least valuable line. Valuing the delay at the average line, CAD 48,167 a week, would give CAD 385,000 and overstate the loss by two thirds. While the fleet is short, the shadow price of an aircraft-week rises from between CAD 22,000 and 29,000 to between CAD 29,000 and 37,000, so any proposal to add flying in those weeks must beat the higher hurdle. In a real network the marginal value comes from a fleet assignment model (Chapter 12), because lines differ in the types that can fly them and feed each other; the logic is the same.

The long-range narrowbody The aircraft that defines network planning in 2026 is the long-range narrowbody.

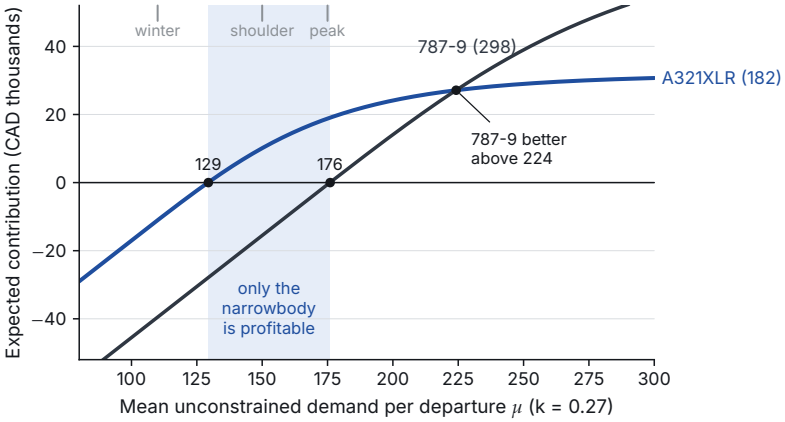


Figure 16.1 Expected contribution per eastbound YUL–SVQ departure for Laurentian's A321XLR (182 seats) and 787-9 (298 seats) as mean demand grows, with demand uncertainty $k = 0.27$, CAD 600 contribution per passenger and illustrative cargo contributions of CAD 3,500 and 10,000. The band marks demand levels at which only the narrowbody is profitable. The ticks at the top mark Laurentian's winter, shoulder and peak mean demands.

DEFINITION 16.2 Long-range narrowbody

A **long-range narrowbody** is a single-aisle aircraft with enough range to fly long-haul sectors, such as transatlantic flights from eastern North America. The Airbus A321XLR, with a published range of 4,700 nautical miles and between about 140 and 240 seats depending on the airline, is the current example (Cirium 2026b).

By the third quarter of 2026, 50 A321XLRs had been delivered and 28 airlines had committed to 440 aircraft. Cirium reports up to 30 percent lower fuel consumption per seat than the Boeing 757, the aircraft this role used to fall to, and lists the trade-offs as lost belly cargo and the passenger preference for widebodies (Cirium 2026b). Boeing's 737-10, certified on 20 October 2026, is rated for up to 3,100 nautical miles (Boeing 2026), well short of the XLR. Chapter 9 develops the aircraft's economics; the example shows the network consequence.

WORKED EXAMPLE 16.2 A thin transatlantic market

Each passenger on Laurentian's YUL–SVQ departure is worth CAD 600: a net fare of CAD 620 plus CAD 30 of ancillary revenue, less CAD 50 of passenger costs. The trip cost excluding passenger costs is $11{,}000 \times 7.0 + 3{,}500 =$ CAD 80,500 on the A321XLR and $15{,}500 \times 7.0 + 7{,}000 =$ CAD 115,500 on the 787-9; per seat, the widebody is cheaper (CAD 388 against 442). Belly cargo contributes CAD 3,500 on the narrowbody, the handbook's figure, and an assumed CAD 10,000 on the widebody, an assumption of this example only (Chapter 9 makes its own).

In the shoulder season demand has a mean of 150 and a standard deviation of 40. The spill formula of Chapter 7 gives an expected load of 145.2 on the XLR (a 79.8 percent load factor) and a contribution of $600 \times 145.2 + 3{,}500 - 80{,}500 =$ CAD 10,115 per departure. The 787-9 carries all 150 expected passengers at a 50.3 percent load factor and loses CAD 15,501. At the peak (mean 175, standard deviation 42) the XLR earns CAD 19,907 and the 787-9 still loses CAD 512; in winter (mean 110, standard deviation 35) both lose, the XLR CAD 11,153 and the 787-9 CAD 39,500.

Figure 16.1 generalizes. The XLR breaks even at a mean demand of about 129 passengers per departure, the 787-9 at about 176, and the 787-9 becomes the better aircraft only above about 224. Between 129 and 176 only the narrowbody can serve the market profitably. The widebody's lower seat cost is irrelevant when its seats cannot be filled; what matters is trip cost against the revenue the market can supply. The winter loss is a reminder that such markets are often seasonal: Air Canada's published Seville schedule pauses between early January and mid-February 2028 (Air Canada 2026f), although nothing in Laurentian's invented numbers explains the real airline's choice.

16.3 Carbon rules that enter route economics

Carbon policy reaches a route through three instruments with different scopes. Do not assign one carbon charge to every tonne on every route, and do not assume that an EU-bound transatlantic flight is treated like a flight within Europe (Network Planning Handbook 2026).

DEFINITION 16.3 Three carbon instruments

The **Carbon Offsetting and Reduction Scheme for International Aviation** (**CORSIA**) of ICAO requires operators to buy emissions units for the growth of international aviation emissions above a baseline, on routes between participating states. **ReFuelEU Aviation** requires fuel suppliers at EU airports to blend rising minimum shares of **sustainable aviation fuel** (**SAF**), and operators to buy most of their fuel there. The **EU Emissions Trading System** (**EU ETS**) requires an allowance for each tonne of CO_2 emitted on covered flights, in 2026 those within the European Economic Area.

CORSIA. Until the end of 2026 only flights between volunteering states carry offsetting obligations. From 2027 the second phase covers all international flights except those to or from exempt states, such as least developed countries, unless they volunteer. As of 1 January 2026, 130 states participated, and the 2024 sector growth factor was 15.4 percent (IATA 2026b). An operator's requirement in a year is its covered emissions multiplied by the sector growth factor, fully sectoral until 2032 (ICAO 2023). Emissions are computed from fuel: 3.16 tonnes of CO_2 per tonne of Jet A or Jet A-1 (Hellenic Civil Aviation Authority n.d.). Canada applies CORSIA to operators emitting more than 10,000 tonnes of CO_2 a year on international flights (Transport Canada 2026); domestic flights are outside it (Network Planning Handbook 2026).

ReFuelEU. Regulation (EU) 2023/2405 sets minimum SAF shares in the fuel supplied at Union airports of 2 percent from 2025, 6 percent from 2030, 20 percent from 2035 and 70 percent from 2050, with a synthetic-fuel sub-mandate averaging 1.2 percent over 2030 and 2031; operators must uplift at each Union airport at least 90 percent of the fuel their flights from it require, so they cannot carry cheaper fuel in (European Parliament and Council of the European Union 2023; EASA 2024). In EASA's 2025 reference prices, fossil jet fuel averaged EUR 640 per tonne, aviation biofuel EUR 1,925 and synthetic fuel an estimated EUR 7,520 (SAF Investor 2026).

EU ETS. Aviation's ETS scope is limited to flights within the European Economic Area until the start of 2027, and the sector moved to full auctioning of allowances in 2026 (European Commission n.d.). On 17 July 2026 the European Commission proposed extending the ETS, from 2029, to departures from the European Economic Area to destinations within 5,000 kilometres of Frankfurt; the proposal

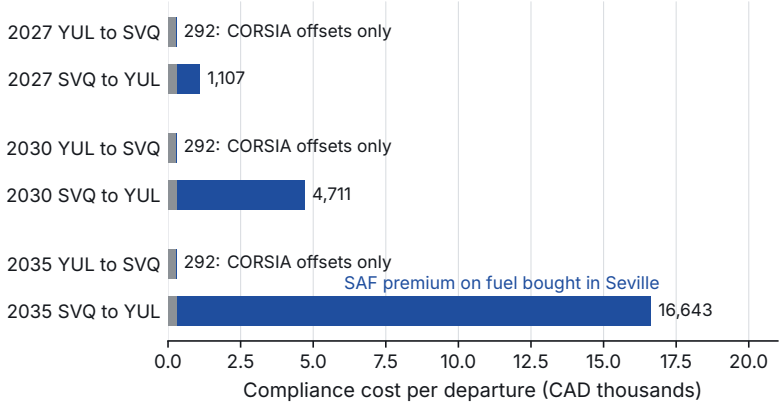


Figure 16.2 Carbon offsetting and SAF-mandate cost per Laurentian YUL–SVQ departure by direction, for 2027, 2030 and 2035 mandate levels, from (16.1). Fuel burn, unit prices and the euro exchange rate are illustrative assumptions, and 2025 reference fuel prices are held constant, which overstates later SAF costs if those prices fall.

needs the agreement of the Parliament and the member states (Legal 500 2026). If the test were applied as reported, departures to Montreal (about 5,850 kilometres from Frankfurt by great circle) and Halifax (about 5,240) would fall outside it and departures to St. John's (about 4,370) inside; this is this book's calculation, not a reading of the legal text.

Canada has no federal SAF blending mandate: its Aviation Climate Action Plan sets an aspirational goal of 10 percent SAF use by 2030, and British Columbia requires 1 percent renewable jet fuel from 2028 rising to 3 percent in 2030 (Transport Canada 2026).

For one flight the compliance cost depends on where the fuel is bought. With F tonnes of fuel, emission factor $e = 3.16$, sector growth factor g (zero outside CORSIA), offset price p_C, allowance price p_E, SAF share s of which synthetic s_x, and price premiums Δ_b and Δ_x of biofuel and synthetic fuel over fossil fuel,

$$K = F\,e\,(g\,p_C + \mathbb{1}_{\mathrm{ETS}}\,p_E) + \mathbb{1}_{\mathrm{EU}}\,F\,[(s - s_x)\,\Delta_b + s_x\,\Delta_x], \tag{16.1}$$

where $\mathbb{1}_{\mathrm{ETS}}$ is 1 if the ETS covers the flight and $\mathbb{1}_{\mathrm{EU}}$ is 1 if the fuel is bought at an EU airport.

WORKED EXAMPLE 16.3 Which direction pays for SAF?

Assume, for illustration only, that Laurentian's A321XLR burns 17 tonnes of fuel eastbound from Montreal and 19 tonnes westbound from Seville, emitting 53.7 and 60.0 tonnes of CO_2.

CORSIA. In 2027 the route is covered both ways. At a growth factor of 15.4 percent the offsets are 8.3 and 9.2 tonnes; at US$25 per unit, the low end of one analyst's 2027 range (Sylvera 2025), and CAD 1.41 per US dollar, Air Canada's 2026 planning rate (Air Canada 2026d), they cost CAD 292 eastbound and CAD 326 westbound.

ReFuelEU. Fuel bought in Montreal carries no mandate; fuel bought in Seville does, and the uplift rule prevents the airline from avoiding it. At the 2 percent share, the premium is $0.02 \times (1{,}925 - 640) =$ EUR 25.70 per tonne of fuel, or CAD 781 for 19 tonnes at an assumed CAD 1.60 per euro. At the 2030 shares, 6 percent including 1.2 percent synthetic, it is $0.048 \times 1{,}285 + 0.012 \times 6{,}880 =$ EUR 144.24 per tonne, or CAD 4,385 per westbound departure, CAD 29 per passenger at 150 passengers. At 2035 shares and unchanged prices it would be CAD 16,317.

EU ETS. The route is outside it both ways; a flight within Europe would pay, at the EUR 80 per allowance the Commission uses to value its SAF support (European Commission n.d.), EUR 253 per tonne of fuel, ten times the 2 percent SAF premium.

The westbound flight carries most of the cost, and by 2030 it is of the same order as the CAD 20 per passenger by which the handbook's Seville candidate falls short of its alternative (Chapter 17). Carbon cost is route- and direction-specific and steps up on known dates; one average price per tonne across the network will misrank routes.

Table 16.1 Which carbon instrument reaches which flight. Sources: IATA (2026b); European Parliament and Council of the European Union (2023); European Commission (n.d.); Transport Canada (2026).

Flight	CORSIA (2027 on)	SAF mandate on fuel bought	EU ETS (2026 rules)
Domestic Canada	no	none (British Columbia from 2028)	no
Canada to United States	yes	none	no
Montreal to Seville	yes	none in Montreal	no
Seville to Montreal	yes	ReFuelEU in Seville	no
Madrid to Seville	no (domestic)	ReFuelEU in Madrid	yes

PITFALL

One carbon price for every tonne. Get the approved fuel and carbon assumptions by region and year from the sustainability and finance teams, and carry fuel burned, the SAF premium, compliance payments and any internal shadow price of carbon as distinct lines (Network Planning Handbook 2026).

16.4 Pricing, offers and distribution

Fares used to be filed in advance, distributed through global distribution systems (GDS) and mapped to booking classes, one letter each, so that an airline could offer at most 26 price points on a flight, shared among its cabins (International Air Transport Association 2021). That constraint is loosening. For a planner the change is in how the seats of a schedule are sold, not in how a seat is valued: the bid price is still the opportunity cost of a seat, and spill and displacement still come from capacity set against demand (Chapters 14 and 15).

DEFINITION 16.4 Dynamic pricing, continuous pricing and dynamic bundling

In IATA's terminology (International Air Transport Association 2021), **dynamic pricing** sets the price of an offer from the context of the request (days to departure, length of stay, competitor activity, remaining seats) without requiring personal data. **Continuous pricing** removes the limit of a fixed number of price points. **Dynamic bundling** assembles the content of the offer (fare, bags, seats, flexibility) at the time of the request.

Lufthansa Group began selling continuously priced fares in October 2020 (Travel Daily 2020), on most European continental routes and only through its NDC interface and own websites (Business Travel News Europe n.d.).

Finer price points are worth less than they sound. A price rounded to a ladder loses contribution roughly in proportion to the square of its distance from the best price, so halving the step between price points cuts the loss by three quarters. In this book's illustrative calculation (willingness to pay spread exponentially on a scale of CAD 350, bid prices between CAD 100 and 600), a ladder with 25 percent steps loses on average 0.84 percent of the attainable contribution and one with 10 percent steps 0.15 percent; 26 price points spanning a tenfold range of fares are about 10 percent apart. The full reference edition derives this in its Chapter 23. In the MIT Passenger Origin-Destination Simulator (PODS), continuous pricing raised revenue by up to 2.0 percent for the first airline to adopt it and up to 1.2 percent when both competitors adopted it (Szymański, Belobaba, and Papen 2021). A segmented version gave the first mover as much as 17 percent, mostly at rivals' expense (Long and Belobaba 2023): large gains come from segmentation and competition, and invite responses.

The **New Distribution Capability (NDC)** is an IATA data standard through which an airline sends its own offers, including dynamic prices and bundles, to sellers through programming interfaces instead of through filed fares in a GDS (IATA 2026c). Adoption is real but gradual: NDC transactions were 21.6 percent of US agency transactions settled through the Airlines Reporting Corporation (ARC) in May 2026, against 21.4 percent a year earlier (Airlines Reporting Corporation 2026). Air Canada launched its NDC program on 14 June 2023; by June 2024 NDC was enabled in points of sale covering 96 percent of its ticket volume, and its fee on tradi-

tional GDS bookings applied only to the lowest fare in each cabin, with NDC and direct bookings exempt (Travelweek 2024). Its pricing uses PROS Real Time Dynamic Pricing Advantage, according to the vendor (PROS 2025); which forecasting and optimization engine its revenue management runs is not public.

> **PITFALL**
>
> Treating a vendor's uplift as a forecast for your airline. Vendors claim gains of 1 to 3 percent from willingness-to-pay data in forecasts (PhocusWire n.d.) and project up to 3.5 percent from a per-request price optimizer (ePlane AI 2025), without a published control group; Delta, pricing about 3 percent of its domestic network with a generative-AI tool by mid-2025, disclosed no revenue result (PhocusWire 2025). Ask what the baseline system was, how the counterfactual was built, over what period, and whether rivals reacted. Price granularity alone is worth tenths of a percent, not whole points.

16.5 The Canadian market in 2026

Structure and the Competition Bureau's study On 19 June 2025 the Competition Bureau published a market study of the domestic airline industry. Air Canada and WestJet together carried 56 to 78 percent of domestic passengers at major airports in 2023, and 2023 domestic passenger shares were 34.1 percent for Air Canada, 30.1 for WestJet, 9.8 for Flair and 9.0 for Porter. One additional competitor on a route lowered average fares by about 9 percent (Competition Bureau Canada 2025). The Herfindahl index of the four named carriers alone is $34.1^2 + 30.1^2 + 9.8^2 + 9.0^2 \approx 2{,}246$ on the 0 to 10,000 scale. The study's recommendations include removing the Minister of Transport's power to override Competition Act review of airline mergers and joint ventures, publishing traffic and slot data, and raising foreign ownership limits (Competition Bureau Canada 2025). No formal government response was found as of October 2026; in March 2026 the limits of 49 percent total and 25 percent for a single foreign investor, and the ban on foreign carriers flying domestic routes, were unchanged (Michell 2026).

Competitors In October 2025 Delta (12.7 percent), Korean Air (10 percent) and Air France-KLM (2.3 percent) bought a combined 25 percent of WestJet, with Onex keeping control (WestJet 2025). In 2026 WestJet retrenched, ending its Copenhagen, Glasgow and

Madrid flying in September (Aviation A2Z 2026). Porter flies 56 Embraer E195-E2s and 29 Dash 8-400s on 131 routes; its five-year plan prioritizes denser service in eastern Canada, including Montreal's Metropolitan Airport, more US flying and sun routes (Travelweek 2026b). Lynx Air and Canada Jetlines left the market in 2024 (Competition Bureau Canada 2025). A federal liquidity facility linked to the rise in jet fuel prices (Department of Finance Canada 2026) approved CAD 150 million each for Porter and Transat and CAD 76 million for Flair (Daily Hive 2026).

Passenger protection The Air Passenger Protection Regulations tie compensation to whether a disruption was within the airline's control, within its control but required for safety, or outside its control (Government of Canada 2019). Amendments pre-published on 21 December 2024 would replace the three categories with a closed list of exceptional circumstances, so that compensation is owed unless one applies, and would require assistance from a two-hour delay (Canadian Transportation Agency 2024); in September 2026 they were still proposed (Canadian Transportation Agency 2026b). Bill C-31, introduced in May 2026, would raise penalties to as much as CAD 1 million (McCarthy Tétrault 2026) and was still at committee stage in late October (Parliament of Canada 2026). These rules convert reliability into money: the compensation and care exposure of a tight connection or rotation belongs in the business case (Chapter 13).

Airports and slots In 2024 only three Canadian airports were fully coordinated for slots (IATA Level 3): Toronto Pearson, Vancouver and Billy Bishop Toronto City; Montreal-Trudeau, Calgary and Quebec City were examples of Level 2 airports, where schedules are adjusted by agreement with a facilitator (Canadian Airports Council 2024). From the summer 2027 season a series of slots at a coordinated airport needs at least seven slots in a northern summer season instead of five (IATA, ACI, and WWACG 2025). Montreal-Trudeau operates around the clock with restrictions (Network Planning Handbook 2026): aircraft certified to the Chapter 3 or 4 noise standards and heavier than 45,000 kilograms may not depart between 00:00 and 07:00 or arrive between 01:00 and 07:00, and louder Chapter 2 aircraft face earlier cut-offs (NAV CANADA 2023), with exemptions granted by the airport operator (Boeing 2011). In December 2025 the operator proposed banning all takeoffs and landings between 01:00 and 06:00 (CBC News 2025); whether and when that will take effect was not known at the time of writing. Never say

casually that YUL closes at night; check the restriction for the aircraft category and the permission that applies (Network Planning Handbook 2026).

16.6 Air Canada in the autumn of 2026

What follows is a public snapshot for orientation, not an internal briefing (Network Planning Handbook 2026). Air Canada describes itself as the largest provider of scheduled passenger services in the Canadian domestic, transborder and international markets, with regional flying bought from Jazz under a capacity purchase agreement and from PAL Airlines, and with Air Canada Rouge as its leisure carrier (Air Canada 2026d).

Second-quarter results Table 16.2 reproduces the main figures of the second-quarter 2026 MD&A; Chapter 2 explains the identities that link them.

Table 16.2 Air Canada, second quarter of 2026 against 2025, as printed in the MD&A, with changes as printed (Air Canada 2026d). Statistics include flying bought from Jazz. The printed RPM level does not reproduce the printed ratios exactly: load factor, RPM growth, the load factor change, yield and yield growth are all consistent with RPMs between about 23,561 and 23,577 million, not 23,597.

Measure	Q2 2026	Q2 2025	Change
Operating revenues (CAD millions)	6,266	5,632	+11%
Passenger revenues (CAD millions)	5,576	5,031	+10.9%
Operating income (CAD millions)	−215	418	−633
Available seat miles (millions)	26,942	26,860	+0.3%
Revenue passenger miles (millions)	23,597	22,796	+3.4%
Passenger load factor	87.5%	84.9%	+2.6 pp
Yield (cents per RPM)	23.7	22.1	+7.2%
PRASM (cents)	20.7	18.7	+10.5%
CASM (cents)	24.1	19.4	+23.9%
Adjusted CASM (cents)	15.5	14.4	+7.4%
Fuel cost per litre (cents)	132.7	88.0	+50.8%

Passenger revenue rose 11 percent on nearly flat capacity, because the load factor rose 2.6 points and yield 7.2 percent; but operating expenses rose 24 percent, driven by fuel (up 49 percent despite a CAD 205 million hedging gain), labour costs and CAD 388 million of pension, labour and legal charges, and the quarter ended with an operating loss. Premium cabins brought about 30 percent of passenger revenue, and sixth-freedom revenue, from passengers connecting between two foreign countries over Canadian hubs, grew 9 per-

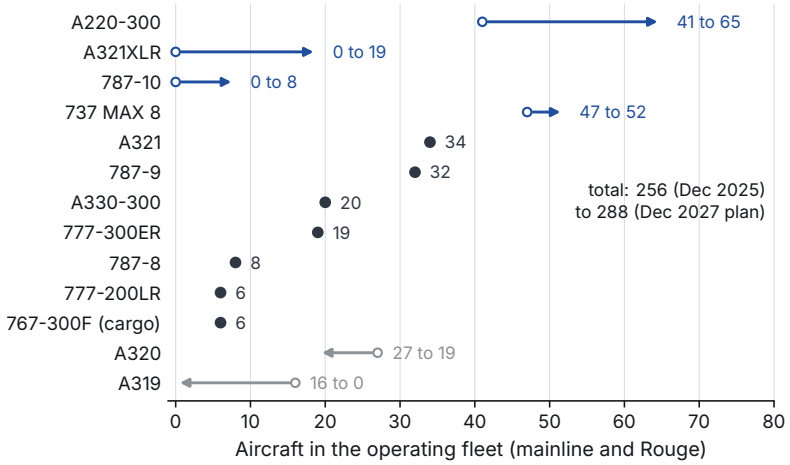


Figure 16.3 Air Canada mainline and Rouge operating fleet by aircraft type, actual on 31 December 2025 against the plan for 31 December 2027, as shown in section 6 of the second-quarter 2026 MD&A (Air Canada 2026d). Brands are combined; the Air Canada Express fleet flown by regional partners (95 aircraft in mid-2026) is planned unchanged and not shown. Planned transfers are subject to approvals.

cent, mostly from the United States to Europe (Air Canada 2026d). Transborder capacity fell 5.7 percent and unit revenue rose 18.5 percent, while Pacific capacity grew 7.9 percent and unit revenue only 3.3 percent; correlation is not causation, since fuel, fares and a weak Canada–US travel market all moved at once.

Fleet and orders In the plan of Figure 16.3, mainline and Rouge together grow from 256 aircraft at the end of 2025 to 288 at the end of 2027: widebodies from 91 to 99 with the arrival of the 787-10, narrowbodies from 165 to 189. The narrowbody fleet is also reshuffled: all 737 MAX 8s move to Rouge during 2026, the Airbus A320 family moves to mainline, and the A319s leave. The order book includes 30 A321XLRs, the first two delivered in the second quarter of 2026 and the last due in 2029; 65 firm A220-300s, of which 46 had been delivered by 30 June 2026; 14 787-10s, ten by 2028 and four by 2030; and eight A350-1000s from 2030 (Air Canada 2026d). At the published layouts of 137 seats for the A220-300 (Air Canada n.d.a) and 182 for the A321XLR (Air Canada 2026f), the 24 additional A220s and 19 XLRs planned by the end of 2027 add 3,288 and 3,458 seats.

These numbers are plans, not capacity. A planned fleet count or an announced order is not the number of aircraft delivered, the number operationally available, or the number assignable to your desk

in a given month (Network Planning Handbook 2026); before building flying on a planned aircraft, ask which constraint binds: tails, engines, maintenance cover, cabin configuration or trained crews.

Network moves Air Canada received its first A321XLR on 24 April 2026 (Air Canada 2026a) and scheduled its first transatlantic XLR service, Montreal–Toulouse, for 15 June (Air Canada 2026b). For summer 2027 it announced Toronto–Oslo (subject to approval), Toronto–Shannon and a daytime Toronto–London Heathrow on the XLR (Air Canada 2026e), and Montreal–Basel-Mulhouse on the XLR, in what it called its largest intercontinental expansion (PAX News 2026).

On 14 October 2026 it announced Montreal–Seville, the only nonstop between North America and Seville: three weekly flights from 8 September 2027, four from 1 October, on the 182-seat A321XLR (14 business and 168 economy seats), departing Montreal at 20:30 and arriving at 09:30 the next morning, returning at 12:45 to arrive at 14:40, and operating until early January and from mid-February to early June 2028 (Air Canada 2026f). Laurentian's YUL–SVQ case mirrors this public schedule; everything it earns and spends is invented.

Two days later Air Canada and United agreed to renew and expand their transborder joint business agreement, in place since 2022, subject to regulatory approval (Air Canada 2026g). The expansion would bring in ten city pairs excluded until now, including Montreal–Chicago and Montreal–Washington, and Transport Canada opened a public consultation on it (Travelweek 2026a). The excluded markets go back to a 2012 consent agreement with the Commissioner of Competition (Lexology n.d.), and in the United States the carriers need the Department of Transportation to extend their antitrust immunity beyond July 2027 (PYMNTS 2026). A joint venture changes which flows count in a route's value (Chapter 10), and this one tests the Competition Bureau's first recommendation. Figure 16.4 collects the dated milestones of this chapter.

16.7 What changed since 2006, and what did not

The courses this book draws on were taught when distribution ran through GDS terminals, the Boeing 757 flew the thinnest transatlantic routes, aircraft supply was taken for granted and nobody priced carbon into a route. Each of those has changed.

What did not change matters more. Seats still perish at departure. Demand still arrives by O&D market while capacity is sup-

2026	2027	2028	2029	2030
Regulation and market access				
Jan EU ETS: allowances for aviation fully auctioned **17 Jul** Commission proposes the ETS for departures to points within 5,000 km, from 2029; not adopted **Oct** Air Canada and United file to renew and widen their transborder JBA	**1 Jan** CORSIA second phase; EU non-CO_2 reporting extends to all flights to and from the EEA **28 Mar** Summer slot series need 7 slots (WASG) **Jul** US antitrust immunity for the JBA ends unless extended	**2028** British Columbia: 1% renewable jet fuel	**2029** Proposed ETS extension would start, if adopted	**2030** ReFuelEU: 6% SAF, at least 1.2% synthetic (two-year average)
Air Canada fleet and network				
24 Apr First A321XLR delivered; in service from 9 Jun **Sep** Summer 2027 expansion announced **Dec (plan)** 9 A321XLR and first 2 787-10s in the fleet	**By end Jan** New chief executive in post **Summer** What the airline calls its largest intercontinental expansion **8 Sep** Montreal–Seville launches on the A321XLR **Dec (plan)** 288 mainline and Rouge aircraft, 19 A321XLR, 65 A220	**2028** 10 of 14 787-10s due by 2028	**2029** Last of 30 A321XLRs	**2030** First A350-1000s; last four 787-10s

Figure 16.4 Milestones of 2026 to 2030 for regulation and market access (top) and for Air Canada's fleet and network (bottom). Year columns are not to scale. Items after October 2026 are plans, proposals or scheduled dates. Sources as cited in this chapter.

plied by flight leg, and spill still governs what an extra seat is worth (Chapter 7). A route's value is still the change it makes to the whole network's contribution, a long-range narrowbody is judged by spill-adjusted contribution like any other gauge, and a carbon charge is one more avoidable cost with a scope.

Read every 2026 development as a change to an input, a constraint or a tool, and then apply the unchanged economics: incremental network contribution, opportunity cost and the value of the marginal seat or aircraft.

16.8 Summary

- Planning systems still chain demand, share, spill and cost models. A fuel shock re-ranks flying by contribution per aircraft-hour, long and thin flying first.

- Engine groundings and delivery backlogs make aircraft time scarce. A missing aircraft costs the marginal line of flying, not the average, and raises the hurdle for new routes.
- The A321XLR opens markets too thin for a widebody: in Laurentian's case only the narrowbody is profitable between about 129 and 176 passengers per departure.
- Carbon costs are route- and direction-specific: CORSIA on nearly all international routes from 2027, SAF mandates on fuel bought in Europe, the EU ETS within Europe.
- Pricing is moving, gradually, toward continuous prices sold through NDC; granularity alone is worth tenths of a percent, and vendor uplifts are self-reported.
- Canada's market remains concentrated, passenger protection reforms are pending, and the Air Canada and United transborder agreement is under review.
- Air Canada pairs strong unit revenue with a fuel and labour cost shock, and plans 288 mainline and Rouge aircraft by the end of 2027, with XLR routes such as Montreal–Seville.

16.9 Further reading

Read the second-quarter MD&A (Air Canada 2026d) and the Competition Bureau's study (Competition Bureau Canada 2025) in full. Belobaba, Odoni and Barnhart (2015) cover distribution, fleet planning and regulation at textbook depth. On pricing technology, start with the PODS study of continuous pricing (Szymański, Belobaba, and Papen 2021); on carbon, the IATA CORSIA fact sheet (IATA 2026b). The full reference edition's Chapter 23 derives the price-grid and marginal-aircraft results stated here.

CHAPTER

17

The Analyst's Toolkit

IN THIS CHAPTER

- The grain of a table, and the data sources an analyst meets, each with its trap.
- A reproducible workflow: data contract, SQL, Python and a spreadsheet.
- Clocks: UTC and local time, and the weeks when Montreal and Seville are five hours apart.
- Decision thresholds and the sensitivity matrix, and the one-page memo that carries a recommendation.
- The audit checks that catch the common errors.
- What to establish with each team in the first weeks.

DRAWS ON Handbook (data catalogue D1 to D23, analytical workflow, templates, reading room and first ninety days); IATA, ACI and WWACG, Worldwide Airport Slot Guidelines, edition 4; Air Canada, Second quarter 2026 MD&A; Bureau of Transportation Statistics; public sources dated up to 27 October 2026

Most errors in planning analysis are not modelling errors. They are passengers counted twice because a join multiplied rows, seats counted three times because codeshares were not removed, a bank that vanishes for a week because a time zone offset was typed in by hand, and a recommendation that cannot be acted on because it does not say what would change it. The models of the earlier chapters are only as good as the tables that feed them and the memo that carries their answer.

This chapter follows the analytical workflow of the handbook (Network Planning Handbook 2026), with a worked example on Laurentian at each step. It recommends a method, not a software stack: which system is authoritative for each number has to be learned inside the team (Chapter 16).

17.1 Data: objects, sources and their traps

Airline data describe four objects that must be kept apart (Network Planning Handbook 2026): aircraft movements (one takeoff or one landing), flight legs, passenger journeys from origin to destination over one or more legs, and tickets, which can cover an outbound and a return. Chapter 2 builds the measures on these objects; here the concern is tables.

DEFINITION 17.1 Grain

The **grain** of a table is the object one row represents, together with the keys that identify it: one row per flight leg and date, per coupon (one passenger on one leg of a ticket), per directional journey, per ticket, or per market and month. Every aggregate must state the grain it sums over, and every join must state the grain of its result.

WORKED EXAMPLE 17.1 One travel day, counted four ways

Laurentian's coupon table for 8 September 2027 holds five illustrative tickets: Boston to Paris via Montreal with a return on 20 September; a one-way Montreal to Paris; two passengers travelling together from Quebec City to Seville via Montreal; and Montreal to Seville with a return on 15 September. One row is one coupon: 11 rows, 8 of them flown on 8 September. Grouping the same rows by different keys gives different answers:

```
SELECT leg_orig, leg_dest, COUNT(*) AS leg_pax
FROM coupons
WHERE flight_date = '2027-09-08'
GROUP BY leg_orig, leg_dest;

SELECT od_orig, od_dest,
       COUNT(DISTINCT journey_id) AS od_pax
FROM coupons
WHERE flight_date = '2027-09-08'
GROUP BY od_orig, od_dest;
```

The first query returns 8 leg passengers: 3 on YUL–SVQ, 2 on YUL–CDG, 2 on YQB–YUL and 1 on BOS–YUL. The second returns 5 directional journeys. The day also has 5 tickets and 4 flights, which make 8 aircraft movements. The five tickets are worth CAD 4,343 together, but joining ticket values to the

coupon table and summing gives CAD 10,533, because each value is repeated on every coupon row.

Each question has its grain: leg passengers for capacity and load factor, journeys for market size and share, tickets for revenue, movements for airport charges. Choose the grain first, then write the query.

No single source covers an airline market. Supply is measured well, because schedules are published; demand is estimated, because no one observes every journey. Table 17.1 summarizes the sources in the handbook's data catalogue, with the trap that most often catches a new analyst.

Table 17.1 Data sources a network analyst meets, adapted from the handbook's catalogue (Network Planning Handbook 2026), with Cirium (2026a) and Bureau of Transportation Statistics (2026a).

Source	What it measures	Main traps
Schedules (OAG, Cirium)	offered flights, seats, times, aircraft and carriers, by leg and date	future schedules change; codeshare duplicates; aircraft type does not fix the seat layout
Booking-based market data (OAG, Cirium)	estimated O&D passengers and fares by carrier and month	estimates, not a census; uneven coverage of direct, NDC and low-cost sales; restatements
BTS T-100 Segment and Market	operated flights, seats and passengers touching the United States	a market record is the trip on one flight number, not the journey
BTS DB1B, then DB1C	sampled tickets: itineraries and fares	10% quarterly sample became 40% monthly in July 2025; international records restricted
Statistics Canada	airport activity, movements, monthly indicators	the domestic O&D survey (2702) is inactive
Airport reports	passengers and movements by month	enplaned, screened, connecting and transit passengers differ
Internal systems	bookings, forecasts, fares, cost definitions	learn which system owns each number

Schedules for future dates are plans that move, so keep the date of each extraction (Network Planning Handbook 2026). **Market estimates** are built mainly from booking records: OAG's chiefly from

GDS bookings (Network Planning Handbook 2026; OAG n.d.), while Cirium blends more than 24 sources, fills gaps by regression and scaling to external control totals, publishes about 55 days after the end of a month and restates at 115 days (Cirium 2026a). For each product, find out what happens to direct bookings, low-cost carriers, NDC sales and poorly covered countries (Network Planning Handbook 2026).

IN PRACTICE **Market data in 2026**

A growing share of agency sales uses NDC rather than traditional GDS connections, 21.6 percent of US agency transactions settled through ARC in May 2026 (Airlines Reporting Corporation 2026), and airlines sell a further share directly. Data built on GDS bookings therefore cover less of the market than they once did, which is why vendors blend them with settlement data, government statistics and models of the gaps (Cirium 2026a). Ask the provider, market by market, what share is observed and what share is modelled.

US government data. The T-100 tables of the Bureau of Transportation Statistics (BTS) report operated traffic on flights touching the United States: a benchmark for Canada–US flying, useless for sizing a Montreal–Europe market (Network Planning Handbook 2026). The **origin and destination survey** samples tickets. Until June 2025 it was the quarterly 10 percent sample known as DB1B; from travel on 1 July 2025 US carriers report a 40 percent sample monthly, and BTS released the first data from the new program, DB1C, on 17 February 2026 (Bureau of Transportation Statistics 2026a, 2026b). Its public tables cover domestic travel only (Network Planning Handbook 2026). **Airport totals**, with their differing definitions of screened, connecting and transit passengers, are a cross-check, not a route forecast (Network Planning Handbook 2026).

What a sample can say A sample of tickets measures a market with error, and the error matters most for the thin markets where new routes are proposed.

RESULT 17.1 Precision of an expanded sample count

If each of a market's N passengers is sampled independently with probability p, the sampled count divided by p is an unbiased estimate of N, with a coefficient of variation of $\sqrt{(1-p)/(pN)}$. Precision grows roughly with the square root of the sample, not in proportion to it.

Real surveys sample tickets, which carry several passengers and both directions, so the true error is larger (Lohr 2021): the formula is a lower bound.

WORKED EXAMPLE 17.2 Thin markets and a broken series

Take a US domestic market with 12,000 passengers a month in both directions together. Under the old 10 percent sample the expanded total has a standard error of 329 passengers, a coefficient of variation of 2.7 percent; under the 40 percent sample, 1.1 percent. For a thin market of 600 passengers a month the figures are 12.2 and 5.0 percent. Quadrupling the sampling rate cuts the standard error by a factor of $\sqrt{6} = 2.45$, not four.

Now the splice. Suppose the new monthly sample records 4,800 passengers. Expanded by $1/0.4 = 2.5$, that is 12,000 passengers, or 200 passengers per day each way (PDEW) in a 30-day month. An analyst who keeps multiplying by ten, as for the old sample, reports 48,000 and an apparent fourfold jump in demand in July 2025. A quarter of the old series with 3,510 sampled passengers expands to 35,100, or 193 PDEW over 91 days, in line with the new series once each is expanded by its own factor.

PITFALL

Splicing a broken series. Do not join DB1B and DB1C, or any series whose sample design changed, by multiplying every new row by the old factor. Record each expansion factor in the data contract and mark the break in every chart that crosses it (Network Planning Handbook 2026).

17.2 A reproducible workflow

The handbook's workflow has six steps (Network Planning Handbook 2026):

1. Frame the decision and its counterfactual: the schedule the airline would actually fly without the change, with the displaced aircraft at its next-best use (Chapter 10). Without it, no number that follows has a meaning.
2. Write a data contract for every input.
3. Join and summarize in SQL.
4. Estimate and test sensitivity in Python.
5. Review scenarios transparently in a spreadsheet.
6. Finish with a short memo.

> **DEFINITION 17.2 Data contract**
>
> A **data contract** records, for one table, its grain; its source; its extraction (snapshot) date; the travel period it covers; its currency; whether fares are gross or net, and of what; whether carriers are marketing or operating carriers; how airports are grouped into city markets; its revision status; and its coverage (Network Planning Handbook 2026).

Table 17.2 An illustrative data contract for two extracts for the Seville case; the fields follow the handbook (Network Planning Handbook 2026).

Field	Schedule extract	Market estimate extract
Grain	one operated flight leg per date	one directional O&D market per carrier and month
Snapshot date	26 October 2026	release of 15 October 2026
Fare treatment	not applicable	average fare; taxes and commissions to confirm
Carriers	operating carrier only, codeshares removed	as defined by the provider
Geography	airports	city markets; is Montreal Metropolitan Airport grouped with YUL?
Revision status	final for past dates, plan for future dates	first release, restated later

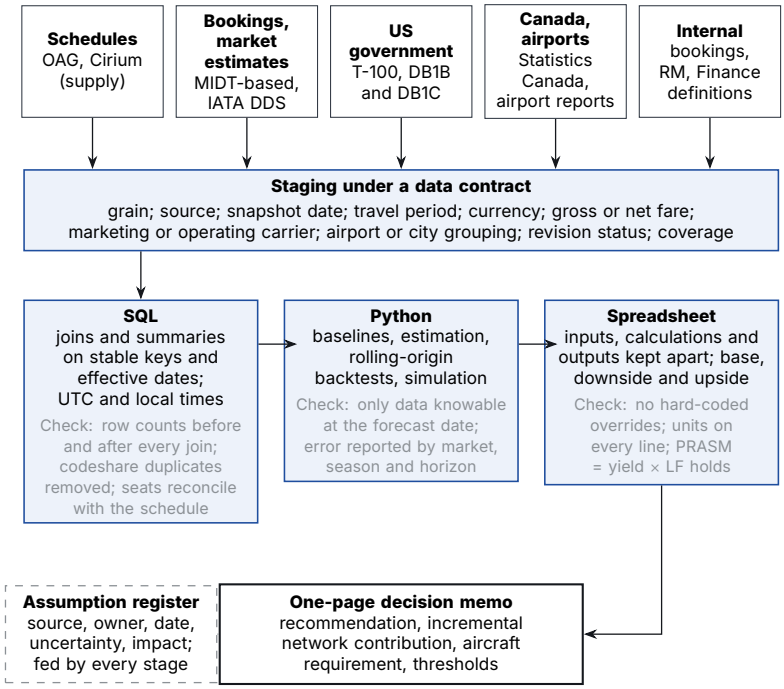


Figure 17.1 The lineage of a route analysis. Each source enters through a data contract; SQL, Python and a spreadsheet each do their own work, with their own audit check; the result reaches a one-page memo, and an assumption register records every assumption.

The contract is the first stage of a lineage that runs from sources to the memo (Figure 17.1). Two habits keep that lineage honest. An **assumption register** holds one line per assumption, with its source, owner, date, uncertainty and the decision it affects (Network Planning Handbook 2026). Vintage discipline keeps the snapshot date of every input, so that a backtest uses only what was knowable on the date of the forecast (Chapter 6).

SQL for joins SQL is the right tool for joins, because a query is explicit and can be rerun when the data are refreshed. Hold airports, carriers and cabin configurations in dimension tables and join through keys that are valid on the right dates (Network Planning Handbook 2026). The first query below counts operated seats once, whatever the number of codes a flight is sold under; the second attaches the seat configuration valid on each date:

```
SELECT SUM(seats) AS seats_operated
FROM schedule
WHERE flight_date = '2027-09-08'
```

```
  AND mkt_carrier = op_carrier
  AND mkt_flight = op_flight;

SELECT s.flight_date, s.op_flight,
       c.seats_j, c.seats_y
FROM schedule AS s
JOIN cabin_config AS c
  ON c.aircraft = s.aircraft
 AND s.flight_date BETWEEN c.valid_from
                       AND c.valid_to
WHERE s.mkt_carrier = s.op_carrier;
```

After every join, compare row counts with the grain you expected: a many-to-many join multiplies traffic or revenue without any error message.

WORKED EXAMPLE 17.3 Seats that exist three times

On 8 September 2027 Laurentian's YUL–CDG flight (298 seats) is sold under its own code and two partner codes, and its YUL–SVQ flight (182 seats) under its own code and one partner code: five rows in a schedule extract. Summing seats over the rows gives 1,258; the first query returns 480, the seats that will fly. The factor of 2.6 would pass unnoticed into seat miles, load factors and market shares.

Python and spreadsheets Python (or R) is the right tool for estimation, backtests and simulation. Start from a simple seasonal or analogue baseline and backtest on rolling windows, so that each forecast sees only earlier data (Chapter 6) (Network Planning Handbook 2026). Formulas used repeatedly belong in small, tested functions; the expected spill of Chapter 7 is one:

```
from scipy.stats import norm

def expected_spill(mu, sigma, C):
    z = (C - mu) / sigma
    return sigma * (norm.pdf(z) - z * norm.sf(z))

expected_spill(160, 40, 180)  # 7.91
```

The call reproduces the 7.91 passengers of expected spill in the handbook's Exercise B.

A spreadsheet is the right tool for the scenario review that colleagues will challenge: inputs, calculations and outputs on separate sheets, a unit on every line, no quiet hard-coded overrides, and cost definitions from Finance (Network Planning Handbook 2026).

When it disagrees with the Python result, find out why before showing either.

17.3 Clocks, seasons and the schedule card

Time is the most error-prone dimension of schedule data. The rules are few (Network Planning Handbook 2026): compute in UTC and keep local time for the commercial reading; carry time zone identifiers, never one UTC offset for the whole year; expect transatlantic flights to cross midnight; and give every weekly pattern an explicit rule for which local date a flight belongs to.

Canada and Europe change their clocks on different dates. Quebec moves its clocks forward on the second Sunday in March and back on the first Sunday in November (National Research Council Canada 2025; Québec 2007); the European Union's summer time runs from the last Sunday in March to the last Sunday in October (European Parliament and Council of the European Union 2001); and IATA's scheduling seasons begin on the last Sundays of March and October (IATA, ACI, and WWACG 2025), following Europe. In software, use the IANA time zone database, which Python's `zoneinfo` module reads.

RESULT 17.2 Clock changes move block time between directions

If a schedule keeps fixed local clock times at both ends, a one-hour fall in the time difference between Seville and Montreal lengthens the implied eastbound block by an hour and shortens the westbound block by an hour; their sum does not change.

WORKED EXAMPLE 17.4 Montreal and Seville, five hours apart

Laurentian's YUL–SVQ schedule mirrors the published Air Canada pattern: depart Montreal 20:30, arrive Seville 09:30 the next morning; depart Seville 12:45, arrive Montreal 14:40. When Seville is six hours ahead, the eastbound block is the elapsed local clock time less the six hours, $(33.5 - 20.5) - 6 = 7.0$, and the westbound block the elapsed clock time plus six, $(14.67 - 12.75) + 6 = 7.92$: 7:00 and 7:55, the blocks of the running case.

In 2026, European summer time ended on Sunday 25 October, but Montreal's daylight time ends only on Sunday 1 November. For that week Seville is five hours ahead, and a database that keeps the published local times implies an 8:00 eastbound block and a 6:55 westbound block. The first wastes an hour; the second is impossible, because the aircraft needs about 7:55 to fly west. The time zone database finds the problem at once:

```
from datetime import datetime
from zoneinfo import ZoneInfo

YUL = ZoneInfo("America/Toronto")  # Montreal
SVQ = ZoneInfo("Europe/Madrid")
UTC = ZoneInfo("UTC")

def to_utc(local, tz):
    t = datetime.fromisoformat(local)
    return t.replace(tzinfo=tz).astimezone(UTC)

dep = to_utc("2026-10-28T20:30", YUL)
arr = to_utc("2026-10-29T09:30", SVQ)
print(arr - dep)  # 8:00:00, an hour too long
```

Figure 17.2 shows the two repairs. Keeping Montreal times moves Seville's arrival to 08:30 and its departure to 11:45; keeping Seville times moves Montreal's departure to 21:30 and its arrival to 15:40. After 1 November the published local times work again, but every UTC time is an hour later than in summer; and a Wednesday 20:30 departure from Montreal is a Thursday departure in UTC, so operating days stored in UTC must shift their weekday flags.

The problem returns every year at both ends of the season: for the real Seville service, from 31 October to 6 November 2027 and from 12 to 25 March 2028. By departure date the affected eastbound flights are those of 30 October to 6 November and 12 to 24 March, because Europe changes its clocks at 21:00 on the Saturday, Montreal time (Table 17.3). The slot guidelines treat such time changes as ad hoc changes and ask coordinators to grant changes of seven days or less where feasible (IATA, ACI, and WWACG 2025). Which repair is better is a network question: keeping Montreal times protects the Montreal banks (Chapter 11).

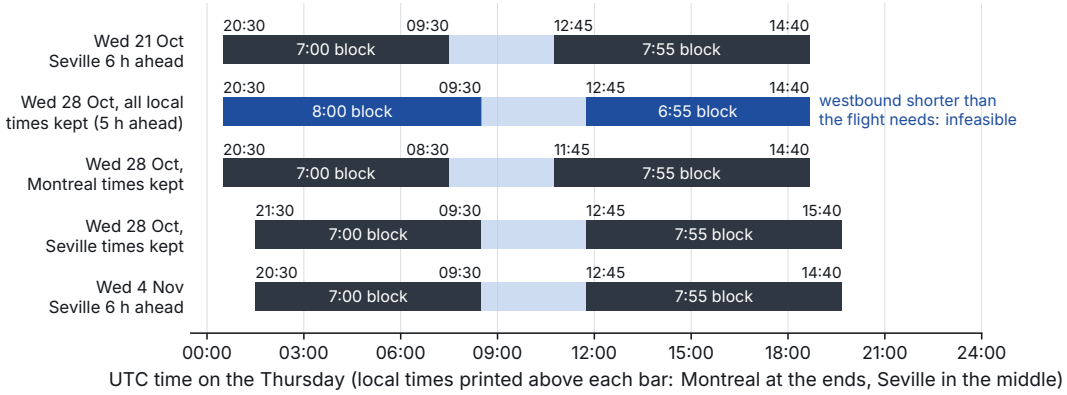


Figure 17.2 Laurentian's YUL–SVQ rotation drawn in UTC on three Wednesdays of autumn 2026, with local times printed above each bar. In the week of 28 October Seville is five hours ahead of Montreal: keeping every published local time implies blocks of 8:00 and 6:55, so one end must move. The light segment is the turn in Seville.

> **PITFALL**
>
> A typed-in offset. "Seville is UTC+2" is true for seven months of the year. Store time zone identifiers and let the database supply the offset for each date; then recompute block times from UTC on both sides of every clock change.

The schedule card Before a schedule is presented, it should fit on a card that a colleague in operations can check, with the owner of each unverified assumption (Network Planning Handbook 2026). Table 17.3 is the card for Laurentian's service; the operating days are invented.

Table 17.3 A schedule card for Laurentian's YUL–SVQ service, with the fields the handbook lists (Network Planning Handbook 2026). Times are computed with the time zone database; operating days are illustrative.

Item	Eastbound LA958	Westbound LA959	Owner
Days	Wed, Fri, Sun from 8 Sep 2027; Mon added from 1 Oct	Thu, Sat, Mon; Tue added	network planning
Local times	YUL 20:30, SVQ 09:30 next day	SVQ 12:45, YUL 14:40	scheduling
UTC, six-hour weeks in summer	00:30 to 07:30	10:45 to 18:40	scheduling
UTC, six-hour weeks in winter	01:30 to 08:30	11:45 to 19:40	scheduling
Five-hour weeks (departure dates)	30 Oct to 6 Nov 2027; 12 to 24 Mar 2028	31 Oct to 6 Nov 2027; 12 to 25 Mar 2028	scheduling
Block	7:00	7:55	flight operations
Turn	3:15 at Seville; minimum to confirm	5:50 at Montreal when a 20:30 departure follows the same day	operations
Aircraft	A321XLR, 182 seats (14 business, 168 economy)	same	fleet
Connections	evening feed from Quebec City and Boston	afternoon onward flights at Montreal	network planning
Unverified	westbound payload by season; Seville handling hours		flight operations, airports

17.4 Sensitivity matrices and decision thresholds

A recommendation is useful when it says what would change it. The handbook asks for coherent base, adverse and favourable cases, a matrix that crosses demand capture with retained fare, and the fuel or block-time threshold that reverses the recommendation (Network Planning Handbook 2026).

DEFINITION 17.3 Decision threshold

The **decision threshold** of an input is the value at which the ranking of two feasible alternatives reverses, with other inputs at their base values. For two inputs varied together, the set of reversal points is the **reversal frontier**.

RESULT 17.3 The reversal frontier of a route

Write the candidate's incremental network contribution as $n\,(f - c) - K$, with n passengers per departure, f the average retained fare, c the passenger-variable cost net of ancillary revenue, and K the remaining net cost, and let the best alternative use of the same aircraft contribute V. The candidate wins when $n\,(f - c) > K + V$. The fare threshold at a given load is

$$f^{*} = c + \frac{K + V}{n},$$

and the load threshold at a given fare is $n^{*} = (K + V)/(f - c)$. Along the frontier, a 1 percent fall in passengers is offset by a 1 percent rise in the fare margin $f - c$.

WORKED EXAMPLE 17.5 The Seville decision, mapped

The handbook's Seville case (Network Planning Handbook 2026) has 150 passengers at an average net fare of CAD 620 with CAD 30 of ancillary revenue each, CAD 3,500 of cargo contribution, an avoidable flight cost of CAD 88,000 at that load, CAD 6,000 gained on other flights, CAD 9,000 lost on the airline's own alternatives and CAD 2,000 of expected disruption cost. Laurentian's cost model splits the CAD 88,000 into CAD 80,500 for the flight and CAD 50 per passenger. Then $c = 50 - 30 = 20$ and $K = 80{,}500 - 3{,}500 - 6{,}000 + 9{,}000 + 2{,}000 = 82{,}000$, so the candidate contributes $150 \times 600 - 82{,}000 =$ CAD 8,000. A feasible alternative use of the aircraft contributes $V =$ CAD 11,000, so the candidate is CAD 3,000 worse.

The thresholds follow at once: $f^{*} = 20 + 93{,}000/150 =$ CAD 640, CAD 20 above the base fare, which is the handbook's threshold,

and $n^* = 93{,}000/600 = 155$ passengers. Table 17.4 maps the surface. Coherent scenarios read the same surface: an adverse case of 135 passengers at CAD 600 leaves the candidate CAD 14,700 behind; a favourable case of 165 passengers at CAD 640 puts it CAD 9,300 ahead.

Other inputs have thresholds too. At CAD 11,000 per A321XLR block hour, the CAD 3,000 gap equals 16.4 minutes of block time. A fuel price change moves both alternatives, so what matters is the difference in fuel exposure: if fuel were CAD 30,800 of the candidate's block cost and CAD 20,000 of the alternative's, the candidate would carry CAD 10,800 more fuel and catch up only if fuel fell by $3{,}000/10{,}800 = 28$ percent.

The memo can now say: use the aircraft for the alternative, unless the retained fare reaches CAD 640 at 150 passengers or the expected load reaches 155 at CAD 620. Whether such fares or loads are attainable is a question for the demand model and for revenue management (Chapter 15).

Table 17.4 Candidate contribution minus that of the best alternative, in CAD per departure, by passengers per departure (rows) and average retained fare (columns), for the handbook's Seville case. The base case is in bold; positive cells favour the candidate.

Passengers	CAD 560	CAD 590	CAD 620	CAD 650	CAD 680
130	−22,800	−18,900	−15,000	−11,100	−7,200
140	−17,400	−13,200	−9,000	−4,800	−600
150	−12,000	−7,500	**−3,000**	1,500	6,000
160	−6,600	−1,800	3,000	7,800	12,600
170	−1,200	3,900	9,000	14,100	19,200

PITFALL

Independent toggles. Varying demand down and fares up at the same time produces scenarios that cannot occur. Build scenarios from a story about the world, keep forecast error and execution risk separate, and report thresholds rather than a single point (Network Planning Handbook 2026).

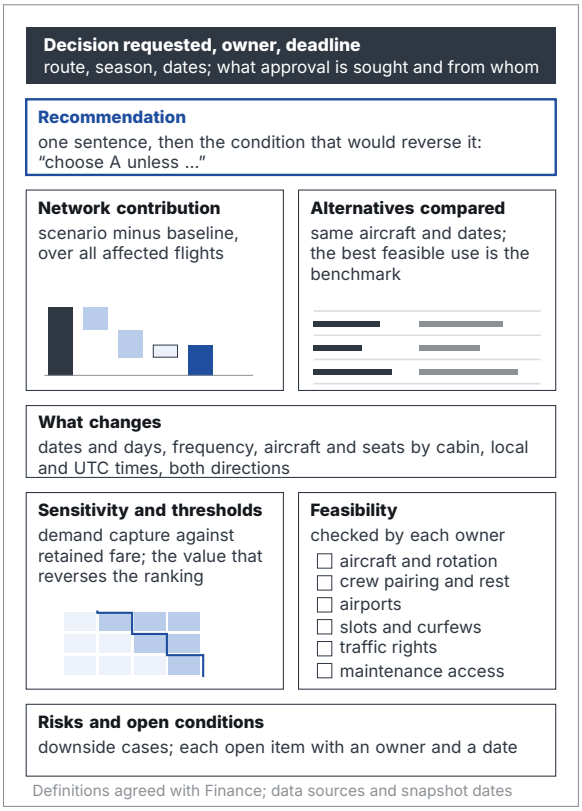


Figure 17.3 The layout of a one-page route decision memo. Each block names what it must contain: the decision and its owner, a one-sentence recommendation with its reversing condition, the incremental contribution bridge, the alternatives on the same aircraft, what changes in the schedule, the sensitivity and thresholds, the feasibility checks with their owners, and the open conditions.

17.5 The one-page decision memo

An analysis ends as a decision memo. The handbook's template asks for the decision and its deadline, the baseline and alternatives, the schedule, the market evidence, revenue by cabin and flow, cost definitions, network effects, feasibility, downside cases, the recommendation and the open conditions, each with an owner (Network Planning Handbook 2026). Figure 17.3 arranges these on one page.

The most important line is the recommendation, and the most important part of it is the condition that would reverse it. The handbook's example recommends one schedule unless westbound payload limits pass a stated level or weaker feed pushes its contribution

below the alternative's (Network Planning Handbook 2026). A sentence of that form tells the reader which assumption to watch; the thresholds of Section 17.4 supply the numbers.

The same discipline applies to two other documents. A **competitor-change note** records the change, when it was seen and the travel dates it touches, counts only capacity that will actually fly, and estimates share and fare effects under each plausible response. A **post-launch review** sets the approved forecast, not a later revision, against outcomes, explains the gap item by item, and tells a slow ramp-up apart from a biased forecast (Network Planning Handbook 2026).

17.6 Audit checks

Checks are cheap and errors are expensive. Table 17.5 expands the handbook's list into the checks worth running on every analysis.

Table 17.5 Audit checks for a route analysis, expanded from the handbook's list (Network Planning Handbook 2026).

Check	What it catches	How to run it
Operated capacity reconciles	codeshare duplicates, missing flights	sum seats over operated flights; compare with the published schedule
Passenger weights reconcile	wrong expansion factors, sample breaks	compare expanded totals with T-100 segment or airport totals
Miles and kilometres	mixed units	one unit per ratio; ASK = 1.609344 × ASM
Directional demand against return capacity	forecasts that fill one direction and empty the other	compare passengers and seats by direction
Actual against scheduled dates	cancelled or added flights treated as operated	join to operated records, not to plans
Bridge from market to carried passengers	jumps from market size to load	show market, share, spill, recapture and stimulation as steps
Sample coverage	false precision in thin markets	report coverage and intervals
Load factor weighting	averages of flight load factors	divide total RPK by total ASK
Identities	arithmetic and transcription errors	PRASM = yield × LF; TRASM − CASM = margin per ASM
Join multiplicity	many-to-many joins	row counts before and after each join
Clocks	typed-in offsets, midnight crossings	recompute blocks from UTC around every clock change
Vintage	look-ahead in backtests	keep snapshot dates; use only what was knowable

Two deserve a number. A 137-seat flight from Montreal to Quebec City (233 kilometres) at 60 percent and a 298-seat flight from Montreal to Paris (5,524 kilometres) at 90 percent average 75 percent, but together they fill 89.4 percent of their seat kilometres, because the long flight dominates (Chapter 2). And the identity check catches errors even in official filings: the printed passenger miles in Air Canada's second-quarter 2026 MD&A do not reproduce the printed load factor and yield exactly (Chapters 2 and 16).

Before you trust a number, recompute it a second way: from another source, from an identity, or from the grain below it. Most planning errors are found by the second computation, not by a better model.

17.7 The first weeks

In the first weeks, establish definitions before using numbers. Table 17.6 lists what to establish with each team.

Table 17.6 What to establish in the first weeks, by team, adapted from the handbook (Network Planning Handbook 2026).

Team	What to establish
Manager and network planning	the markets and horizons you own; the planning calendar; what makes a recommendation ready for approval
Forecasting	the agreed baseline; the inputs an analyst may change; how the models are tested
Revenue management	where scarce capacity is most valuable; the fare and cabin assumptions planning should adopt; how a schedule change shifts availability and displacement
Finance	the official definition of contribution; the treatment of connecting, partner, cargo and loyalty revenue; which costs are avoidable at each horizon
Scheduling and fleet	the system of record; publication and freeze dates; how tails, maintenance and reserves are checked
Crew and operations	which changes create step costs or infeasibility; the reliability assumptions and recovery options
Airports and slots	the binding gates, handling windows, restrictions and slots; who validates dates and exceptions
Alliances, legal and regulatory	which partner flows and commercial rights are in scope; how joint-venture effects enter the case

Three habits complete the method (Network Planning Handbook 2026). Ask to see a route paper that was approved, one that was rejected and one post-launch review. Rebuild a known result from its source inputs, and account for every difference, before trying to improve a model. And hold on to the distinctions that matter: a demand forecast versus passengers carried; allocated profit versus incremental value; the airline's own contribution versus the joint venture's; a connection that is valid versus one that is reliable; a model's output versus an approved operating plan.

For reading, the handbook's highest-priority book is The Global Airline Industry, in the order of its chapters 3, 6, 7, 8, 5 and 10 (Belobaba, Odoni, and Barnhart 2015); Appendix C lists the rest.

17.8 Summary

- Keep movements, legs, journeys and tickets apart: every table has a grain, and every aggregate and join must respect it. Market data are estimates with uneven coverage.
- The US ticket survey changed in July 2025 from a 10 percent quarterly sample to a 40 percent monthly one: expand each series by its own factor, and give thin markets intervals, not points.
- Write a data contract for every table and keep an assumption register; use SQL for joins, Python for estimation, a spreadsheet for transparent scenarios.
- Store UTC with time zone identifiers. When clocks change on different dates, fixed local times move block time from one direction to the other.
- Report thresholds: in the handbook's Seville case the decision reverses at CAD 640 at 150 passengers, or 155 passengers at CAD 620. The memo states the recommendation and the condition that would reverse it.
- Recompute every important number a second way before trusting it.

17.9 Further reading

The handbook (Network Planning Handbook 2026) is the source of the workflow, data catalogue and templates of this chapter, and repays rereading after the first month. For data, read the providers' own descriptions (Cirium 2026a; OAG n.d.) and the BTS pages on the new survey (Bureau of Transportation Statistics 2026b); Lohr (2021) covers the sampling theory. The Worldwide Airport Slot Guidelines (IATA, ACI, and WWACG 2025) are short enough to read in full. The full reference edition's Chapter 24 derives the sampling, clock and frontier results stated here.

APPENDIX

A

Mathematical Notes

The chapters of this edition state a few formulas and explain them in words. This appendix collects the derivations behind them, one section per result. Each section names the chapters that use the result, sets up the model, derives the result in full and checks it against the running-case numbers that the chapters quote. Nothing here is needed to follow the chapters; it is for the reader who wants to see why a rule holds and where it stops holding. Belobaba, Odoni, and Barnhart (2015), Talluri and van Ryzin (2004b) and Train (2009) are the standard references.

Notation follows the chapters. Demand D has mean μ, standard deviation σ and k-factor $k = \sigma/\mu$; C is a capacity in seats; φ and Φ are the standard normal density and distribution function; fares f_j are numbered from the highest; and $[x]^+ = \max(x, 0)$. Laurentian numbers are illustrative, and numbers from the MIT lectures are historical teaching values.

A.1 Why unit cost falls with stage length

Chapter 3 states that the cost per available seat-kilometre (CASK) falls with stage length, steeply for short legs and gently for long ones, and Chapter 2 relies on the same shape when it adjusts unit costs for stage length. This section derives the curve and its elasticity from the trip-cost model of the running case.

Setup The trip cost of a departure with block time BH carrying N passengers is

$$\mathrm{TC} = c_{\mathrm{BH}}\,\mathrm{BH} + c_{\mathrm{dep}} + c_{\mathrm{pax}}\,N, \tag{A.1}$$

where c_{BH} is the cost per block hour of the aircraft type, c_{dep} the cost per departure and c_{pax} the cost per passenger; the first two terms are the **capacity cost** of the departure. Block time grows linearly with the stage length d in kilometres,

$$\mathrm{BH}(d) = \tau_0 + \frac{d}{s}, \tag{A.2}$$

with a fixed allowance τ_0 for taxi, climb and approach and an effective speed s; the running case uses $\tau_0 = 0.65$ hours and $s = 850$ km per block hour. An aircraft with C seats produces $C\,d$ available seat-kilometres per departure.

Result Let $F = c_{\mathrm{dep}} + c_{\mathrm{BH}}\tau_0$ be the capacity cost incurred once per departure and $v = c_{\mathrm{BH}}/s$ the capacity cost per kilometre flown. Then

$$\mathrm{CASK}(d) = \frac{F + v\,d}{C\,d} = \frac{F}{C\,d} + \frac{v}{C}, \tag{A.3}$$

and the elasticity of unit cost with respect to stage length is

$$\frac{\mathrm{d}\ln \mathrm{CASK}}{\mathrm{d}\ln d} = -\frac{F}{F + v\,d}.$$

Derivation Substituting (A.2) into the capacity cost gives $c_{\mathrm{BH}}(\tau_0 + d/s) + c_{\mathrm{dep}} = F + v\,d$: a fixed amount per departure plus an amount proportional to distance. Dividing by the $C\,d$ seat-kilometres gives (A.3). The first term falls as $1/d$; the second, the cruise cost per seat-kilometre, does not depend on distance, so CASK falls towards v/C and never below it.

For the elasticity, take logarithms, $\ln \mathrm{CASK} = \ln(F + v\,d) - \ln C - \ln d$, and differentiate with respect to $\ln d$:

$$\frac{\mathrm{d}\ln \mathrm{CASK}}{\mathrm{d}\ln d} = \frac{v\,d}{F + v\,d} - 1 = -\frac{F}{F + v\,d}.$$

The elasticity is minus the share of the departure's capacity cost that does not vary with distance. It is close to -1 on very short legs, where almost all of the cost is incurred once per departure, and close to zero on long ones. Because it changes with d, no single exponent describes the curve.

Passenger costs have the same shape: with $N = \mathrm{LF} \cdot C$ passengers, the passenger cost per ASK is $c_{\mathrm{pax}}\mathrm{LF}/d$, because a passenger is checked in and fed once per flight, not once per kilometre.

The stage-length adjustment of Chapter 2 follows. To restate a unit cost observed at stage length d at a reference length d_0, move along the cost curve:

$$\mathrm{CASK}_{\mathrm{adj}} = \mathrm{CASK}(d)\,\frac{v + F/d_0}{v + F/d}.$$

A constant-elasticity rule, $\mathrm{CASK}(d)\,(d/d_0)^{\theta}$, is exact only if θ is the arc elasticity of the curve between d and d_0, which depends on both

distances; an exponent borrowed from another comparison can reverse a conclusion.

Check Laurentian's 737-8 has 169 seats, costs CAD 9,000 per block hour and CAD 2,500 per domestic departure. Hence $F = 2{,}500 + 9{,}000 \times 0.65 =$ CAD 8,350 and $v = 9{,}000/850 =$ CAD 10.59 per kilometre, so the per-seat departure cost F/C is CAD 49.4 and the cruise cost v/C is 6.27 cents per ASK, in line with Chapter 3. Montreal–Toronto (507 km) has a block time of 1.246 hours and a capacity cost of $9{,}000 \times 1.246 + 2{,}500 =$ CAD 13,714; Montreal–Vancouver (3,682 km) has 4.982 hours and CAD 47,338. Dividing by 169×507 and $169 \times 3{,}682$ seat-kilometres gives 16.0 and 7.6 cents. The Vancouver flight is 7.3 times as long, costs 3.5 times as much and has less than half the unit cost. The elasticity is $-8{,}350/(8{,}350 + 10.59 \times 500) = -0.61$ at 500 km and -0.18 at 3,682 km. Across the fleet, F/C is close to CAD 50 for every type, while v/C runs from 6.12 cents (787-9) to 7.11 cents (A321XLR): types differ mainly in cruise cost, routes mainly in distance.

A.2 The S-curve of market share

Chapter 4 states that in a competitive market an airline's share of passengers rises more than proportionally with its share of frequencies, and Chapter 11 uses that curve, together with the effect of frequency on market size, to price an added frequency. This section derives both effects: the S-curve as a logit model in log frequency, and the schedule delay that makes frequency grow the market.

Setup In a market served by nonstop carriers $b = 1, \dots, n$ with daily frequencies f_b, carrier a's **frequency share** is $\mathrm{FS}_a = f_a / \sum_b f_b$. Belobaba's S-curve writes its market share as

$$\mathrm{MS}_a = \frac{\mathrm{FS}_a^{\alpha}}{\sum_b \mathrm{FS}_b^{\alpha}}, \qquad \alpha > 1, \tag{A.4}$$

with historical estimates of α between 1.3 and 1.7 (Belobaba 2006d). Frequency also changes market size. A traveller who would like to leave at t^* and takes departure d suffers a **schedule delay** $|t^* - d|$; total trip time T is access and processing time t_0, plus block time t_b, plus expected schedule delay (Douglas and Miller 1974); and market demand is $D = M\,P^{\eta_P}\,T^{\eta_T}$, with time elasticity $\eta_T < 0$.

Result Under this model:

1. If traveller n's utility for carrier a is $U_{na} = \alpha \ln f_a + \gamma' x_a + \varepsilon_{na}$, with independent Gumbel errors (Appendix A.3) and other attributes x_a, then $\ln(\mathrm{MS}_a/\mathrm{MS}_b) = \alpha \ln(f_a/f_b) + \gamma'(x_a - x_b)$, and (A.4) holds when the x_b are equal.
2. In a duopoly, $\partial \mathrm{MS}_1/\partial f_1 = \alpha\, \mathrm{MS}_1(1 - \mathrm{MS}_1)/f_1$.
3. With preferred times uniform over an operating day of H hours and f equally spaced departures, expected schedule delay is K/f with $K = H/4$, and the frequency elasticity of market demand is $\eta_f = |\eta_T|\ (K/f)/T(f)$, where $T(f) = t_0 + t_b + K/f$.

Derivation Part 1. The logit formula of Appendix A.3 gives $\mathrm{MS}_a = e^{V_a}/\sum_b e^{V_b}$ with $V_a = \alpha \ln f_a + \gamma' x_a$, and $e^{\alpha \ln f_a} = f_a^\alpha$, so

$$\mathrm{MS}_a = \frac{f_a^\alpha\, e^{\gamma' x_a}}{\sum_b f_b^\alpha\, e^{\gamma' x_b}}.$$

When the x_b are equal, dividing above and below by $(\sum_b f_b)^\alpha$ gives (A.4); in the ratio of two shares the denominator cancels, leaving log odds linear in log frequency.

The same model says what α means. If each of carrier a's f_a flights were a separate, equally attractive alternative in a plain logit, the probability of choosing one of them would be $f_a e^{V_a}/\sum_b f_b e^{V_b}$: shares proportional to frequency, $\alpha = 1$. If a carrier's flights were close substitutes, grouped in a nest with parameter $\lambda < 1$ (Appendix A.3), α would be λ, below one. An exponent above one therefore says that each flight becomes more attractive when the carrier has more of them (more time windows covered, the option of changing flights, loyalty programs), effects that differ by market.

Part 2. In a duopoly write $x = \mathrm{FS}_1$ and $g(x) = x^\alpha/(x^\alpha + (1-x)^\alpha)$, so that $\ln[g/(1-g)] = \alpha \ln[x/(1-x)]$: the log odds of market share are α times those of frequency share. Differentiating both sides,

$$\frac{g'(x)}{g(1-g)} = \frac{\alpha}{x(1-x)},$$

which at $x = \frac{1}{2}$ gives $g' = \alpha$: at parity the S-curve multiplies the proportional effect by α. Since $\partial x/\partial f_1 = f_2/(f_1+f_2)^2 = x(1-x)/f_1$, the chain rule gives $\partial \mathrm{MS}_1/\partial f_1 = \alpha\, g(1-g)/f_1$, which is part 2.

The frequency contest of Chapter 11 follows. If two carriers split a market worth R a day by the S-curve and each daily frequency costs c, carrier 1 maximizes $R\,\mathrm{MS}_1 - cf_1$, with first-order condition $R\alpha\,\mathrm{MS}_1(1-\mathrm{MS}_1)/f_1 = c$. At a symmetric point $\mathrm{MS}_1 = \frac{1}{2}$, so each flies $f^* = \alpha R/(4c)$ and together they spend $2cf^* = \alpha R/2$ on frequency, three quarters of the market's revenue when $\alpha = 1.5$ (an equilibrium for $0 < \alpha < 2$).

Part 3. With departures at $d_j = (2j-1)H/(2f)$, each serves the travellers whose preferred time lies within $H/(2f)$ of it, and their distance to it is uniform on $[0, H/(2f)]$, with mean $H/(4f)$. Unequal spacing does worse: a catchment of length ℓ contributes at least $\ell^2/4$ to the integral of $|t^* - d|$ over the day, and $\sum_j \ell_j^2/4$ with $\sum_j \ell_j = H$ is smallest, at $H^2/(4f)$, when every $\ell_j = H/f$. Hence $T(f) = t_0 + t_b + K/f$ with $K = H/4$, and with $\partial T/\partial f = -K/f^2$ the chain rule gives

$$\eta_f = \frac{\partial D}{\partial f}\,\frac{f}{D} = \eta_T\,\frac{f}{T}\Big(-\frac{K}{f^2}\Big) = |\eta_T|\,\frac{K/f}{T(f)},$$

the time elasticity times the share of schedule delay in total trip time. Going from f to $f+1$ departures saves $K/(f(f+1))$ hours, so the effect of each added frequency fades roughly as $1/f^2$.

Check On Montreal–Toronto, Laurentian flies 8 daily flights against a competitor's 10, with $\alpha = 1.5$. Its frequency share is $8/18 = 0.444$ and its market share

$$\frac{0.444^{1.5}}{0.444^{1.5} + 0.556^{1.5}} = 41.7 \text{ percent},$$

against 44.4 percent under proportional shares. Against a competitor with three daily nonstops, one, two, three and four Laurentian flights win 16.1, 35.2, 50.0 and 60.6 percent: the second frequency adds 19.1 points where proportional shares would add 15.0. For market size, take $H = 16$ hours ($K = 4$), $t_0 = 2.5$ and $t_b = 1.25$ hours on YUL–YYZ, and $\eta_T = -1$. Then $T(1) = 7.75$ and $T(2) = 5.75$ hours, so a second daily flight raises market demand by $7.75/5.75 - 1 = 34.8$ percent. On YUL–CDG, with $t_0 = 4.0$ and $t_b = 6.9$ hours and an eastbound evening window of about six hours ($K = 1.5$), T falls only from 12.4 to 11.65 hours and a second flight adds 6.4 percent.

A.3 Logit choice probabilities, IIA and the logsum

Chapter 5 states the logit share formula, explains the independence of irrelevant alternatives (IIA) in words and uses the logsum to value a schedule change in dollars; the S-curve of Chapter 4 and the QSI recapture rates of Chapter 12 rest on the same formula. This section derives it, then IIA, the elasticities and the logsum, and states how nested logit relaxes IIA.

Setup Traveller n chooses from a choice set C_n (the itineraries on offer plus an outside option, not booking, whose utility is normalized to zero) the alternative with the highest utility

$$U_{nj} = V_{nj} + \varepsilon_{nj}, \qquad V_{nj} = \beta' x_{nj},$$

where x_{nj} holds the observed attributes (fare, elapsed time, departure time, connections) and ε_{nj} everything the analyst does not see. Logit assumes that the ε_{nj} are independent across alternatives with the standard **Gumbel** distribution, $F(\varepsilon) = \exp(-e^{-\varepsilon})$, whose mean is Euler's constant $\gamma_{\mathrm{E}} = 0.5772$ (McFadden 1974; Train 2009, ch. 3).

Result With independent Gumbel errors:

1. The choice probability is

$$P_{ni} = \frac{e^{V_{ni}}}{\sum_{j \in C_n} e^{V_{nj}}}. \tag{A.5}$$

2. (IIA) $P_{ni}/P_{nk} = e^{V_{ni} - V_{nk}}$ whatever the other alternatives. A new alternative m takes from each existing alternative j the amount $P^0_{nj} P^1_{nm}$, where superscripts 0 and 1 denote before and after.
3. The logsum $I_n = \ln \sum_j e^{V_{nj}}$ satisfies $\mathbb{E}[\max_j U_{nj}] = I_n + \gamma_{\mathrm{E}}$ and $\partial I_n / \partial V_{ni} = P_{ni}$.

Derivation Two properties of independent standard Gumbel variables do all the work. The first is **max-stability**: the maximum of $V_j + \varepsilon_j$ over j is again Gumbel, with location $I = \ln \sum_j e^{V_j}$. The maximum lies below u exactly when every term does, so

$$\Pr\Big(\max_j (V_j + \varepsilon_j) \le u\Big) = \prod_j \exp(-e^{-(u - V_j)}) = \exp(-e^{-(u - I)}).$$

The second is that the difference of two independent Gumbel variables is **logistic**. Conditioning on $\varepsilon_1 = s$ and substituting $w = e^{-s}$,

$$\begin{aligned} \Pr(\varepsilon_2 - \varepsilon_1 < x) &= \int_{-\infty}^{\infty} \exp(-e^{-s}(1 + e^{-x}))\, e^{-s}\, \mathrm{d}s \\ &= \int_0^{\infty} e^{-w(1+e^{-x})}\, \mathrm{d}w = \frac{1}{1 + e^{-x}}. \end{aligned}$$

For part 1, alternative i must beat the best of the others. By max-stability that composite rival has utility $I_{-i} + \varepsilon^*$, with $I_{-i} = \ln \sum_{j \neq i} e^{V_j}$ and ε^* standard Gumbel and independent of ε_i. By the logistic property,

$$P_i = \Pr(\varepsilon^* - \varepsilon_i < V_i - I_{-i}) = \frac{1}{1 + e^{I_{-i} - V_i}} = \frac{e^{V_i}}{e^{V_i} + \sum_{j \neq i} e^{V_j}}.$$

Every multinomial logit is thus a binary logit between one alternative and a composite of all the others.

For part 2, the common denominator cancels in the ratio. Adding alternative m raises the denominator from $S = \sum_j e^{V_j}$ to $S + e^{V_m}$, so every existing probability is multiplied by $S/(S + e^{V_m}) = 1 - P_m^1$, and alternative j loses $P_j^0 P_m^1$: the new alternative draws from each existing one in proportion to its share, whether or not it is a close substitute. Elasticities show the same pattern. Differentiating (A.5), an attribute x_k with coefficient β_k (for the fare, $\beta_f < 0$) has own elasticity $\beta_k \, x_{ik}(1 - P_i)$, and the cross elasticity of P_i with respect to an attribute of j is $-\beta_k \, x_{jk} P_j$, the same for every $i \neq j$.

For part 3, max-stability says that $\max_j U_j$ is Gumbel with location I, whose mean is $I + \gamma_{\mathrm{E}}$, and the derivative of I with respect to V_i is $e^{V_i} / \sum_j e^{V_j} = P_i$. If utility is linear in money with marginal utility $\alpha = -\beta_f$ (unrelated to the S-curve exponent), the change in a traveller's expected consumer surplus between situations 0 and 1 is

$$\Delta \mathbb{E}[\mathrm{CS}] = \frac{1}{\alpha}(I^1 - I^0).$$

A small fare cut Δf on alternative i raises V_i by $\alpha \, \Delta f$ and the logsum by $P_i \, \alpha \, \Delta f$, so surplus rises by $P_i \, \Delta f$: the probability of buying times the saving (Small and Rosen 1981). For larger changes, averaging the probabilities before and after gives the **rule of a half**, $\Delta \mathbb{E}[\mathrm{CS}] \approx \frac{1}{2} \sum_j (P_j^0 + P_j^1) \, \Delta V_j / \alpha$.

Nested logit relaxes IIA by grouping close substitutes into nests B_k with dissimilarity parameters $0 < \lambda_k \leq 1$. With the inclusive value $I_k = \ln \sum_{j \in B_k} e^{V_j / \lambda_k}$, the probability of alternative i in nest B_k factors into a logit among nests and a logit within the nest:

$$P_i = \frac{e^{\lambda_k I_k}}{\sum_l e^{\lambda_l I_l}} \cdot \frac{e^{V_i / \lambda_k}}{\sum_{j \in B_k} e^{V_j / \lambda_k}}.$$

Within a nest, utilities are divided by λ_k, so small differences produce large shifts; IIA holds inside a nest but not across nests, and $\lambda_k = 1$ for every nest returns plain logit. The formula comes from a generalized extreme value distribution in which the errors of two alternatives in the same nest have correlation $1 - \lambda_k^2$ (Train 2009, ch. 4); the full reference edition derives it.

Quality of service index (QSI) models are logit models too. If an itinerary's score multiplies factors $a_k > 0$ for its attributes, $\mathrm{QSI}_i = \prod_k a_k^{x_{ik}}$, the QSI share $\mathrm{QSI}_i / \sum_j \mathrm{QSI}_j$ is (A.5) with $V_i = \ln \mathrm{QSI}_i$. The recapture rate $\mathrm{QSI}_r / (\mathrm{QSI}_r + \mathrm{msc})$ of spill and fleet assignment models, with msc the competitors' total QSI share, is the logit probability of choosing r when the spilled passenger's choice set shrinks to r and the competitors (Barnhart 2003).

Check On YUL–CDG a representative traveller has utilities 0 for not booking, 0.40 for Laurentian's 18:30 nonstop, 0.25 for competitor B's 20:45 and -0.60 for carrier C's connection via London Heathrow. The exponentials 1, 1.492, 1.284 and 0.549 sum to 4.325, so the shares are 23.1, 34.5, 29.7 and 12.7 percent. Laurentian adds a 21:15 nonstop with utility 0.40: the sum rises to 5.816, every existing share is multiplied by $4.325/5.816 = 0.744$, Laurentian's total rises to 51.3 percent, and the new flight draws from the four existing alternatives in exactly the base proportions. Calibrated to the same base shares with $\lambda = 0.5$, nests by carrier cut Laurentian's gain from 16.8 to 8.2 points, and nests by time of day raise it to 21.1. With $\beta_f = -0.004$ per dollar and a CAD 760 fare, the 18:30's own-fare elasticity under logit is $-0.004 \times 760 \times 0.655 = -1.99$. For the logsum, $V = (1, 0.5, -0.2)$ gives $I = 1.646$, an expected maximum utility of $1.646 + 0.577 = 2.223$ and probabilities 0.524, 0.318 and 0.158. A same-day alternative with a QSI of 0.18, against competitors holding 42 percent of the market, recaptures $0.18/(0.18 + 0.42) = 30$ percent of spill.

A.4 The normal loss function and the spill formula

Chapter 7 states the expected spill formula $\mathbb{E}[S] = \sigma L(z)$ with its intuition and uses its derivative, the probability that a seat is used, to value capacity; Chapters 11 and 12 put the same formula inside gauge comparisons and fleet assignment costs. This section derives the formula, the identities of the loss function and the value of the marginal seat.

Setup Demand D for a departure is normal with mean μ and standard deviation σ, and the aircraft has C seats. The load is $Y = \min(D, C)$ and the spill is $S = [D - C]^+$. Capacity in standard units is $z = (C - \mu)/\sigma$: $z = 0$ means as many seats as mean demand, $z = 1$ one standard deviation of spare room.

Result With $L(z) = \varphi(z) - z(1 - \Phi(z))$, the standard normal loss function,

$$\begin{aligned} \mathbb{E}[S] &= \sigma\, L(z), \qquad \mathbb{E}[Y] = \mu - \sigma\, L(z), \\ \frac{\mathrm{d}\,\mathbb{E}[Y]}{\mathrm{dC}} &= \Pr(D > C). \end{aligned} \tag{A.6}$$

Derivation By definition $\mathbb{E}[S] = \int_C^\infty (x - C) f(x)\,\mathrm{d}x$, where f is the density of demand. Substitute $x = \mu + \sigma t$, so that $f(x)\,\mathrm{d}x = \varphi(t)\,\mathrm{d}t$ and $x - C = \sigma(t - z)$:

$$\mathbb{E}[S] = \sigma \int_z^\infty (t - z)\,\varphi(t)\,\mathrm{d}t = \sigma\Big[\int_z^\infty t\,\varphi(t)\,\mathrm{d}t - z(1 - \Phi(z))\Big].$$

Since $\varphi'(t) = -t\,\varphi(t)$, the remaining integral is $\big[-\varphi(t)\big]_z^\infty = \varphi(z)$, which gives (A.6). The load follows from $Y = D - S$.

The loss function $L(z) = \mathbb{E}\big[[Z - z]^+\big]$ is the expected amount by which a standard normal variable Z exceeds z (Belobaba 2006b). Three identities make it easy to use. Its slope is $L'(z) = -(1 - \Phi(z))$, since differentiating the formula gives $-z\varphi(z) - (1 - \Phi(z)) + z\varphi(z)$; with $L''(z) = \varphi(z) > 0$ it is decreasing and convex. Next, $L(-z) = L(z) + z$, because $[Z - z]^+ - [z - Z]^+ = Z - z$ and $-Z$ is also standard normal; so expected empty seats are $\sigma(L(z) + z)$, and load plus empty seats equals C. Finally, $L(0) = \varphi(0) = 0.3989$: an aircraft sized at mean demand spills on average 40 percent of a standard deviation, while far from $z = 0$, $L(z)$ approaches $\max(-z, 0)$, the answer that ignores variability (Figure A.1).

Differentiating $\mathbb{E}[Y] = \mu - \sigma L(z)$ with $\mathrm{d}z/\,\mathrm{d}C = 1/\sigma$ gives the marginal seat, $\mathrm{d}\mathbb{E}[Y]/\,\mathrm{d}C = -L'(z) = \Pr(D > C)$. This needs no normality: seat x is occupied exactly when demand exceeds x, so for any demand distribution

$$\mathbb{E}[\min(D, C)] = \int_0^C \Pr(D > x)\,\mathrm{d}x,$$

and expected load is concave in capacity, each extra seat being used less often than the one before.

Two consequences are used in the chapters. Dividing by C, with the demand factor $\mathrm{DF} = \mu/C$, gives $\sigma/C = k\,\mathrm{DF}$ and $z = (1 - \mathrm{DF})/(k\,\mathrm{DF})$, so the load factor and the spill rate $\mathbb{E}[S]/\mu = k\,L(z)$ depend only on the demand factor and the k-factor: one spill table serves every aircraft size (Boeing Commercial Airplane Company 1978). And if each passenger contributes r and each seat costs c, expected contribution $r\,\mathbb{E}[\min(D, C)] - c\,C$ has the falling derivative $r\Pr(D > C) - c$, so the best continuous capacity satisfies the newsvendor rule

$$\Pr(D > C^*) = \frac{c}{r}, \qquad C^* = \mu + \sigma\,\Phi^{-1}\Big(1 - \frac{c}{r}\Big).$$

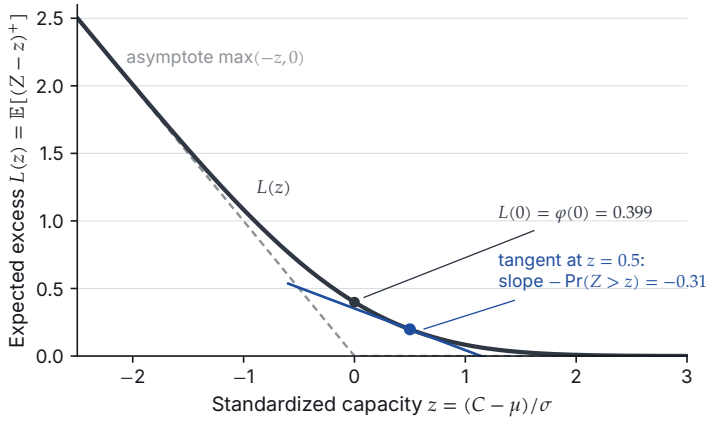


Figure A.1 The standard normal loss function. With capacity z standard deviations above mean demand, $L(z)$ is the expected spill in standard deviations. It approaches the deterministic answer $\max(-z, 0)$ when capacity is far from mean demand and departs from it most at $z = 0$; its slope at any z is minus the probability that demand exceeds capacity.

Table A.1 The standard normal density, tail probability and loss function. For negative z, read $L(-z)$ in the last column. In a spreadsheet, $L(z)$ is `NORM.S.DIST(z,FALSE)-z*(1-NORM.S.DIST(z,TRUE))`.

z	$\varphi(z)$	$1-\Phi(z)$	$L(z)$	$L(-z)$
0.00	0.3989	0.5000	0.3989	0.3989
0.25	0.3867	0.4013	0.2863	0.5363
0.50	0.3521	0.3085	0.1978	0.6978
0.75	0.3011	0.2266	0.1312	0.8812
1.00	0.2420	0.1587	0.0833	1.0833
1.25	0.1826	0.1056	0.0506	1.3006
1.50	0.1295	0.0668	0.0293	1.5293
2.00	0.0540	0.0228	0.0085	2.0085

Check The handbook's Exercise B has $\mu = 160$ and $\sigma = 40$ (Network Planning Handbook 2026). With 180 seats, $z = 0.5$ and $L(0.5) = 0.3521 - 0.5 \times 0.3085 = 0.1978$ (Table A.1), so $\mathbb{E}[S] = 40 \times 0.1978 = 7.91$, $\mathbb{E}[Y] = 152.09$ and the expected load factor is 84.5 percent; expected empty seats are $40 \times 0.6978 = 27.91$, and $152.09 + 27.91 = 180$. With 200 seats, $z = 1$, $L(1) = 0.0833$, $\mathbb{E}[S] = 3.33$ and $\mathbb{E}[Y] = 156.67$. The twenty extra seats carry $156.67 - 152.09 = 4.58$ passengers: the average of $\Pr(D > x)$ over seats 181 to 200 is 0.229, between the 0.31 of the first and the 0.16 of the last. For the newsvendor rule, moving from the A220-300 to the 737-8 on YUL–YVR costs CAD 5,700 for 32 more seats, CAD 178

a seat. With $r = 400$ the critical ratio is $1 - 178/400 = 0.555$ and $\Phi^{-1}(0.555) = 0.138$, so at $\mu = 150$ and $k = 0.30$ the ideal continuous capacity is $150 + 0.138 \times 45 \approx 156$ seats, nearer the 737-8, which is indeed the better type.

A.5 The fleet assignment model in symbols

Chapter 12 explains in words how fleet assignment chooses an aircraft type for every flight, so that aircraft balance at every station and no type is used more than it exists. This section writes the model in symbols, derives the count line that turns flows into aircraft, and shows why leg-based spill cannot be exact on a network.

Setup For each fleet type $k \in \mathcal{K}$, with N_k aircraft, the **time-space network** has one line per station. Each flight $i \in \mathcal{F}$ has a departure node at its departure time and an arrival node at its ready time (arrival plus the type's minimum turn), joined by a flight arc. Ground arcs join consecutive nodes on a station line, a wrap-around arc joins each station's last node to its first, and a **count line** at a chosen time crosses some ground and flight arcs.

The variables are $x_{ik} = 1$ if flight i is flown by type k (0 otherwise) and $y_g \geq 0$, the number of aircraft on ground arc g. At node v of type k's network, $I(v)$ and $O(v)$ are the flights arriving and departing, and y_{v^-} and y_{v^+} the ground arcs entering and leaving; $\mathcal{G}_k^0$ and $\mathcal{F}_k^0$ are the ground arcs and flights of type k that cross the count line. The fleet assignment model is (Hane et al. 1995; Barnhart 2003)

$$
\begin{aligned}
\min\ & \textstyle\sum_k \sum_i c_{ik}\, x_{ik} \\
\text{s.t. } & \textstyle\sum_k x_{ik} = 1, \quad i \in \mathcal{F}, \\
& y_{v^-} + \textstyle\sum_{i \in I(v)} x_{ik} = y_{v^+} + \sum_{i \in O(v)} x_{ik}, \quad v \in \mathcal{V}_k, \\
& \textstyle\sum_{g \in \mathcal{G}_k^0} y_g + \sum_{i \in \mathcal{F}_k^0} x_{ik} \leq N_k, \quad k \in \mathcal{K}, \\
& x_{ik} \in \{0,1\}, \quad y_g \geq 0.
\end{aligned}
\tag{A.7}
$$

The **cover** constraints fly every flight with exactly one type (pairs that are not allowed are left out); the **balance** constraints say that aircraft are neither created nor destroyed at any node; the **count** constraints limit the aircraft in use to those available. The cost of flying flight i with type k is its operating cost plus the revenue it loses to spill, net of recapture at rate b_i:

$$
c_{ik} = \mathrm{OC}_{ik} + (1 - b_i)\, r_i\, \sigma_i\, L\Big(\frac{C_k - \mu_i}{\sigma_i}\Big), \tag{A.8}
$$

where C_k is the seat count of type k, μ_i and σ_i the mean and standard deviation of the flight's demand, and r_i its contribution per passenger. Contribution is $r_i\mu_i - c_{ik}$; the first term does not depend on the type, so minimizing total c_{ik} is maximizing contribution.

Result If the flows of type k satisfy balance at every node, the number of type-k aircraft crossing a vertical line at time t (on ground arcs that span t plus on flight arcs whose departure-to-ready interval contains t) is the same for every t. The count constraint therefore bounds the aircraft actually in use, wherever the line is drawn.

Derivation Take two times $t < t'$ and add up the balance equations of all nodes with times in $(t, t']$. Every arc that crosses t but not t' ends at one of these nodes, and every arc that crosses t' but not t starts at one of them; arcs that start and end inside the window appear on both sides and cancel. Inflow equals outflow at every node, so the flow crossing t equals the flow crossing t'. With wrap-around arcs the same argument runs on the cyclic day.

The same bookkeeping says how many aircraft an assignment needs. If $A_s(t)$ and $D_s(t)$ count the ready arrivals and departures of type k at station s from the count time up to t, the aircraft on the ground there number $g_s + A_s(t) - D_s(t)$, which must never be negative. The smallest feasible g_s is the deepest deficit of departures over arrivals, $\max_t [D_s(t) - A_s(t)]$, so a balanced assignment of type k needs $\sum_s \max_t [D_s(t) - A_s(t)] + \left|\mathcal{F}_k^0\right|$ aircraft: those waiting at stations at the count time plus those in the air.

The model solves quickly because each type's balance constraints alone form a network flow problem, whose linear relaxation has integral solutions; only cover, count and side constraints can make it fractional, and in practice it is nearly integral (Hane et al. 1995; Barnhart 2006). The dual price of a binding count constraint values one more aircraft of that type to this timetable, within the range where the price holds.

The model's weakness is that it prices passengers leg by leg, while a connecting passenger needs a seat on every leg. Take Barnhart's legs A–B and B–C, with 80 local A–B passengers at USD 200, 90 local B–C passengers at USD 250 and 50 A–C connections at USD 400 (Barnhart 2006). Let $S(C_1, C_2)$ be the revenue spilled by the best allocation of passengers to C_1 and C_2 seats. With 80 seats on both legs, the best plan carries the locals and spills the 50 connections and 10 B–C locals, so $S(80, 80) = 22{,}500$. With 150 seats on both legs nothing is spilled. With 80 seats on A–B only, the plan carries the 50 connections and 30 A–B locals, so $S(80, 150) = 10{,}000$; with 80 seats on

B–C only, $S(150,80) = 15{,}000$. Costs that add up by leg would need $S(80,80)+S(150,150) = S(80,150)+S(150,80)$, but $22{,}500 \neq 25{,}000$. Itinerary-based fleet assignment, built on a passenger mix model that chooses which passengers to carry, removes this error at the price of a larger model (Barnhart, Kniker, and Lohatepanont 2002).

Check Laurentian's 09:00 YUL–YVR has mean demand 172 with $k = 0.30$, so $\sigma = 51.6$, and a prorated leg fare of CAD 455.65; net of CAD 20 of passenger cost, $r = 435.65$, and spill is assumed lost ($b = 0$). For the A220-300, $z = (137 - 172)/51.6 = -0.678$ and $L(z) = 0.8265$, so expected spill is 42.647 passengers and the spill cost CAD 18,579. With an operating cost of $8{,}000 \times 5.2 + 2{,}000 =$ CAD 43,600 and unconstrained revenue $435.65 \times 172 =$ CAD 74,932, the A220 contributes CAD 12,753. For the 737-8, $z = -0.058$, $L = 0.4287$, spill is 22.12 (CAD 9,637) and the operating cost CAD 49,300, so it contributes CAD 15,995: CAD 3,243 more, although mean demand almost fits the smaller aircraft. On the sixteen-flight bank through Montreal, the model built from such coefficients earns CAD 213,803 a day against CAD 211,972 for the planner's hand-built lines of flying; giving every flight its best type regardless of balance would earn CAD 216,985 but needs six narrowbodies instead of three. The count line at 01:00 crosses four aircraft (two A220s at Quebec City and Boston, the 737-8 at LaGuardia, the 787-9 in the air to Paris), exactly the fleet available; a third A220-300 would add CAD 1,301 a day to this timetable.

A.6 Delay propagation and slack

Chapter 13 explains how one late arrival travels down an aircraft's day, why delays grow towards the evening, and why the first minutes of slack are worth more than later ones. This section derives the propagation recursion, its closed form and the rule for how much slack to plan.

Setup Flight j follows flight i on the same aircraft. The planned turn is the planned departure time of j minus the planned arrival time of i, and the **slack** s_{ij} is the planned turn minus the minimum turn time (35 minutes for Laurentian's A220). If i arrives TAD_i minutes late, the **propagated delay** into j and j's total departure and arrival delays are

$$\begin{aligned} \mathrm{PD}_{ij} &= \max(\mathrm{TAD}_i - s_{ij},\ 0), \\ \mathrm{TDD}_j = \mathrm{PD}_{ij} + \mathrm{IDD}_j, &\qquad \mathrm{TAD}_j = \mathrm{PD}_{ij} + \mathrm{IAD}_j, \end{aligned} \tag{A.9}$$

where the **independent** departure and arrival delays IDD_j and IAD_j are the delays that j would have suffered with an on-time aircraft (Lan, Clarke, and Barnhart 2006; Barnhart and Clarke 2003). Crews and passengers obey the same construction, with the minimum sit or connecting time in place of the minimum turn.

Result Along a rotation and at a single turn:

1. Number one aircraft's flights $1, \dots, n$, let s_k be the slack after flight k, $a_k = \mathrm{IAD}_k - s_k$, and W_k the propagated delay into flight k, with $W_1 = 0$. Then $W_{k+1} = \max(W_k + a_k, 0)$ and

$$W_{k+1} = \max\Big\{0, \max_{1 \le h \le k} \sum_{m=h}^{k} a_m\Big\}.$$

2. For an inbound delay X and slack s, $\mathbb{E}\,[X - s]^+ = \int_s^\infty \Pr(X > x)\,\mathrm{d}x$, which is decreasing and convex in s with slope $-\Pr(X > s)$. If a minute of slack costs c_s and a minute of propagated delay $c_d > c_s$, the best slack satisfies $\Pr(X > s^*) = c_s / c_d$.

Derivation Flight k arrives $W_k + \mathrm{IAD}_k$ minutes late by (A.9), so the delay it passes to flight $k+1$ is $\max(W_k + \mathrm{IAD}_k - s_k, 0) = \max(W_k + a_k, 0)$. This is the recursion Lindley wrote for the waiting time in a single-server queue (Lindley 1952): the aircraft is the server and each flight a customer. For the closed form, use induction. $W_2 = \max(a_1, 0)$. If the formula holds for W_k, then

$$W_k + a_k = \max\Big\{a_k, \max_{h \le k-1} \sum_{m=h}^{k} a_m\Big\} = \max_{h \le k} \sum_{m=h}^{k} a_m,$$

and taking the maximum with zero gives the formula for W_{k+1}. The delay inherited by an evening flight is set by the worst stretch of the day before it, the largest excess of independent delay over slack on any run of consecutive flights; it rises with every upstream delay and falls with every upstream slack.

For part 2, write $[X - s]^+ = \int_s^\infty \mathbb{1}\{X > x\}\,\mathrm{d}x$ and take expectations inside the integral, which is allowed because the integrand is nonnegative. The derivative with respect to the lower limit, $-\Pr(X > s)$, rises towards zero as s grows, so the function is convex. The total cost $c_s\, s + c_d\, \mathbb{E}\,[X - s]^+$ has derivative $c_s - c_d \Pr(X > s)$, negative while $\Pr(X > s) > c_s / c_d$ and positive afterwards, which gives the rule: a newsvendor condition, like the capacity rule of Appendix A.4 and Littlewood's rule of Appendix A.7. The first minute of slack saves about $\Pr(X > 0)$ minutes of expected delay, the thirtieth only $\Pr(X > 30)$.

Arrival delays are skewed and can be negative; a shifted log-normal fits them well (Barnhart and Clarke 2003): $\ln(X-\theta) \sim \mathcal{N}(\ln m, \delta^2)$ with a shift $\theta < 0$, a median m of $X-\theta$ and a shape δ. For $s > \theta$ let $z_s = (\ln(s-\theta) - \ln m)/\delta$, so that $\Pr(X > s) = 1 - \Phi(z_s)$. Writing $X - \theta = m\,e^{\delta Z}$ with Z standard normal and completing the square, $e^{\delta z}\varphi(z) = e^{\delta^2/2}\varphi(z-\delta)$, so

$$\mathbb{E}\,[X-s]^+ = m\,e^{\delta^2/2}[1-\Phi(z_s-\delta)] - (s-\theta)[1-\Phi(z_s)].$$

Setting $1-\Phi(z_s) = c_s/c_d$ and solving for s gives the best slack,

$$s^* = \theta + m\,\exp\!\Big(\delta\,\Phi^{-1}(1-c_s/c_d)\Big).$$

Check Line B of the Laurentian A220 day has eight turns with slacks of 5, 5, 5, 5, 10, 5, 10 and 0 minutes. If its first flight arrives 60 minutes late and nothing else goes wrong, $a_1 = 60 - 5$ and every later a_k is minus its slack, and the recursion gives propagated delays of 55, 50, 45, 40, 30, 25, 15 and 15 minutes, 275 in all. With 10 more minutes on every turn they are 45, 30, 15 and then zero: 90 minutes. For LaGuardia inbounds with $\theta = -15$, $m = 14$ and $\delta = 0.9$, the formula gives expected propagated delay of 9.6 minutes with no slack, 7.6 with 5 minutes, 6.1 with 10 and 2.9 with 30. At CAD 20 per minute of slack and CAD 80 per minute of delay the critical ratio is 0.25, so

$$s^* = -15 + 14\,e^{0.9\times 0.674} = 10.7 \text{ minutes},$$

and the expected cost of the turn is CAD 689 a day, against CAD 771 with no slack and CAD 830 with 30 minutes.

A.7 Littlewood's rule and EMSR-b

Chapter 14 states Littlewood's rule (keep a seat for the higher fare while that fare, times the probability of selling it, exceeds the lower fare) and its extension to many fare classes, and Chapter 15 reads the resulting booking limits. This section derives the rule, its normal form and the EMSR-b heuristic.

Setup A cabin of C seats sells two fares, $f_1 > f_2$. Low-fare demand books first, as advance-purchase rules encourage, and high-fare demand D_1 is random and independent of low-fare demand D_2. The airline sets a **protection level** y for class 1: class 2 may buy at most $C - y$ seats, its **booking limit**. With n classes, the protection level y_j reserves seats for classes 1 to j together, $y_1 \le \ldots \le y_{n-1}$, and the nested booking limits are $b_1 = C$ and $b_{j+1} = C - y_j$.

Result The optimal protection level for class 1 is

$$y^* = \max\{y : f_1 \Pr(D_1 \geq y) \geq f_2\}, \tag{A.10}$$

and for normal demand $y^* = \mu_1 + \sigma_1 \Phi^{-1}(1 - f_2/f_1)$. It depends neither on class 2 demand nor on capacity, as long as $y^* \leq C$ (Littlewood 1972).

Derivation Let $R(y)$ be expected revenue with protection level y. Raising y by one changes the outcome only on departures where class 2 would have used the extra seat, that is, when $D_2 \geq C - y$. There the airline loses f_2 on one low-fare sale and gains f_1 if high-fare demand reaches $y + 1$. By independence,

$$R(y+1) - R(y) = \Pr(D_2 \geq C - y)\,[f_1 \Pr(D_1 \geq y + 1) - f_2].$$

The first factor is non-negative and does not affect the sign, which is why low-fare demand and capacity drop out. The bracket decreases in y, so R rises while it is positive and falls afterwards, and the last seat worth protecting is (A.10). For continuous demand the condition holds with equality, $\Pr(D_1 > y^*) = f_2/f_1$, and for normal demand $\Pr(D_1 > y) = 1 - \Phi((y - \mu_1)/\sigma_1)$ inverts to the normal form.

The rule is a newsvendor condition: protecting one seat too few costs $f_1 - f_2$, protecting one too many costs f_2, and balancing the two gives $\Pr(D_1 \leq y^*) = (f_1 - f_2)/f_1$. When the low fare is more than half the high fare, $\Phi^{-1}(1 - f_2/f_1) < 0$ and the airline protects fewer seats than mean high-fare demand; when it is less than half, more.

With more classes, **EMSR-b** applies the rule at every cut (Belobaba 1989, 2006l). To set y_j it merges classes 1 to j into one artificial class and protects it against class $j + 1$:

$$y_j = \mu_{1:j} + \sigma_{1:j}\,\Phi^{-1}\Big(1 - \frac{f_{j+1}}{\bar{f}_{1:j}}\Big), \tag{A.11}$$

where the merged class has, for independent demands, mean, standard deviation and demand-weighted fare

$$\mu_{1:j} = \sum_{k \leq j} \mu_k, \qquad \sigma_{1:j} = \Big(\sum_{k \leq j} \sigma_k^2\Big)^{1/2}, \qquad \bar{f}_{1:j} = \frac{1}{\mu_{1:j}} \sum_{k \leq j} f_k\, \mu_k.$$

For two classes this is Littlewood's rule. For more it is a heuristic, because it values the y-th protected seat at $\bar{f}_{1:j} \Pr(D_1 + \cdots + D_j \geq y)$. The exact value comes from a dynamic program: the **marginal value of a seat**, the expected revenue lost if one remaining seat is taken away, equals $f_1 \Pr(D_1 \geq y)$ for the first cut, falls as more seats remain, and

makes the optimal policy a set of nested protection levels (Brumelle and McGill 1993; Talluri and van Ryzin 2004b). That marginal value is the single-leg bid price.

Check On YUL–SVQ the economy cabin has 168 seats. Class Y sells at CAD 1,249 with demand $\mathcal{N}(12, 5^2)$ and class B at CAD 919. The fare ratio is $919/1{,}249 = 0.736$ and $\Phi^{-1}(1 - 0.736) = -0.630$, so

$$y^* = 12 - 0.630 \times 5 = 8.85.$$

In whole seats, with the continuity correction $\Pr(D_Y \geq y) \approx 1 - \Phi((y - 0.5 - 12)/5)$, the ninth protected seat earns $1{,}249 \times \Pr(D_Y \geq 9) = 947$ in expectation, more than the B fare, and the tenth 864, less: Laurentian protects 9 seats, below mean Y demand because B pays 74 percent of Y, and B's booking limit is $168 - 9 = 159$. At the last cut, classes Y, B, M and Q have mean demand $12 + 18 + 28 + 44 = 102$, standard deviation $\sqrt{25 + 49 + 121 + 289} = 22.0$ and weighted fare $75{,}538/102 = 740.6$. Against the V fare of CAD 419 the ratio is 0.5658, $\Phi^{-1}(1 - 0.5658) = -0.1656$, and

$$y_4 = 102 - 0.1656 \times 22.0 = 98.36.$$

The four cuts protect 9, 26, 54 and 98 seats, so the nested booking limits are 168, 159, 142, 114 and 70; the exact dynamic program gives the same protection levels on this flight.

A.8 The cost-based overbooking limit

Chapter 14 explains why airlines accept more reservations than seats and how far they go: accept another booking while the chance that the cabin is already full stays below a ratio of two costs. This section derives that rule from the binomial show-up model and shows what a service target implies about the cost of a denied boarding.

Setup A cabin has C seats. If b bookings are held at departure and each passenger shows up independently with probability q, the number of show-ups $N(b)$ is binomial, $\mathrm{Bin}(b, q)$, with mean bq and variance $bq(1 - q)$. A **spoiled seat**, an empty seat on a flight that closed for sale, costs c_{SP}: the contribution of the passenger who could have filled it, which on a flight controlled by revenue management is a low fare. A **denied boarding** costs c_{DB}: compensation, care, rebooking and goodwill. The expected cost of authorizing b bookings is

$$K(b) = c_{\mathrm{DB}}\, \mathbb{E}[[N(b) - C]^+] + c_{\mathrm{SP}}\, \mathbb{E}[[C - N(b)]^+]. \qquad \text{(A.12)}$$

Result One more authorized booking changes the expected cost by

$$K(b+1) - K(b) = q\Big[(c_{\mathrm{DB}} + c_{\mathrm{SP}})\Pr(N(b) \geq C) - c_{\mathrm{SP}}\Big].$$

K is convex in b, and the cost-minimizing authorization level AU^* is the smallest b with

$$\Pr(N(b) \geq C) \;\geq\; \rho = \frac{c_{\mathrm{SP}}}{c_{\mathrm{SP}} + c_{\mathrm{DB}}}. \tag{A.13}$$

Derivation Couple the two situations: $N(b+1) = N(b) + X$, where X is the show-up indicator of the extra booking, equal to 1 with probability q and independent of $N(b)$. If $X = 0$ nothing changes. If $X = 1$, the extra passenger either fills an empty seat, when $N(b) < C$, which saves c_{SP}, or is denied boarding, when $N(b) \geq C$, which costs c_{DB}. Hence

$$K(b+1) - K(b) = q\Big[c_{\mathrm{DB}}\Pr(N(b) \geq C) - c_{\mathrm{SP}}\Pr(N(b) < C)\Big],$$

which rearranges to the result. The coupling also shows that $N(b)$ grows stochastically with b, so the increment rises with b: K is discretely convex, and its minimum is at the first b whose increment is non-negative, which is (A.13) (Belobaba 2006m). The show-up probability of the extra booking cancels, because that passenger matters only by showing up. Maximizing contribution, $r\,\mathbb{E}[\min(N, C)] - c_{\mathrm{DB}}\mathbb{E}\big[[N - C]^+\big]$, is the same problem with $c_{\mathrm{SP}} = r$, since $\min(N, C) = C - [C - N]^+$. Random booking demand does not move the limit: if bookings are $\min(D, b)$, the increment is multiplied by $\Pr(D > b)$, which does not change its sign.

For large b, the normal approximation with a continuity correction, $\Pr(N(b) \geq C) \approx 1 - \Phi\big((C - \tfrac{1}{2} - bq)/\sqrt{bq(1-q)}\big)$, turns the rule into a formula. Setting the standardized value equal to $z = \Phi^{-1}(1-\rho)$ and replacing $bq(1-q)$ by its value $C(1-q)$ near $b = C/q$ gives

$$\mathrm{AU}^* \approx \frac{C - \tfrac{1}{2} - z\sqrt{C(1-q)}}{q}\,: \tag{A.14}$$

start from the deterministic limit C/q and pull back by z standard deviations of the show-ups. When a denied boarding costs more than a spoiled seat, $\rho < \tfrac{1}{2}$, $z > 0$ and the optimal limit lies below C/q. If the show-up rate itself varies across departures with variance σ_q^2, the variance of show-ups becomes $bq(1-q) + b(b-1)\sigma_q^2$, which grows with b^2, and the limit falls.

A service target, such as at most a 5 percent chance of any denied boarding on a departure, is a cost rule in disguise. With $p_b = \Pr(N(b) \geq C)$, the limit b is optimal exactly when $p_{b-1} < \rho \leq p_b$. Substituting $\rho = c_{\mathrm{SP}}/(c_{\mathrm{SP}} + c_{\mathrm{DB}})$ and solving each inequality for c_{DB},

$$c_{\mathrm{SP}} \frac{1 - p_b}{p_b} \leq c_{\mathrm{DB}} < c_{\mathrm{SP}} \frac{1 - p_{b-1}}{p_{b-1}},$$

the range of denied-boarding costs that a given limit implicitly assumes.

Check Treat the 182 seats of the A321XLR on YUL–SVQ as one cabin, with $q = 0.93$, $c_{\mathrm{SP}} = \text{CAD } 550$ and $c_{\mathrm{DB}} = \text{CAD } 1{,}650$, so $\rho = 550/2{,}200 = 0.25$. With 192 bookings $\Pr(N \geq 182) = 0.206$, below 0.25, so the 193rd booking is accepted; with 193 it is 0.295, so the 194th is not. The optimal authorization level is 193, an overbooking factor of 1.06; (A.14) gives 192.6, hence the same 193, while the deterministic limit $182/0.93 = 195.7$ would accept two or three more. At 193 bookings the flight boards 179.0 passengers on average, denies boarding to 0.46 and spoils 2.97 seats, an expected cost of CAD 2,391 against CAD 7,007 without overbooking. The limit is optimal for any denied-boarding cost between CAD 1,314 and CAD 2,117. A 5 percent service rule allows 190 bookings ($p_{189} = 0.042$, $p_{190} = 0.079$), as if a denied boarding cost CAD 6,396 to 12,423. If the show-up rate varies by 2.5 percentage points across departures, the cost-based limit falls to 191.

A.9 Bid prices as shadow prices

Chapter 14 states the bid-price rule (accept a request when its fare covers the sum of the bid prices of the legs it uses), and Chapters 10 and 15 use bid prices and displacement costs to value capacity and connections. This section derives bid prices as the dual prices of a network linear program and explains why a bid price values a few seats but not an aircraft.

Setup Legs $l \in \mathcal{L}$ have capacities C_l. An origin-destination fare product $j \in \mathcal{J}$, an itinerary sold at a fare, has fare f_j and mean demand μ_j, and $a_{lj} = 1$ if product j uses leg l. Replacing random demand by its mean, the **deterministic linear program** (DLP) allocates x_j seats to each product (Talluri and van Ryzin 2004b; Belobaba 2006n):

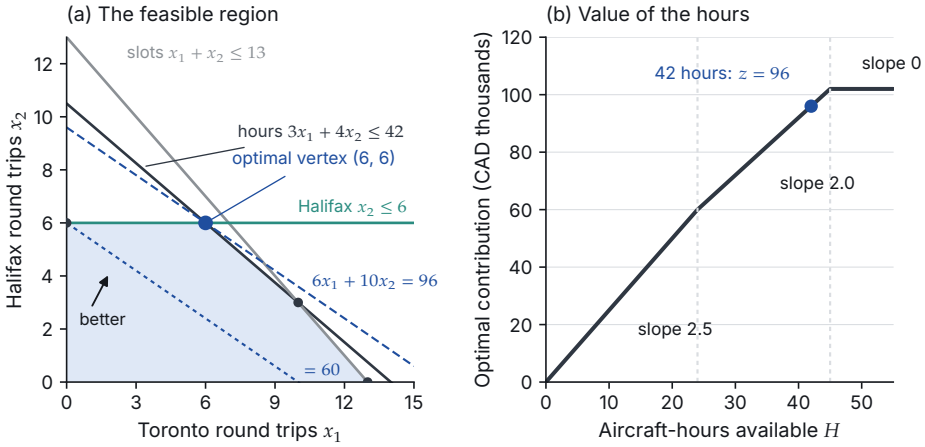


Figure A.2 Shadow prices on a small linear program: 42 spare A220-300 aircraft-hours a day shared between Montreal–Toronto round trips (3 hours, CAD 6,000) and Montreal–Halifax round trips (4 hours, CAD 10,000), with 13 slots and Halifax demand for at most 6 trips (illustrative). (a) The optimum flies six of each, for CAD 96,000 a day. (b) The optimal contribution against the hours available is concave and piecewise linear; its slopes, 2.5, 2.0 and 0 thousand CAD per hour, are the shadow price of an hour over successive ranges.

$$\begin{aligned} V(C) = \max_{x} \ & \textstyle\sum_j f_j\, x_j \\ \text{s.t.} \ & \textstyle\sum_j a_{lj}\, x_j \le C_l, \quad l \in \mathcal{L}, \\ & 0 \le x_j \le \mu_j, \quad j \in \mathcal{J}. \end{aligned} \tag{A.15}$$

Result Let $\pi_l \ge 0$ be optimal dual prices of the capacity constraints. At an optimal allocation:

1. if $f_j > \sum_l a_{lj}\pi_l$, product j is fully accepted, $x_j = \mu_j$;
2. if $f_j < \sum_l a_{lj}\pi_l$, it is rejected, $x_j = 0$;
3. if $\pi_l > 0$, leg l is full.

Moreover, $V(C') \le V(C) + \sum_l \pi_l\,(C'_l - C_l)$ for every capacity vector C'.

Derivation Attach a price $\pi_l \ge 0$ to each capacity constraint and a rent $\beta_j \ge 0$ to each demand bound. The Lagrangian

$$\sum_j f_j x_j + \sum_l \pi_l\Big(C_l - \sum_j a_{lj}x_j\Big) + \sum_j \beta_j(\mu_j - x_j)$$

is bounded above over $x \ge 0$ only if the coefficient of every x_j is non-positive, $f_j - \sum_l a_{lj}\pi_l - \beta_j \le 0$, and its supremum is then $\sum_l C_l\pi_l + \sum_j \mu_j\beta_j$, an upper bound on $V(C)$. The best such bound is the dual:

$$\begin{aligned} \min_{\pi,\beta} \ & \textstyle\sum_l C_l \pi_l + \sum_j \mu_j \beta_j \\ \text{s.t.} \ & \textstyle\sum_l a_{lj} \pi_l + \beta_j \geq f_j, \quad j \in \mathcal{J}, \qquad \pi, \beta \geq 0. \end{aligned} \tag{A.16}$$

The dual splits network revenue into a price for every seat and a rent for every unit of demand, each fare being covered by the prices of its seats plus its rent; by strong duality the two optimal values are equal (Bertsimas and Tsitsiklis 1997).

At optimal solutions of the two programs, **complementary slackness** holds:

$$x_j\Big(\sum_l a_{lj}\pi_l + \beta_j - f_j\Big) = 0, \qquad \beta_j(\mu_j - x_j) = 0,$$
$$\pi_l\Big(C_l - \sum_j a_{lj}x_j\Big) = 0.$$

If $f_j > \sum_l a_{lj}\pi_l$, dual feasibility forces $\beta_j > 0$, and the second condition gives $x_j = \mu_j$: statement 1. If $x_j > 0$, the first condition gives $f_j = \sum_l a_{lj}\pi_l + \beta_j \geq \sum_l a_{lj}\pi_l$, so a fare below its path bid price is rejected: statement 2. Statement 3 is the third condition. Products whose fare equals their path bid price are **marginal**: the LP may accept part of their demand, and the bid price alone does not say how much, one reason bid prices must be recomputed often.

Displacement cost gives the same rule. The network value of product j on leg l is its fare minus the displacement costs of its other legs, and accepting it when that value exceeds the leg's own price is accepting it when its fare covers the path:

$$f_j - \sum_{l' \neq l} a_{l'j}\,\pi_{l'} > \pi_l \iff f_j > \sum_l a_{lj}\,\pi_l.$$

For the marginal value of capacity, note that the dual's constraints do not involve C. By strong duality $V(C')$ is the minimum over that fixed set of $\sum_l C'_l\pi'_l + \sum_j \mu_j\beta'_j$, a minimum of affine functions of C', and therefore concave and piecewise linear in capacity. Evaluating the minimum at the particular (π, β) that is optimal for C gives

$$V(C') \leq \sum_l C'_l\pi_l + \sum_j \mu_j\beta_j = V(C) + \sum_l \pi_l\,(C'_l - C_l).$$

The bid price is the slope of V at C when it is unique, and multiplying it by a large change in capacity can only overstate the value of seats added and understate the loss from seats removed; Figure A.2 shows the shape on a small planning problem. Finally, the DLP bounds what any booking control can earn: whatever a policy sells, X_j, satisfies the capacity constraints and $0 \leq X_j \leq D_j$ on every realization,

so $\mathbb{E}[X]$ is feasible for (A.15) and expected revenue cannot exceed $V(C)$.

Check On the two-leg network of Appendix A.5 with 100 seats per leg, the DLP carries all 80 A–B locals, 80 B–C locals and 20 connections, for USD 44,000. B–C locals and connections are carried below demand, so their rents are zero and their fares equal their path prices: $\pi_{\mathrm{BC}} = 250$ and $\pi_{\mathrm{AB}} = 400 - 250 = 150$. The dual value $100 \times 150 + 100 \times 250 + 80 \times (200 - 150) = 44{,}000$ equals the revenue. Laurentian's bank of four legs and sixteen products earns CAD 365,949 in the DLP, with bid prices of CAD 159 on YQB–YUL, 109 on BOS–YUL, 490 on YUL–CDG and 0 on YUL–SVQ. They can be read off the marginal products: a Quebec City seat is worth the local restricted fare of 159; a Paris seat is worth the YQB–CDG restricted connection's 649 minus its feeder seat, 490; and a Boston seat is worth the BOS–CDG restricted connection's 599 minus its Paris seat, 109. For an upgauge of YQB–YUL from 137 to 169 seats, the naive value $159 \times 32 =$ CAD 5,088 per bank compares with CAD 459 from re-solving the DLP and CAD 1,257 under random demand: the bid price values the first extra seat, not thirty-two.

APPENDIX

B

Glossary

This glossary defines the terms and acronyms used in this edition and in a network planning department. The chapter in brackets develops each idea; where a term has a common trap, the entry names it.

A

A15 The share of flights arriving within 15 minutes of their scheduled time; A14, the share arriving no more than 14 minutes late, is essentially the same measure under another name. Flight punctuality is not passenger punctuality: a late feeder can strand passengers whose onward flight arrives on time (Chapter 13).

ACMI Aircraft, crew, maintenance and insurance supplied by one airline to another for a fee per block hour (a wet lease); a dry lease supplies the aircraft only.

Advance purchase A fare rule requiring the ticket to be bought a minimum number of days before departure: the classic fence between leisure travellers, who plan early, and business travellers, who often cannot (Chapter 14).

Aircraft routing The sequence of flight legs assigned to an individual aircraft (a tail), chosen so that every aircraft reaches a maintenance station at the required intervals (Chapter 12).

Alliance A multilateral cooperation among airlines (Star Alliance, oneworld, SkyTeam) covering codeshares, loyalty programs, lounges and connections. It does not by itself permit joint pricing, which requires a joint venture with antitrust approval (Chapter 8).

Ancillary revenue Revenue from products sold alongside the fare, such as bags, seats and upgrades. Check whether a revenue figure already includes it before adding it to a route case.

ASK, ASM Available seat kilometre and available seat mile: one seat, occupied or not, flown one kilometre or one statute mile; the basic unit of capacity. Never mix kilometres and miles in one ratio; Air Canada reports in miles (Chapter 2).

Authorization level (AU) The number of reservations an airline will accept on a flight or cabin, set above physical capacity to allow for no-shows (Chapter 14 and Appendix A.8).

Avoidable cost A cost that disappears if the decision under study is not taken, over the horizon of that decision; the relevant cost for incremental route and schedule decisions (Chapter 3).

B

Bank A cluster of arrivals at a hub followed by a cluster of departures, so that arriving passengers can connect to many departing flights, at the price of peaks in gate, runway and staff use (Chapter 11).

Bid price The opportunity value of one seat on a flight leg, used as a threshold: a request is accepted if its fare is at least the sum of the bid prices of the legs it uses. It is a marginal value, valid for a few seats, not a fare or a cost (Chapters 14 and 15 and Appendix A.9).

Block time, block hour The time from the aircraft leaving its departure gate to arriving at its destination gate; the main driver of crew, maintenance and fuel costs (Chapter 3).

Booking class An inventory bucket identified by a letter to which fare products are mapped. Revenue management opens and closes booking classes, not individual fares (Chapter 14).

Booking curve Cumulative bookings for a future flight plotted against days before departure; the basis of pickup forecasting (Chapter 6).

Booking limit The maximum number of seats that may be sold in a booking class (and, under nesting, in all lower classes); the capacity minus the protection level of the classes above (Chapter 14).

Branded fare A fare family sold under a name (basic, standard, flexible), each brand bundling attributes such as bags, seat choice and change rules (Chapter 14).

Break-even load factor The load factor at which revenue equals cost, approximately unit cost divided by yield. It depends on the cost concept used and ignores the fact that yield falls as load factor rises (Chapter 2).

Buffer Time added to a scheduled block or turn beyond what is needed on average, to absorb delays. Buffers improve punctuality and reduce aircraft utilization (Chapter 13).

Buy-down A passenger buying a cheaper fare product than the one he or she would have bought if the cheaper one had been unavailable (Chapter 14).

C

CASK, CASM Operating cost per available seat kilometre or mile. It falls with stage length and aircraft size, so comparisons need a stage-length adjustment (Chapter 3 and Appendix A.1).

Codeshare An airline (the marketing carrier) selling seats under its own flight number on a flight operated by another (the operating carrier). Remove duplicate marketing records when counting flights.

Connection window The range of acceptable connecting times at a hub, from the minimum connecting time to a maximum beyond which the connection is unattractive or not offered (Chapter 11).

CORSIA ICAO's Carbon Offsetting and Reduction Scheme for International Aviation, under which airlines offset growth in international emissions above a baseline (Chapter 16).

Crew pairing A sequence of duty periods, starting and ending at the same crew base, that covers a set of flights within the legal and contractual rules (Chapter 12).

Curfew A period, usually at night, during which an airport restricts or prohibits operations (Chapter 11).

D

DB1B, DB1C The US Department of Transportation's origin and destination survey: DB1B, a quarterly ten percent ticket sample until mid-2025, and its successor DB1C, a monthly forty percent sample. The series cannot be spliced by rescaling (Network Planning Handbook 2026).

Deadhead A crew member travelling as a passenger to position for a later duty, at a cost in money and seats (Chapter 12).

Decision window The range of departure and arrival times a traveller considers acceptable, used in the Boeing path preference model and the PODS simulator to decide which itineraries enter a traveller's choice set (Chapter 5).

Demand factor Unconstrained demand divided by capacity. Unlike the load factor, it can exceed one (Chapter 7).

Denied boarding Refusing to carry a passenger holding a confirmed reservation because more passengers showed up than there are seats; voluntary if the passenger accepts compensation to give up the seat (Chapter 14).

Displacement cost The revenue lost on other passengers when a seat is used for a given booking. On a full leg a connection displaces a local, so it is worth its fare minus the displacement costs of its legs (Chapters 10 and 14).

Duty period A crew member's working day, from report to release, containing one or more flights and limited by fatigue rules (Chapter 12).

E

EDTO, ETOPS Extended diversion time operations (the ICAO term) and extended-range twin-engine operations: rules for routes beyond a threshold time from a diversion airport, requiring approval of the type, operator and operation (Chapter 9).

EMSR-b Expected marginal seat revenue, version b: the standard rule for setting nested protection levels for several fare classes on one flight, which applies Littlewood's rule at every cut with the higher classes merged into one (Chapter 14 and Appendix A.7).

Endogeneity Correlation between an explanatory variable and the unobserved part of demand. Fares are endogenous because revenue management raises them when demand is high, which biases naive estimates of price elasticity toward zero (Chapter 5).

EU ETS The European Union Emissions Trading System, under which airlines surrender allowances for covered flights; check its scope for flights outside Europe each planning year (Chapter 16).

F

Fare basis The code identifying a fare and its rules on a ticket. Several fare bases can share one booking class.

Fare product A fare level together with its rules (advance purchase, refundability, changes, bags) for a market. In network revenue management the unit of control is the origin-destination fare product (Chapter 14).

Fence A restriction that stops customers with a high willingness to pay from buying a low fare, such as advance purchase or nonrefundability. Fences make price discrimination possible (Chapter 14).

Fleet assignment The assignment of an aircraft type to each flight of a schedule so as to maximize contribution, with revenue lost to spill counted as a cost, subject to aircraft availability and flow balance (Chapter 12 and Appendix A.5).

Freedoms of the air The rights a country may grant to foreign airlines. The third and fourth allow carriage to and from the airline's home country; the fifth, between two foreign countries on a service starting or ending at home; the sixth, between two foreign countries through the home hub (Chapter 8).

Frequency share An airline's share of the departures in a market. Market share rises more than proportionally with it, the S-curve effect (Chapter 4 and Appendix A.2).

Fully allocated cost A cost per flight that includes a share of every company cost, including overheads that would not change if the flight were cancelled; misleading for incremental decisions (Chapters 3 and 10).

G

Gauge The size of the aircraft on a route, in seats. Upgauging replaces an aircraft with a larger one; frequency against gauge is the central schedule design choice (Chapter 11).

GDS A global distribution system (Amadeus, Sabre, Travelport) through which travel agents book flights; the main source of MIDT data, not the whole market.

Go-show A passenger who arrives for a flight without a reservation and is accommodated.

H

Hub premium The higher average fare an airline earns in markets touching its hub, from loyalty, schedule dominance, airport constraints and a business-rich passenger mix (Chapter 8).

Hub-and-spoke network A network in which most routes join outlying airports (spokes) to a few hubs, where passengers between spokes connect. With n spokes, one hub connects about $n^2/2$ markets with n routes (Chapter 8).

I

IATA season The two scheduling seasons used worldwide: the northern summer season, starting on the last Sunday of March, and the northern winter season, starting on the last Sunday of October (Chapter 11).

IIA Independence of irrelevant alternatives: in a logit model, the ratio of the probabilities of two alternatives does not depend on the others, so a new itinerary draws share from every existing one in proportion, which is unrealistic for near-identical itineraries (Chapter 5 and Appendix A.3).

Interline An agreement that lets an airline sell one ticket including another airline's flights, with baggage checked through and revenue prorated (Chapter 8).

Itinerary The sequence of flight legs a passenger uses to travel in one direction from origin to destination.

J

Joint venture A partnership in which airlines share revenue or profit on a set of markets whichever partner operates, usually with antitrust immunity to coordinate prices and capacity (Chapter 8).

K

k-factor The coefficient of variation of demand, $k = \sigma / \mu$, typically between 0.20 and 0.40 for flight demand (Chapter 7).

L

Leg A single nonstop operation of an aircraft between two airports, also called a flight leg or segment. Capacity is sold by leg; demand is generated by origin and destination.

Littlewood's rule Protect a seat for the higher of two fares while that fare times the probability of selling it exceeds the lower fare (Chapter 14 and Appendix A.7).

Load factor Revenue passenger kilometres divided by available seat kilometres. For a system it must be computed as a ratio of sums, not an average of ratios (Chapter 2).

Logit The discrete choice model in which the probability of choosing alternative i is $e^{V_i} / \sum_j e^{V_j}$, derived from independent extreme value errors (Chapter 5 and Appendix A.3).

Logsum The expected maximum utility of a choice set, $\ln \sum_j e^{V_j}$ up to a constant. Divided by the marginal utility of money, it measures consumer surplus (Chapter 5).

Loss function The normal loss function $L(z)$, the expected amount by which a standard normal variable exceeds z; expected spill is $\sigma L(z)$ (Chapter 7 and Appendix A.4).

M

MCT Minimum connecting time: the shortest interval between an arrival and a departure for which a connection may be sold, set by airport, terminal and type of connection (Chapter 11).

MIDT Marketing Information Data Tapes: booking records collected from global distribution systems. Coverage of direct and new distribution channels is incomplete, so market totals built from MIDT are estimates (Network Planning Handbook 2026).

MTOW, MZFW, MLW, OEW Maximum take-off weight, maximum zero-fuel weight, maximum landing weight and operating empty

weight: the structural weights that, with fuel capacity, define an aircraft's payload-range diagram (Chapter 9).

N

NDC New Distribution Capability: an IATA standard that lets airlines send offers, including ancillaries and dynamic prices, directly to sellers rather than through filed fares in a GDS (Chapter 16).

Nested logit A choice model that groups close substitutes into nests, so that alternatives in the same nest compete more with each other than with alternatives elsewhere (Chapter 5).

Nesting The inventory rule that a seat available to a lower booking class is also available to every higher class (Chapter 14).

Network contribution The change in the whole network's revenue minus avoidable cost caused by a decision, including effects on other flights through feed, displacement and recapture (Chapter 10).

No-show A passenger holding a reservation who does not turn up for the flight (Chapter 14).

O

O&D Origin and destination: the true start and end points of a passenger's journey, as opposed to the legs it uses. Demand is generated by O&D markets; capacity is supplied by legs.

ODF An origin-destination fare product: an itinerary combined with a fare class, the unit of control in network revenue management (Chapter 14).

Overbooking Accepting more reservations than seats to compensate for no-shows, balancing the cost of empty seats against the cost of denied boardings (Chapter 14 and Appendix A.8).

P

Payload-range diagram A chart of the maximum payload an aircraft can carry as a function of range, bounded by the maximum zero-fuel weight, the maximum take-off weight and the fuel capacity (Chapter 9).

PDEW Passengers per day each way: a market's annual passengers divided by 365 and by two directions. Check whether a source has already split directions before dividing by two (Network Planning Handbook 2026).

PODS The Passenger Origin-Destination Simulator, a competitive network simulation used by MIT and a consortium of airlines to test revenue management methods (Chapters 5 and 14).

POS Point of sale: the country or channel where a ticket was sold, not necessarily where the traveller lives or where the journey starts.

PRASK, PRASM Passenger revenue per available seat kilometre or mile, equal to yield times load factor (Chapter 2).

Propagated delay Delay passed from one flight to the next flight of the same aircraft, crew or passengers when the slack between them is too small to absorb it (Chapter 13 and Appendix A.6).

Prorate The allocation of a ticket's revenue among the legs or carriers of its itinerary, for example in proportion to distance. A prorated leg revenue is an accounting allocation, not a marginal value (Chapter 10).

Protection level The number of seats reserved for a booking class and the classes above it, which lower classes may not use (Chapter 14).

Q

QSI A share model in which each itinerary's score reflects its stops, departure time, elapsed time and aircraft, and shares are proportional to scores; a logit model in disguise (Chapters 4 and 5).

R

RASK, RASM Total operating revenue, including cargo and other revenue, per available seat kilometre or mile.

Recapture Spilled demand that the airline retains on another of its own flights or itineraries (Chapter 7).

ReFuelEU Aviation The EU regulation requiring fuel suppliers at EU airports to blend rising shares of sustainable aviation fuel, and airlines to refuel there (Chapter 16).

Revenue management Controlling which fare products are available on each flight over the booking period to maximize revenue from fixed, perishable capacity (Chapter 14).

Rostering Assigning pairings, reserve days and days off to individual crew members, usually monthly (Chapter 12).

Rotation A sequence of flights that returns an aircraft to its starting point, such as YUL–SVQ–YUL. Rotation time, turns included, sets how many aircraft a schedule needs (Chapter 11).

RPK, RPM Revenue passenger kilometre and mile: one paying passenger carried one kilometre or one statute mile; the basic unit of traffic.

S

S-curve The more-than-proportional relationship between frequency share and market share in a competitive market (Chapter 4 and Appendix A.2).

SAF Sustainable aviation fuel, from biomass, waste or synthetic sources, blended with conventional jet fuel at several times its cost (Chapter 16).

Schedule delay The gap between the time a traveller would like to depart and the time an available flight departs. It falls as frequency rises, which gives frequency value beyond seats (Chapter 11).

Sell-up A passenger buying a more expensive fare product when the cheaper one he or she wanted is unavailable (Chapter 14).

Sixth-freedom traffic Passengers between two foreign countries through an airline's home hub, such as Boston to Paris through Montreal on a Canadian airline (Chapter 8).

Slack The time between when an aircraft, crew or passenger is ready for the next flight and that flight's scheduled departure. Positive slack absorbs delay (Chapter 13).

Slot A permission to land or take off on a specific date and at a specific time at a coordinated airport, allocated twice a year under the Worldwide Airport Slot Guidelines (Chapter 11).

Spill Unconstrained demand above capacity, turned away during the sale; a denied boarding, by contrast, happens at the gate (Chapter 7).

SSIM IATA's Standard Schedules Information Manual, the industry standard for exchanging schedule data (Chapter 11).

Stage length The distance flown per departure. Unit costs and yields fall as it rises (Chapter 3).

Stimulation New traffic created by a better product, typically a new nonstop service, as opposed to traffic diverted from existing itineraries (Chapter 4).

T

T-100 The US Department of Transportation's monthly operated-traffic data. The market table reports passengers between boarding and deplaning points on one flight number, not necessarily their full journey (Network Planning Handbook 2026).

Time-space network A network whose nodes are events (an arrival or departure at an airport at a time) and whose arcs are flights and ground waits: the standard picture for fleet assignment and aircraft routing (Chapter 12).

Turn time The ground time between an aircraft's arrival and its next departure. The minimum turn time depends on the aircraft, the airport and the type of flight (Chapter 11).

U

Unconstraining Estimating true demand from booking data that capacity and booking limits have capped. Loads of full flights are lower bounds, not measurements (Chapter 7).

Utilization Average block hours flown per aircraft per day. Higher utilization spreads ownership and other fixed costs over more output (Chapter 3).

V

Value of time The money a traveller would pay to save one hour of travel time, the ratio of the time coefficient to the fare coefficient of a choice model (Chapter 5).

W

WASG The Worldwide Airport Slot Guidelines of ACI, IATA and the Worldwide Airport Coordinators Group, the rules for slot allocation at coordinated airports (Chapter 11).

Willingness to pay The highest price a traveller would pay for a given itinerary and fare product; its distribution across travellers limits what price discrimination can extract (Chapter 14).

Y

Yield Passenger revenue per revenue passenger kilometre or mile. It falls with distance and with load factor, so compare it at like-for-like stage lengths (Chapter 2).

APPENDIX

C

Sources and Reading List

This book was written from the Network Planning Reading Library, a collection of course material, industry documents and company filings assembled on 27 October 2026, supplemented by current public sources. This appendix lists the library documents and the chapters that use them, then the books and papers to read next.

C.1 The reading library

C.1.1 MIT 16.75J Airline Management, Spring 2006

A graduate course at the Massachusetts Institute of Technology taught by Peter P. Belobaba, with lectures by Cynthia Barnhart, published on MIT OpenCourseWare under a Creative Commons BY-NC-SA licence (Belobaba 2006c). Some decks also appear in the library as separate files under descriptive names.

Table C.1 Lectures and readings of MIT 16.75J used in this book.

Lecture	Title	Used in
1	Introduction to the airline planning process	Chapters 1 and 8
2	Airline economics review	Chapters 2, 4, 8 and 14
3	Operating costs and productivity measures	Chapter 3
4a	Boeing, Load factor analysis (1978)	Chapter 7
4b	Airline demand analysis and spill modelling	Chapter 7
5	Airline schedule development	Chapter 11
6a	Introduction to the PODS passenger choice model	Chapters 4 and 5
6b	Boeing, Decision window path preference methodology (1993)	Chapter 5

Lecture	Title	Used in
7	The ePODS airline management game	Chapters 5 and 14
9	Airline fleet assignment (Barnhart)	Chapter 12
11	Route planning and evaluation	Chapter 10
13a	Airline operations control	Chapter 13
16	Airline fleet planning models	Chapter 9
18, 19, 20	Flight leg revenue management; overbooking; network origin-destination control	Chapters 14 and 15
23, 24, 26	Pricing challenges; O&D control lessons; willingness to pay and competitive revenue management	Chapter 14
Reading	Boeing, Demand driven dispatch	Chapter 12

C.1.2 MIT 1.206J Airline Schedule Planning, Spring 2003

A graduate course taught by Cynthia Barnhart on the optimization models behind schedule planning, also on MIT OpenCourseWare (Barnhart 2003). It is the source for the fleet assignment, crew and robustness material, and its problem sets supply several of the small instances of Chapter 12.

Table C.2 Lectures and assignments of MIT 1.206J used in this book.

Lecture	Title	Used in
1	Schedule planning and airline networks	Chapters 1, 8 and 12
2–4	Multicommodity flows and their solution	Appendix A.5
5–9	Passenger mix, fleet assignment, crew scheduling, maintenance routing, crew pairing and aircraft routing	Chapter 12
10	Schedule design	Chapter 11
11–14	Airline operations, recovery and robustness (Barnhart and Clarke)	Chapter 13
Problem sets 1–3	Express network design, fleet assignment, crew pairing	Chapter 12

C.1.3 Discrete choice

Kenneth Train's Discrete Choice Methods with Simulation, second edition (2009), chapters 2, 3, 4, 6 and 13: the standard graduate text on the models of Chapter 5. The book paraphrases its results; it does not reproduce its text.

C.1.4 Industry, regulatory and company documents

Table C.3 Industry, regulatory and company documents in the library.

Document	Date	Used in
Network planning preparation handbook	27 October 2026	throughout
Air Canada, second-quarter 2026 MD&A	11 August 2026	Chapters 2, 3 and 16
Competition Bureau, Cleared for take-off, executive summary	2025	Chapters 8 and 16
IATA, ACI and WWACG, Worldwide Airport Slot Guidelines, edition 4	effective 1 August 2025	Chapter 11
EUROCONTROL, Arriving on time, Think Paper 6	January 2020	Chapter 13
Airbus, A321 aircraft characteristics, revision 33	1 June 2024	Chapters 9 and 11
Lufthansa Systems, Air Canada and NetLine/Ops ++	21 January 2025	Chapters 13 and 16

The handbook is a thirty-page preparation guide for a new network planning analyst: vocabulary, unit discipline, route economics, data sources and their traps, a worked Montreal–Seville case and a ninety-day plan. Its Air Canada facts are cited to public sources, and this book reproduces its exercises with attribution.

C.1.5 Current public sources

Facts about the industry in 2026 (fleet programs, network announcements, regulation, technology and market structure) come from public sources consulted on 27 October 2026, cited where they are used, mostly in Chapter 16. Public facts change quickly; recheck any of them before relying on it at work.

C.2 Reading the older material

The 2003 and 2006 courses predate branded fares, the New Distribution Capability, continuous pricing, the A321XLR and the current Canadian passenger protection rules. Their methods are not dated: the spill model, the S-curve, Littlewood's rule, fleet assignment, crew pairing and bid prices are what planning and revenue management systems still implement. Their numbers are dated, and this book labels them historical. Treat a lecture's rule of thumb (a spill rate above 5 percent is too high) as an order of magnitude to test against current data.

C.3 Where to go next

The full reference edition This Network Planning edition explains mechanisms and keeps mathematics to the few formulas a planner computes; Appendix A derives those. The full reference edition of Perishable Capacity, in 24 chapters, contains what was left out: the complete derivations and proofs, the estimation methods (maximum likelihood choice models, instruments and endogeneity, forecasting models, unconstraining), and the optimization models written out in full (fleet assignment and its itinerary-based extension, crew pairing by column generation, the dynamic programs of seat inventory control and network revenue management), with their worked examples.

Textbooks Belobaba, Odoni and Barnhart's The Global Airline Industry (2015), by the authors of the two MIT courses, is the closest single book to this one. Doganis (2019) gives the commercial and strategic view. For revenue management, Talluri and van Ryzin (2004b) remains the reference for the theory, and Gallego and Topaloglu (2019) covers pricing and choice-based methods. For operations research across the planning chain, see Barnhart and Smith (2012) and Bazargan (2010). For demand modelling, read Train (2009) and Garrow (2010); for forecasting, Hyndman and Athanasopoulos (2021).

Papers by topic

- Spill and unconstraining: Swan (2002); Weatherford and Pölt (2002); Abramovich (2013) on spill under revenue management.
- Itinerary choice: Coldren and coauthors (2003); Coldren and Koppelman (2005).
- Airline demand and competition: Berry, Carnall and Spiller (1996); Berry, Levinsohn and Pakes (1995).
- Schedule design and fleet assignment: Hane and coauthors (1995); Barnhart, Kniker and Lohatepanont (2002); Lohatepanont and Barnhart (2004); Barnhart and Cohn (2004) for a survey.
- Crew and routing: Vance and coauthors (1997); Gopalan and Talluri (1998); Cohn and Barnhart (2003).
- Robustness and recovery: Lan, Clarke and Barnhart (2006); Clausen and coauthors (2010).
- Revenue management: Littlewood (1972); Belobaba (1987, 1989); Brumelle and McGill (1993); Talluri and van Ryzin (1998, 2004a); Gallego and van Ryzin (1994); Cooper, Homem-de-Mello and Kleywegt (2006); Fiig and coauthors (2010); McGill and van Ryzin (1999) for a survey.

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Index

Symbols

180 seats, 115
200 seats, 116
80 percent test, 202

A

additive pickup, 103
adjusted CASM, 25
Air against no air, 82
aircraft and crew, 205
aircraft balance, 212
aircraft movement, 20
aircraft productivity, 44
aircraft rotation planning, 3, 192
aircraft time, 199
aircraft utilization, 44
airline model, 99
airline O&D, 97
airport slot, 201
airport totals, 312
allocated, 166
alternative, 10
ancillaries, 251
ARIMA models, 99
assignment cost, 211
assumption register, 315
authorization level, 258
available seat-kilometres, 23
available seat-miles (ASM), 23
average load factor, 117
average passenger trip length, 27
average stage length, 27
avoidable, 11
avoidable cost, 46

B

balance, 338
bank, 15, 137, 194
bank peak, 205
barriers and recommendations, 143
behind, 135
behind the gateway, 134
between two spokes, 133
beyond, 135
bias, 101
bid price, 124, 168, 264, 274
Bid price and DLP, 278
bid price times seats, 276
bidline, 223
bid-price control, 264
binding resources, 10
block hours, 37
Blue Sky policy, 142
booking class, 250
booking curve, 103
booking limit, 253, 342
booking profile, 119
booking simulations, 276
bottom-up, 98, 149
branded fares, 251
break errors down, 102
break-even, 29, 108
break-even load factor, 48
break-even rent, 159
breguet range equation, 154
business, 58
buy down, 124, 281
buy-down, 267

C

cabotage, 142
calendar time, 37
capacity, 112
capacity cost, 38, 328
capacity gap, 150
capital and strategic case, 166
censored, 113
censored estimation, 119
check the intervals, 102
choice probability, 75
choice-based, 269
circuity, 131
coefficient, 75
coherent scenarios, 181
committed, 46
company, 16
competition, 8
competitor-change note, 324
complementary slackness, 348
concave, 115
concentration, 143
connection window, 194
connections, 204
connections displace locals, 263
constrained demand, 284

continuous pricing, 269, 301
contribution, 280
contribution value, 274
controls, 277
coordinator, 201
CORSIA, 297, 299
cost driver, 37
Cost per ASK (CASK), 25
cost per passenger, 42
count, 338
count line, 198, 209, 338
count passes, 219
counterfactual, 10
counting markets, 138
coupon, 20
cover, 212, 338
coverage state, 85
credit, 221
crew, 7, 225
crew scheduling, 4
curve, 156

D

data, 100, 104, 105
data contract, 314
dates, 199
deadheads, 220
decision, 278
decision threshold, 321
decision window, 85
decompose, 5
delay, 239
Delta-T, 85
demand, 190
demand and spill, 204
demand capture, 182
demand driven dispatch, 213
demand factor, 117
denied boarding, 258, 344
departures, 37
deterministic linear program, 274, 346
dichotomy of demand and supply, 56
difficult 2026, 17
direction, 179
displacement cost, 170, 275
disrupted, 239
dissimilarity parameter, 82
distance, 37
distance and winds, 156
diverted, 70
DLP re-solved, 276
double marginalization, 140
duty 1, 222
duty 2, 222
duty period, 220
dynamic bundling, 301
dynamic pricing, 301
dynamic regression, 99

E

economies of density, 132
economies of scope, 132
eighth, 142
elastic, 58
EM algorithm, 119
EMSR-b, 255, 343
entry, 84
entry and fares, 143
equivalent still-air distance, 156
estimating feed, 138
EU ETS, 297, 299
ex post bid price, 275
ex post optimum, 276
expected schedule delay, 187
exponential smoothing, 99
extended diversion time operations, 158
extra seats, 116

F

facilitator, 201
fan chart, 108
fare adjustment, 269
fare product, 250
fare response, 84
feasible, 9
feasible connection, 194
fence, 249
fenced, 281
five-hour weeks and winter, 199
fleet, 7
fleet assignment, 3, 192
fleet assignment problem, 211
fleet assignment with time windows, 203
fleet commonality, 159
fleet count, 213
fleet plan, 148
fleet planning, 3
fleet transition, 17
flexible fare, 60
flight, 20
flight contribution, 166
flight leg, 3, 20, 129
flow, 135
forecasts, 104
frequency and demand, 160
frequency elasticity, 160

frequency planning, 3, 192
frequency share, 65, 330
fuel-limited segment, 154
fully allocated cost, 46
functional, 37

G
gateway, 129
generalized cost, 63, 78
grain, 310
granularity, 273
gravity model, 105
ground arcs, 209
Gumbel, 333

H
historic precedence, 201
historical, 103
Historics Baseline Date, 201
holdout, 100
Holt–Winters, 99
horizon, 285
hub premium, 144
hub-and-spoke network, 129
hubs, 129
hubs and network, 16

I
IATA scheduling season, 200
increment, 10
incremental, 9, 166, 192
incremental cost, 46
incremental network contribution, 11, 166
incremental schedule design, 203
independence of irrelevant alternatives, 81
independent, 230, 341
indivisibility, 8
inelastic, 58
information, 273
inputs, 160
instruments, 93
involuntary, 260
itinerary, 3, 20, 56
itinerary and fare product, 97
itinerary-based fleet assignment model, 216

J
journey, 10, 11, 20

K
k-factor, 112

L
latent class, 83
Laurentian's assumptions, 177
layover, 220
leg, 97
leg arithmetic, 277
leg demand, 56
leg view, 280
leg/class control, 263
leisure, 58
level 1, 201
level 2, 201
level 3, 201
lines as built fail, 219
lines of flying, 217
load, 112, 125
load consolidation, 130
load factor, 23
local, 135
logistic, 333
longer route, 157
long-range narrowbody, 295
loss function, 114

M
maintenance, 225
marginal, 348
marginal value of a seat, 343
market, 97
market characteristics, 57
market elasticity, 79
market estimates, 311
market share, 65
master flight list, 203
maximum landing weight, 154
maximum take-off weight, 154
maximum zero-fuel weight, 154
max-stability, 333
mean absolute error, 101
mean absolute percentage error, 101
mean absolute scaled error, 101
mean error, 101
median, 192
metal neutrality, 140
mileage prorate, 168
minimum connecting time, 136, 194
missions, 156
mixed logit, 83
model finds it, 219
model's plan, 214

money value, 78
Montreal to Seville, 17
multiplicative pickup, 103

N

naive, 99
natural experiments, 93
nested logit, 82
nests, 82
nests by carrier, 82
nests by time of day, 82
network, 277, 286
network at mean demand, 279
network contribution, 170
network displacement cost, 169
network effects, 8, 215
network value, 168
New Distribution Capability (NDC), 301
new entrants, 201
new entrants and the calendar, 201
ninth freedoms, 142
normally distributed, 112
no-shows, 258

O

objective, 37, 273
O&D and leg, 284
one 787-9, 191
one input at a time, 181
one market, one season, 161
only differences in utility matter, 75
operating empty weight, 154
operations control, 4
opportunity cost, 46
origin and destination survey, 312
origin-destination control, 263
origin-destination (O&D) market, 20, 56, 129
outside option, 75
overbooking, 257
overbooking factor, 258

P

pairing, 220, 222
Paris, 63
passenger mix model, 216
Passenger revenue per ASK (PRASK), 25
passengers, 37, 225
passengers rank the routings differently, 235
path quality index, 85
pattern, 180
payload, 154
peak, 176
per round trip, 178
perishable, 7
perishable seats, 7
pickup, 103
planner's plan, 214
point-to-point network, 129
pool, 201
post-launch review, 4, 324
prediction, 106
prediction interval, 107
price, 57
price discrimination, 248
price elasticity, 58
price of real aircraft, 214
pricing, 3
pricing and revenue management, 4
product, 58
product differentiation, 248
program, 161
propagated delay, 230, 340
propagated delay barely moves, 235
prorate, 168
protection level, 253, 342

Q

Q-forecasting, 268
quality of service index, 89
quality of service index (QSI), 68

R

ramp-up, 107
random demand and revenue management, 279
reading, 77
ready time, 209
real schedule, 177
recapture, 212, 215
recapture rate, 122, 211
reconciliation, 98
ReFuelEU, 297, 299
ReFuelEU Aviation, 297
regression, 103
regression with calendar effects, 99
regulation and access, 8
release, 220
reliability, 204
repair, 225
report, 220
representative utility, 75
restricted fare, 60

results, 100
retiming, 234
returns to scale, 132
revenue management, 3
revenue passenger-kilometres, 23
revenue passenger-miles (RPM), 23
Revenue per ASK (RASK), 25
revenue value, 274
reversal frontier, 321
RM, 7
RM rejections, 284
robust, 229
robust routing, 234
rolling, 196
rolling origins, 102
rolling-origin evaluation, 100
rostering, 223
rotation, 197
route evaluation, 3
route planning, 3
routing, 7
rule of a half, 334

S

saturation frequency, 188
scenarios, 107, 108
schedule card, 192
schedule delay, 61, 187, 330
schedule design, 203
schedule development, 3, 192
schedule tolerance, 85
schedules, 311
scores, 71
S-curve, 65, 189
season, 180
seasonal naive, 99
seat cost, 42
sell-up, 267
sensitivity, 71, 162
serially nested, 253
series of slots, 201
shadow price, 172, 293
share and stimulation, 71
share model, 88
short connect, 223
short haul blocks long haul, 263
shoulder, 176
show-up rate, 258
simulated controls, 278
sits, 220
sixth-freedom traffic, 135
slack, 230, 340
sources, 71
spill, 112, 284
spill factor, 117
spill model, 276, 278
spill rate, 117
spill tables, 117
spiral-down, 285
spiral-down effect, 267
spoiled seat, 258, 344
spoke, 129
SSIM, 200
stage-length adjustment, 27
staging problem, 148
step 2: locate the inconsistency, 31
step 3: decompose growth, 32
step 4: margin and break-even, 32
step 5: regions, 32
stimulated, 70
stochastic delay, 187
straight-rate prorate, 168
strings, 218
structural plateau, 154
summer time, 199
sustainable aviation fuel (SAF), 297
swap points decide, 219

T

their value, 116
thin flow, 102
threshold, 182
ticket, 20
time, 57
time away from base, 220
time-space network, 209, 338
timetable, 6
timetable development, 3, 192
top-down, 98, 149
Toronto, 63
total cost, 87
total trip time, 61
track bias separately, 102
trap, 106
trip cost, 38, 42
Two A220s, 191

U

uncertainty, 8, 100
unconstrain, 117, 118
unconstrained demand, 112, 284
unit cost falls with distance, 27
unit measures, 29
United transborder agreement, 17
upgauge, 119
US government data, 312
use it or lose it, 201

utilization, 198

V

value of integration, 5
value of time, 78
visiting friends and relatives (VFR), 58
voluntary, 260

W

weight-limited segment, 154
where the difference comes from, 161
willingness to pay, 267
winter, 176
without fences, 281
wrap-around arc, 209

Y

yield, 25
yield falls with distance, 27
YUL–SVQ, 15

Z

zero-fuel weight, 154